Configuration Summary

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DHW ANALYSIS - CONFIGURATION SUMMARY
Project: oisst_regression
Generated: 2026-10-14 04:07:39
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-- Data Preparation --

  input_dir.............................. /g/data/ia39/aus-ref-clim-data-nci/oisst/data/yearly/
  file_pattern........................... oisst-avhrr-v02r01_{year}.nc
  year_range............................. (1981, 2024)
  variable............................... sst
  detrend_method......................... global_linear
  standardise_method..................... global
  fill_method............................ diffusion
  fill_diffusion_iterations.............. 500
  fill_fallback_value.................... 0
  downscale.............................. True
  target_lat_size........................ 256
  target_lon_size........................ 512
  temporal_average....................... none
  subset_region.......................... False
  lat_bounds............................. None
  lon_bounds............................. None
  diagnostic_year........................ 2020
  diagnostic_pdf......................... diagnostics/data_preparation_diagnostics_256x512.pdf
  output_dir............................. data/processed_256x512/
  save_intermediates..................... True
  downscaled_file........................ sst_downscaled.nc
  temporally_averaged_file............... sst_temporally_averaged.nc
  detrended_file......................... sst_detrended.nc
  standardised_file...................... sst_standardised.nc
  filled_file............................ sst_filled.nc
  final_file............................. sst_final.nc
  subset_file............................ sst_subset.nc
  trend_params_file...................... trend_params.nc
  standardise_params_file................ std_params.nc
  resume_after_stage..................... 0
  chunks................................. {'time': 366, 'lat': 720, 'lon': 720}
  dask_scheduler......................... threads
  dask_n_workers......................... 8
  dask_threads_per_worker................ 2
  dask_memory_limit...................... 12GB
  dask_num_threads....................... 16

-- Model Training --

  model_name............................. sfno
  input_history.......................... 10
  output_channels........................ 1
  embed_dim.............................. 70
  num_layers............................. 5
  num_modes_lat.......................... 48
  num_modes_lon.......................... 96
  mlp_ratio.............................. 2.5
  activation............................. gelu
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norm_type.............................. instance_norm
dropout_rate........................... 0.2
use_residual........................... True
spectral_bias.......................... True
add_grid_features...................... True
add_seasonal_encoding.................. False
forecast_horizon....................... 90
initial_rollout_steps.................. 1
max_rollout_steps...................... 5
rollout_epoch_start.................... 50
rollout_epoch_end...................... 90
rollout_loss_decay..................... 0.95
train_years............................ (1982, 2015)
val_years.............................. (2016, 2019)
test_years............................. (2020, 2024)
learning_rate.......................... 0.0002
min_learning_rate...................... 1e-06
batch_size............................. 10
num_epochs............................. 120
weight_decay........................... 1e-05
optimizer.............................. adamw
scheduler.............................. cosine
scheduler_patience..................... 10
scheduler_factor....................... 0.5
warmup_epochs.......................... 5
gradient_clip_norm..................... 1
loss_function.......................... huber
early_stopping_patience................ 20
early_stopping_metric.................. val_loss
input_data_path........................ data/processed_256x512/sst_final.nc
checkpoint_dir......................... checkpoints_256x512/
best_model_path........................ checkpoints_256x512/best_model.pt
training_log_path...................... results_256x512/training_log.csv
random_seed............................ 42
num_workers............................ 4
distributed............................ True
dist_backend........................... nccl
find_unused_parameters................. False
sync_batchnorm......................... False
master_port............................ 29500

-- Model Inference --

model_path............................. checkpoints_256x512/best_model.pt
input_data_path........................ data/processed_256x512/sst_final.nc
run_validation......................... True
run_test............................... True
forecast_lead_steps.................... 90
chain_stride_steps..................... 90
output_dir............................. results_256x512/
val_predictions_file................... predictions_val.nc
test_predictions_file.................. predictions_test.nc

-- Concat Chains --
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  predictions_dir..... /scratch/um09/xl5758/DHW-project/results_256x512
  predictions_file_template..... predictions_{split}.nc
  std_params_path..... data/processed_256x512/std_params.nc
  trend_params_path..... data/processed_256x512/trend_params.nc
  output_dir..... /scratch/um09/xl5758/DHW-project/results_256x512
  actual_file_template..... sst_actual_{split}_continuous.nc
  predicted_file_template..... sst_predicted_{split}_chains.nc
  splits..... ['test', 'val']
  lon_correction_threshold..... 0.6

-- Calc DHW --

  mmm_path..... /home/272/xl5758/ct5km_climatology_v3.1.nc
  results_dir..... /scratch/um09/xl5758/DHW-project/results_256x512
  predicted_file_template..... sst_predicted_{split}_chains.nc
  actual_file_template..... sst_actual_{split}_continuous.nc
  dhw_output_dir..... /scratch/um09/xl5758/DHW-project/dhw_output_256x512
  output_file_template..... dhw_products_{split}.nc
  run_test..... True
  run_validation..... True
  window_days..... 84
  hotspot_threshold..... 1

-- DHW Plots --

  dhw_dir..... /scratch/um09/xl5758/DHW-project/dhw_output_256x512
  output_pdf..... dhw_analysis_combined.pdf
  output_csv..... dhw_metrics_combined.csv
  spatial_lead_steps..... [1, 7, 14, 30, 60, 90]
  n_sample_chains..... 6
  plot_init_dates..... ['2016-05-20', '2016-10-15', '2017-02-10', '2017-07-09', '20
  dhw_vmax..... 20
  dhw_diff_vmax..... 8
  tropic_lat..... 30
  dhw_cmap..... ['#CCe6F0', '#441C94', '#6F47BF', '#815CCC', '#A383E6', '#ED
  error_cmap..... ['#053061', '#2166AC', '#92C5DE', '#F7F7F7', '#F4A582', '#D6
  rmse_cmap..... YlOrRd
  bias_cmap..... RdBu_r
  bleaching_thresholds..... [(0, 4, 'No Stress', '#27AE60'), (4, 8, 'Warning', '#F39C12'
  gbr_lat_bounds..... (-25.0, -10.0)
  gbr_lon_bounds..... (142.0, 157.0)
  reef_boxes..... [{'name': 'Great Barrier Reef', 'lat': (-25.0, -10.0), 'lon'
```

DHW Analysis - Summary

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DEGREE HEATING WEEKS (DHW) - COMBINED ANALYSIS
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METHODOLOGY  (NOAA Coral Reef Watch v3.1)
  SSTA      = SST - DC              (DC = daily climatology)
  HotSpot   = max(SST - MMM, 0)     (MMM = Maximum Monthly Mean)
  DHW_i     = (1/7) x sum(HotSpot_j) for j=i-83..i, HotSpot_j >= 1.0C

SLIDING WINDOW CONSTRUCTION  (always 84 days ending on valid date)
  Lead L:  (84-L) actual + L predicted SST days
  Lead 1:  83 actual +  1 predicted
  Lead 30: 54 actual + 30 predicted
  Lead 84:  0 actual + 84 predicted
  Lead 90: previous 84 predicted days (days 7-90)

VERIFICATION
  Actual DHW  : 84 observed days ending on the forecast valid date
  Persistence : observed DHW on the day before lead 1, held constant
  Sample      : chains defined at every lead step (same chains for
                model and persistence at all leads)

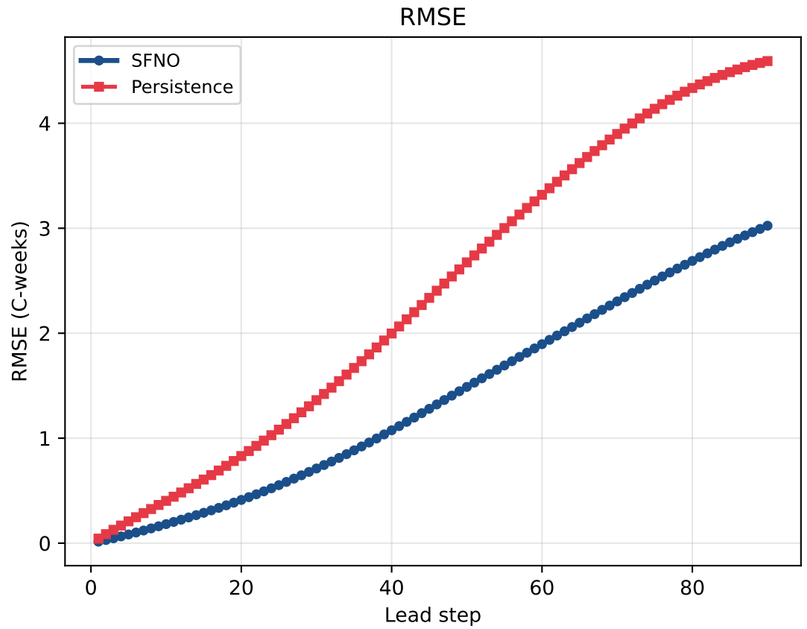
BLEACHING ALERT THRESHOLDS
  DHW  0-4  : No Stress
  DHW  4-8  : Warning
  DHW  8-12 : Alert Level 1
  DHW 12-16 : Alert Level 2
  DHW 16-20 : Alert Level 3
  DHW 20->= : Alert Level 4-5

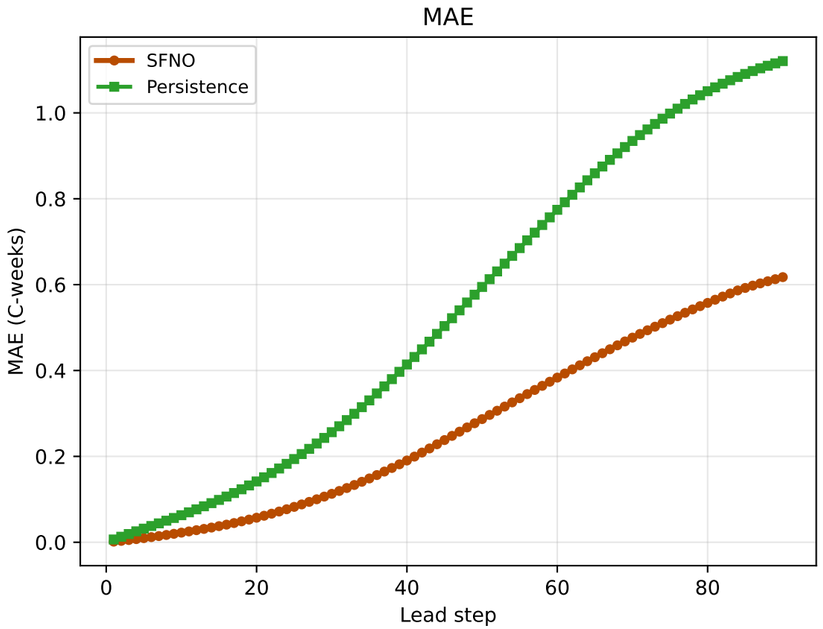
DATASETS
  Test          19 chains x 90 leads x 256 lat x 512 lon
  Val           16 chains x 90 leads x 256 lat x 512 lon
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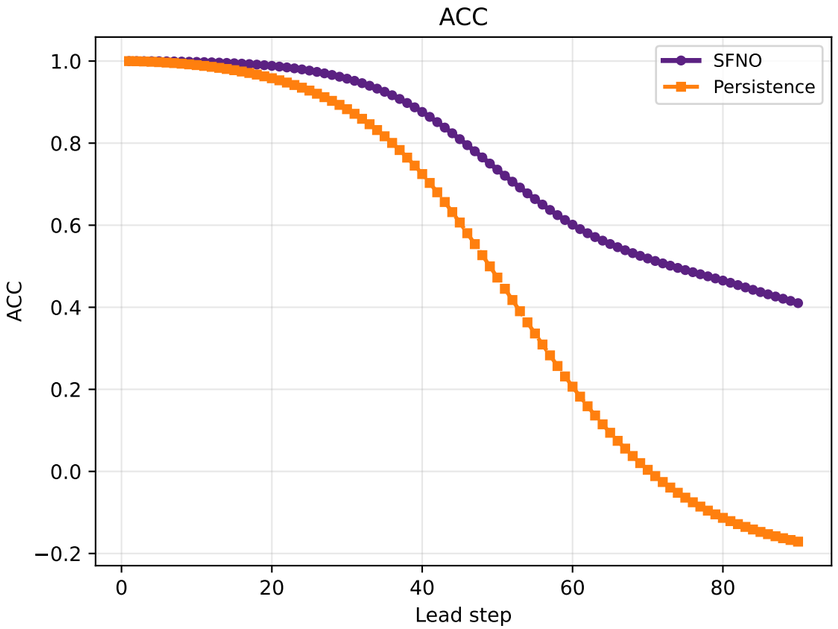
DHW Metrics by Lead Step - Test (18 chains scored at every lead)

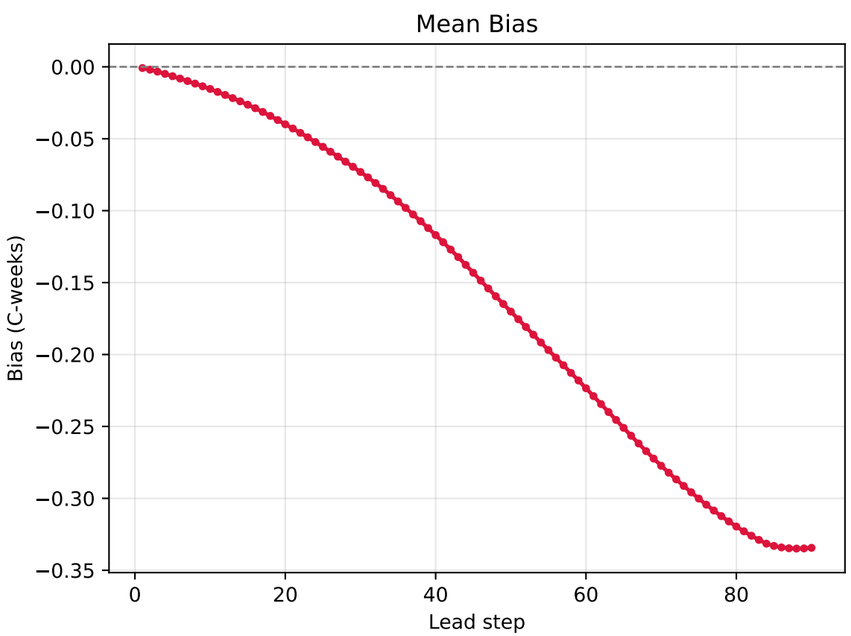
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0147	0.0437	66.3%	0.0015	0.0065	-0.0008	1.0000	0.9999	1.0000
2	0.0302	0.0858	64.9%	0.0032	0.0129	-0.0020	0.9999	0.9996	0.9999
3	0.0468	0.1272	63.2%	0.0052	0.0192	-0.0034	0.9999	0.9990	0.9998
4	0.0644	0.1678	61.6%	0.0074	0.0255	-0.0049	0.9998	0.9983	0.9996
5	0.0828	0.2079	60.2%	0.0096	0.0317	-0.0065	0.9996	0.9974	0.9993
6	0.1015	0.2473	58.9%	0.0120	0.0378	-0.0082	0.9994	0.9964	0.9990
7	0.1208	0.2863	57.8%	0.0144	0.0440	-0.0099	0.9991	0.9951	0.9985
8	0.1406	0.3252	56.8%	0.0170	0.0502	-0.0117	0.9988	0.9937	0.9980
9	0.1609	0.3640	55.8%	0.0196	0.0565	-0.0135	0.9984	0.9921	0.9974
10	0.1815	0.4030	55.0%	0.0224	0.0630	-0.0154	0.9980	0.9902	0.9966
11	0.2024	0.4426	54.3%	0.0252	0.0697	-0.0174	0.9975	0.9882	0.9958
12	0.2236	0.4827	53.7%	0.0281	0.0767	-0.0196	0.9969	0.9860	0.9948
13	0.2450	0.5232	53.2%	0.0311	0.0838	-0.0217	0.9963	0.9835	0.9937
14	0.2670	0.5644	52.7%	0.0342	0.0911	-0.0240	0.9955	0.9808	0.9924
15	0.2894	0.6060	52.2%	0.0375	0.0986	-0.0263	0.9947	0.9778	0.9910
16	0.3123	0.6484	51.8%	0.0410	0.1064	-0.0288	0.9937	0.9746	0.9894
17	0.3361	0.6920	51.4%	0.0447	0.1147	-0.0314	0.9926	0.9710	0.9876
18	0.3607	0.7367	51.0%	0.0487	0.1233	-0.0341	0.9914	0.9671	0.9856
19	0.3859	0.7827	50.7%	0.0528	0.1323	-0.0370	0.9900	0.9628	0.9833
20	0.4119	0.8299	50.4%	0.0571	0.1416	-0.0400	0.9883	0.9582	0.9806
21	0.4384	0.8777	50.1%	0.0617	0.1513	-0.0429	0.9866	0.9532	0.9777
22	0.4657	0.9267	49.8%	0.0665	0.1613	-0.0459	0.9845	0.9477	0.9745
23	0.4937	0.9771	49.5%	0.0716	0.1718	-0.0490	0.9823	0.9417	0.9709
24	0.5227	1.0286	49.2%	0.0769	0.1826	-0.0523	0.9797	0.9352	0.9668
25	0.5527	1.0812	48.9%	0.0824	0.1938	-0.0556	0.9768	0.9281	0.9623
26	0.5836	1.1352	48.6%	0.0881	0.2054	-0.0590	0.9736	0.9204	0.9572
27	0.6149	1.1903	48.3%	0.0940	0.2175	-0.0624	0.9701	0.9121	0.9517
28	0.6465	1.2466	48.1%	0.1001	0.2300	-0.0659	0.9662	0.9030	0.9456
29	0.6787	1.3041	48.0%	0.1063	0.2430	-0.0695	0.9618	0.8933	0.9390
30	0.7113	1.3622	47.8%	0.1128	0.2563	-0.0731	0.9571	0.8828	0.9318
31	0.7442	1.4212	47.6%	0.1194	0.2702	-0.0769	0.9519	0.8715	0.9240
32	0.7776	1.4816	47.5%	0.1263	0.2845	-0.0808	0.9462	0.8593	0.9155
33	0.8121	1.5431	47.4%	0.1334	0.2993	-0.0849	0.9399	0.8461	0.9063
34	0.8478	1.6057	47.2%	0.1408	0.3146	-0.0892	0.9329	0.8319	0.8961
35	0.8844	1.6692	47.0%	0.1486	0.3304	-0.0936	0.9252	0.8167	0.8851
36	0.9213	1.7333	46.8%	0.1565	0.3466	-0.0981	0.9168	0.8006	0.8733
37	0.9589	1.7984	46.7%	0.1646	0.3631	-0.1026	0.9077	0.7832	0.8606
38	0.9972	1.8644	46.5%	0.1730	0.3799	-0.1073	0.8978	0.7647	0.8470
39	1.0361	1.9307	46.3%	0.1816	0.3969	-0.1121	0.8873	0.7452	0.8327
40	1.0754	1.9973	46.2%	0.1904	0.4141	-0.1169	0.8760	0.7247	0.8176
41	1.1151	2.0645	46.0%	0.1995	0.4315	-0.1219	0.8640	0.7030	0.8017
42	1.1555	2.1322	45.8%	0.2089	0.4492	-0.1270	0.8513	0.6802	0.7852
43	1.1966	2.2003	45.6%	0.2184	0.4671	-0.1323	0.8380	0.6565	0.7681
44	1.2383	2.2685	45.4%	0.2282	0.4852	-0.1377	0.8241	0.6319	0.7505
45	1.2799	2.3367	45.2%	0.2380	0.5035	-0.1432	0.8097	0.6065	0.7327
46	1.3216	2.4048	45.0%	0.2478	0.5218	-0.1486	0.7951	0.5805	0.7147
47	1.3636	2.4729	44.9%	0.2576	0.5401	-0.1541	0.7802	0.5539	0.6966
48	1.4056	2.5407	44.7%	0.2674	0.5584	-0.1595	0.7652	0.5269	0.6785
49	1.4471	2.6080	44.5%	0.2771	0.5765	-0.1649	0.7502	0.4997	0.6607
50	1.4883	2.6747	44.4%	0.2868	0.5946	-0.1701	0.7354	0.4724	0.6432
51	1.5293	2.7409	44.2%	0.2966	0.6127	-0.1755	0.7207	0.4450	0.6260
52	1.5703	2.8067	44.1%	0.3063	0.6308	-0.1808	0.7061	0.4177	0.6091
53	1.6111	2.8720	43.9%	0.3160	0.6488	-0.1862	0.6917	0.3904	0.5927
54	1.6519	2.9370	43.8%	0.3257	0.6670	-0.1916	0.6776	0.3632	0.5765
55	1.6928	3.0018	43.6%	0.3354	0.6851	-0.1969	0.6637	0.3361	0.5606
56	1.7335	3.0662	43.5%	0.3451	0.7031	-0.2022	0.6502	0.3093	0.5451
57	1.7740	3.1300	43.3%	0.3547	0.7211	-0.2075	0.6372	0.2828	0.5301
58	1.8145	3.1931	43.2%	0.3643	0.7389	-0.2128	0.6246	0.2569	0.5157
59	1.8549	3.2559	43.0%	0.3739	0.7566	-0.2181	0.6126	0.2314	0.5018
60	1.8955	3.3183	42.9%	0.3834	0.7743	-0.2234	0.6013	0.2065	0.4885
61	1.9365	3.3806	42.7%	0.3930	0.7918	-0.2289	0.5905	0.1823	0.4758
62	1.9776	3.4423	42.6%	0.4026	0.8092	-0.2344	0.5805	0.1588	0.4638
63	2.0185	3.5030	42.4%	0.4121	0.8263	-0.2400	0.5711	0.1363	0.4525
64	2.0591	3.5624	42.2%	0.4216	0.8430	-0.2455	0.5624	0.1148	0.4417
65	2.0999	3.6209	42.0%	0.4309	0.8593	-0.2510	0.5542	0.0942	0.4315
66	2.1407	3.6784	41.8%	0.4402	0.8751	-0.2564	0.5464	0.0745	0.4216
67	2.1816	3.7350	41.6%	0.4495	0.8906	-0.2618	0.5390	0.0556	0.4120
68	2.2225	3.7906	41.4%	0.4587	0.9057	-0.2672	0.5320	0.0375	0.4027
69	2.2631	3.8450	41.1%	0.4677	0.9204	-0.2724	0.5253	0.0202	0.3938
70	2.3033	3.8978	40.9%	0.4766	0.9346	-0.2773	0.5189	0.0039	0.3851
71	2.3434	3.9488	40.7%	0.4852	0.9483	-0.2822	0.5128	-0.0114	0.3766
72	2.3832	3.9981	40.4%	0.4937	0.9615	-0.2868	0.5070	-0.0258	0.3683
73	2.4230	4.0462	40.1%	0.5021	0.9742	-0.2914	0.5014	-0.0394	0.3603
74	2.4624	4.0927	39.8%	0.5103	0.9865	-0.2958	0.4959	-0.0520	0.3525
75	2.5017	4.1379	39.5%	0.5185	0.9984	-0.3002	0.4907	-0.0640	0.3448
76	2.5405	4.1815	39.2%	0.5266	1.0099	-0.3044	0.4855	-0.0752	0.3371
77	2.5785	4.2232	38.9%	0.5345	1.0208	-0.3085	0.4805	-0.0857	0.3296
78	2.6156	4.2628	38.6%	0.5423	1.0312	-0.3123	0.4754	-0.0956	0.3220
79	2.6523	4.3003	38.3%	0.5499	1.0411	-0.3160	0.4703	-0.1047	0.3143
80	2.6886	4.3359	38.0%	0.5574	1.0505	-0.3196	0.4650	-0.1132	0.3063
81	2.7247	4.3699	37.6%	0.5649	1.0595	-0.3229	0.4597	-0.1212	0.2981
82	2.7607	4.4017	37.3%	0.5722	1.0680	-0.3260	0.4541	-0.1285	0.2896
83	2.7964	4.4318	36.9%	0.5794	1.0760	-0.3289	0.4484	-0.1352	0.2807
84	2.8317	4.4602	36.5%	0.5865	1.0836	-0.3315	0.4426	-0.1415	0.2716
85	2.8659	4.4869	36.1%	0.5923	1.0907	-0.3331	0.4370	-0.1475	0.2627
86	2.8991	4.5116	35.7%	0.5977	1.0974	-0.3341	0.4316	-0.1529	0.2537
87	2.9314	4.5346	35.4%	0.6029	1.1038	-0.3347	0.4262	-0.1580	0.2445
88	2.9627	4.5555	35.0%	0.6078	1.1097	-0.3349	0.4209	-0.1628	0.2351
89	2.9934	4.5746	34.6%	0.6127	1.1153	-0.3348	0.4155	-0.1671	0.2255
90	3.0238	4.5922	34.2%	0.6175	1.1206	-0.3344	0.4101	-0.1711	0.2155

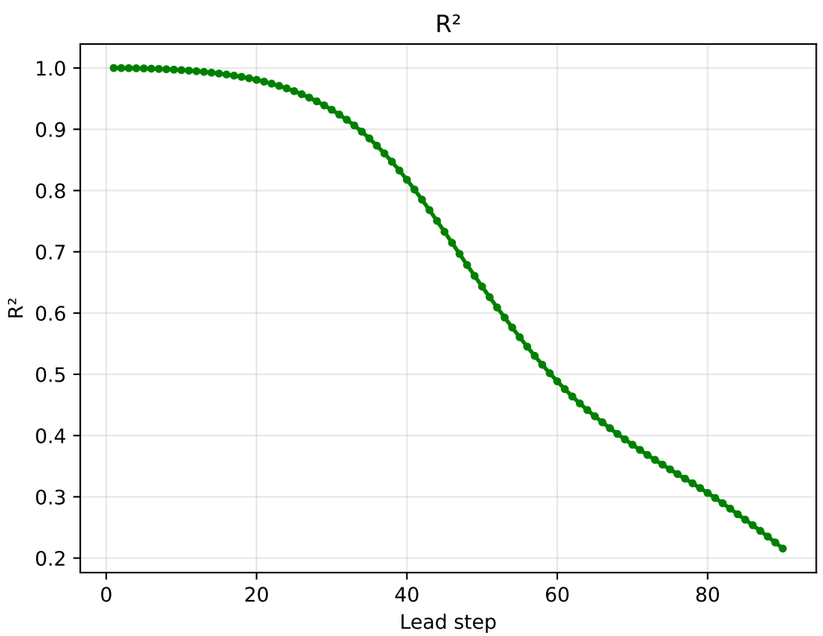
DHW Forecast Skill vs Lead Step - Test

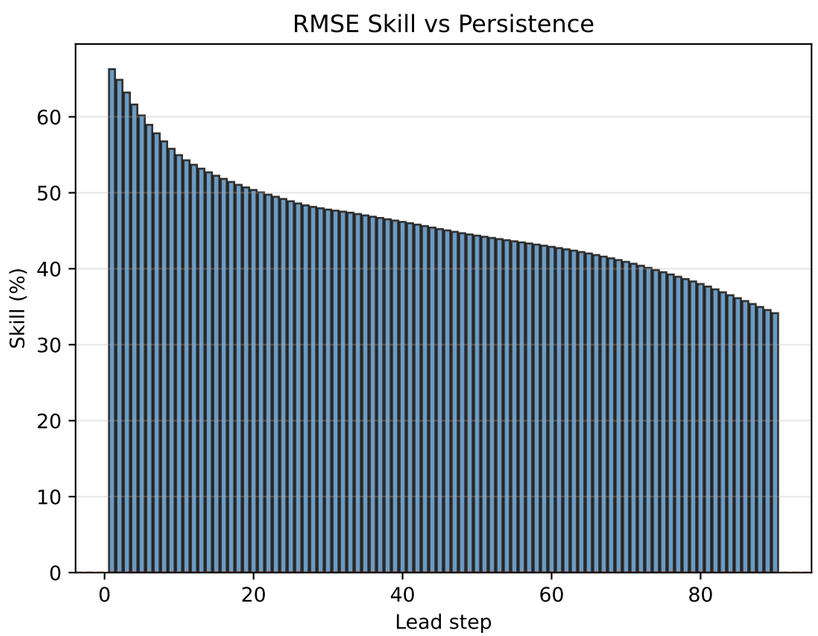


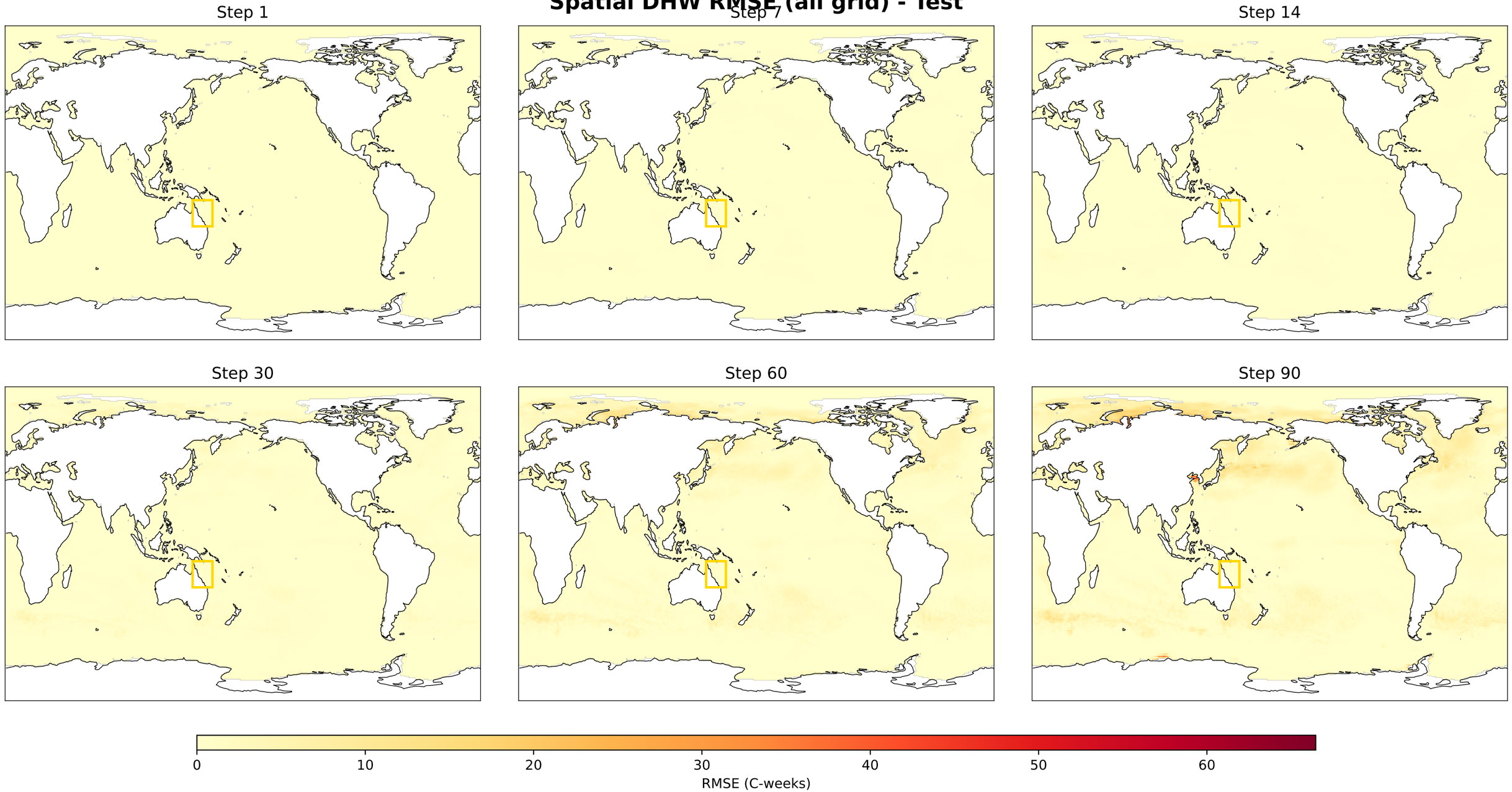


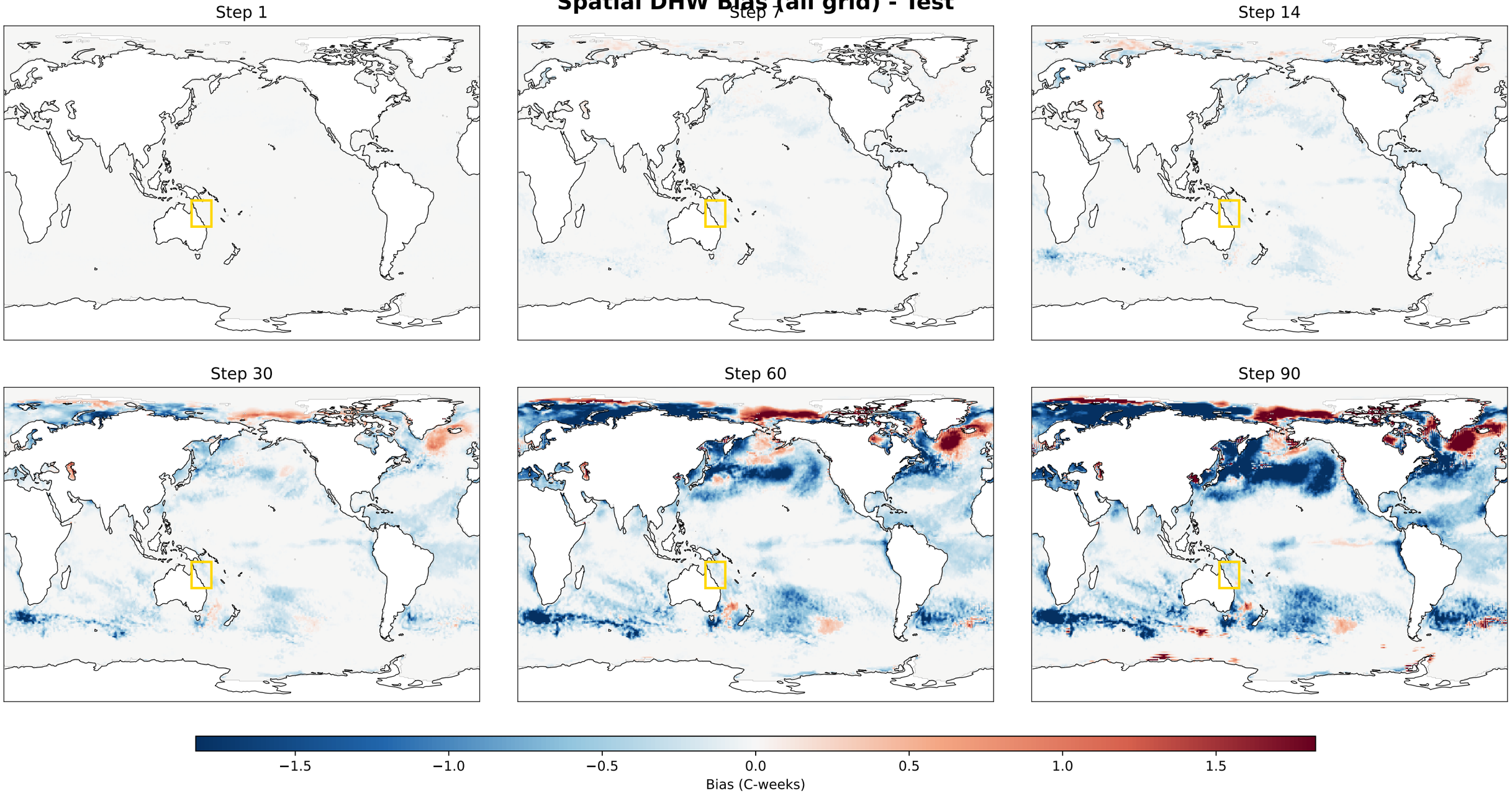


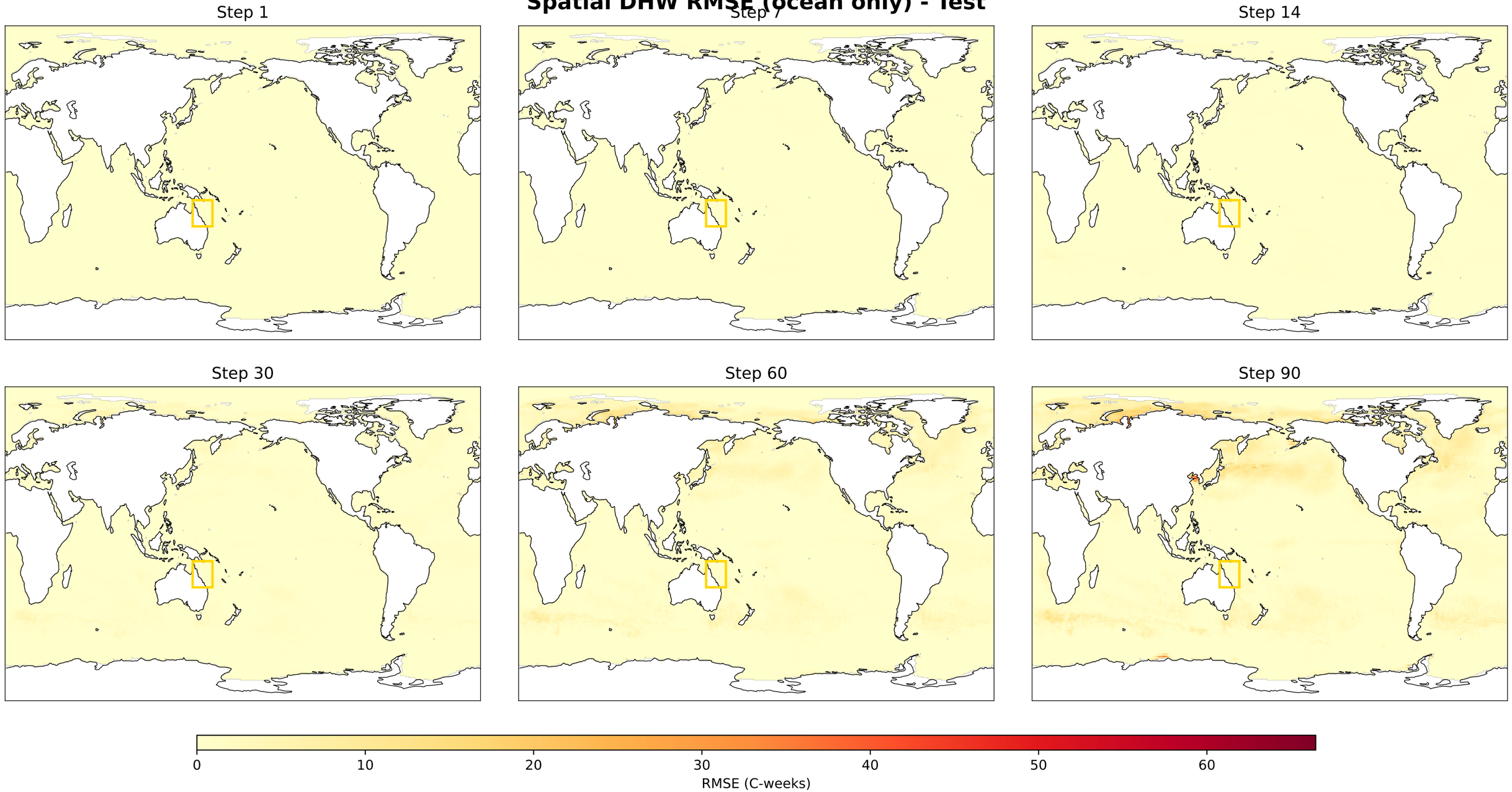


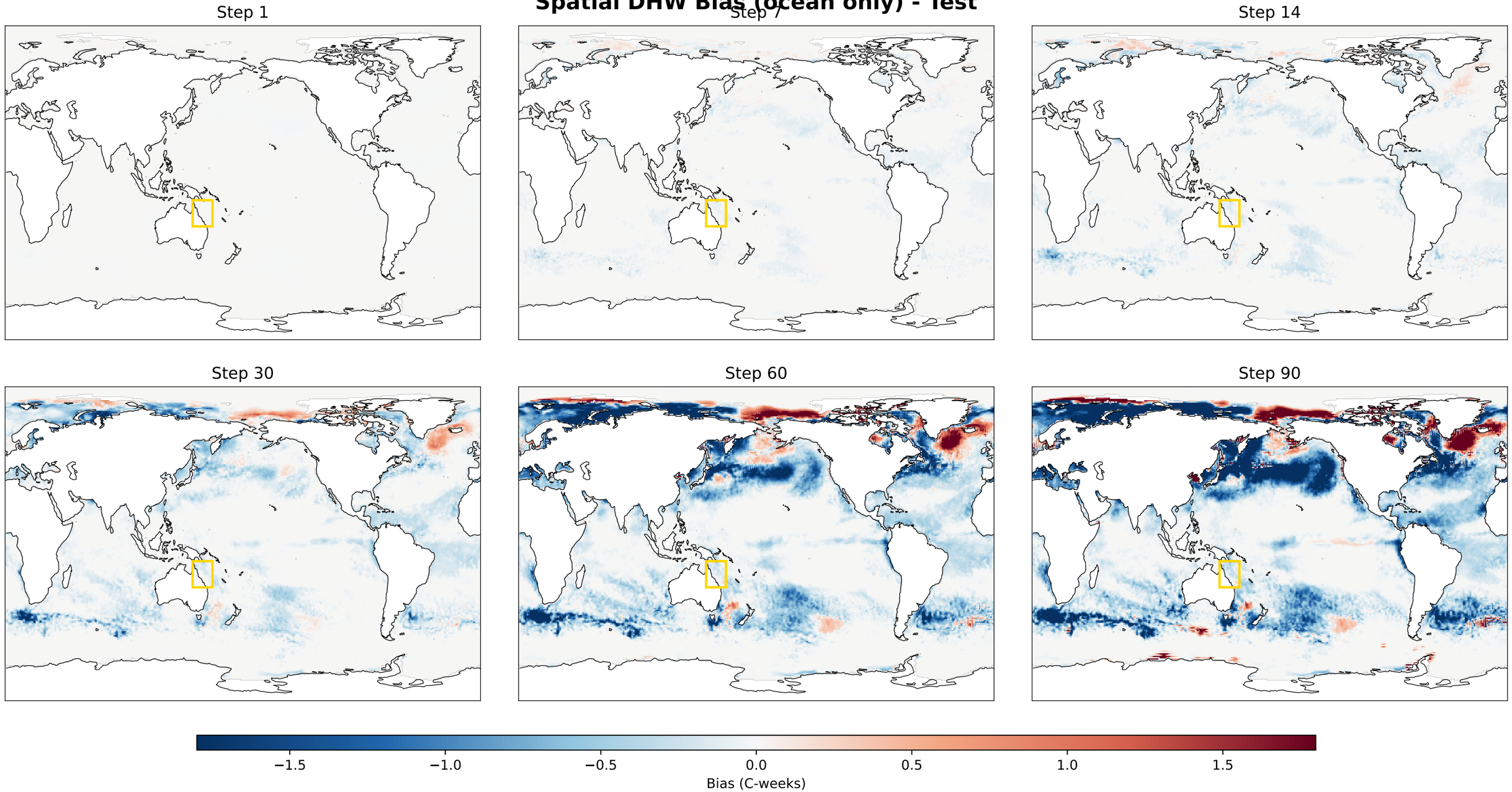





Spatial DHW RMSE (all grid) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
10
20
30
40
50
60
RMSE (C-weeks)

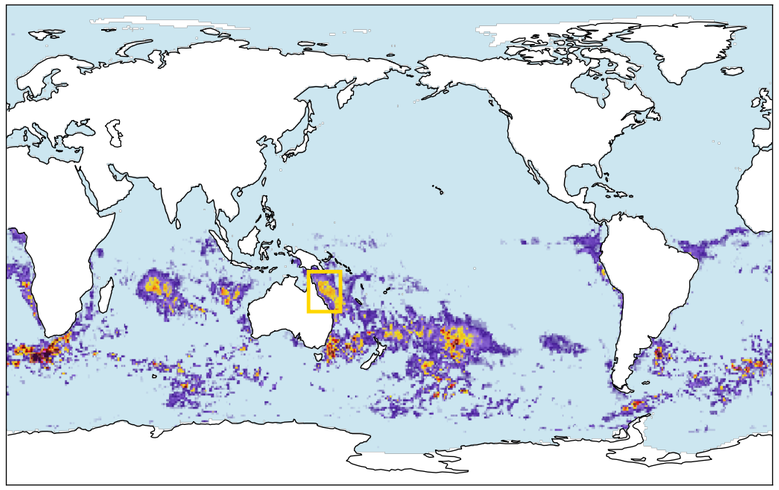

Spatial DHW Bias (all grid) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
10
20
30
40
50
60
RMSE (C-weeks)

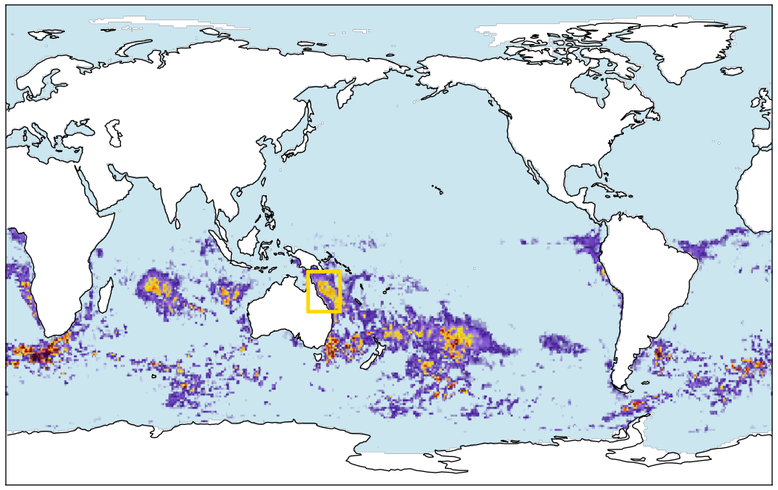

Spatial DHW Bias (ocean only) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Bias (C-weeks)

DHW Chain init: 2020-04-10 - Test (C-weeks)

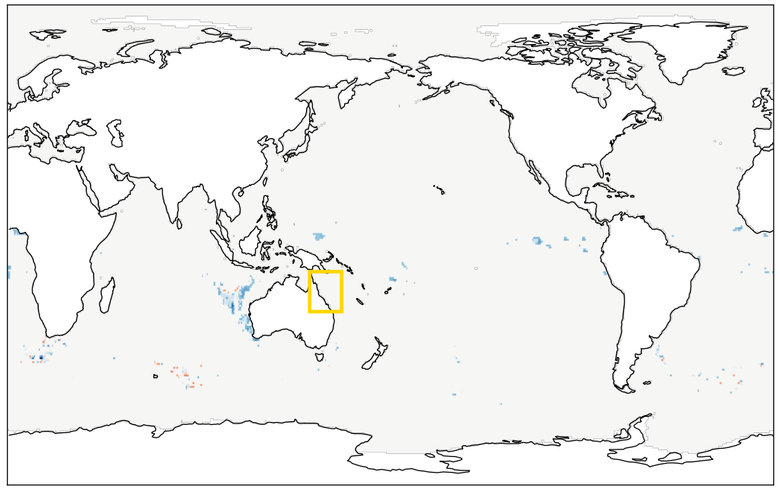
Actual DHW - Step 1



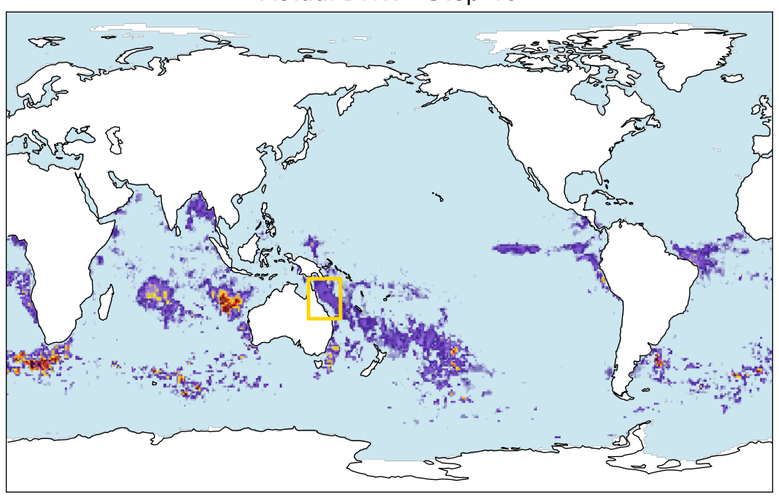
Predicted DHW - Step 1



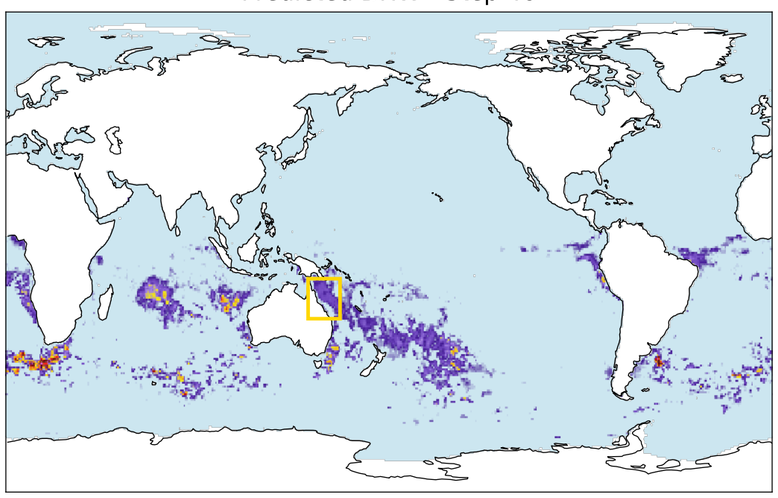
Error (Pred - Actual) - Step 1 [scale ±0.47]



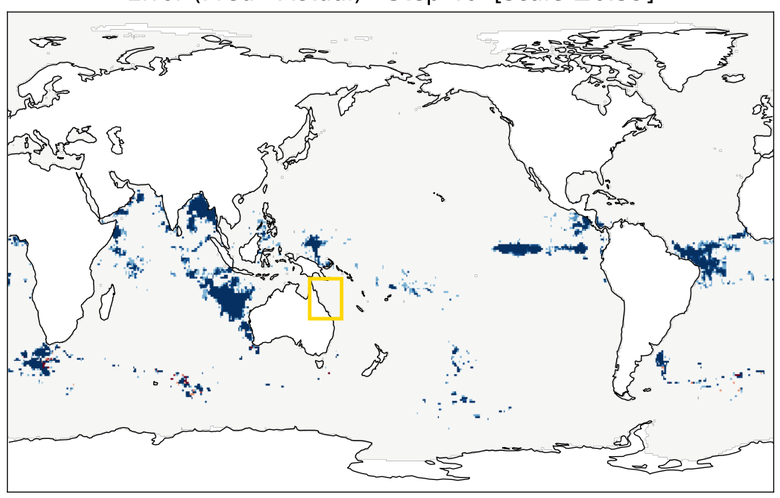
Actual DHW - Step 46



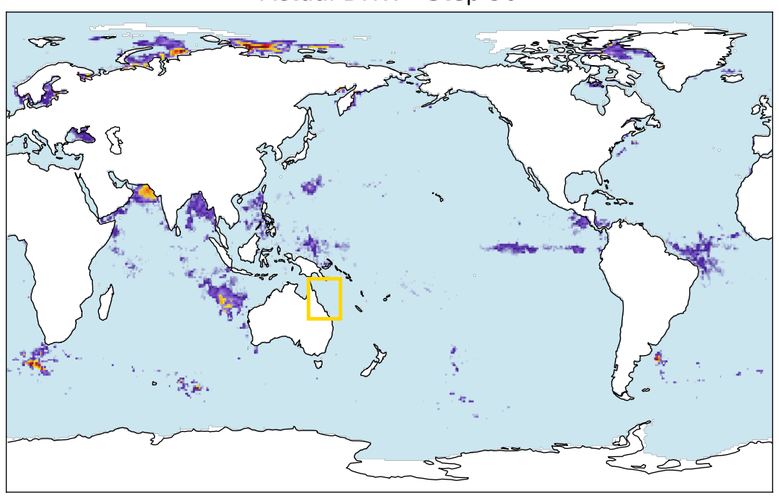
Predicted DHW - Step 46



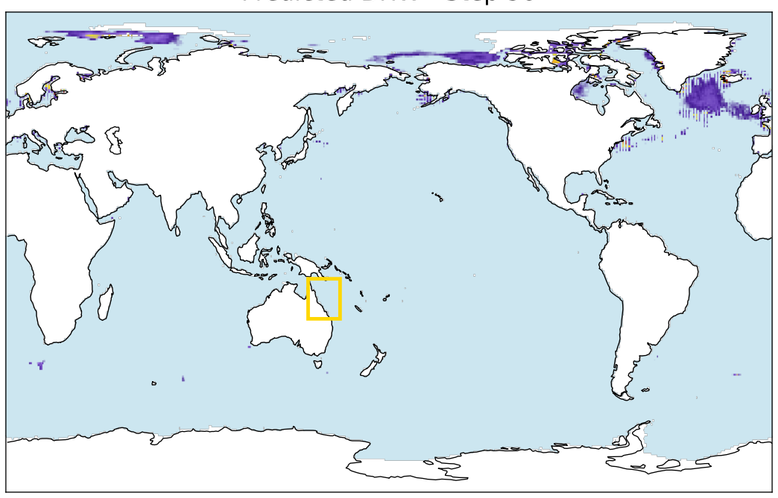
Error (Pred - Actual) - Step 46 [scale ±0.39]



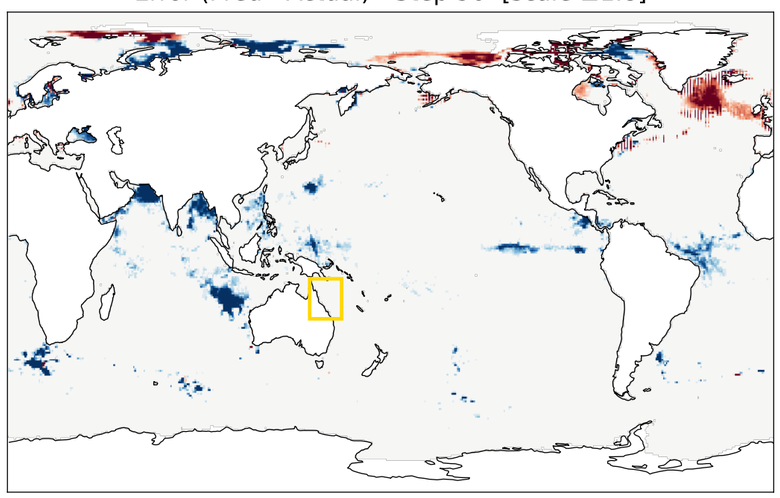
Actual DHW - Step 90

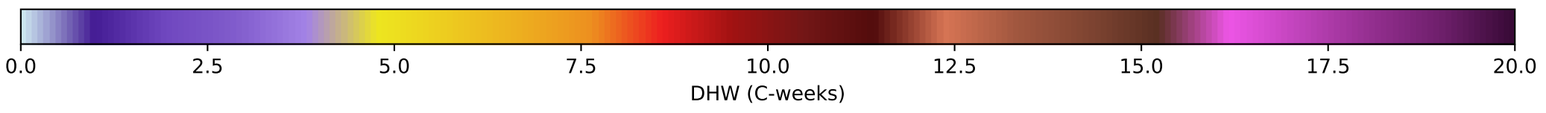


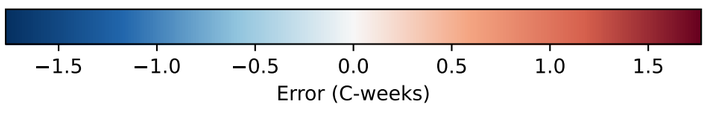
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±1.8]

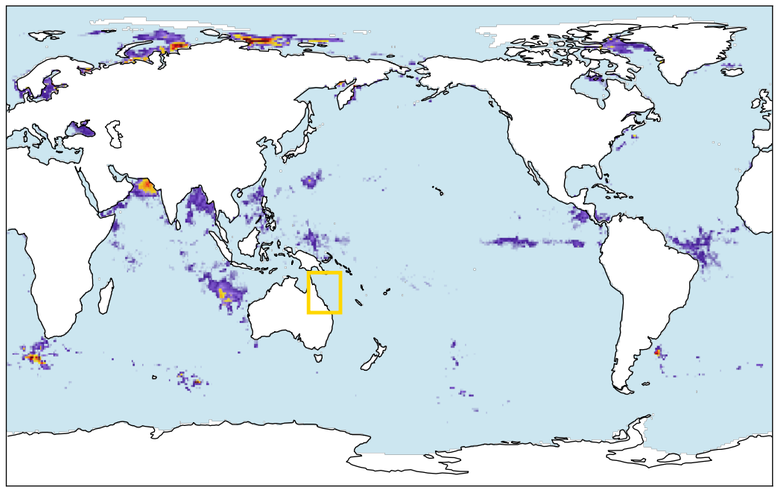




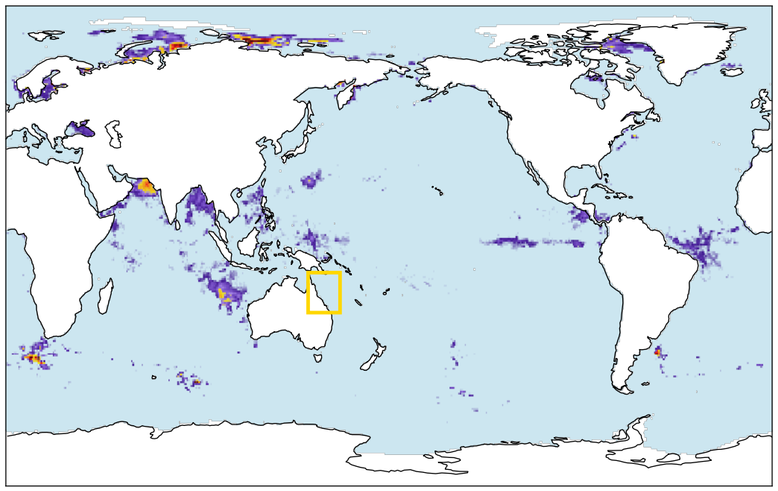


DHW Chain init: 2020-07-09 - Test (C-weeks)

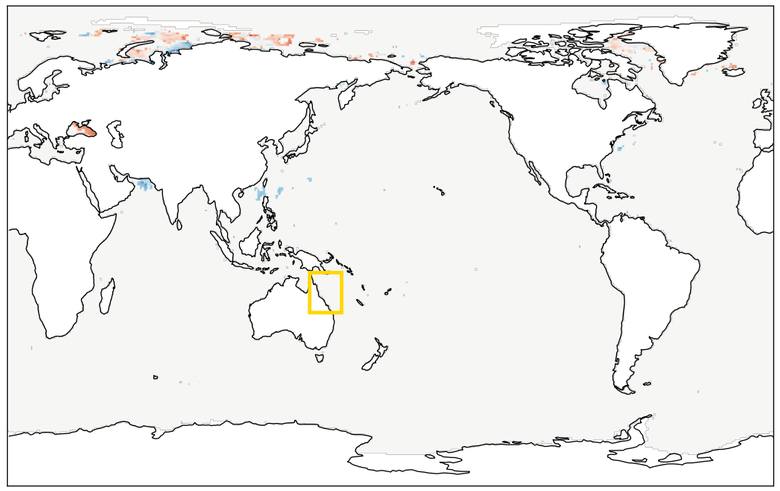
Actual DHW - Step 1



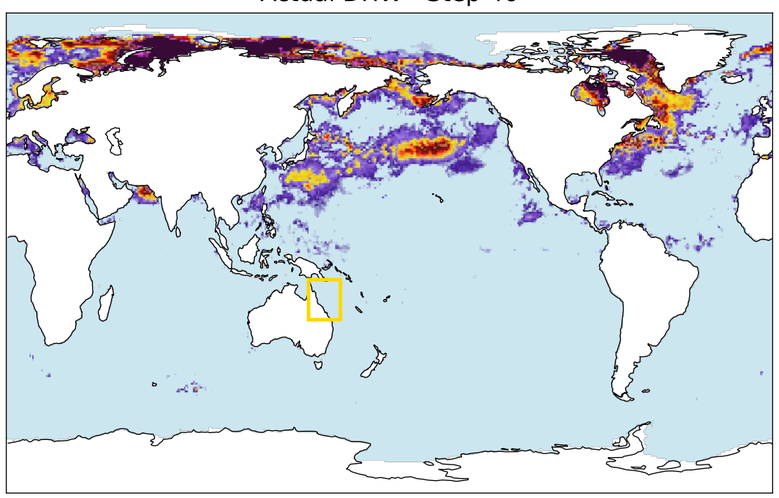
Predicted DHW - Step 1



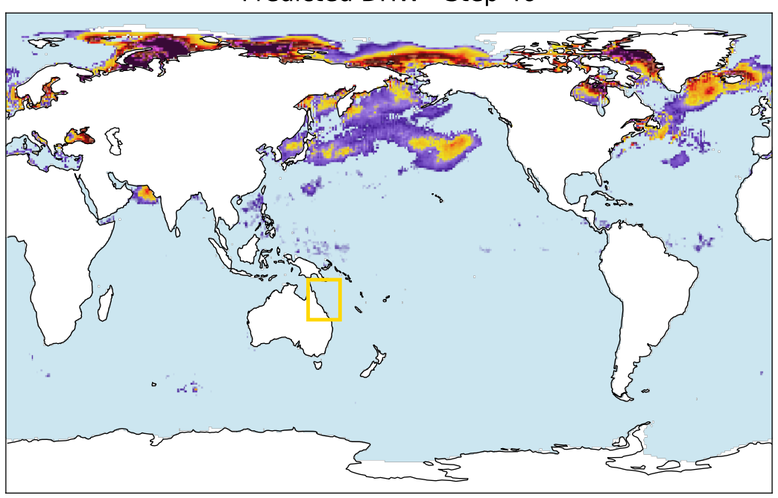
Error (Pred - Actual) - Step 1 [scale ±0.54]



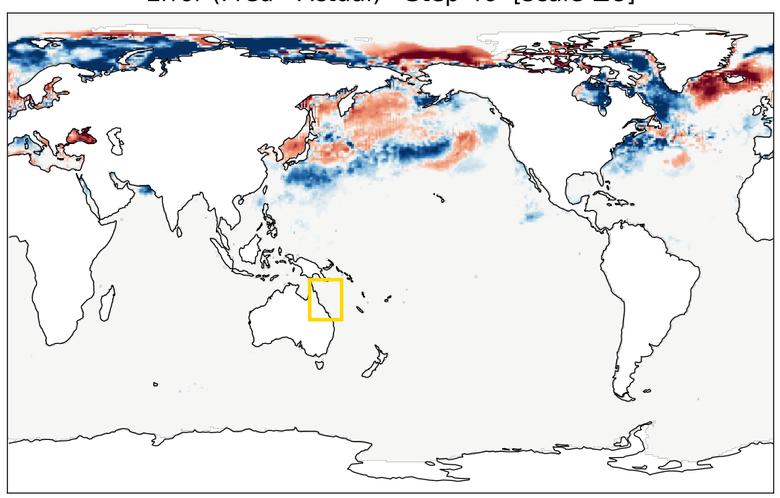
Actual DHW - Step 46



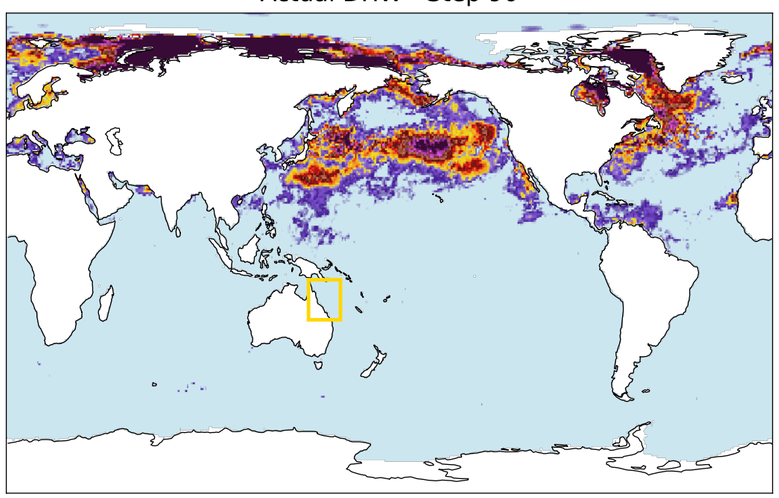
Predicted DHW - Step 46



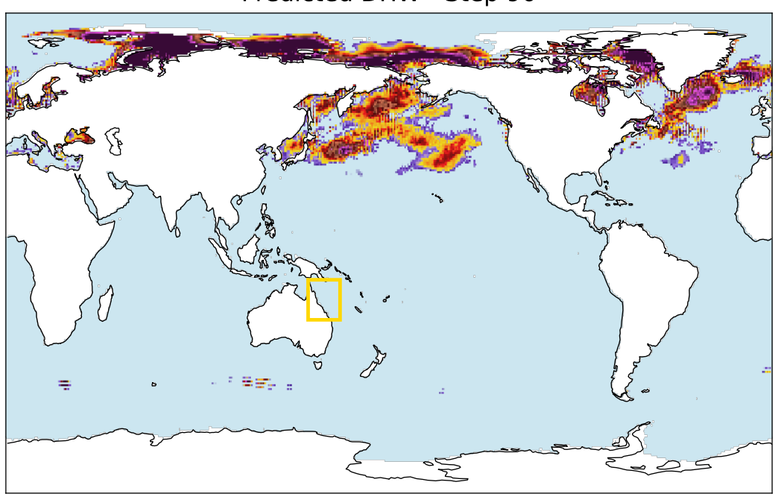
Error (Pred - Actual) - Step 46 [scale ±8]



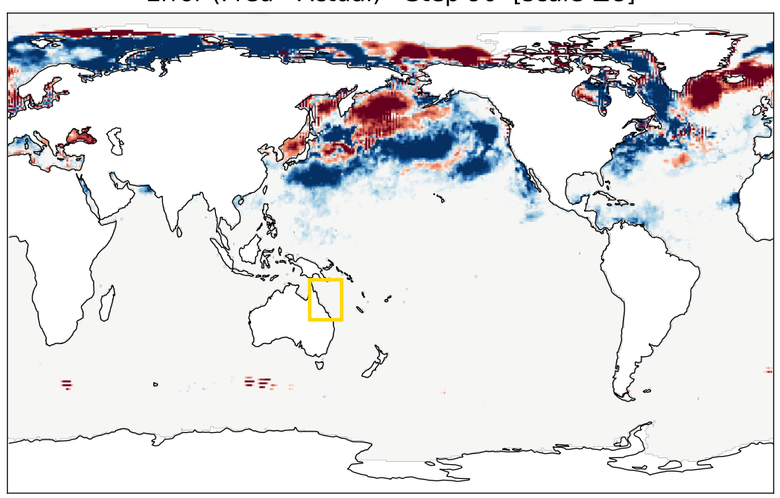
Actual DHW - Step 90

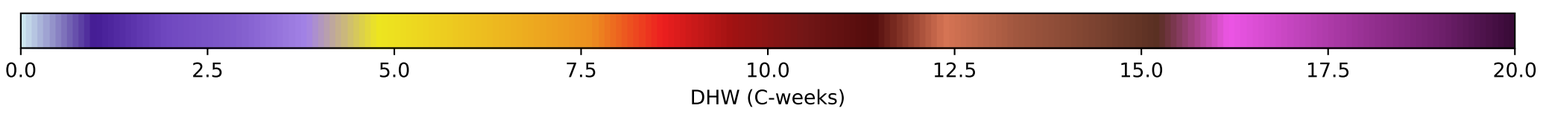


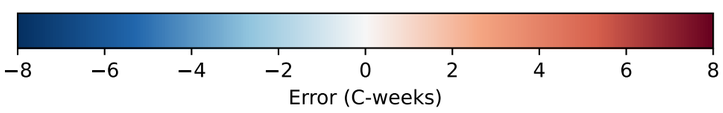
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

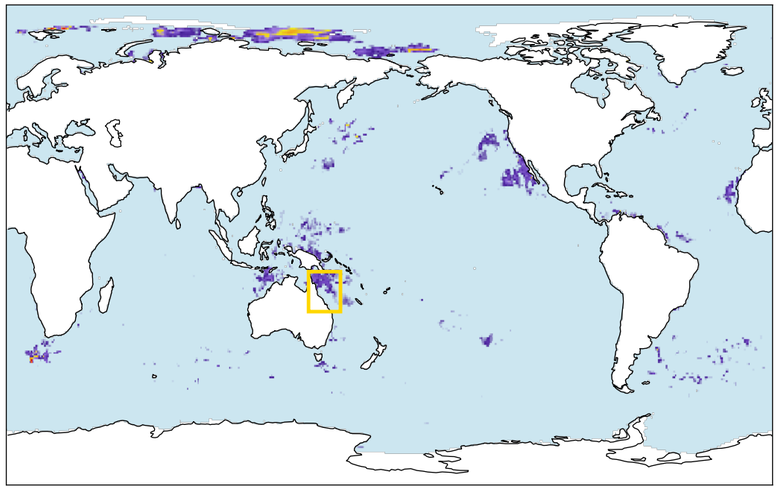




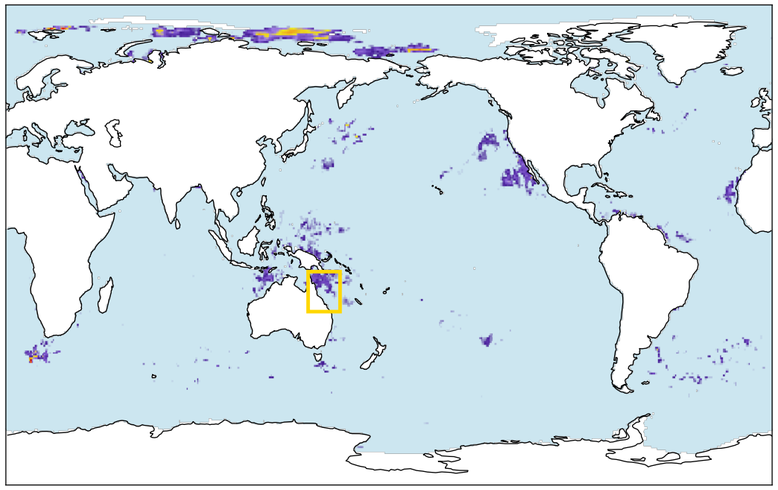


DHW Chain init: 2021-01-06 - Test (C-weeks)

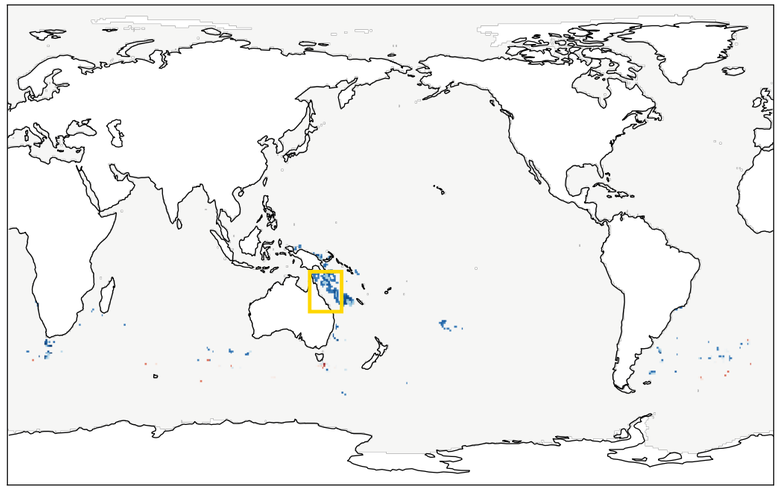
Actual DHW - Step 1



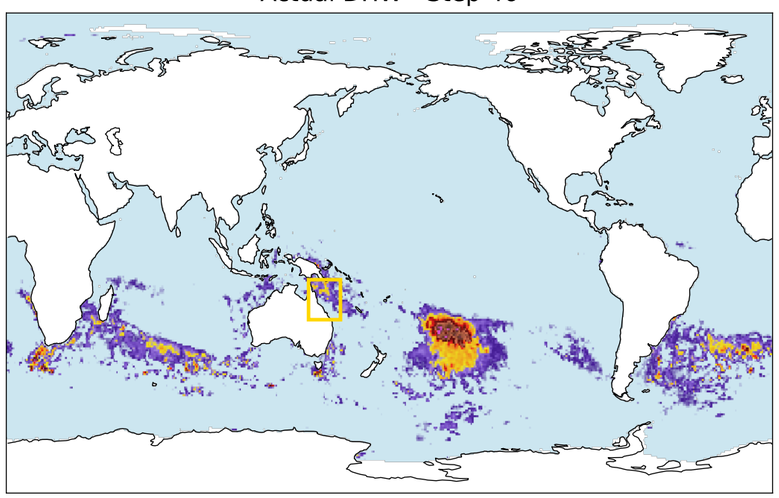
Predicted DHW - Step 1



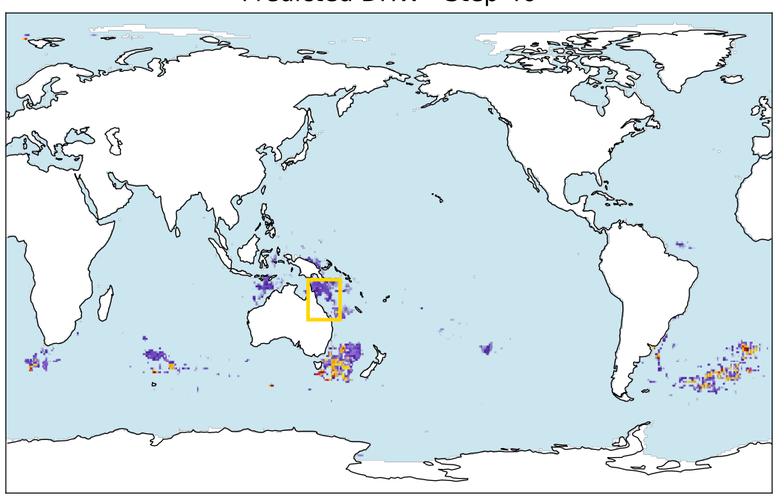
Error (Pred - Actual) - Step 1 [scale ±0.26]



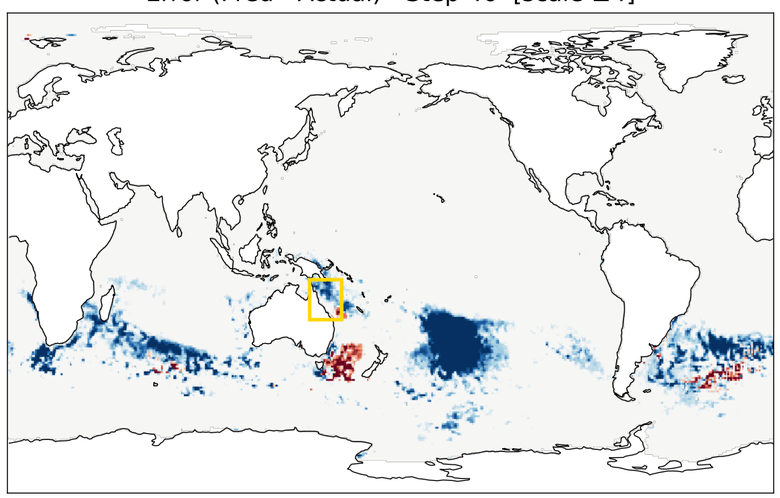
Actual DHW - Step 46



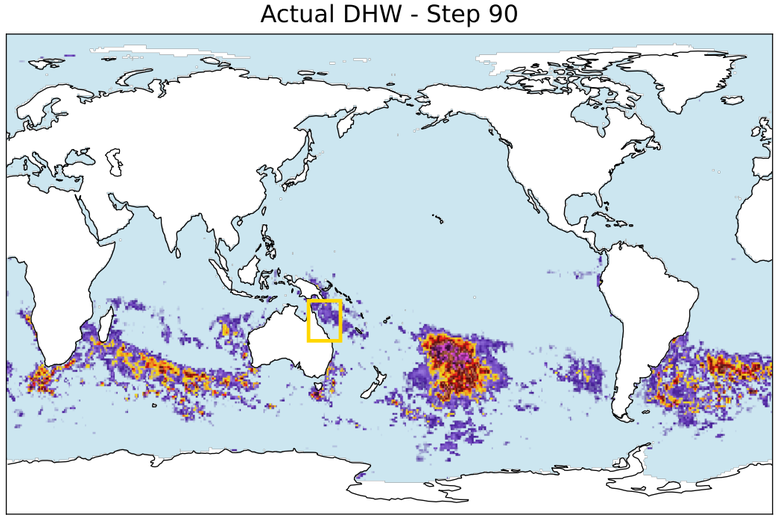
Predicted DHW - Step 46



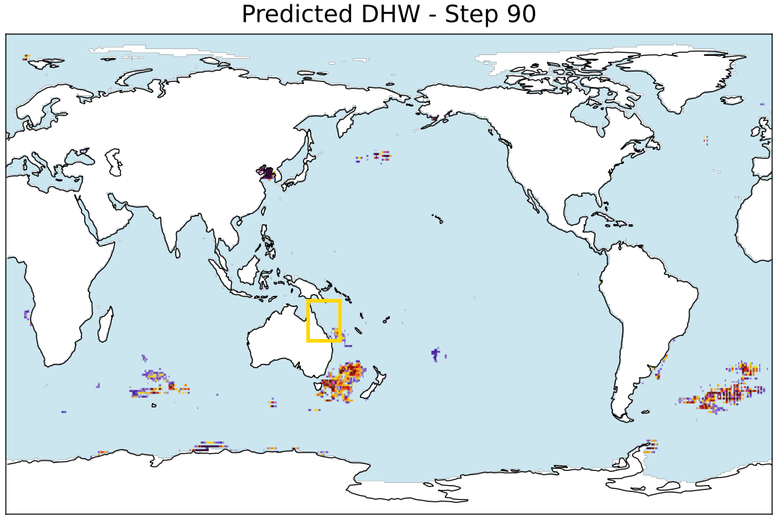
Error (Pred - Actual) - Step 46 [scale ±4]



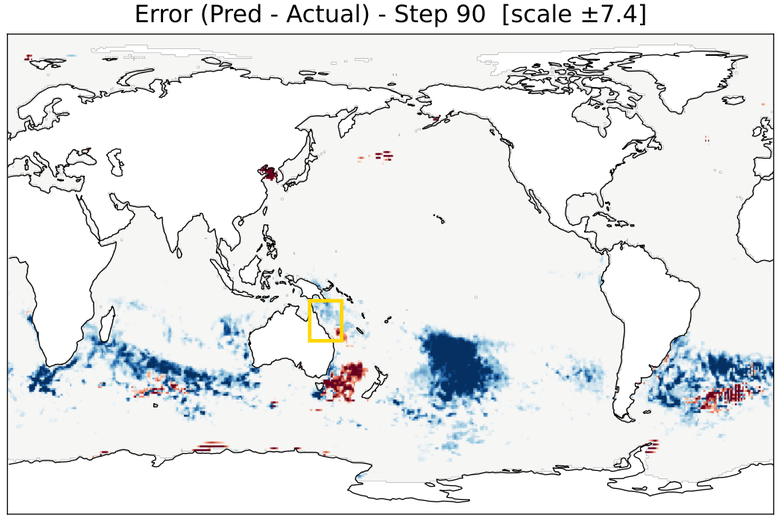
Actual DHW - Step 90

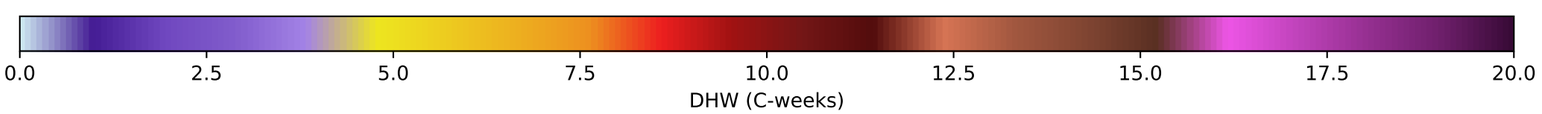


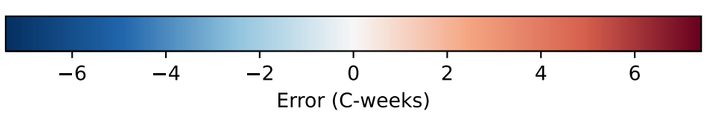
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±7.4]

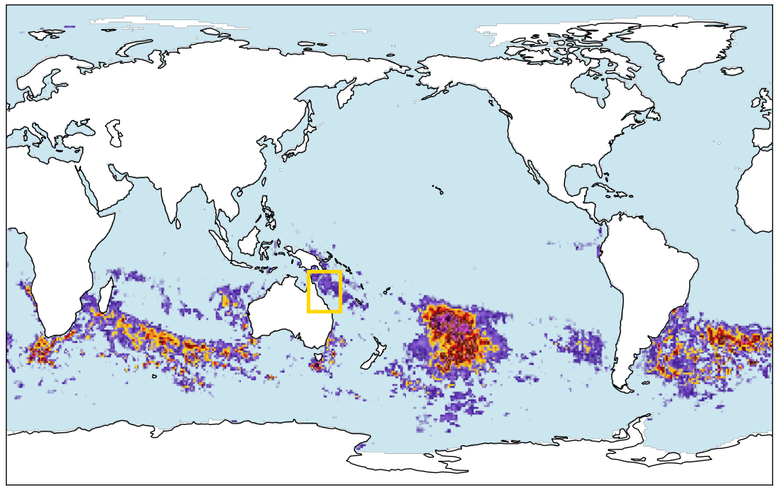




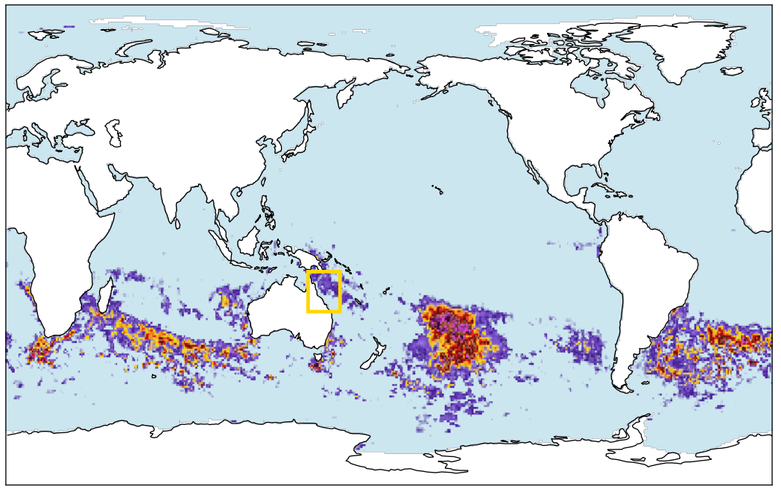


DHW Chain init: 2021-04-06 - Test (C-weeks)

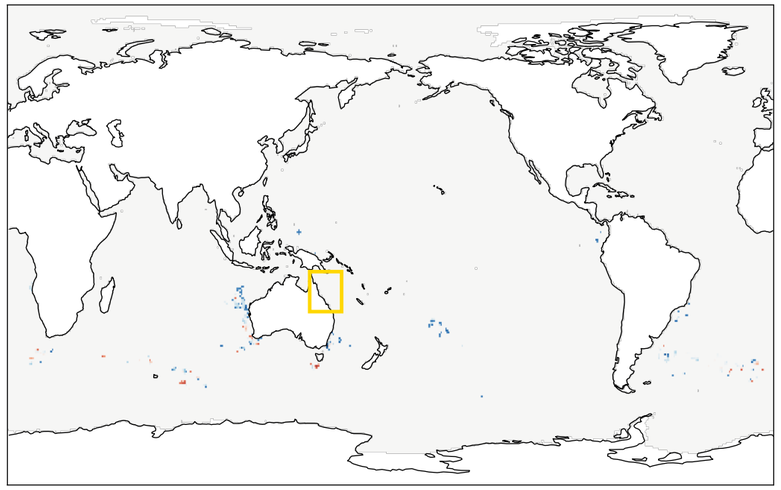
Actual DHW - Step 1



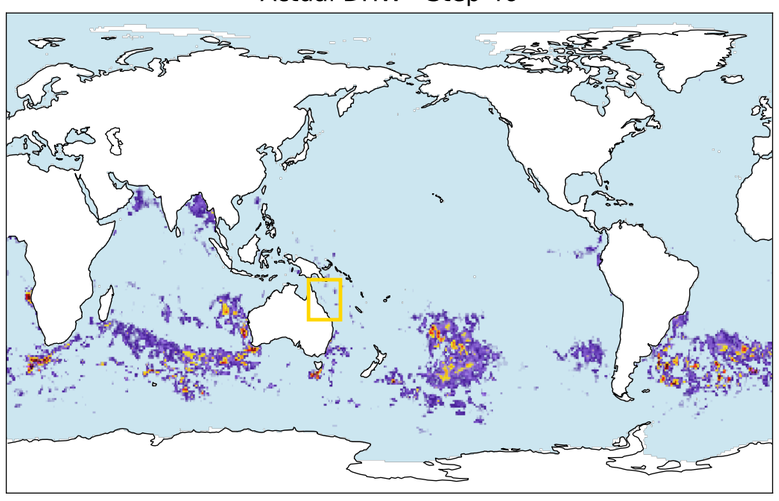
Predicted DHW - Step 1



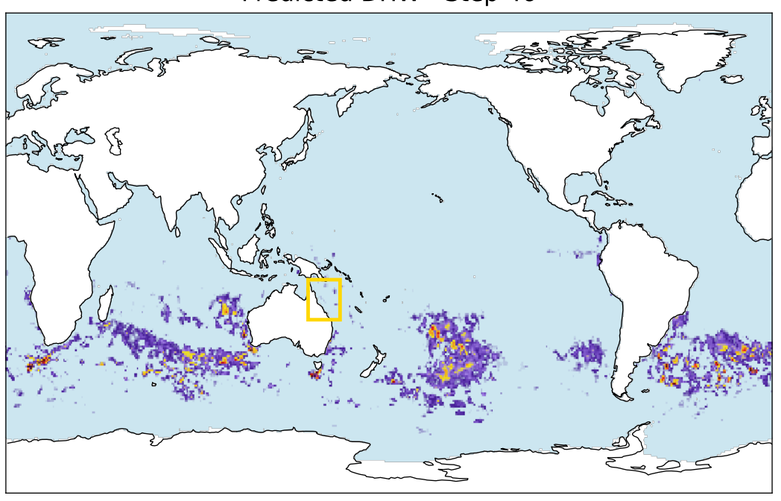
Error (Pred - Actual) - Step 1 [scale ±0.28]



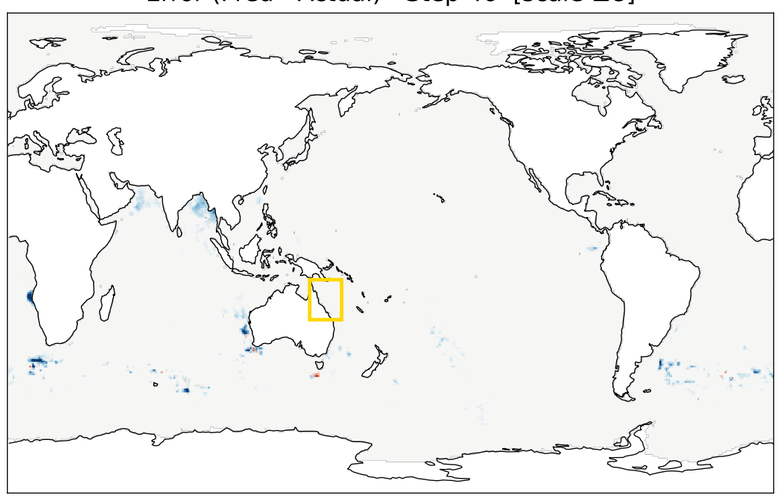
Actual DHW - Step 46



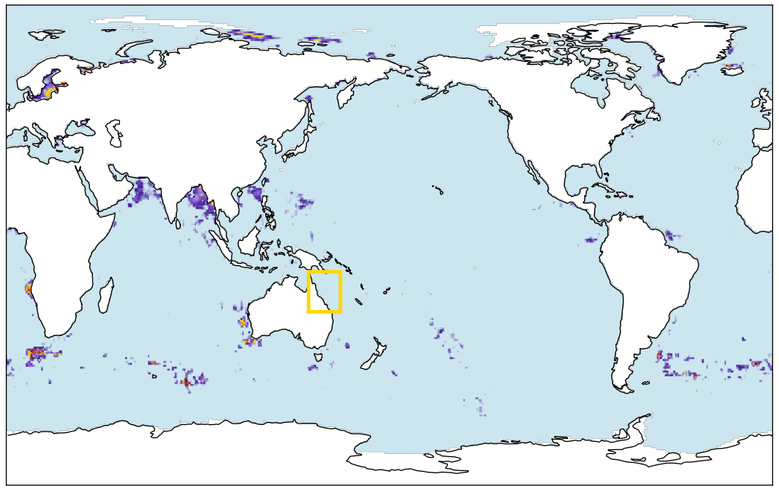
Predicted DHW - Step 46



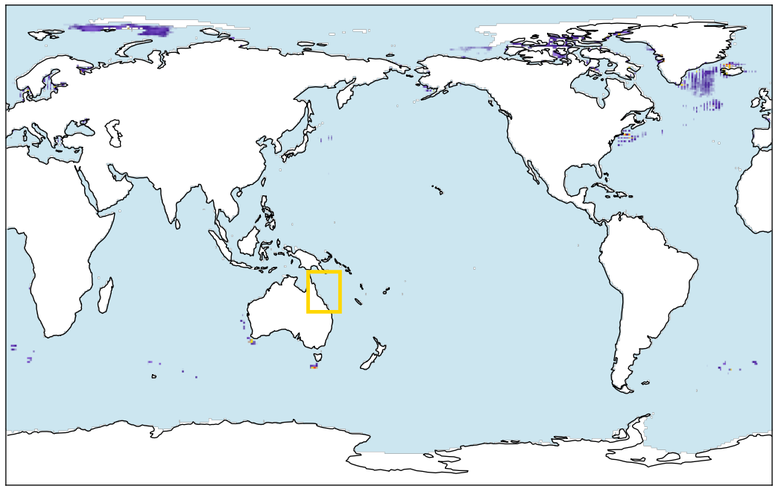
Error (Pred - Actual) - Step 46 [scale ±8]



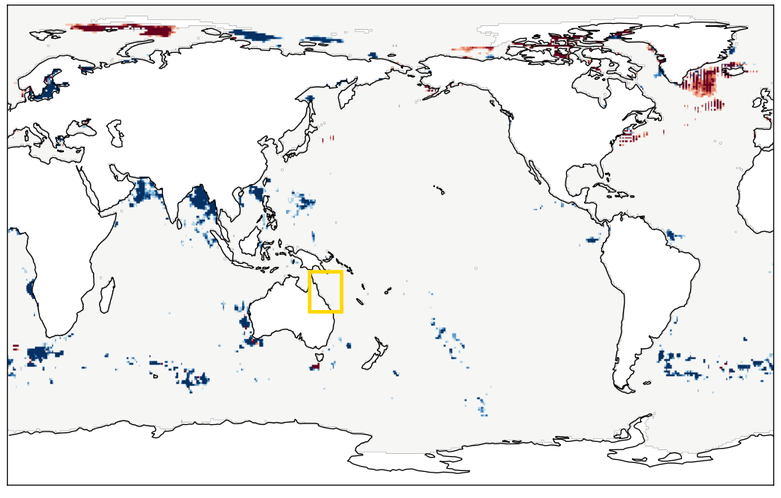
Actual DHW - Step 90

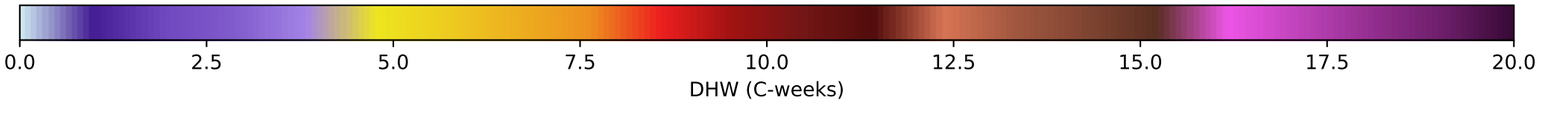


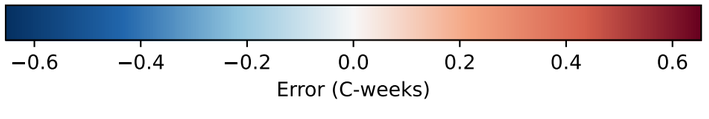
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.65]

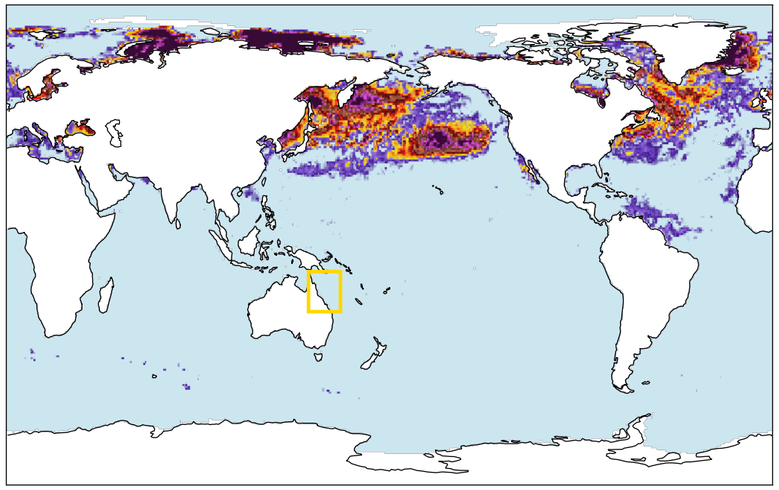




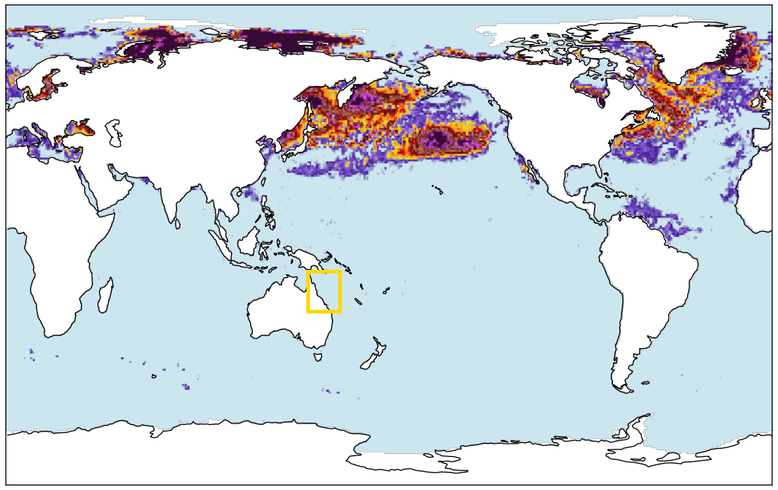


DHW Chain init: 2021-10-03 - Test (C-weeks)

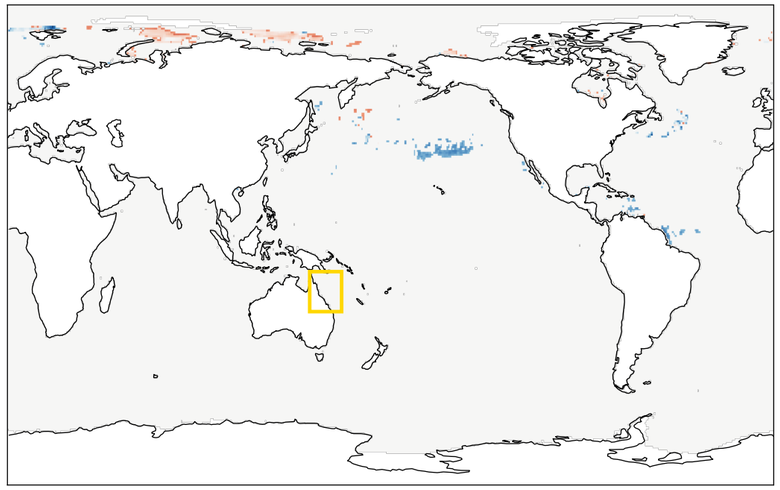
Actual DHW - Step 1



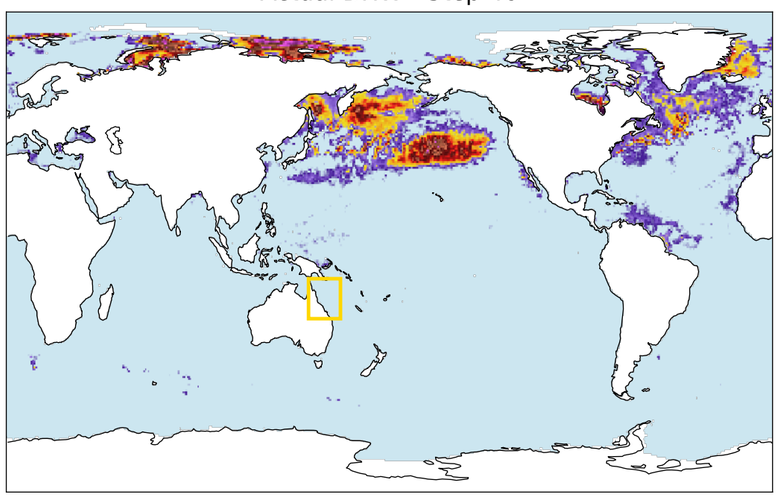
Predicted DHW - Step 1



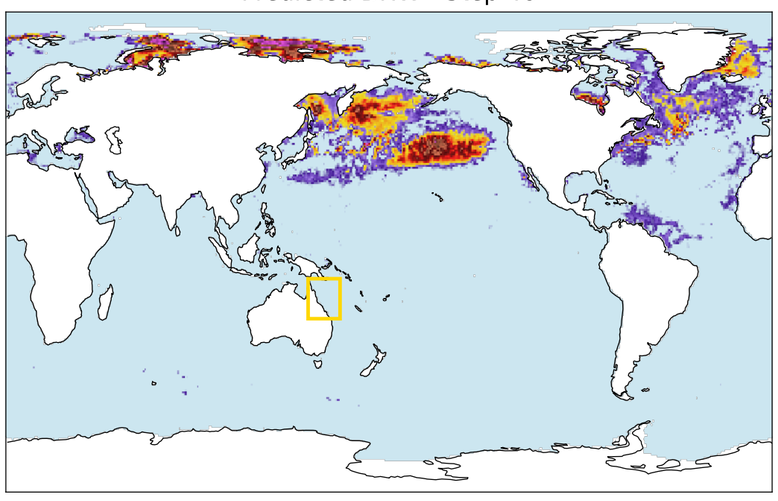
Error (Pred - Actual) - Step 1 [scale ±0.38]



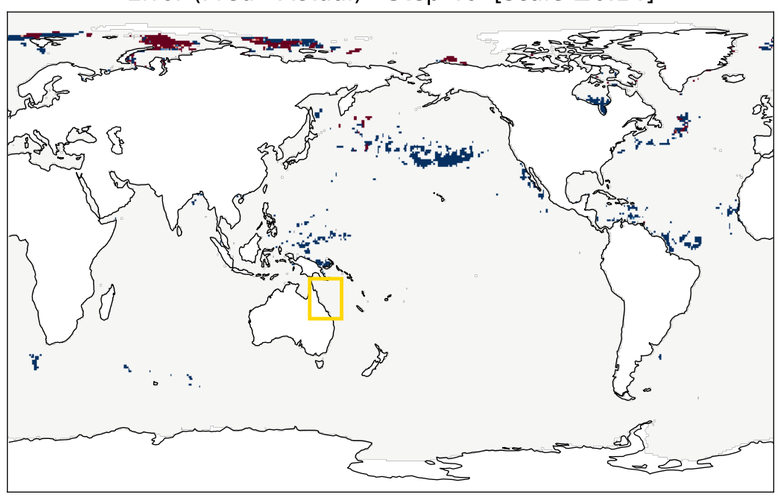
Actual DHW - Step 46



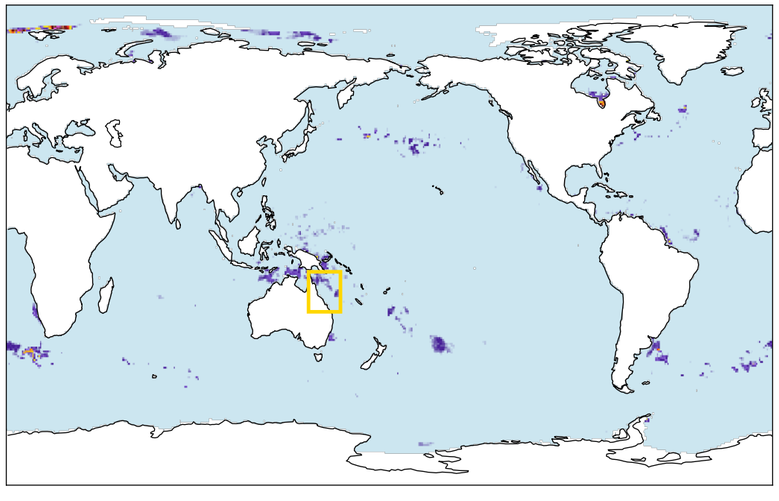
Predicted DHW - Step 46



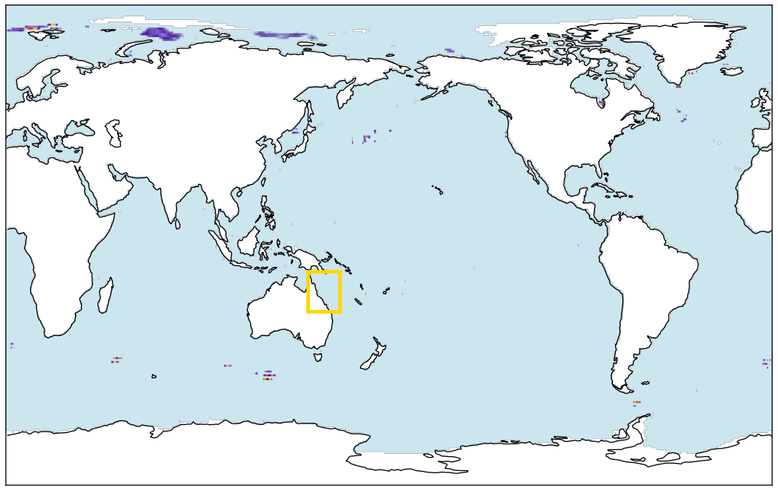
Error (Pred - Actual) - Step 46 [scale ±0.14]



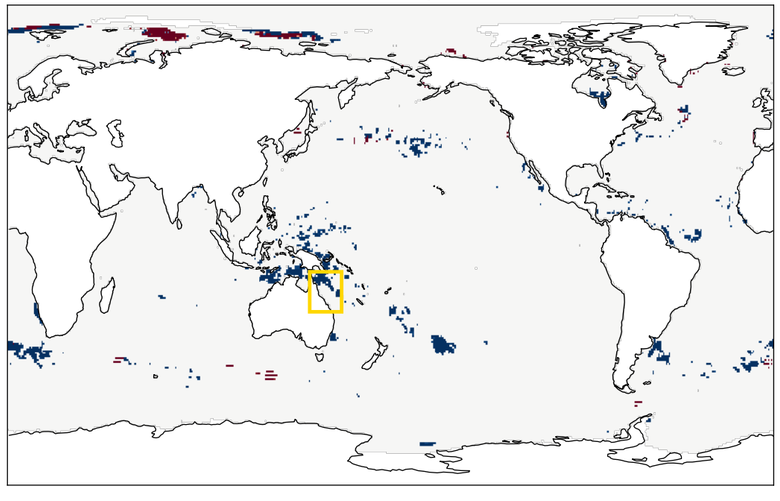
Actual DHW - Step 90

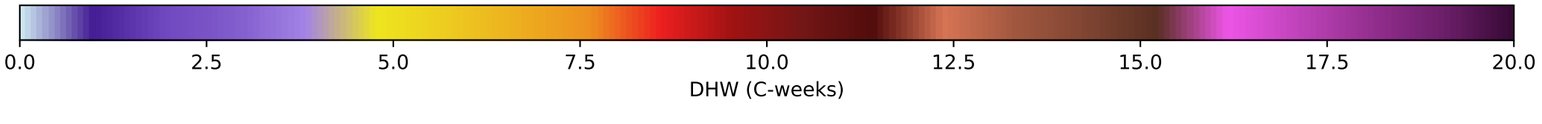


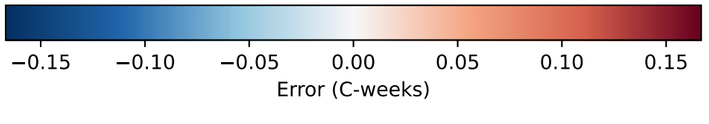
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.17]

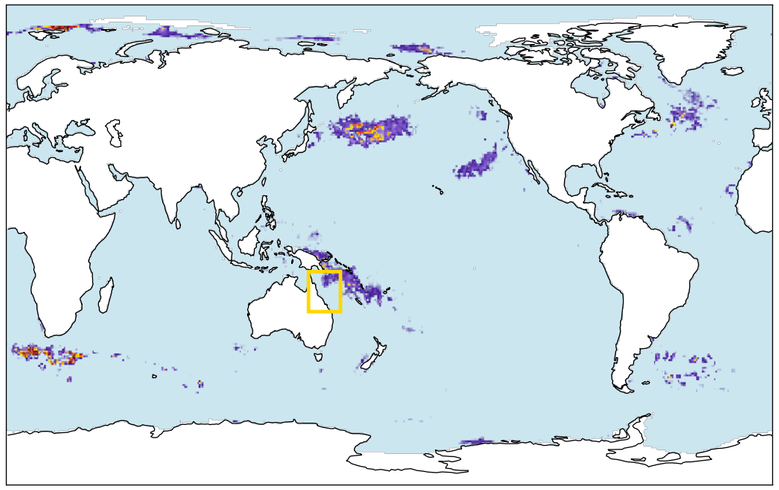




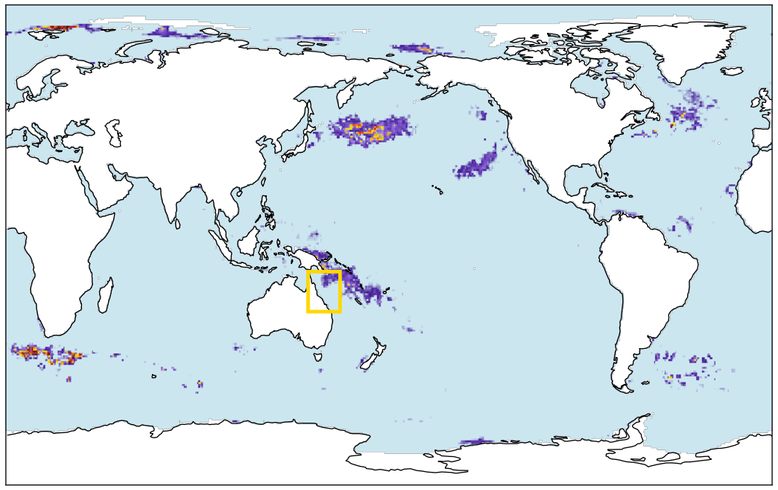


DHW Chain init: 2022-12-27 - Test (C-weeks)

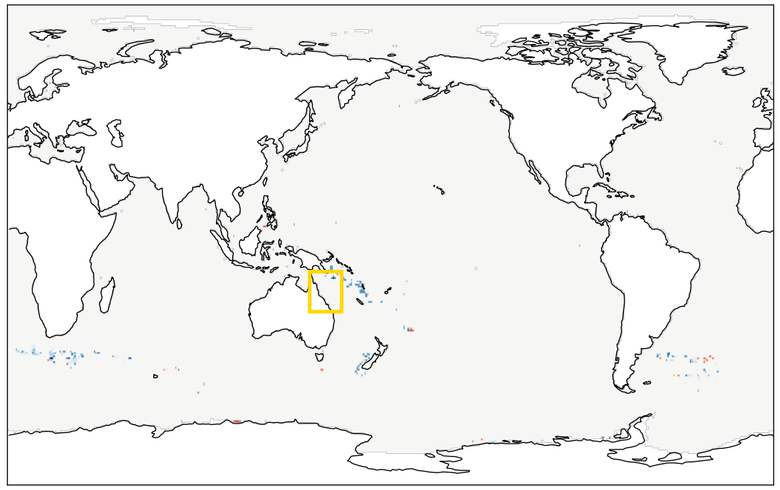
Actual DHW - Step 1



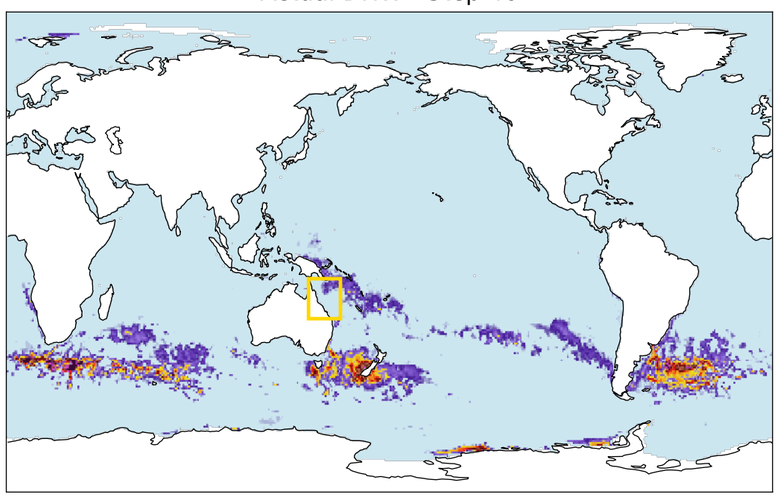
Predicted DHW - Step 1



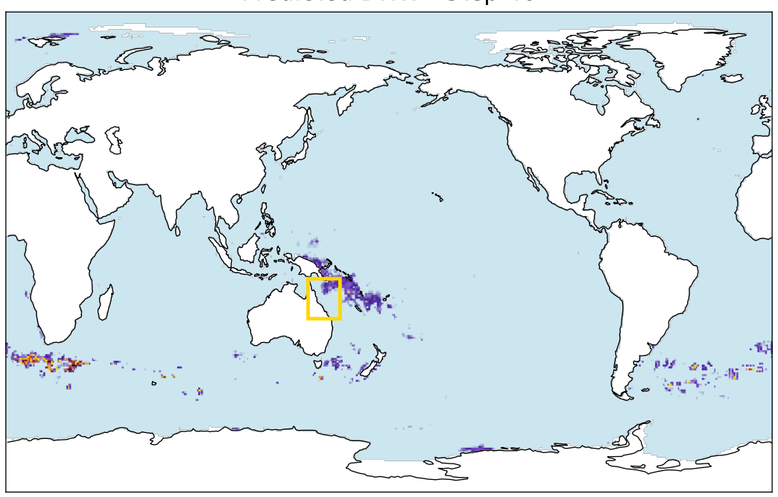
Error (Pred - Actual) - Step 1 [scale ±0.32]



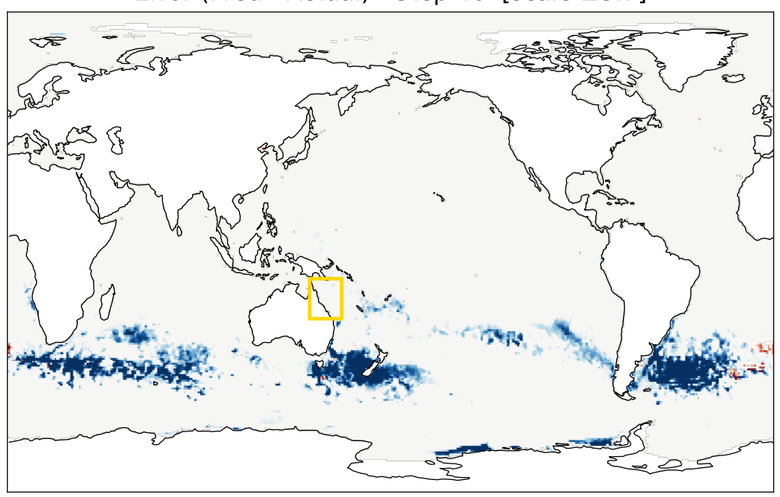
Actual DHW - Step 46



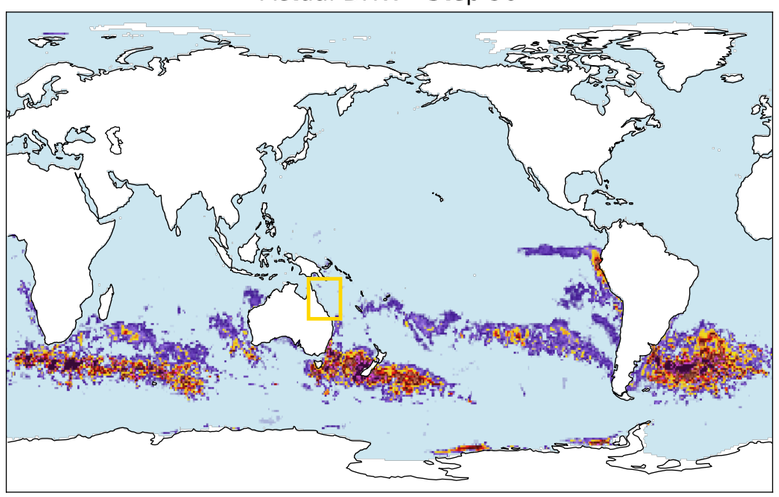
Predicted DHW - Step 46



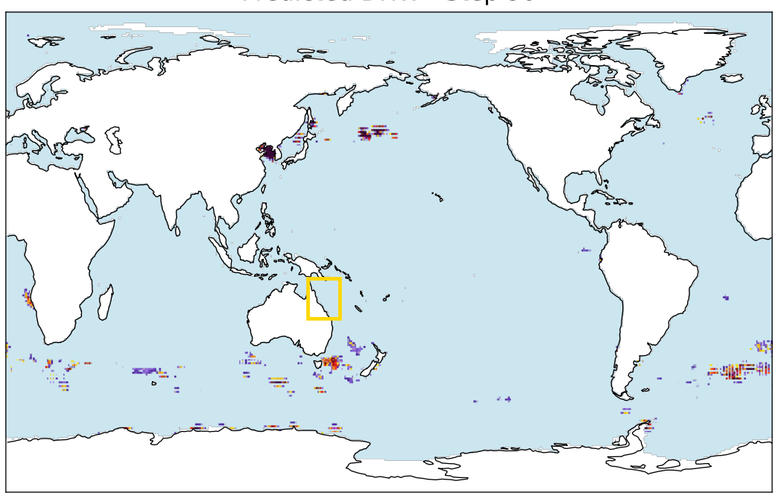
Error (Pred - Actual) - Step 46 [scale ±3.7]



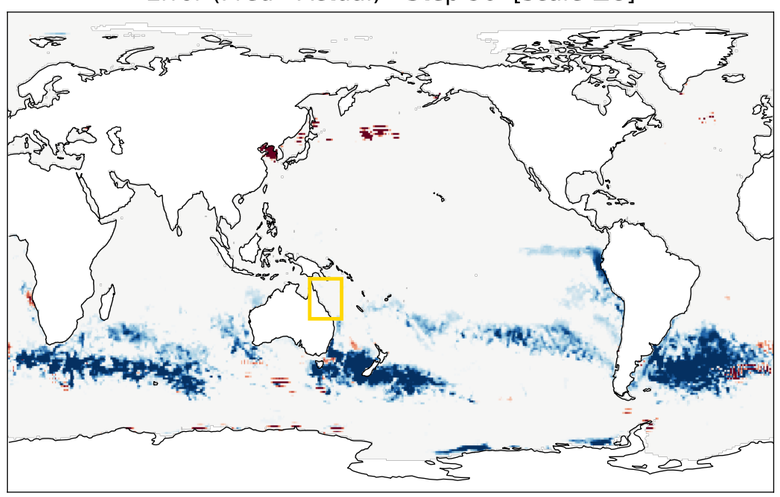
Actual DHW - Step 90

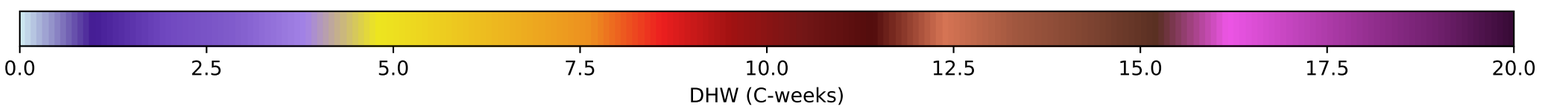


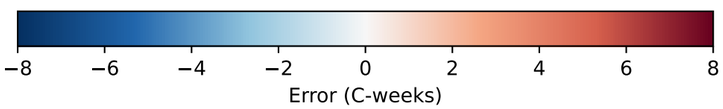
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

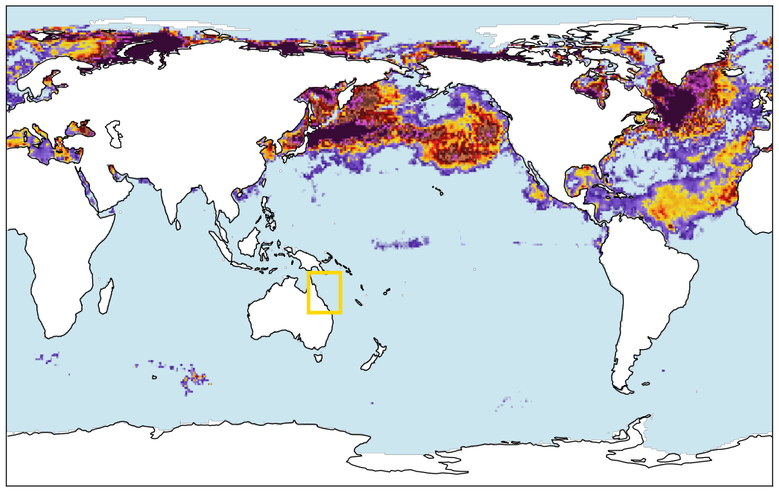




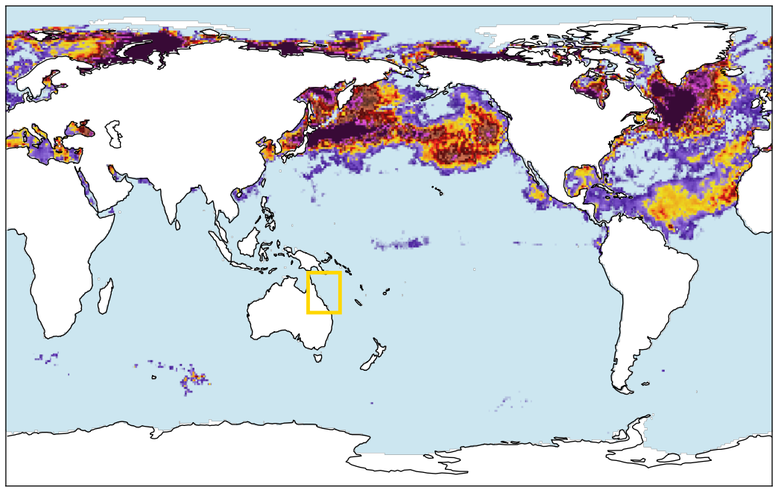


DHW Chain init: 2023-09-23 - Test (C-weeks)

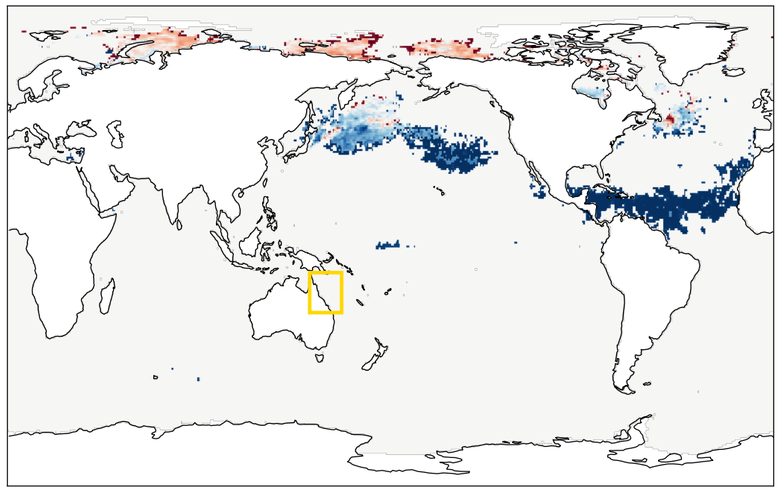
Actual DHW - Step 1



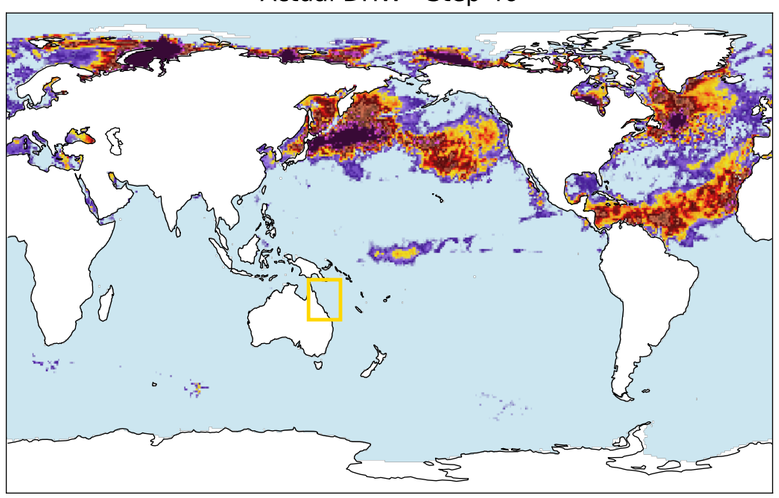
Predicted DHW - Step 1



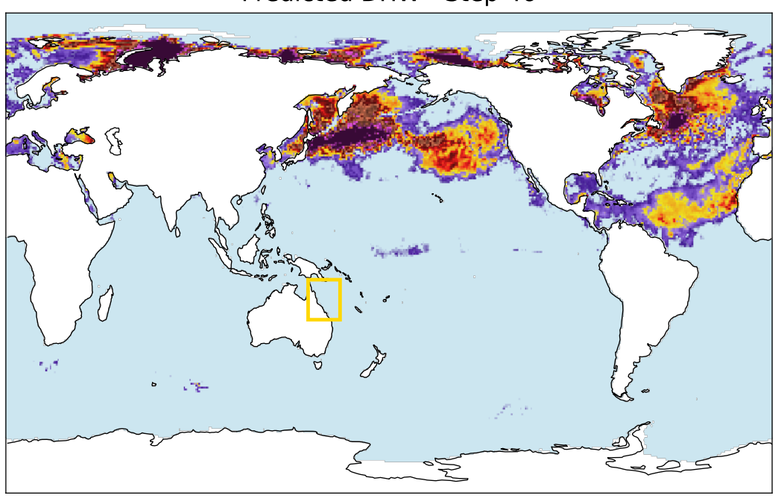
Error (Pred - Actual) - Step 1 [scale ±0.17]



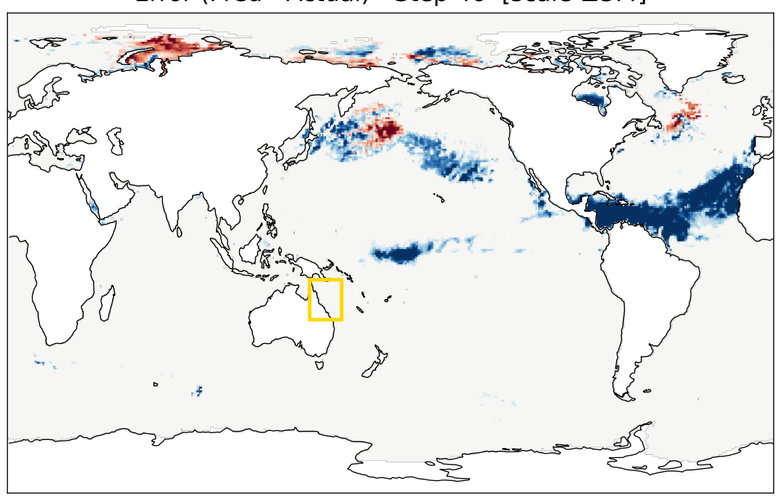
Actual DHW - Step 46



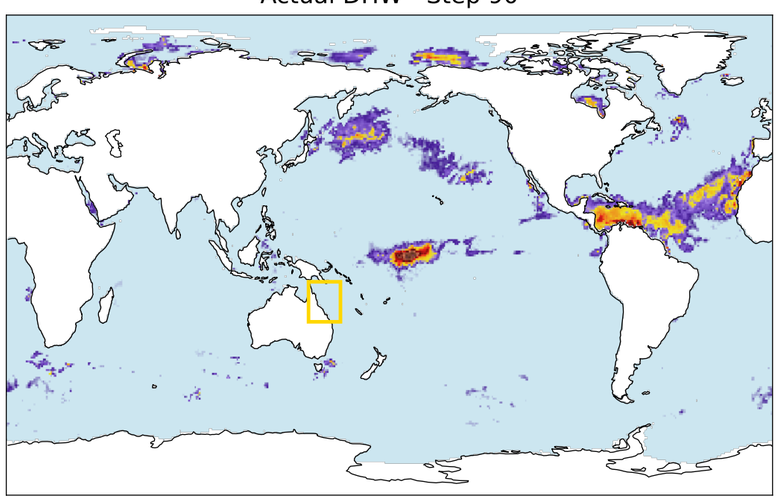
Predicted DHW - Step 46



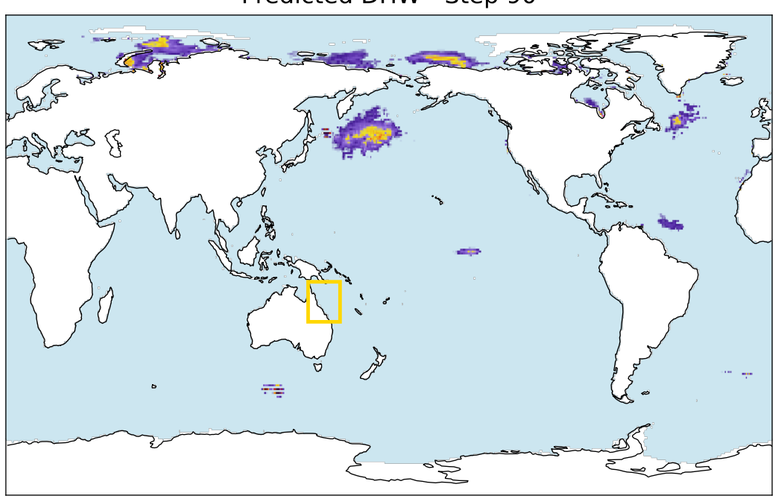
Error (Pred - Actual) - Step 46 [scale ±3.4]



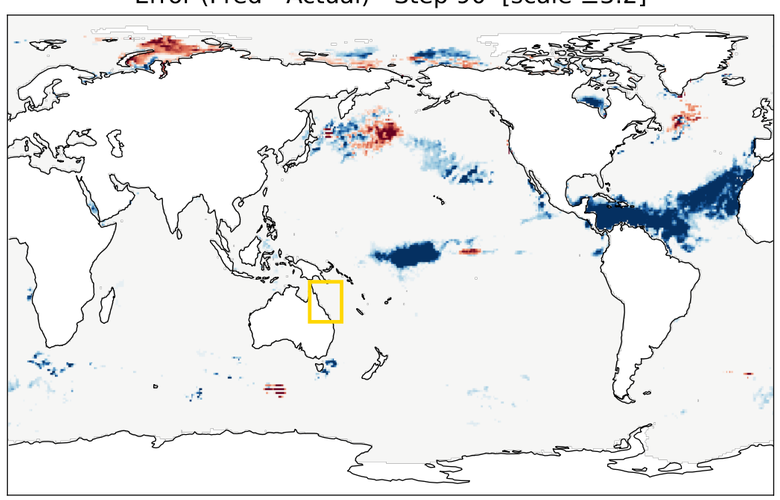
Actual DHW - Step 90

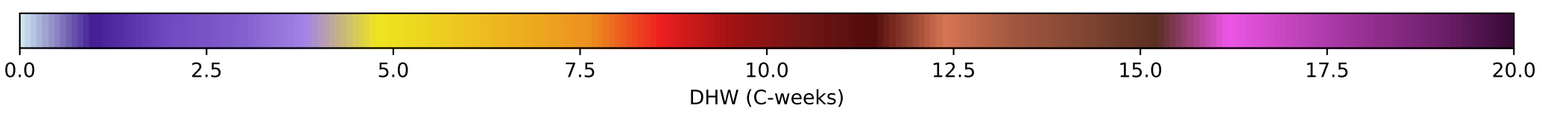


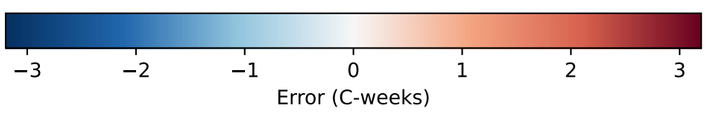
Predicted DHW - Step 90



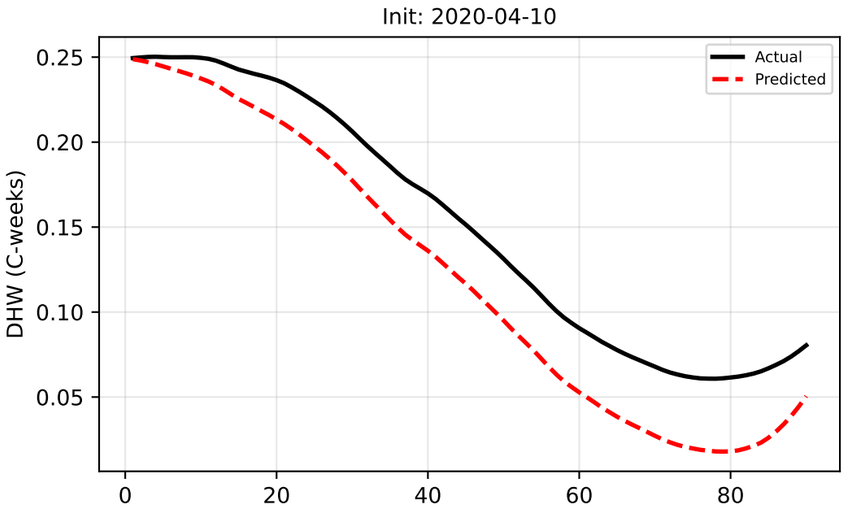
Error (Pred - Actual) - Step 90 [scale ±3.2]

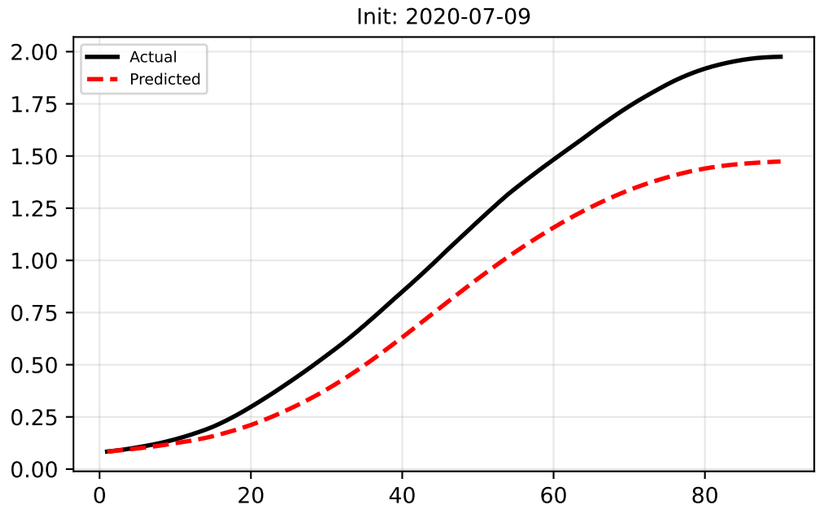


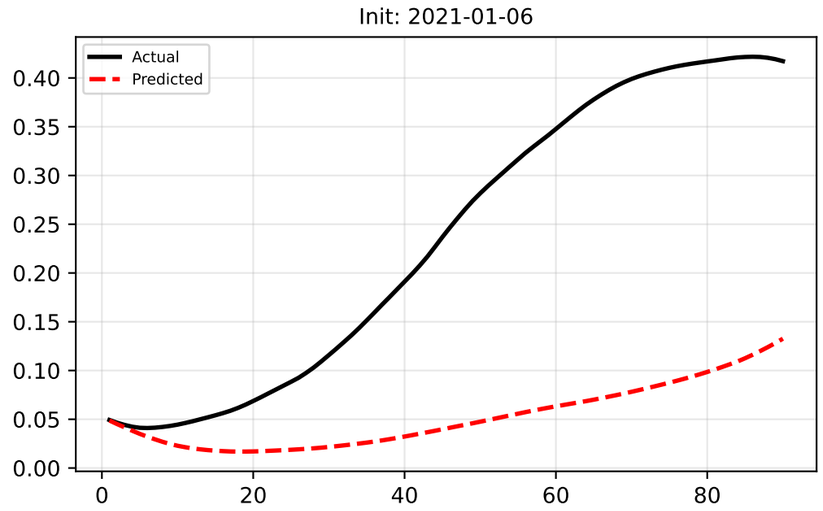


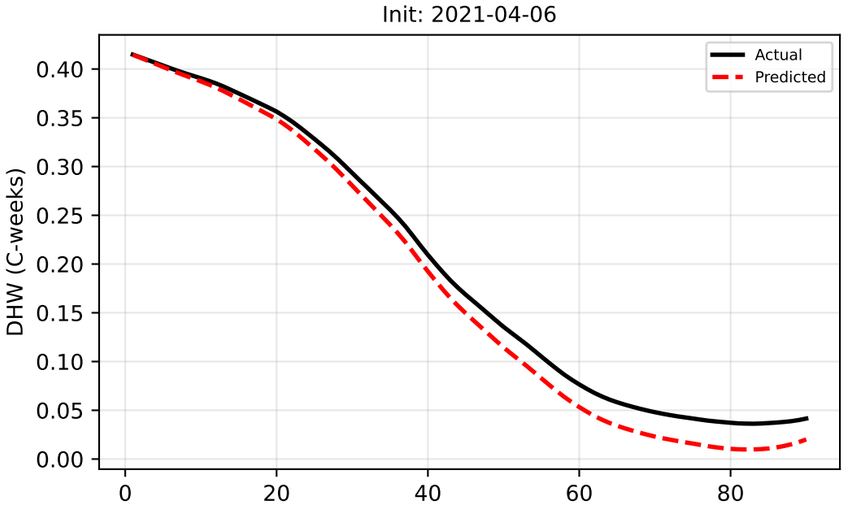


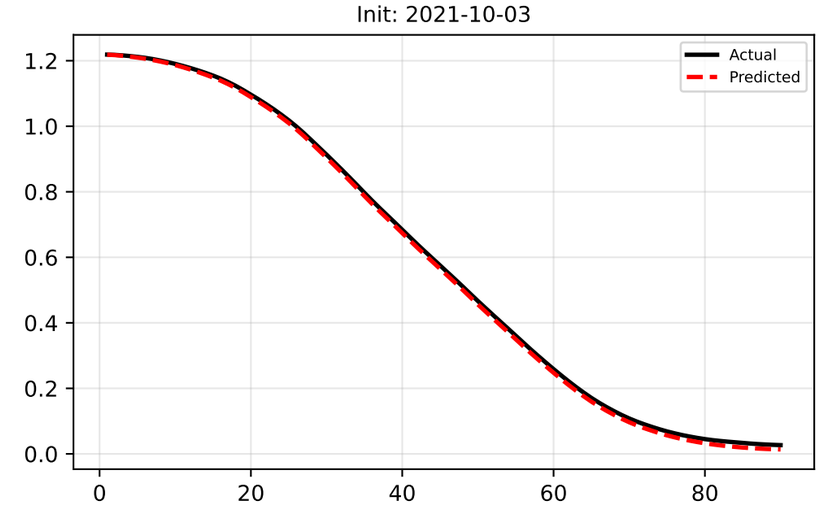
Global-Mean DHW - Test (C-weeks)

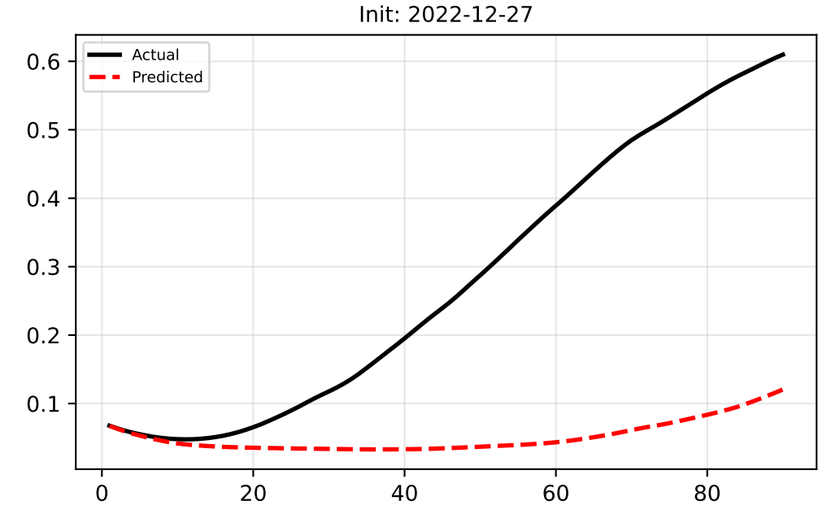


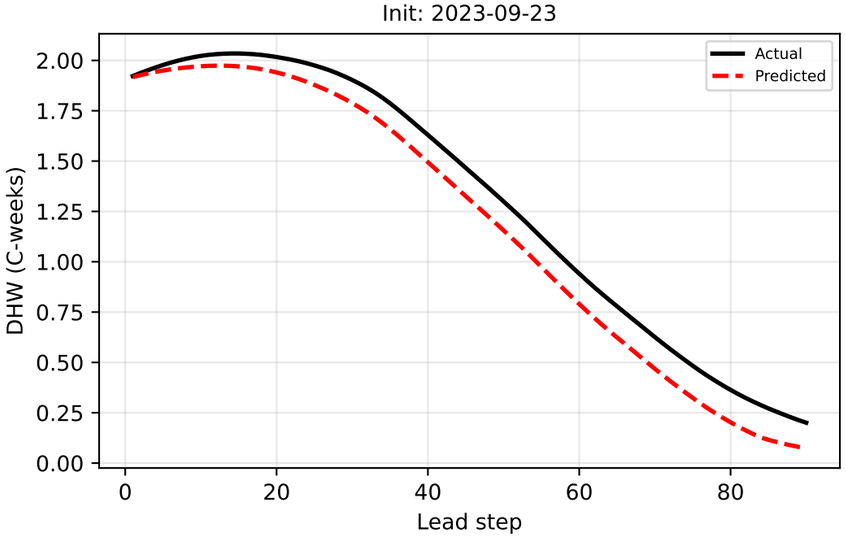




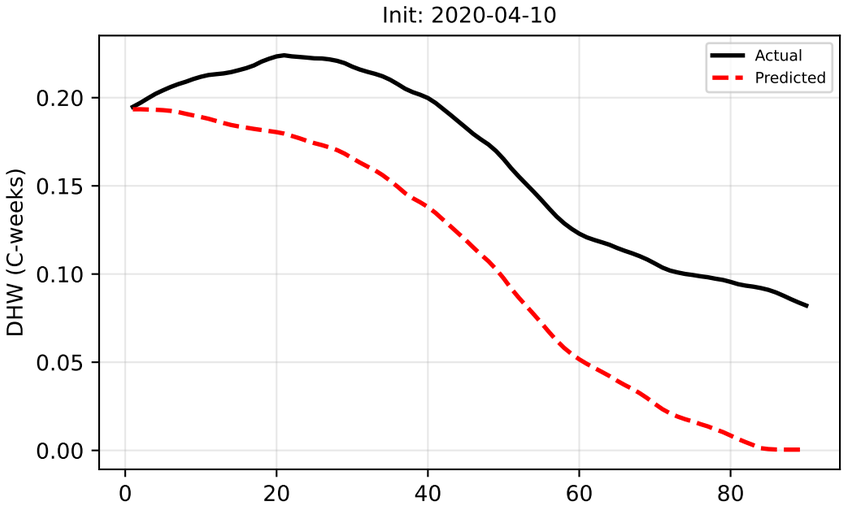


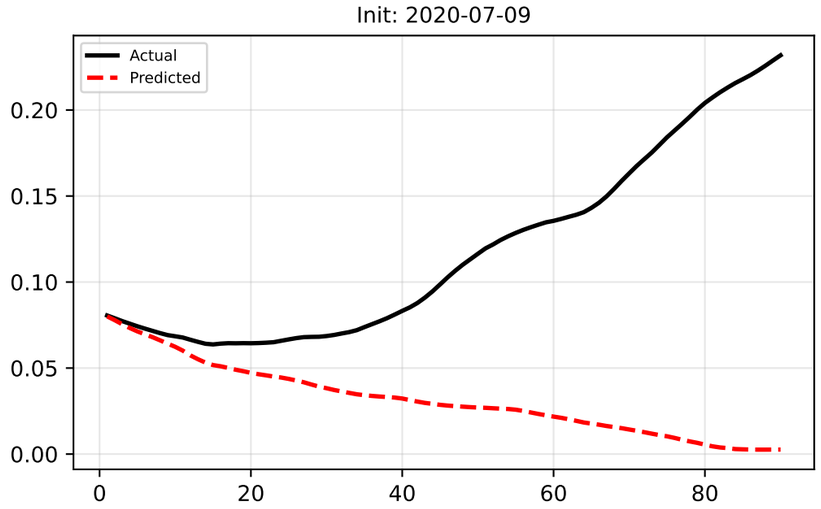


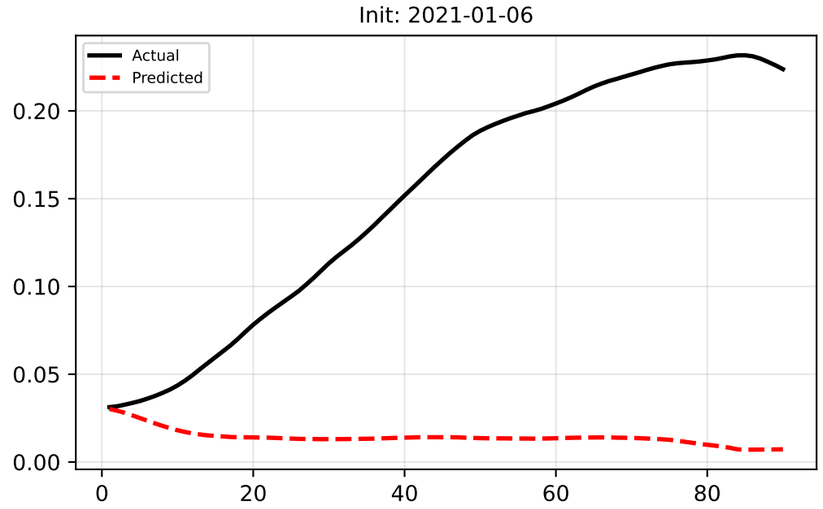


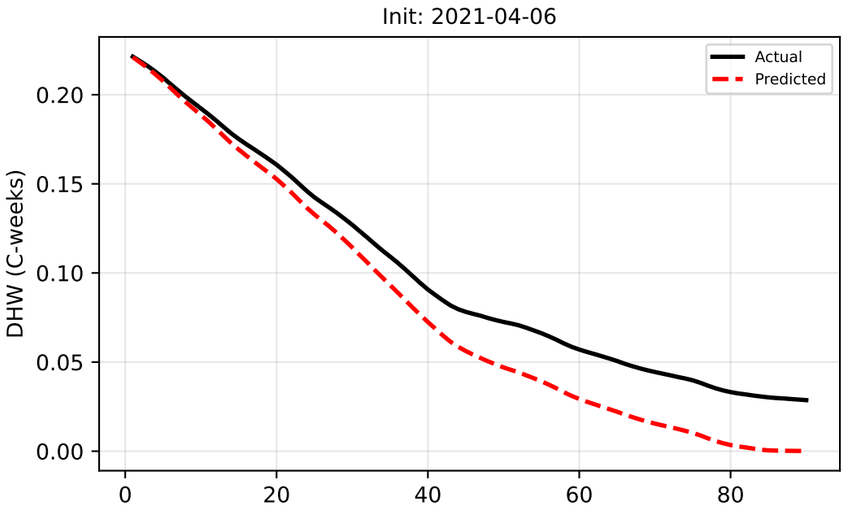


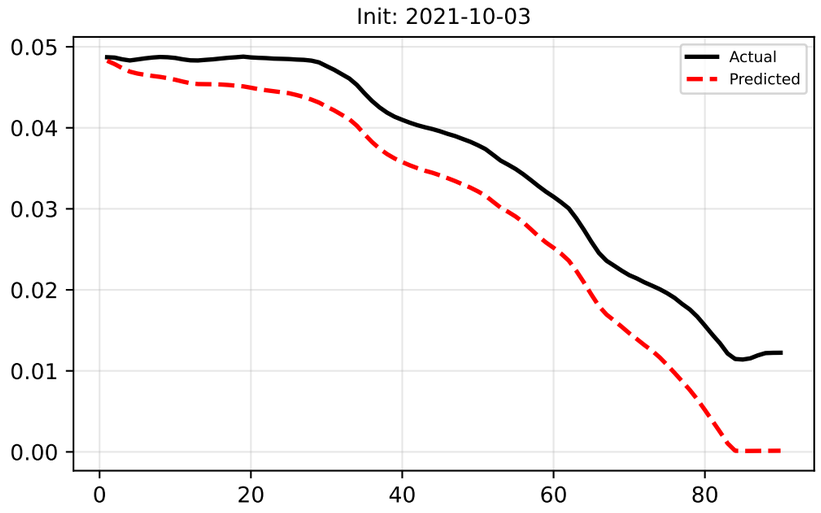
Tropical-Mean DHW (|lat|<=30) - Test (C-weeks)

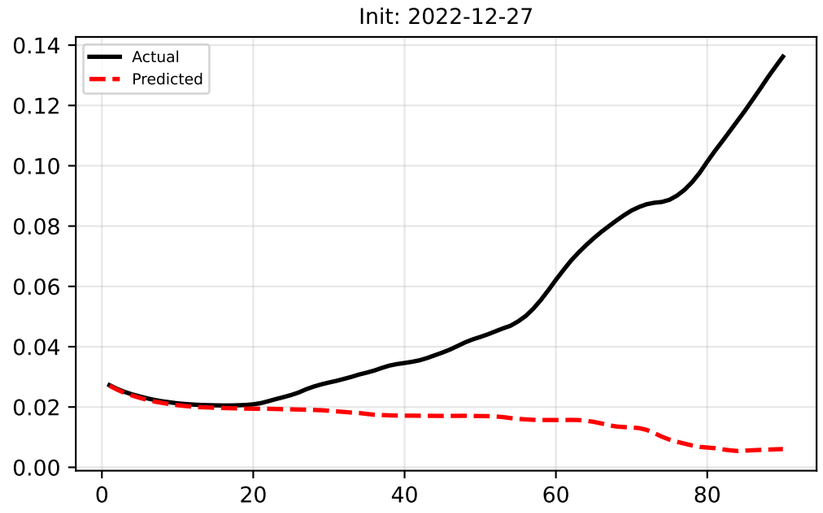


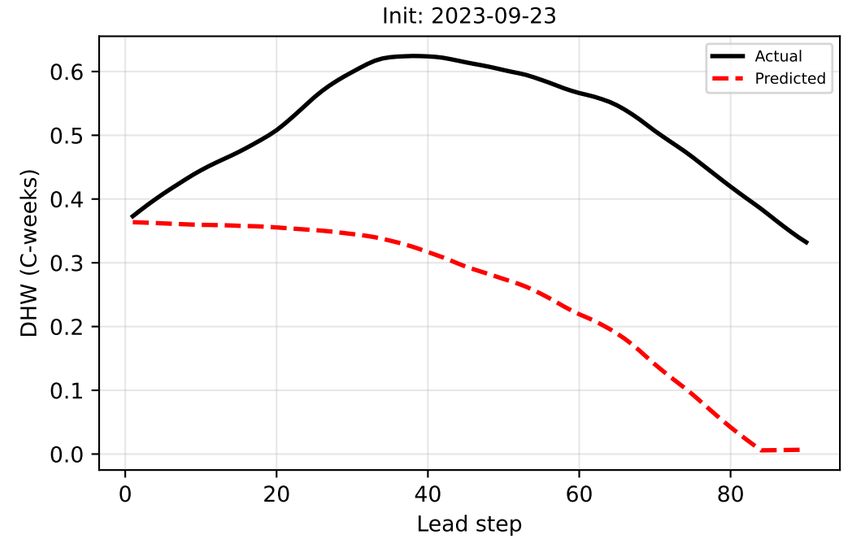


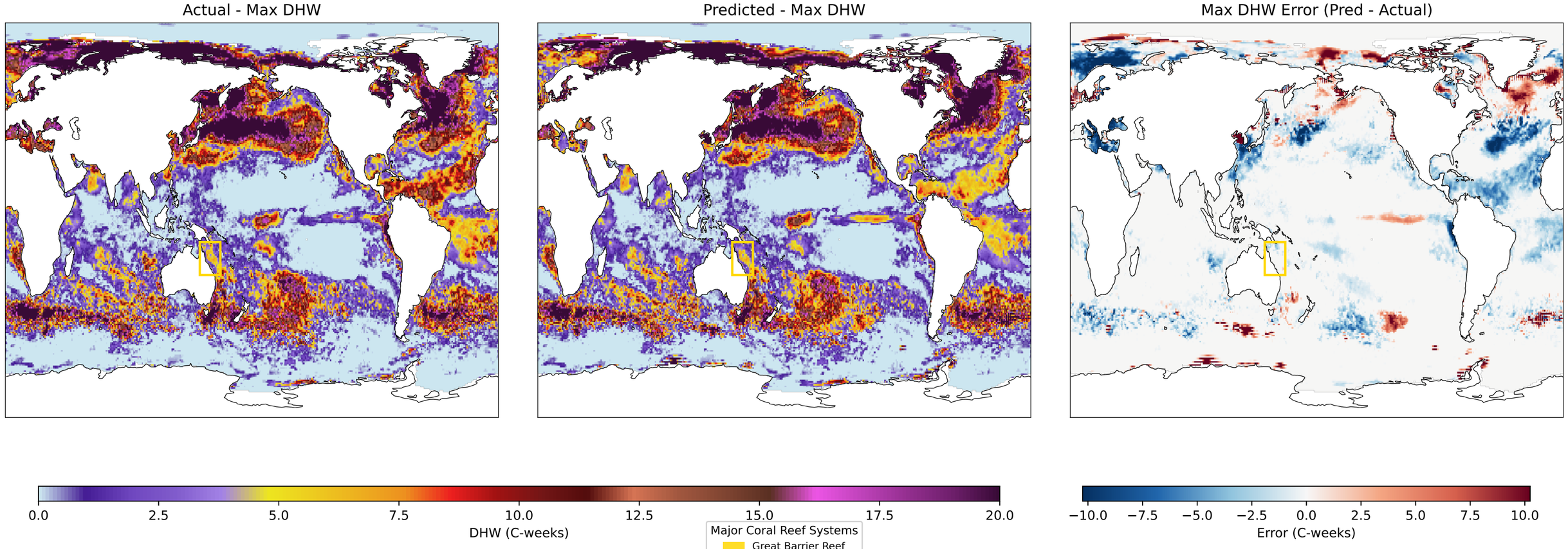


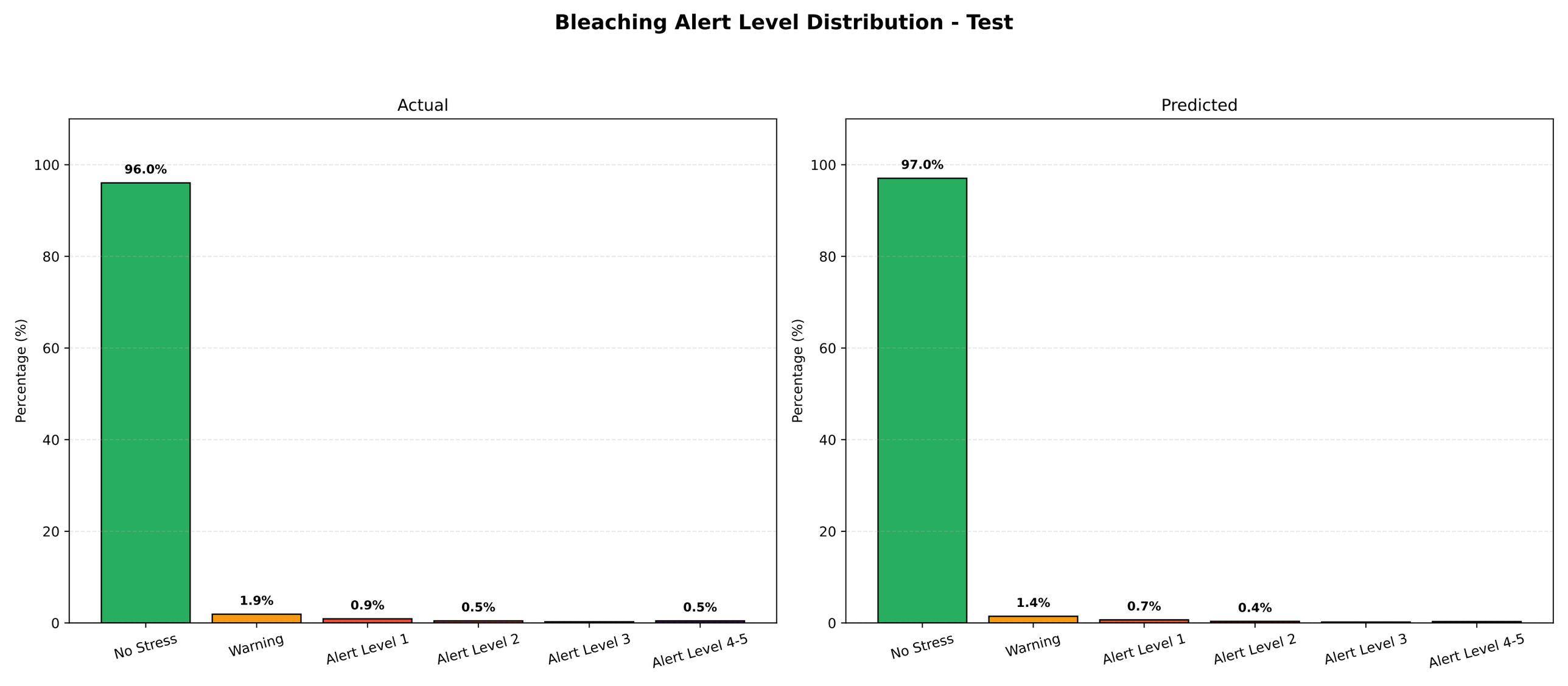






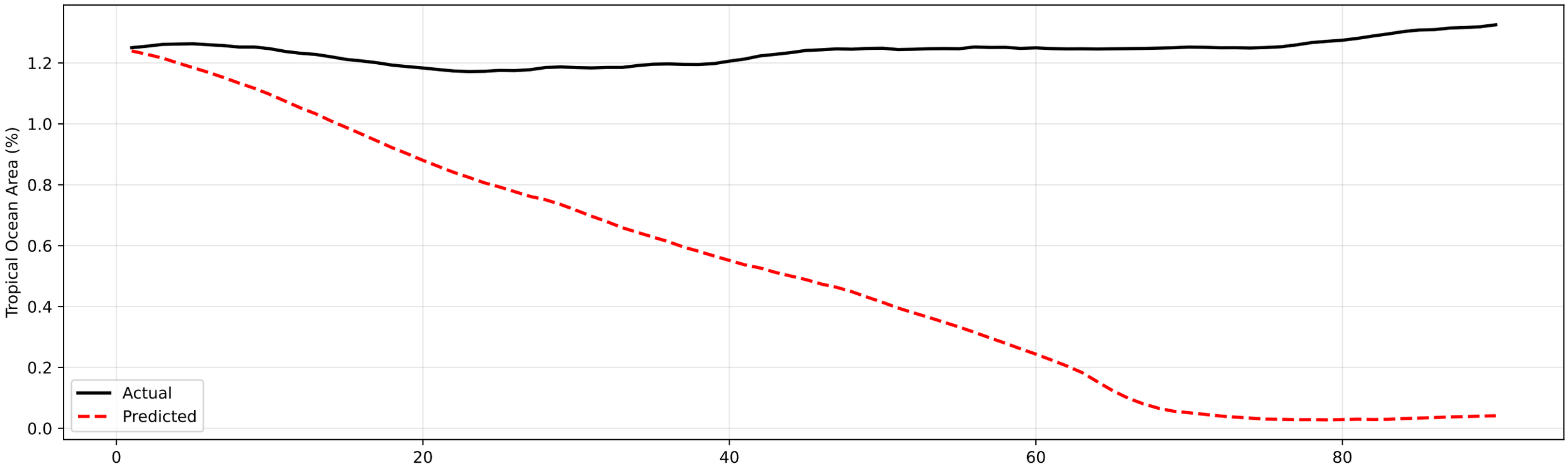



Peak DHW (max over all chains & leads) - Test (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
Error (C-weeks)

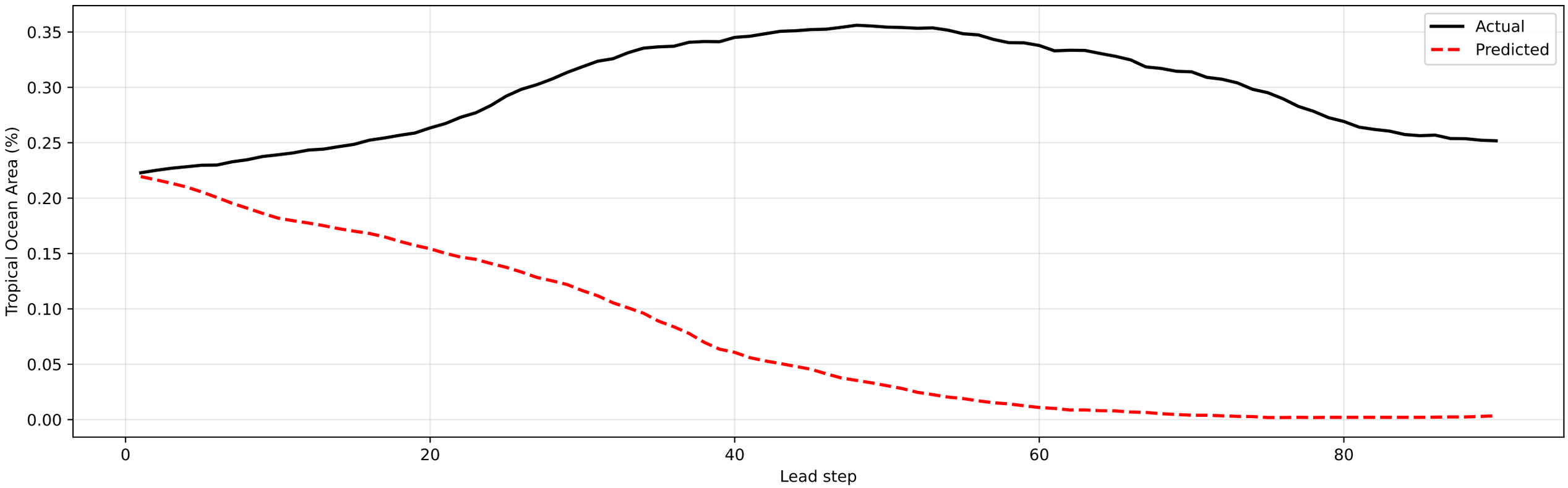

Bleaching Alert Level Distribution - Test
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
96.0%
1.9%
0.9%
0.5%
0.5%
97.0%
1.4%
0.7%
0.4%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Test

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)



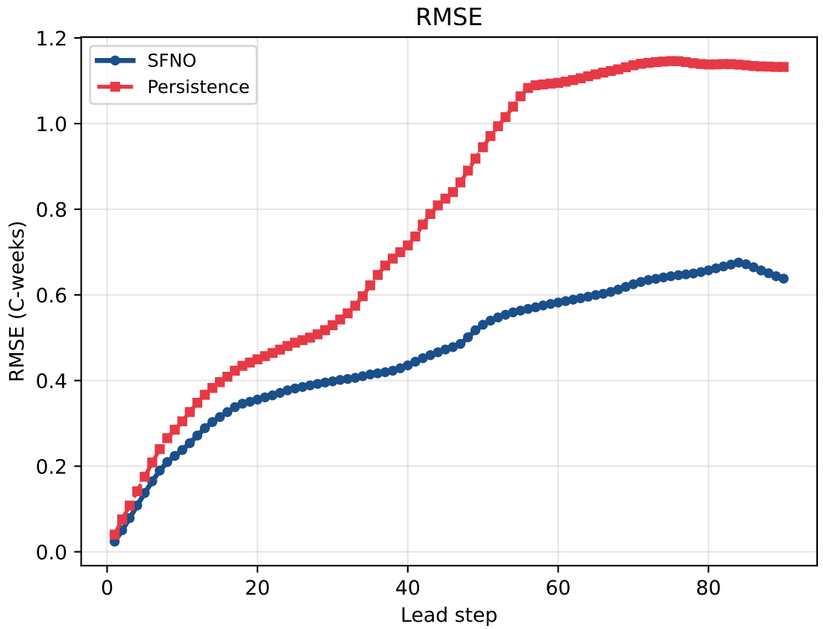
Great Barrier Reef Analysis - Test

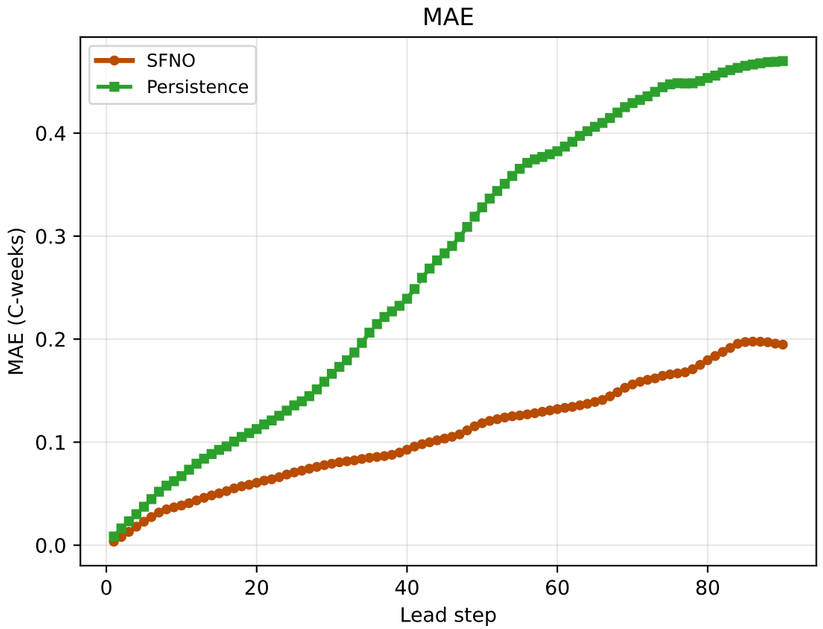
```
                    GREAT BARRIER REEF - TEST
Lat -25.0 deg to -10.0 deg  |  Lon 142.0 deg to 157.0 deg  (0-360)
                    Grid: 21 lat x 21 lon points
```

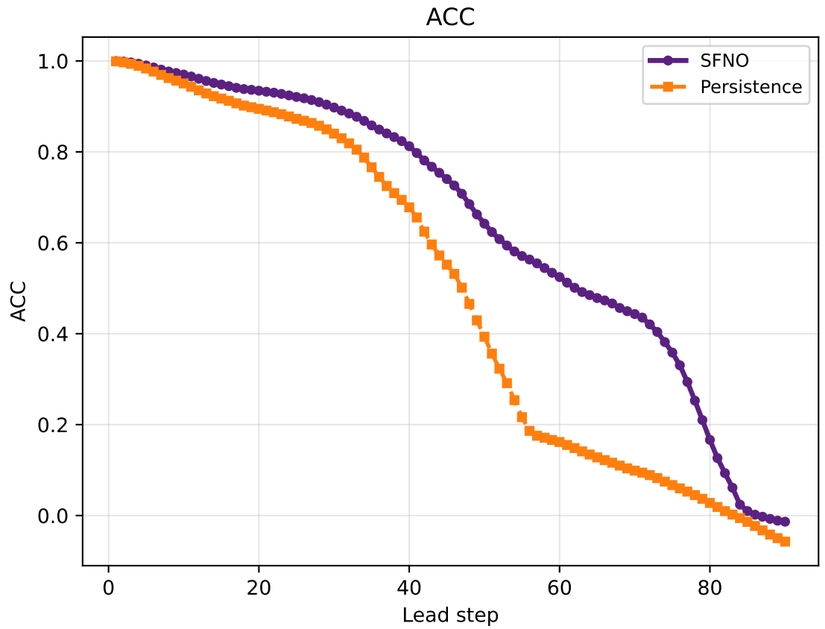
DHW Metrics by Lead Step - Test - GBR (18 chains scored at every lead)

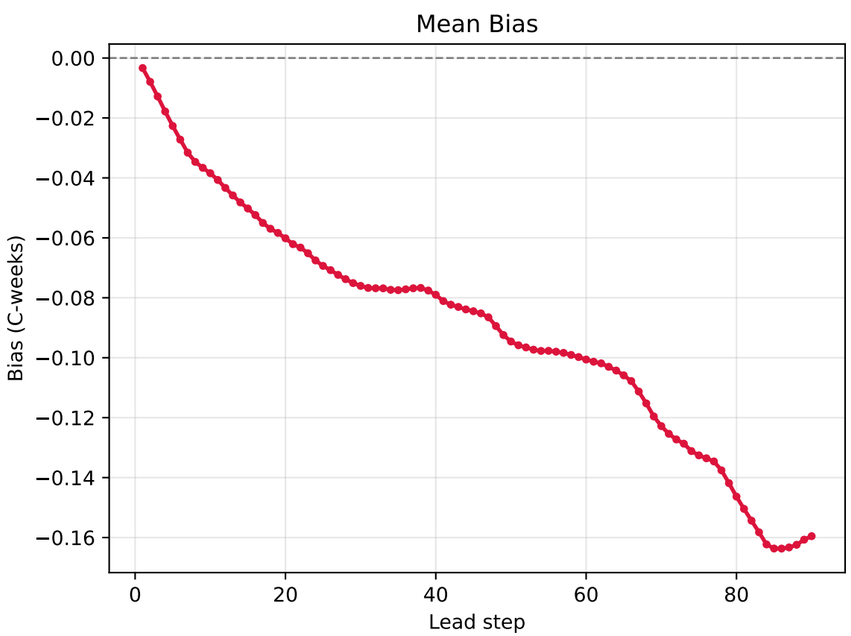
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0238	0.0400	40.6%	0.0034	0.0084	-0.0033	0.9997	0.9991	0.9994
2	0.0502	0.0757	33.6%	0.0080	0.0163	-0.0079	0.9986	0.9969	0.9973
3	0.0787	0.1080	27.2%	0.0129	0.0232	-0.0128	0.9967	0.9937	0.9935
4	0.1082	0.1414	23.5%	0.0179	0.0301	-0.0178	0.9937	0.9892	0.9879
5	0.1373	0.1751	21.6%	0.0228	0.0373	-0.0227	0.9899	0.9835	0.9806
6	0.1646	0.2084	21.0%	0.0273	0.0447	-0.0272	0.9856	0.9767	0.9723
7	0.1897	0.2397	20.9%	0.0317	0.0518	-0.0316	0.9809	0.9693	0.9634
8	0.2096	0.2654	21.0%	0.0348	0.0578	-0.0347	0.9767	0.9624	0.9555
9	0.2240	0.2852	21.5%	0.0367	0.0621	-0.0366	0.9735	0.9567	0.9494
10	0.2377	0.3047	22.0%	0.0385	0.0670	-0.0384	0.9701	0.9505	0.9429
11	0.2537	0.3264	22.3%	0.0408	0.0733	-0.0407	0.9658	0.9431	0.9348
12	0.2715	0.3480	22.0%	0.0434	0.0791	-0.0433	0.9608	0.9354	0.9255
13	0.2886	0.3666	21.3%	0.0459	0.0839	-0.0458	0.9559	0.9286	0.9163
14	0.3030	0.3827	20.8%	0.0483	0.0885	-0.0482	0.9517	0.9225	0.9083
15	0.3147	0.3962	20.6%	0.0503	0.0925	-0.0502	0.9481	0.9173	0.9017
16	0.3263	0.4089	20.2%	0.0525	0.0959	-0.0524	0.9446	0.9123	0.8952
17	0.3378	0.4228	20.1%	0.0551	0.1006	-0.0550	0.9409	0.9067	0.8884
18	0.3459	0.4343	20.4%	0.0571	0.1051	-0.0570	0.9382	0.9018	0.8834
19	0.3504	0.4419	20.7%	0.0586	0.1088	-0.0584	0.9365	0.8983	0.8803
20	0.3555	0.4495	20.9%	0.0606	0.1127	-0.0602	0.9345	0.8947	0.8767
21	0.3607	0.4569	21.1%	0.0627	0.1172	-0.0621	0.9323	0.8909	0.8728
22	0.3653	0.4642	21.3%	0.0640	0.1211	-0.0632	0.9302	0.8872	0.8690
23	0.3710	0.4722	21.4%	0.0660	0.1256	-0.0651	0.9275	0.8829	0.8642
24	0.3771	0.4807	21.6%	0.0686	0.1306	-0.0675	0.9242	0.8779	0.8585
25	0.3814	0.4884	21.9%	0.0706	0.1357	-0.0694	0.9209	0.8727	0.8527
26	0.3849	0.4943	22.1%	0.0723	0.1396	-0.0708	0.9180	0.8684	0.8477
27	0.3886	0.5002	22.3%	0.0742	0.1446	-0.0723	0.9141	0.8636	0.8410
28	0.3921	0.5078	22.8%	0.0760	0.1512	-0.0738	0.9094	0.8574	0.8331
29	0.3952	0.5177	23.7%	0.0777	0.1587	-0.0751	0.9039	0.8494	0.8237
30	0.3981	0.5293	24.8%	0.0791	0.1663	-0.0760	0.8976	0.8401	0.8131
31	0.4014	0.5425	26.0%	0.0805	0.1731	-0.0767	0.8908	0.8298	0.8019
32	0.4036	0.5568	27.5%	0.0814	0.1795	-0.0768	0.8843	0.8187	0.7911
33	0.4062	0.5747	29.3%	0.0823	0.1870	-0.0768	0.8772	0.8049	0.7794
34	0.4101	0.5968	31.3%	0.0837	0.1964	-0.0773	0.8682	0.7873	0.7650
35	0.4142	0.6223	33.4%	0.0848	0.2063	-0.0775	0.8582	0.7661	0.7491
36	0.4168	0.6465	35.5%	0.0856	0.2146	-0.0772	0.8491	0.7450	0.7350
37	0.4194	0.6685	37.3%	0.0865	0.2215	-0.0768	0.8406	0.7248	0.7222
38	0.4229	0.6847	38.2%	0.0877	0.2269	-0.0767	0.8326	0.7092	0.7101
39	0.4286	0.6997	38.7%	0.0899	0.2324	-0.0776	0.8236	0.6944	0.6966
40	0.4354	0.7155	39.2%	0.0926	0.2392	-0.0790	0.8125	0.6780	0.6798
41	0.4439	0.7363	39.7%	0.0958	0.2487	-0.0811	0.7975	0.6556	0.6574
42	0.4521	0.7643	40.8%	0.0980	0.2595	-0.0823	0.7811	0.6245	0.6332
43	0.4593	0.7888	41.8%	0.0999	0.2685	-0.0831	0.7673	0.5963	0.6131
44	0.4659	0.8089	42.4%	0.1018	0.2764	-0.0839	0.7540	0.5720	0.5938
45	0.4724	0.8247	42.7%	0.1035	0.2833	-0.0845	0.7403	0.5520	0.5742
46	0.4780	0.8400	43.1%	0.1052	0.2904	-0.0852	0.7260	0.5316	0.5540
47	0.4854	0.8626	43.7%	0.1076	0.2992	-0.0865	0.7078	0.5012	0.5289
48	0.5009	0.8902	43.7%	0.1114	0.3089	-0.0894	0.6851	0.4654	0.4978
49	0.5175	0.9181	43.6%	0.1154	0.3188	-0.0924	0.6624	0.4293	0.4676
50	0.5302	0.9447	43.9%	0.1184	0.3280	-0.0946	0.6421	0.3933	0.4411
51	0.5398	0.9710	44.4%	0.1206	0.3364	-0.0958	0.6237	0.3562	0.4174
52	0.5473	0.9938	44.9%	0.1223	0.3438	-0.0966	0.6080	0.3229	0.3974
53	0.5536	1.0151	45.5%	0.1239	0.3508	-0.0973	0.5940	0.2911	0.3796
54	0.5591	1.0396	46.2%	0.1252	0.3584	-0.0977	0.5809	0.2538	0.3630
55	0.5632	1.0637	47.1%	0.1259	0.3654	-0.0977	0.5708	0.2164	0.3499
56	0.5672	1.0831	47.6%	0.1270	0.3711	-0.0980	0.5632	0.1862	0.3393
57	0.5710	1.0895	47.6%	0.1281	0.3745	-0.0984	0.5547	0.1755	0.3280
58	0.5753	1.0914	47.3%	0.1295	0.3768	-0.0991	0.5444	0.1712	0.3146
59	0.5788	1.0933	47.1%	0.1308	0.3795	-0.0998	0.5340	0.1663	0.3013
60	0.5821	1.0951	46.8%	0.1321	0.3824	-0.1006	0.5246	0.1618	0.2891
61	0.5855	1.0980	46.7%	0.1333	0.3869	-0.1013	0.5123	0.1553	0.2738
62	0.5889	1.1016	46.5%	0.1343	0.3916	-0.1019	0.5010	0.1484	0.2600
63	0.5921	1.1058	46.5%	0.1358	0.3973	-0.1030	0.4916	0.1411	0.2488
64	0.5957	1.1105	46.4%	0.1372	0.4019	-0.1043	0.4849	0.1343	0.2408
65	0.5996	1.1150	46.2%	0.1390	0.4061	-0.1059	0.4784	0.1279	0.2328
66	0.6025	1.1190	46.2%	0.1410	0.4099	-0.1078	0.4733	0.1219	0.2269
67	0.6069	1.1228	45.9%	0.1445	0.4146	-0.1113	0.4665	0.1162	0.2195
68	0.6123	1.1267	45.7%	0.1484	0.4199	-0.1152	0.4566	0.1098	0.2082
69	0.6187	1.1316	45.3%	0.1528	0.4252	-0.1196	0.4495	0.1037	0.2006
70	0.6248	1.1362	45.0%	0.1560	0.4292	-0.1228	0.4432	0.0987	0.1942
71	0.6301	1.1396	44.7%	0.1586	0.4323	-0.1254	0.4355	0.0943	0.1861
72	0.6348	1.1418	44.4%	0.1605	0.4357	-0.1273	0.4205	0.0888	0.1702
73	0.6374	1.1435	44.3%	0.1619	0.4400	-0.1287	0.4039	0.0823	0.1523
74	0.6407	1.1449	44.0%	0.1644	0.4446	-0.1311	0.3816	0.0746	0.1278
75	0.6435	1.1460	43.8%	0.1658	0.4474	-0.1326	0.3585	0.0670	0.1023
76	0.6459	1.1457	43.6%	0.1668	0.4484	-0.1335	0.3306	0.0598	0.0713
77	0.6479	1.1435	43.3%	0.1678	0.4482	-0.1346	0.2940	0.0527	0.0294
78	0.6500	1.1409	43.0%	0.1708	0.4483	-0.1376	0.2528	0.0450	-0.0193
79	0.6533	1.1392	42.7%	0.1751	0.4505	-0.1419	0.2100	0.0368	-0.0714
80	0.6573	1.1381	42.2%	0.1795	0.4535	-0.1463	0.1664	0.0277	-0.1269
81	0.6620	1.1382	41.8%	0.1836	0.4558	-0.1504	0.1265	0.0187	-0.1792
82	0.6664	1.1390	41.5%	0.1876	0.4588	-0.1544	0.0933	0.0098	-0.2257
83	0.6707	1.1389	41.1%	0.1915	0.4612	-0.1582	0.0612	0.0022	-0.2734
84	0.6754	1.1377	40.6%	0.1955	0.4634	-0.1623	0.0240	-0.0056	-0.3336
85	0.6715	1.1361	40.9%	0.1972	0.4653	-0.1637	0.0099	-0.0139	-0.3664
86	0.6645	1.1345	41.4%	0.1976	0.4666	-0.1637	0.0019	-0.0230	-0.3927
87	0.6570	1.1335	42.0%	0.1975	0.4678	-0.1633	-0.0026	-0.0326	-0.4137
88	0.6505	1.1331	42.6%	0.1970	0.4689	-0.1624	-0.0073	-0.0418	-0.4334
89	0.6436	1.1323	43.2%	0.1956	0.4692	-0.1607	-0.0113	-0.0498	-0.4518
90	0.6379	1.1322	43.7%	0.1947	0.4699	-0.1595	-0.0135	-0.0574	-0.4659

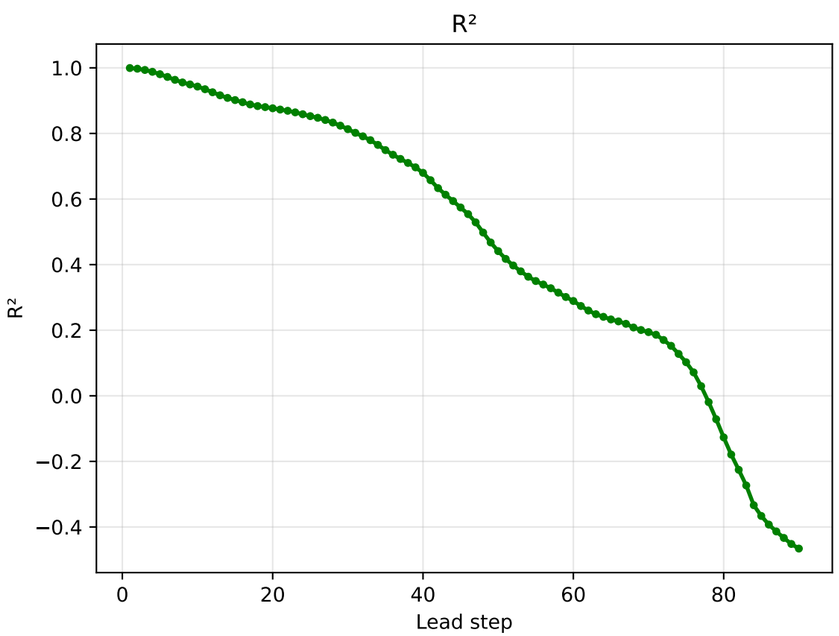
DHW Forecast Skill vs Lead Step - Test - GBR

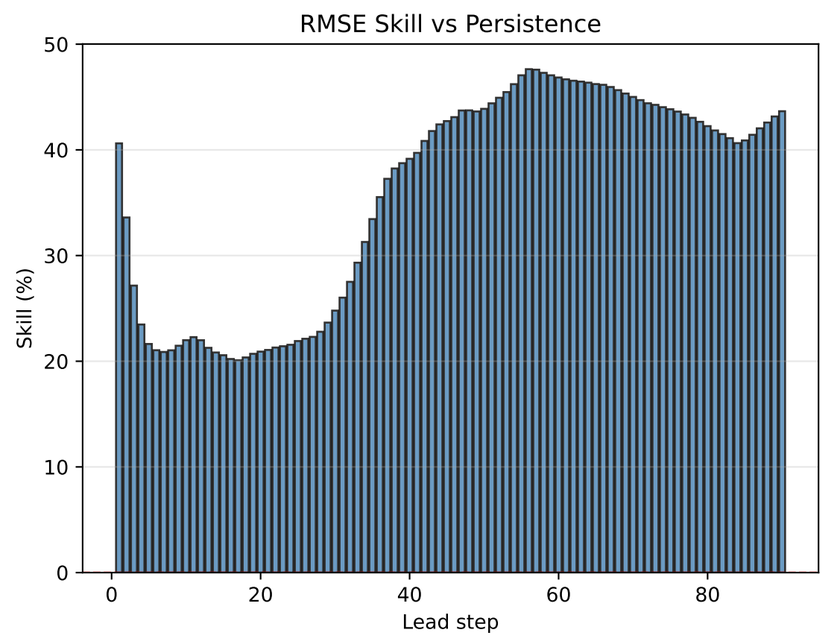


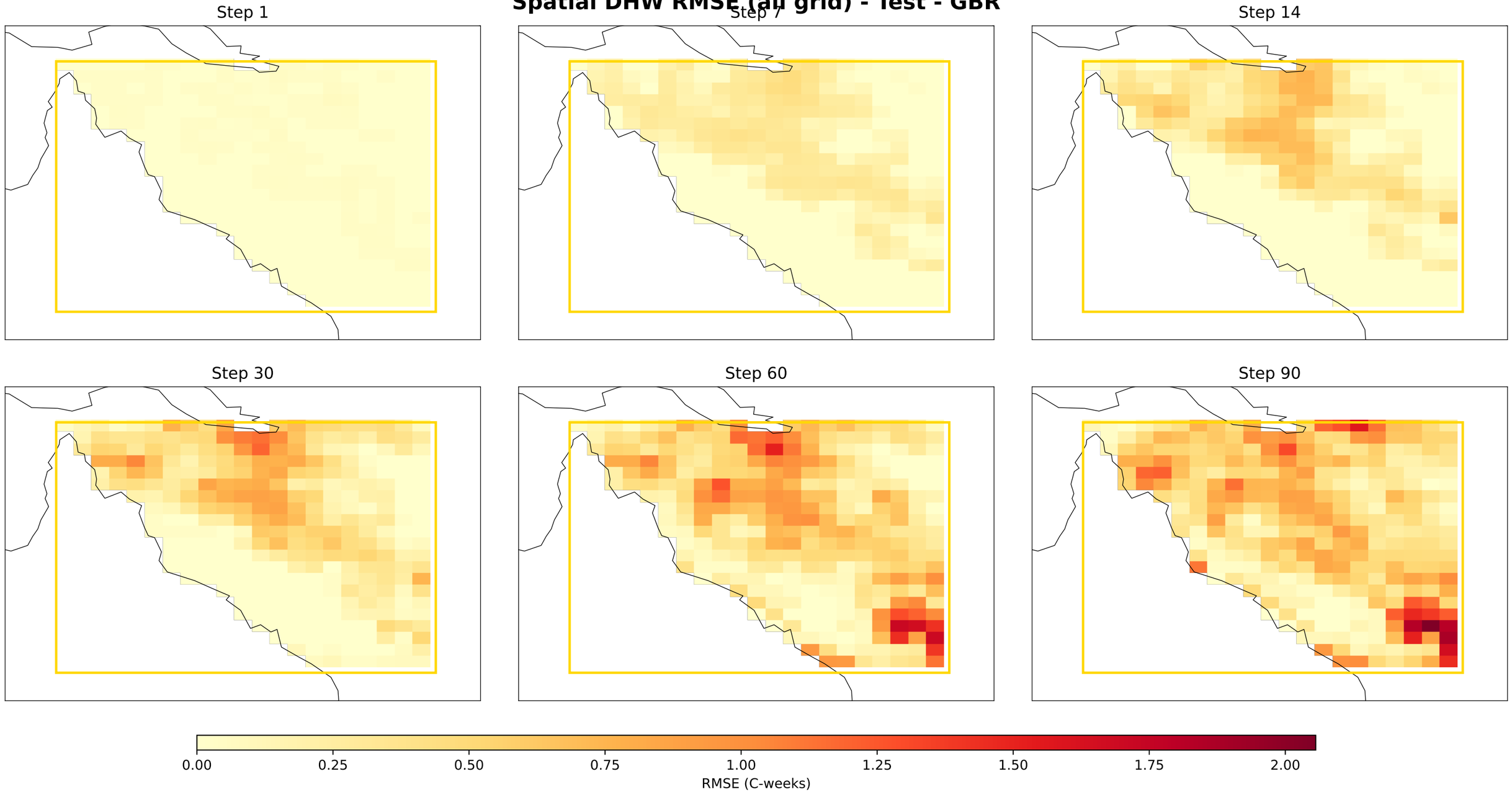


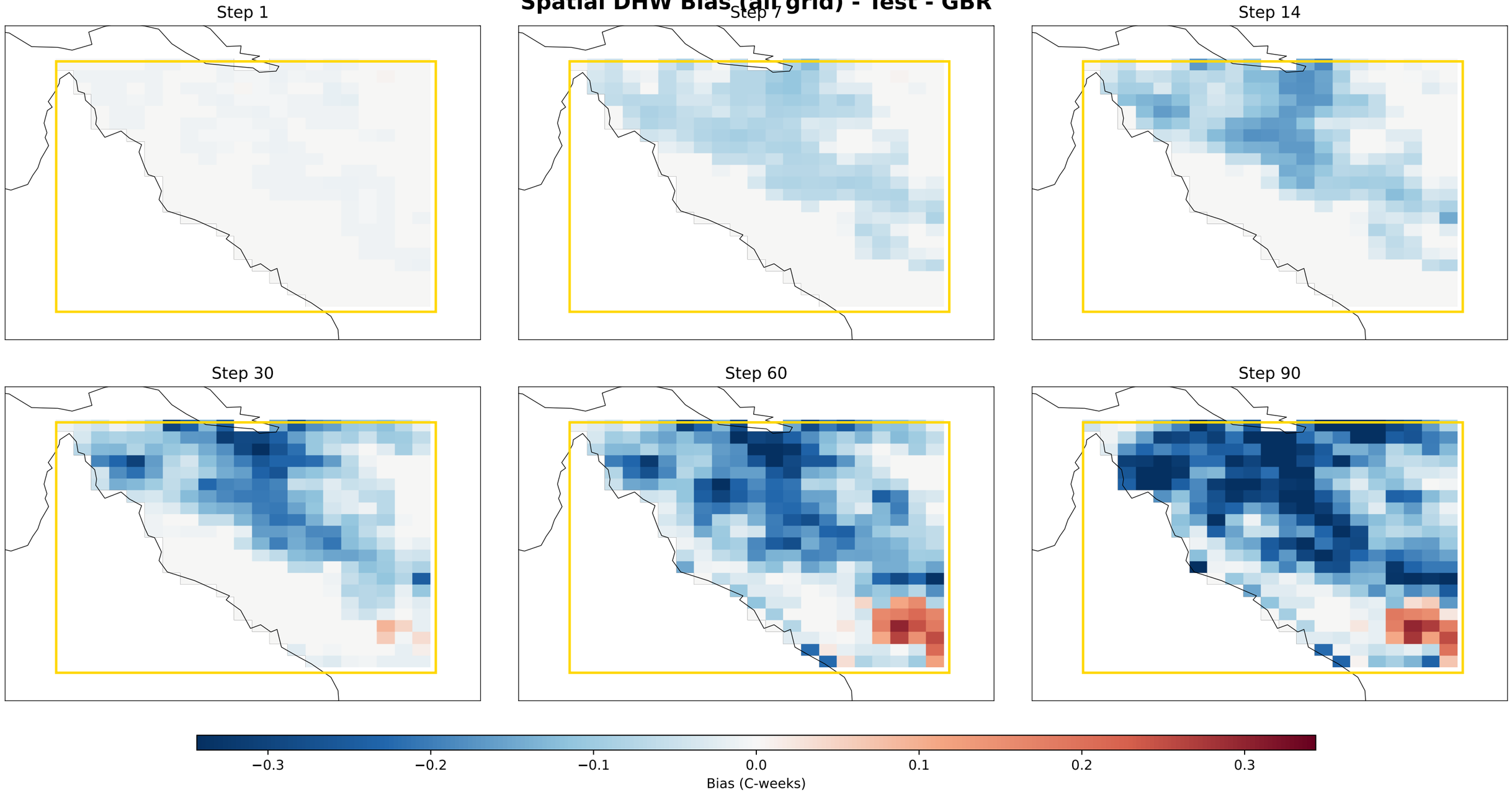


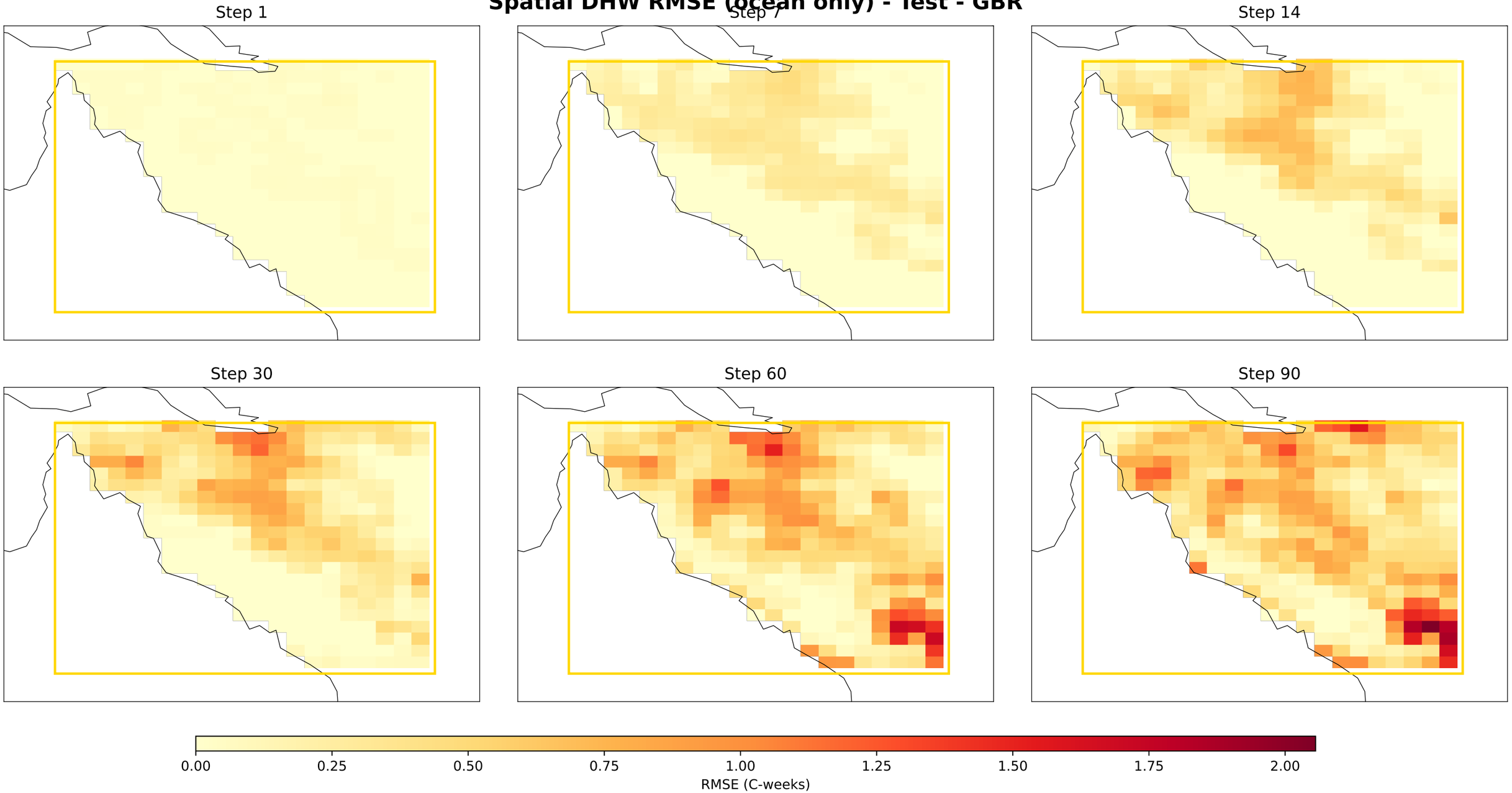


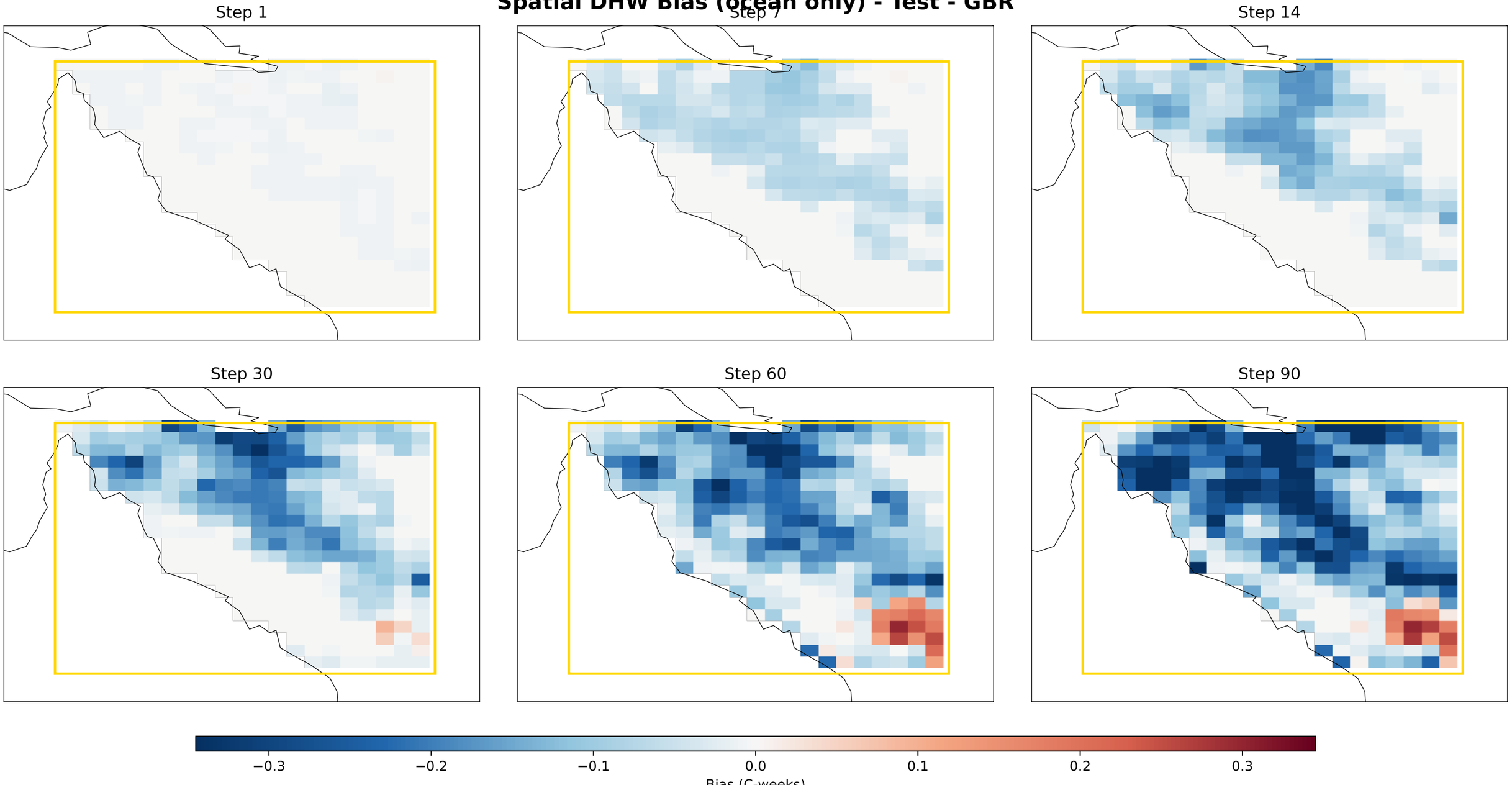





Spatial DHW RMSE (all grid) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
RMSE (C-weeks)

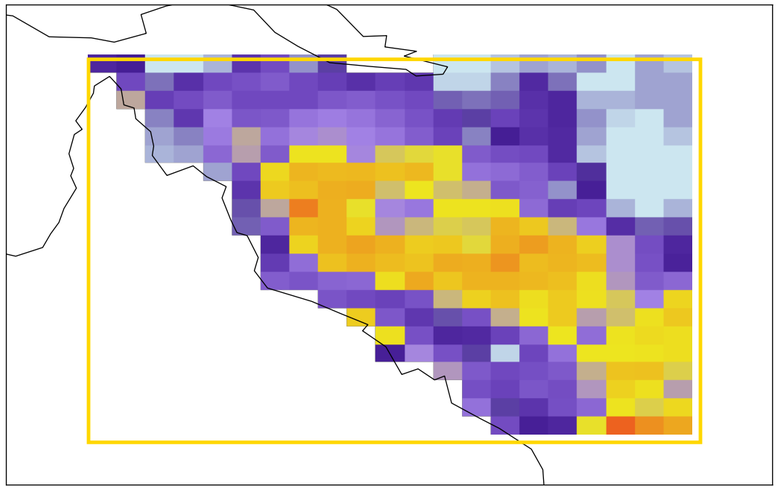

Spatial DHW Bias (all grid) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
RMSE (C-weeks)

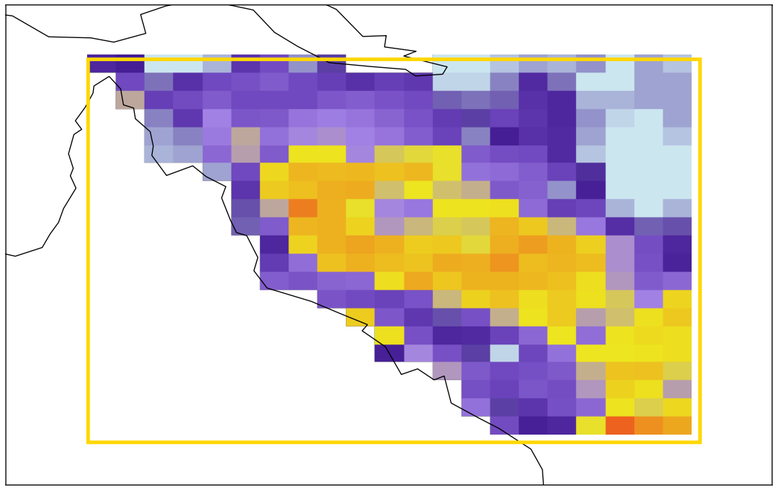

Spatial DHW Bias (ocean only) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Bias (C-weeks)

DHW Chain init: 2020-04-10 - Test - GBR (C-weeks)

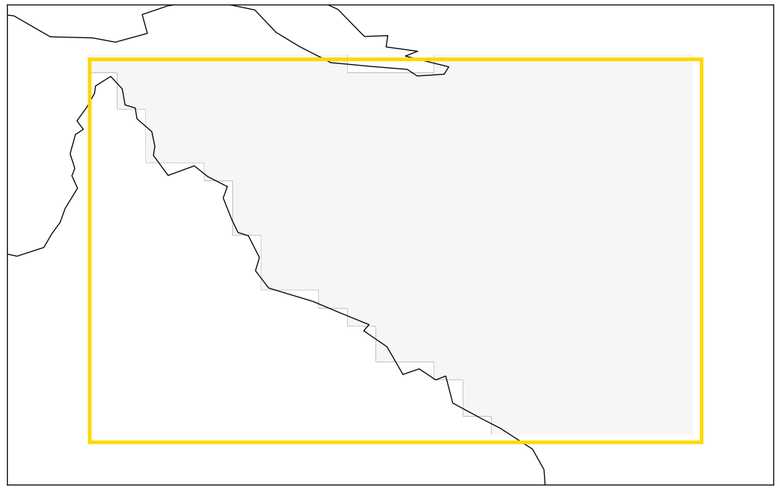
Actual DHW - Step 1



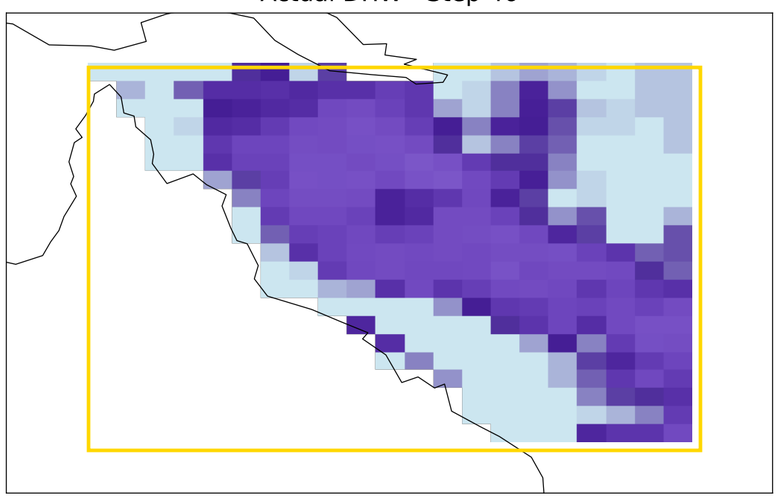
Predicted DHW - Step 1



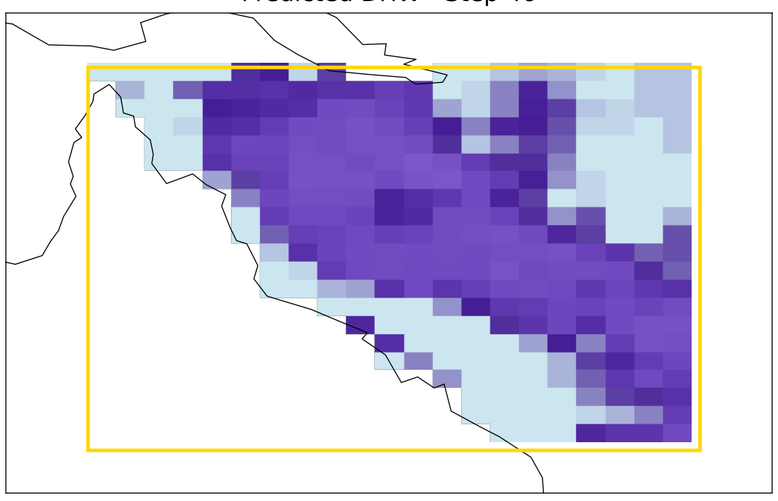
Error (Pred - Actual) - Step 1 [scale ±8]



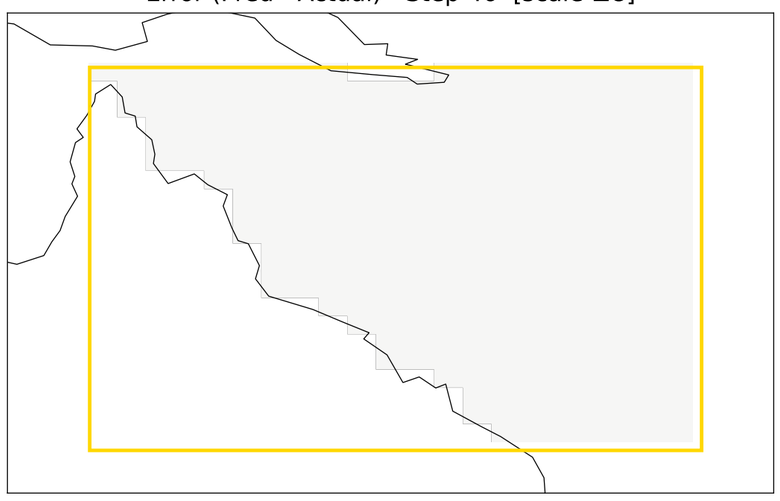
Actual DHW - Step 46



Predicted DHW - Step 46



Error (Pred - Actual) - Step 46 [scale ±8]



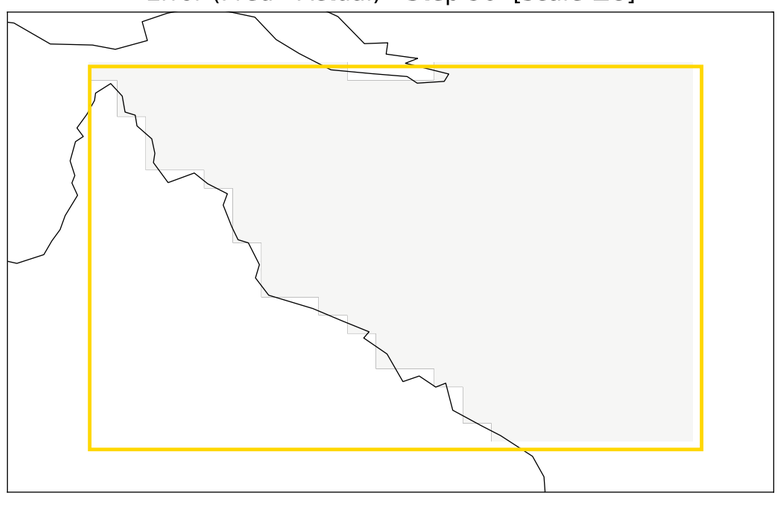
Actual DHW - Step 90

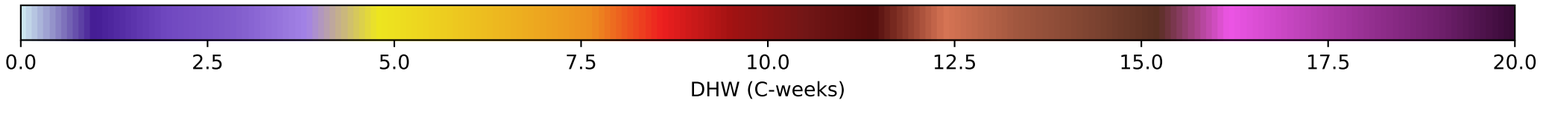


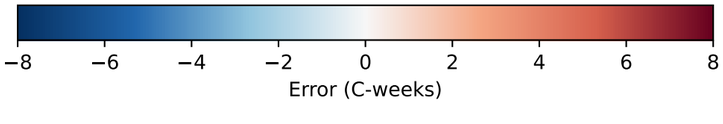
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

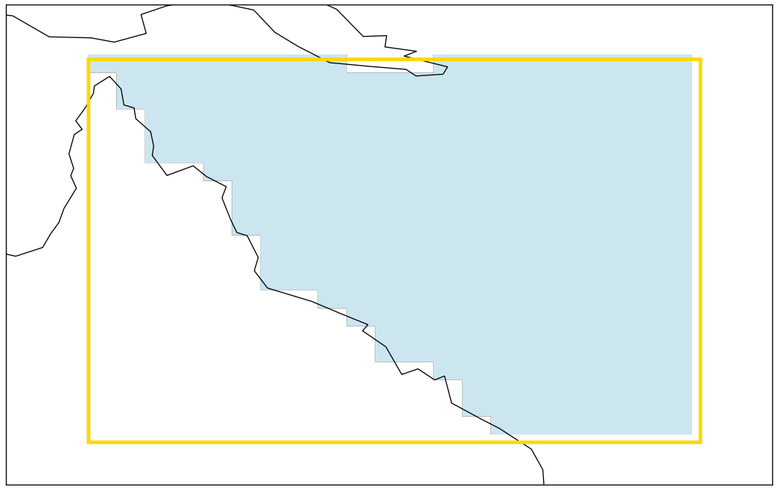




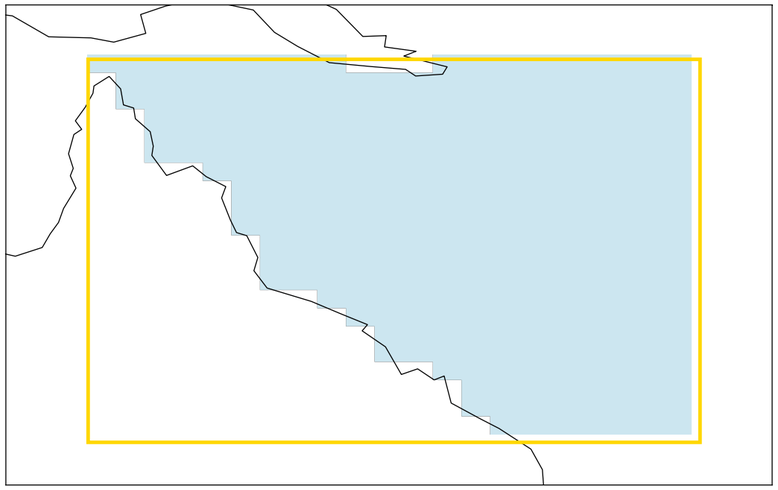


DHW Chain init: 2020-07-09 - Test - GBR (C-weeks)

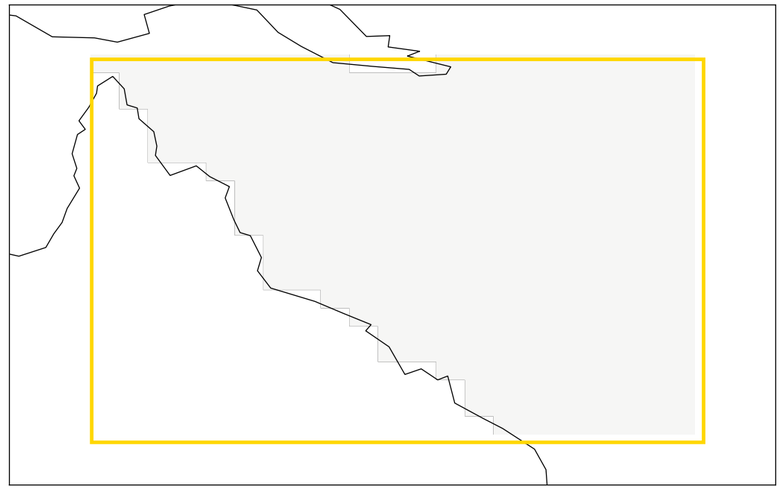
Actual DHW - Step 1



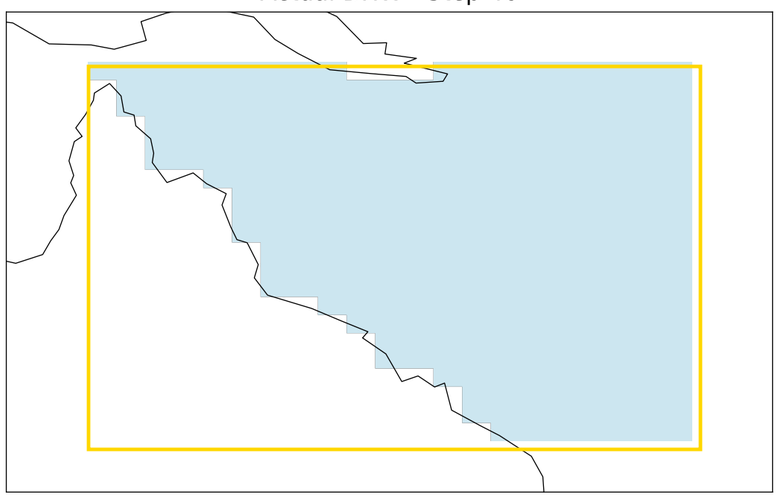
Predicted DHW - Step 1



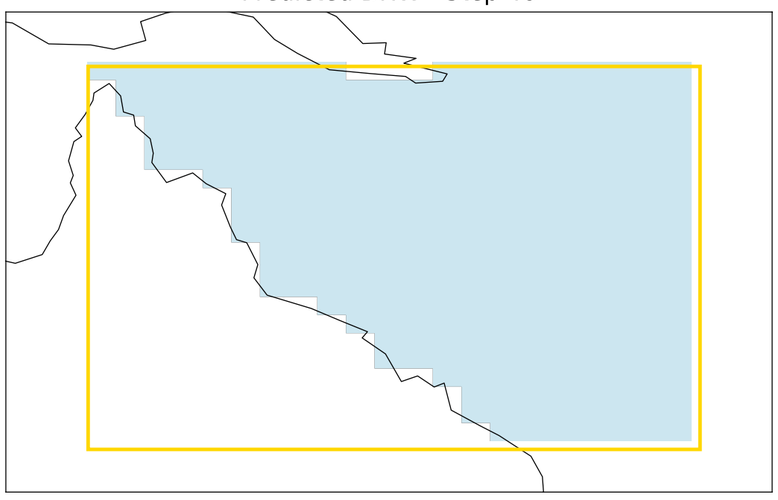
Error (Pred - Actual) - Step 1 [scale ±8]



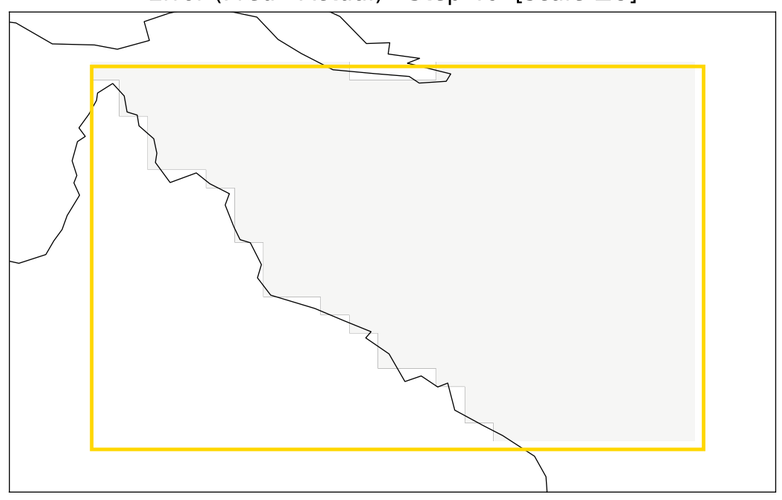
Actual DHW - Step 46



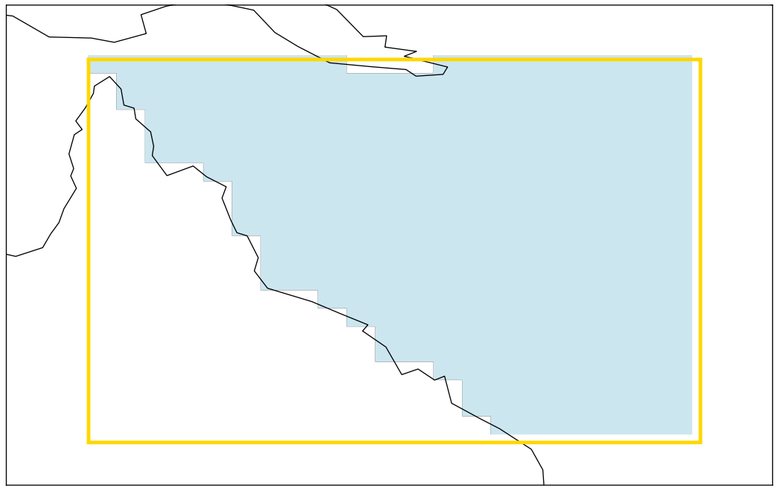
Predicted DHW - Step 46



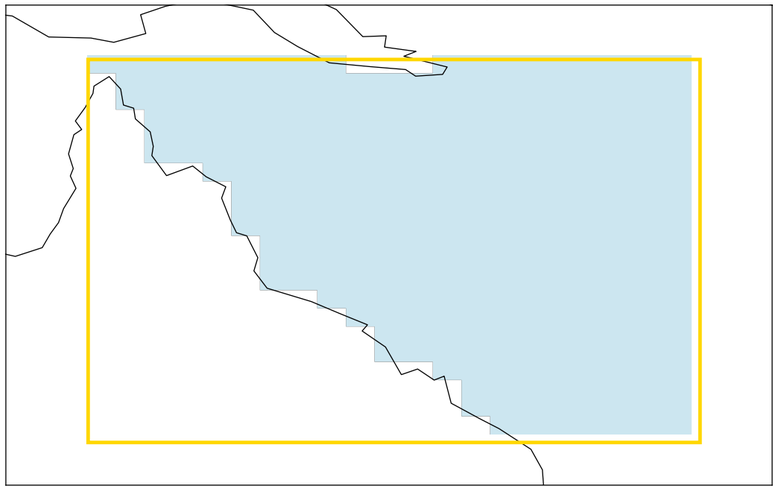
Error (Pred - Actual) - Step 46 [scale ±8]



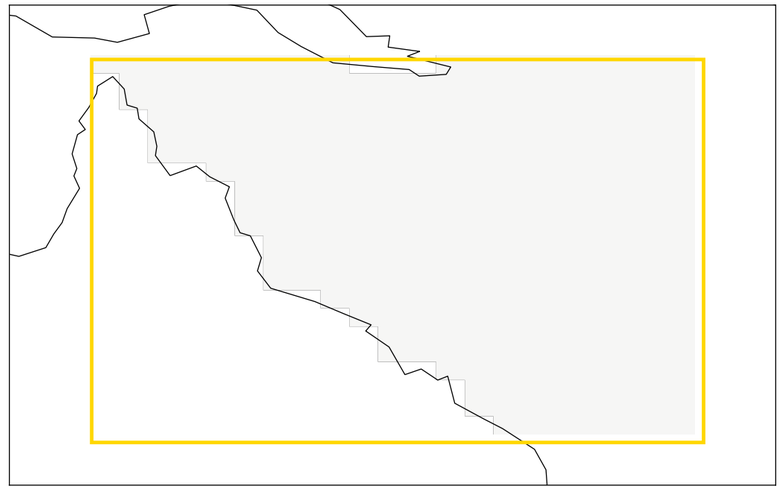
Actual DHW - Step 90

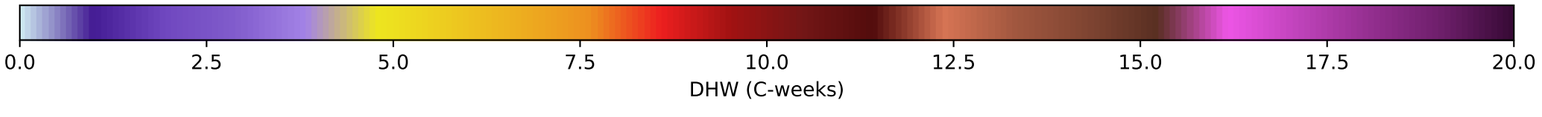


Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

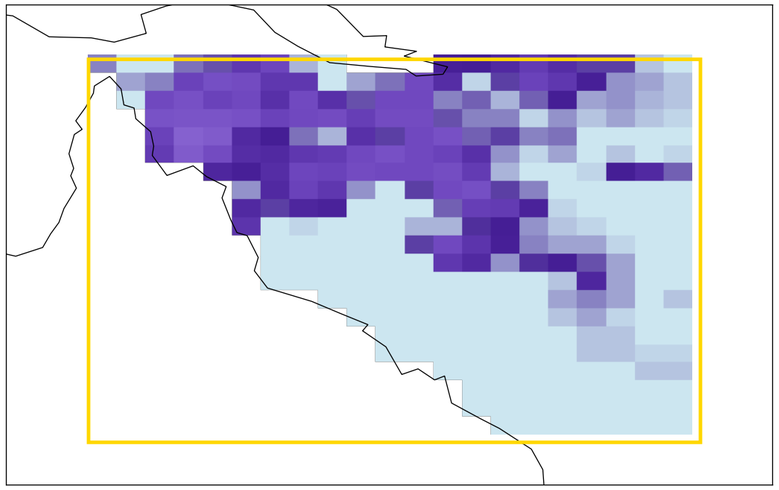




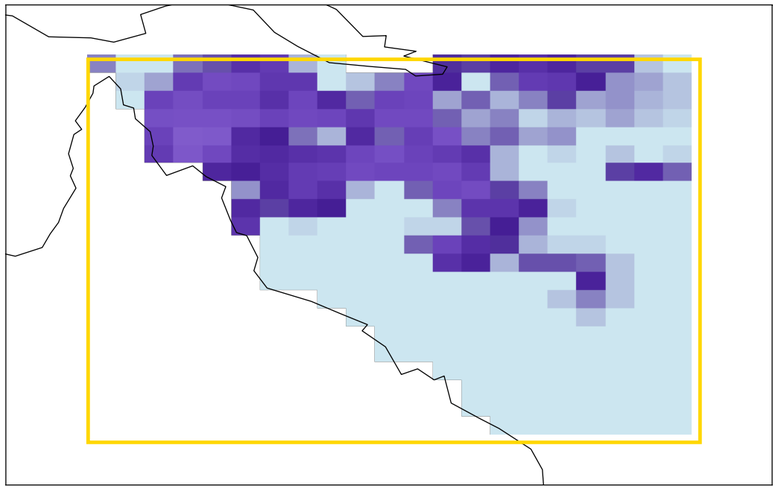


DHW Chain init: 2021-01-06 - Test - GBR (C-weeks)

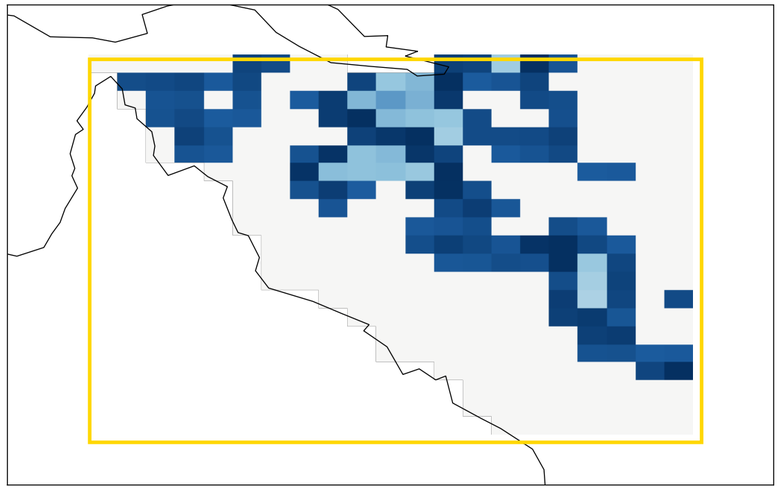
Actual DHW - Step 1



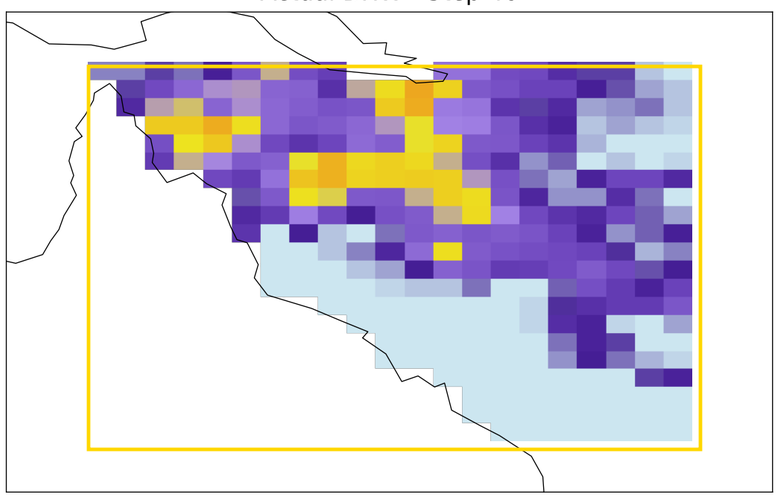
Predicted DHW - Step 1



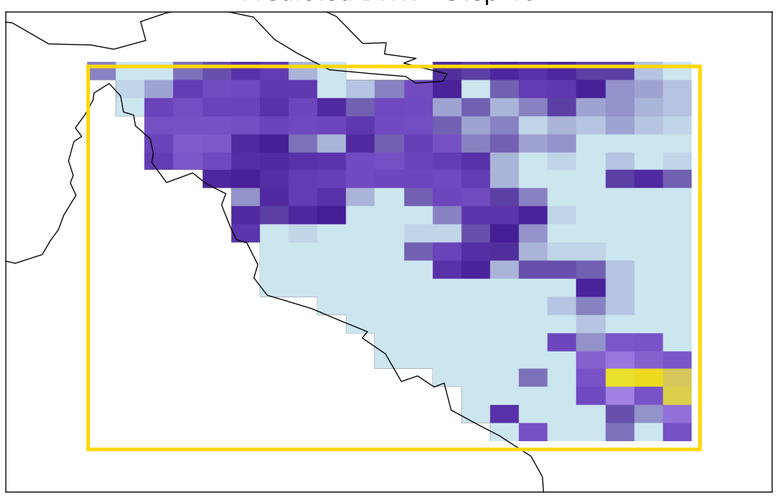
Error (Pred - Actual) - Step 1 [scale ±0.2]



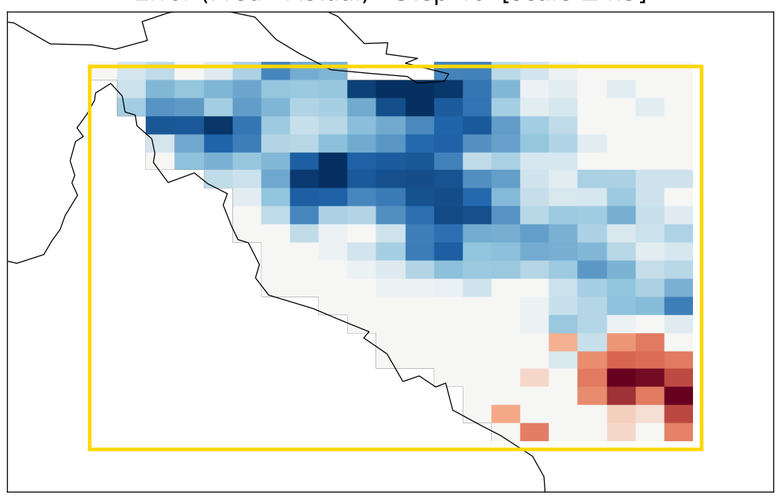
Actual DHW - Step 46



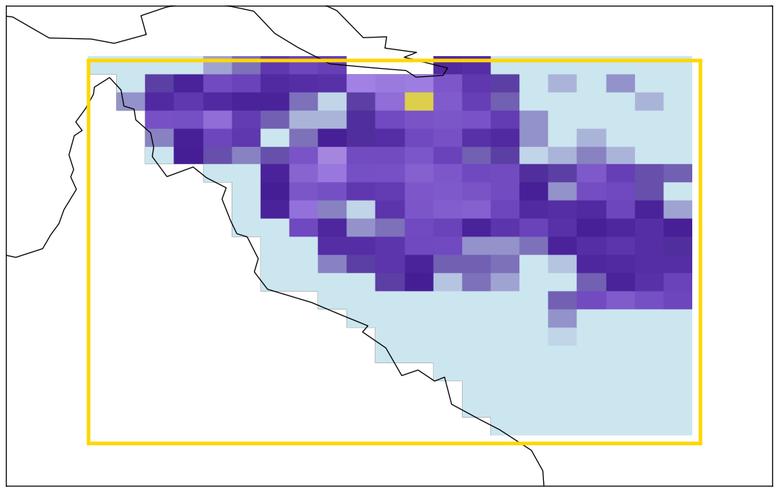
Predicted DHW - Step 46



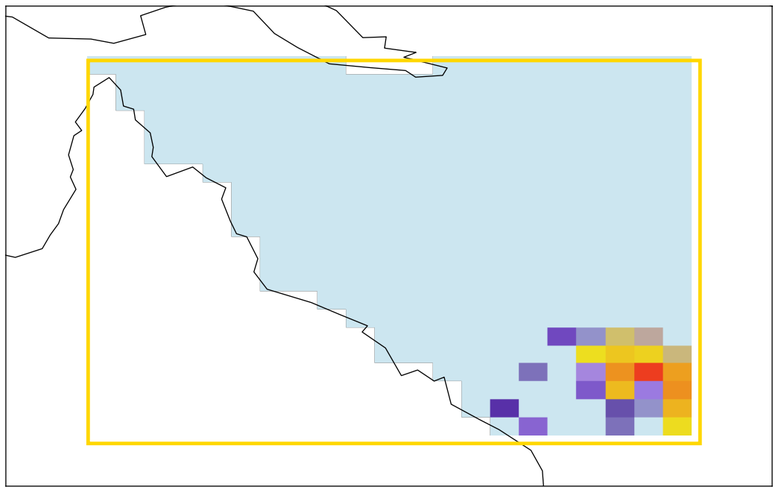
Error (Pred - Actual) - Step 46 [scale ±4.5]



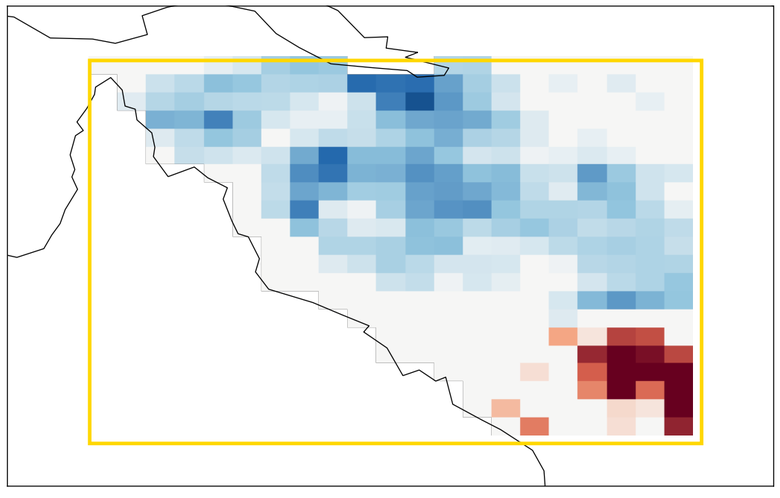
Actual DHW - Step 90

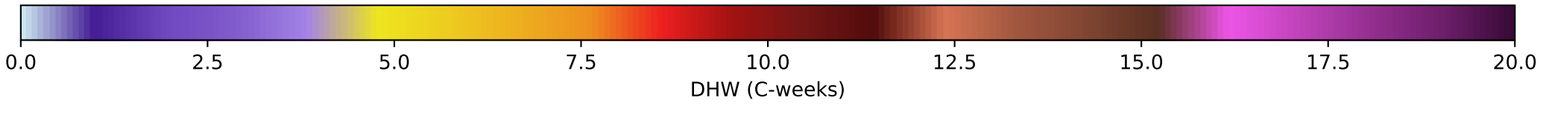


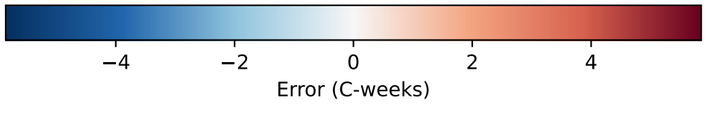
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±5.9]

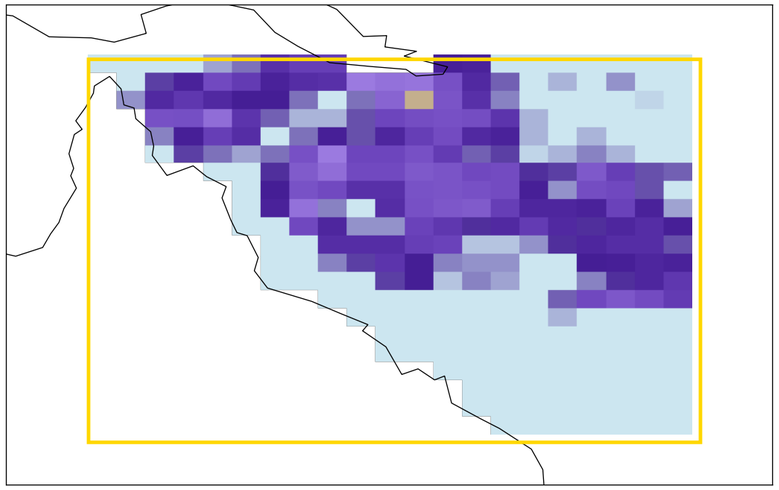




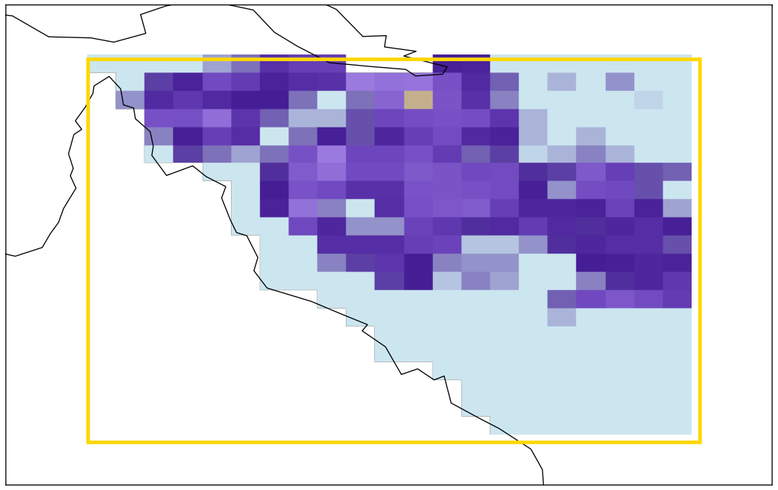


DHW Chain init: 2021-04-06 - Test - GBR (C-weeks)

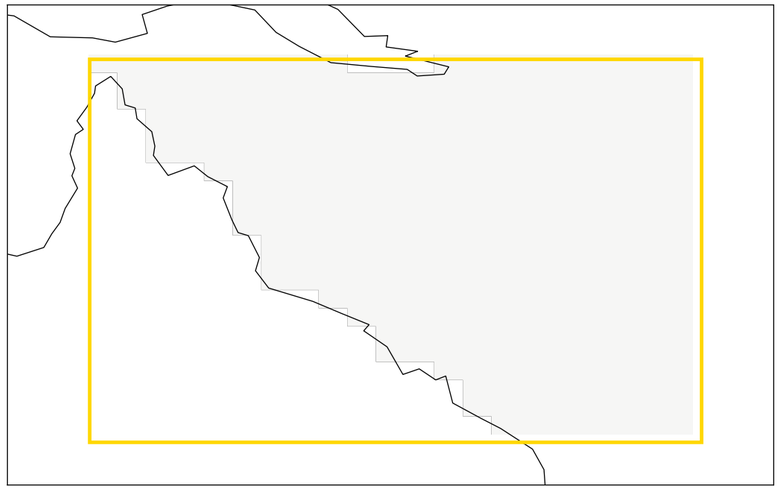
Actual DHW - Step 1



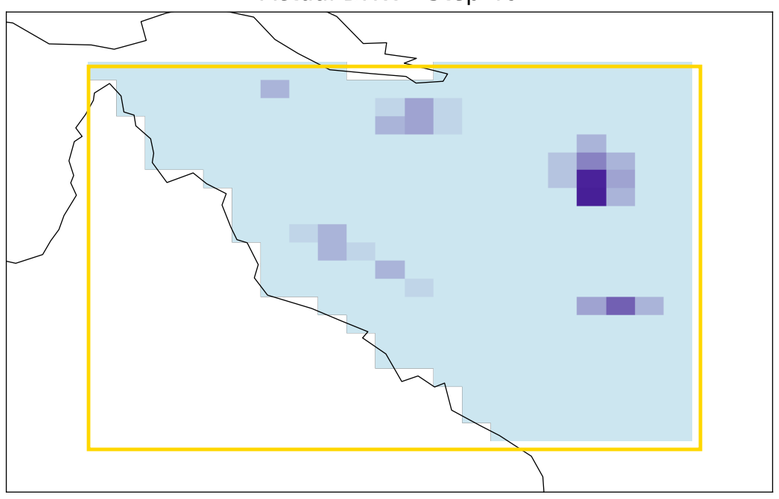
Predicted DHW - Step 1



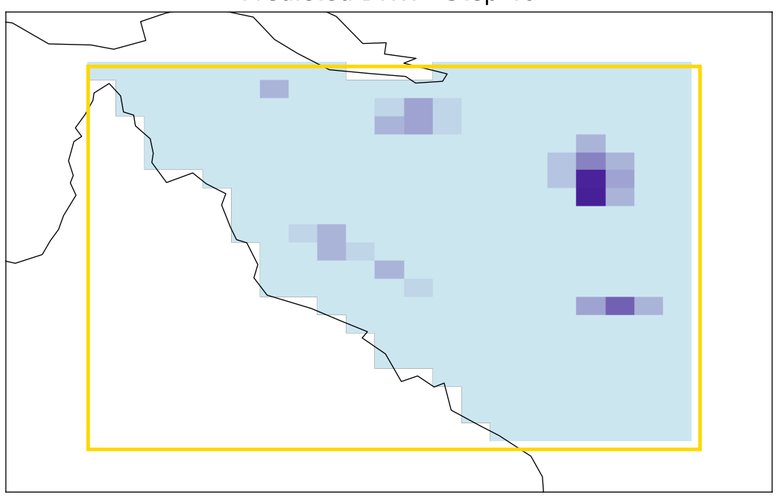
Error (Pred - Actual) - Step 1 [scale ±8]



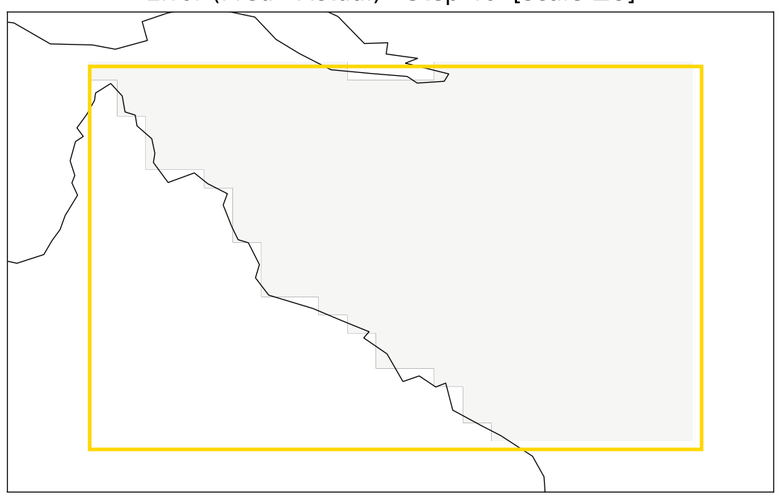
Actual DHW - Step 46



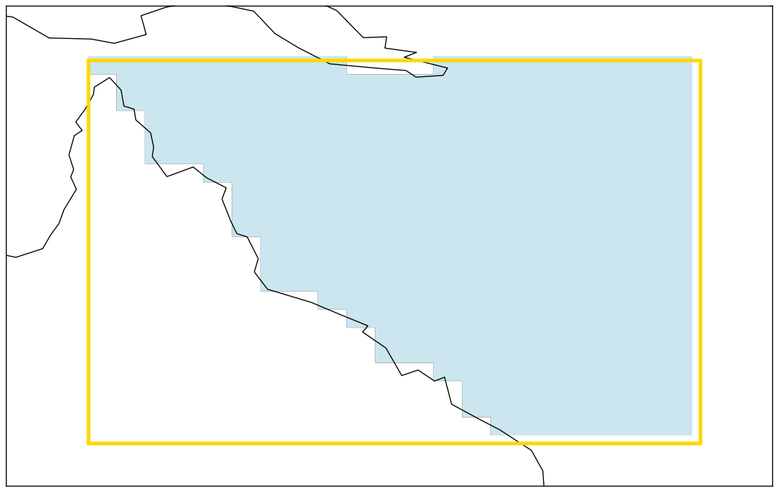
Predicted DHW - Step 46



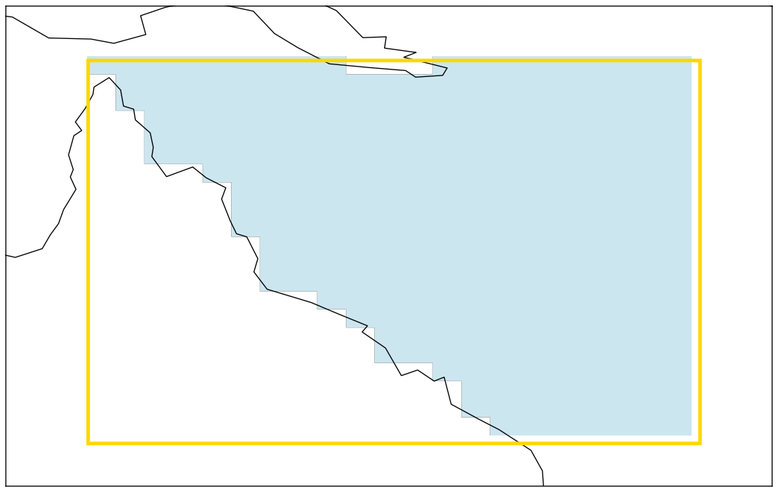
Error (Pred - Actual) - Step 46 [scale ±8]



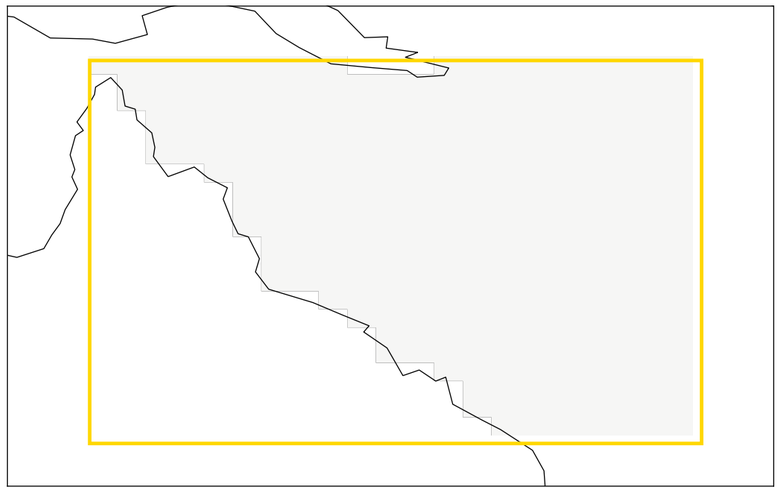
Actual DHW - Step 90

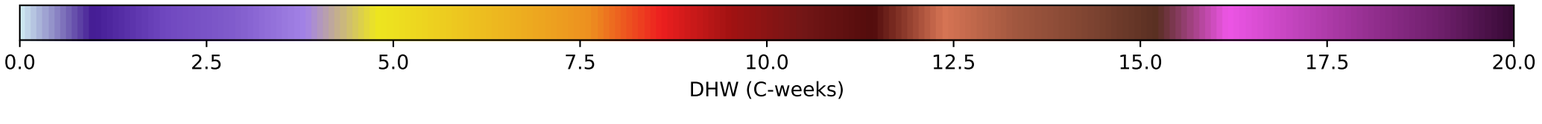


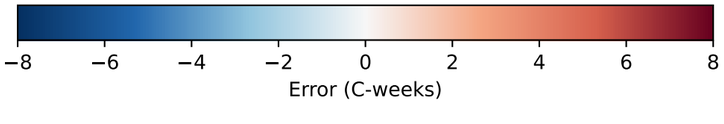
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

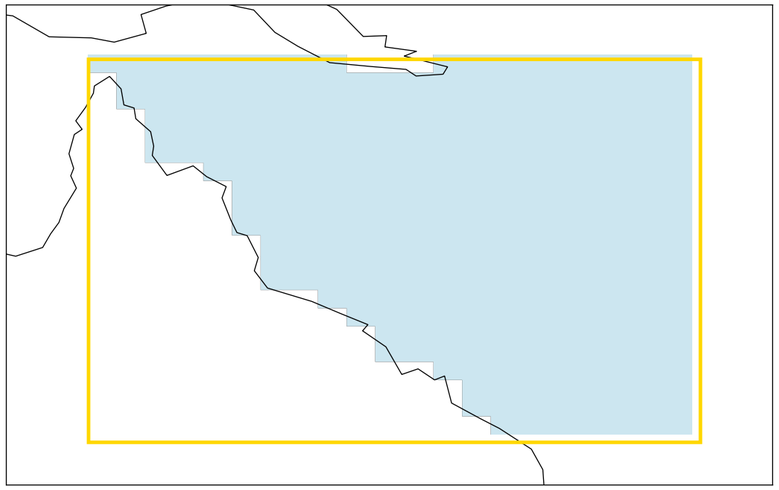




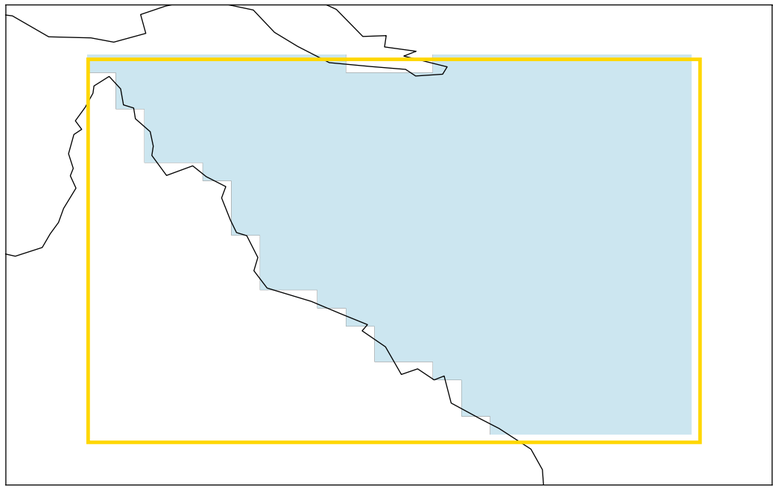


DHW Chain init: 2021-10-03 - Test - GBR (C-weeks)

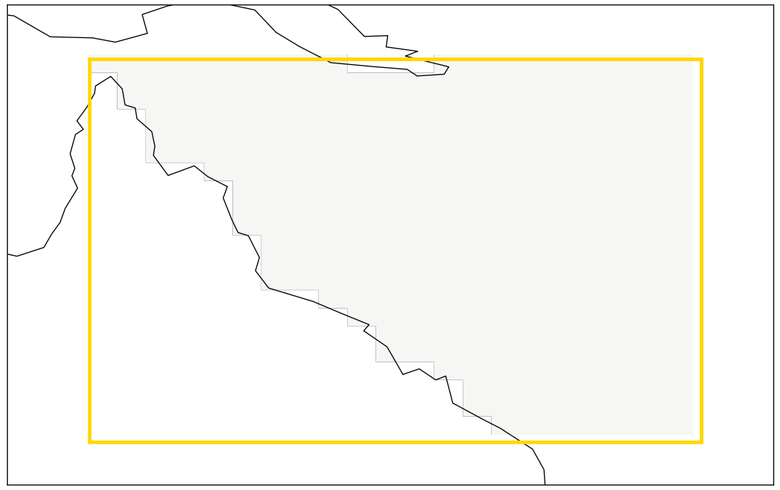
Actual DHW - Step 1



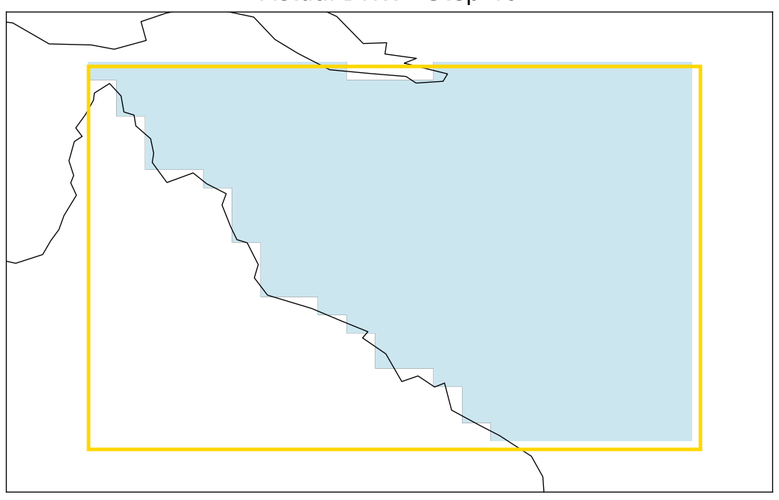
Predicted DHW - Step 1



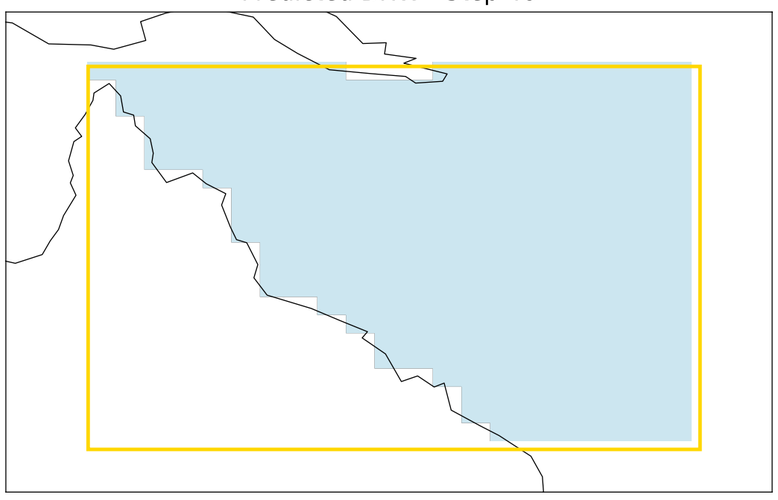
Error (Pred - Actual) - Step 1 [scale ±8]



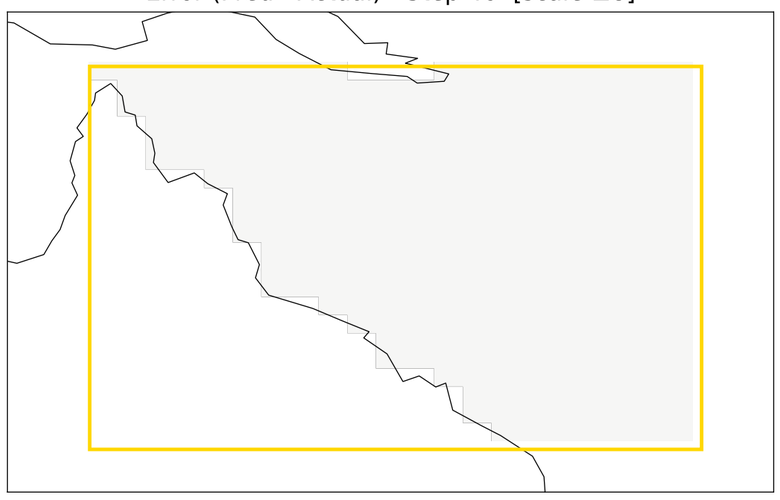
Actual DHW - Step 46



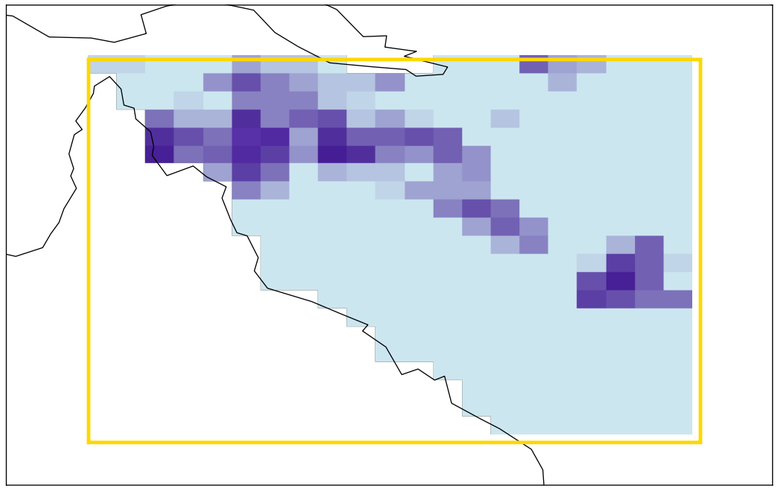
Predicted DHW - Step 46



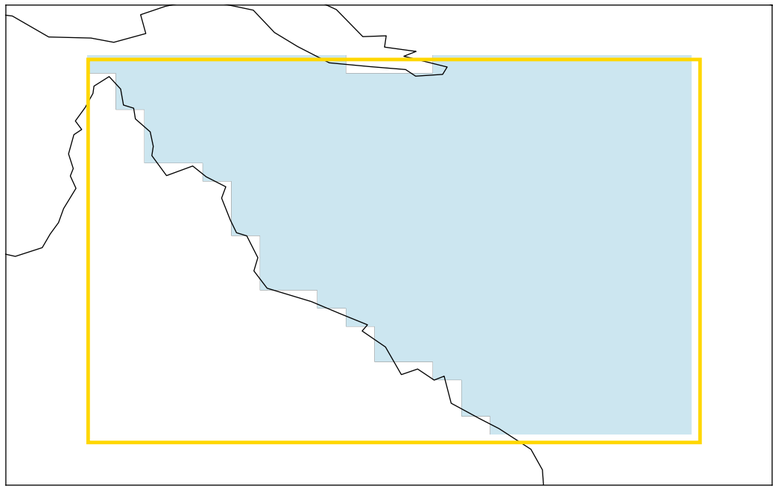
Error (Pred - Actual) - Step 46 [scale ±8]



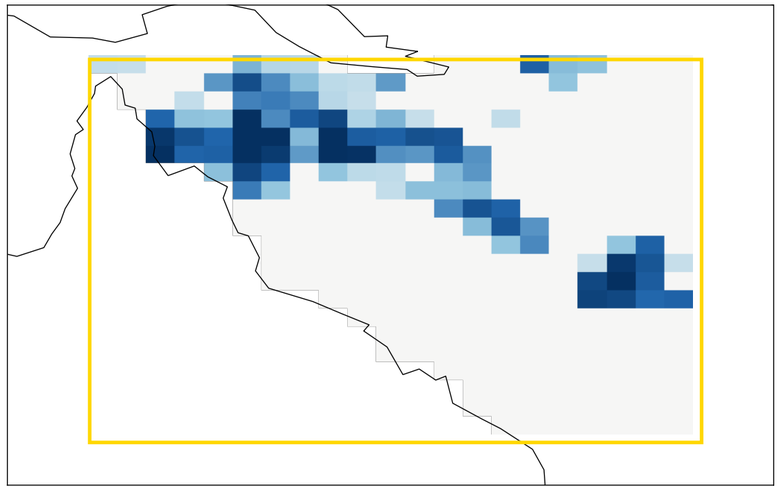
Actual DHW - Step 90

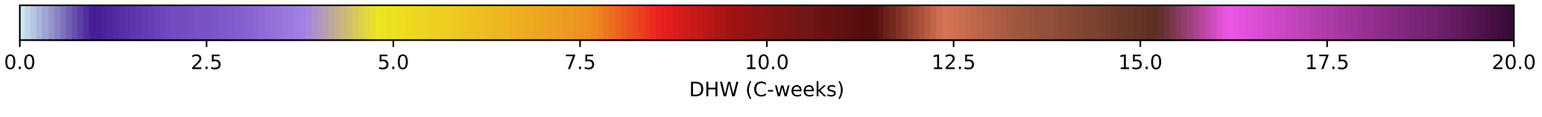


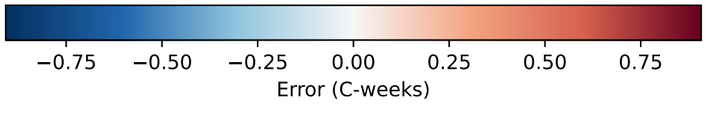
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.91]

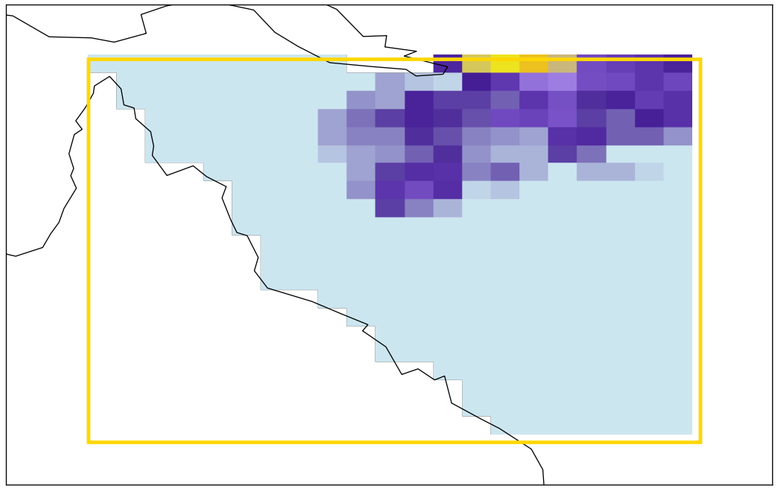




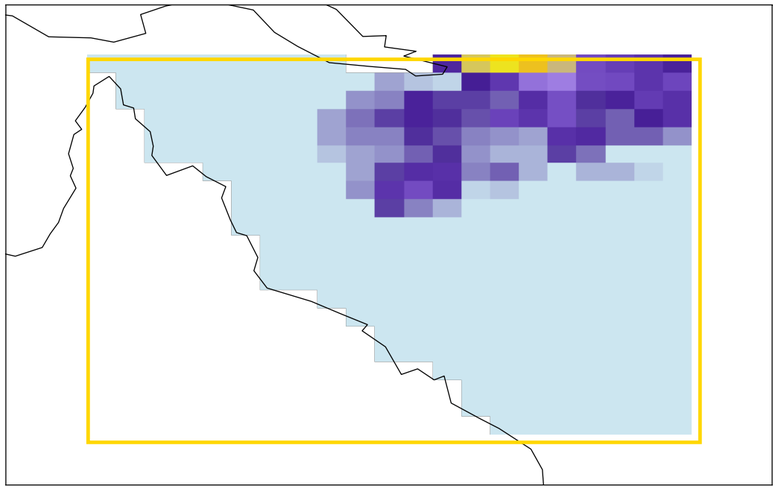


DHW Chain init: 2022-12-27 - Test - GBR (C-weeks)

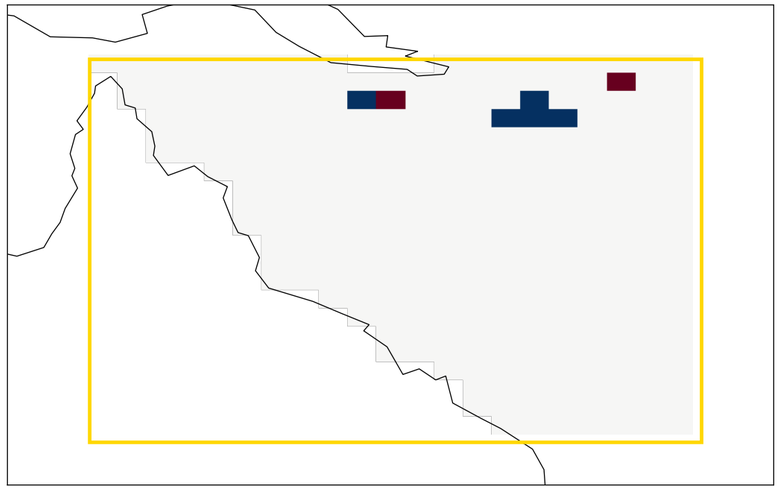
Actual DHW - Step 1



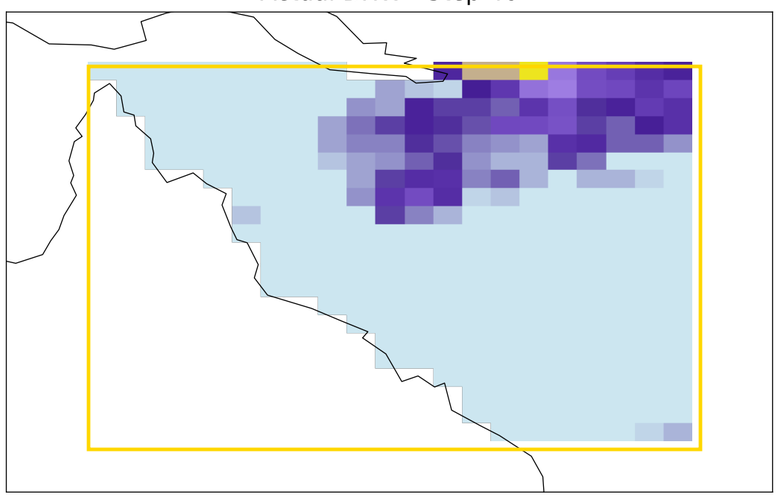
Predicted DHW - Step 1



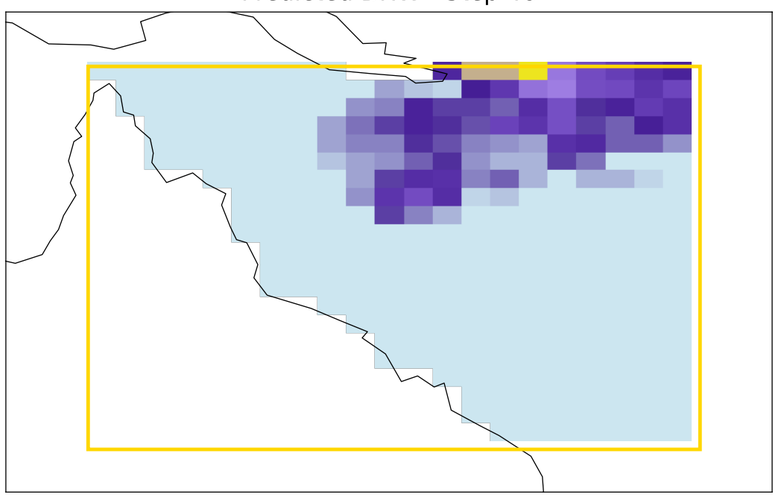
Error (Pred - Actual) - Step 1 [scale ±0.011]



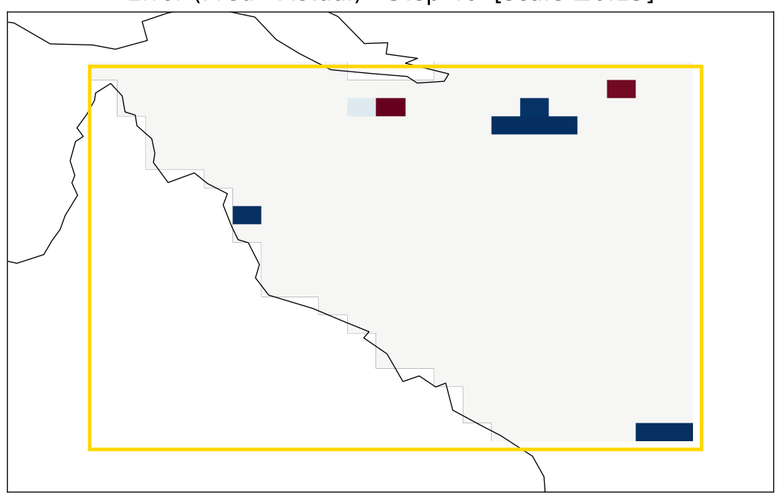
Actual DHW - Step 46



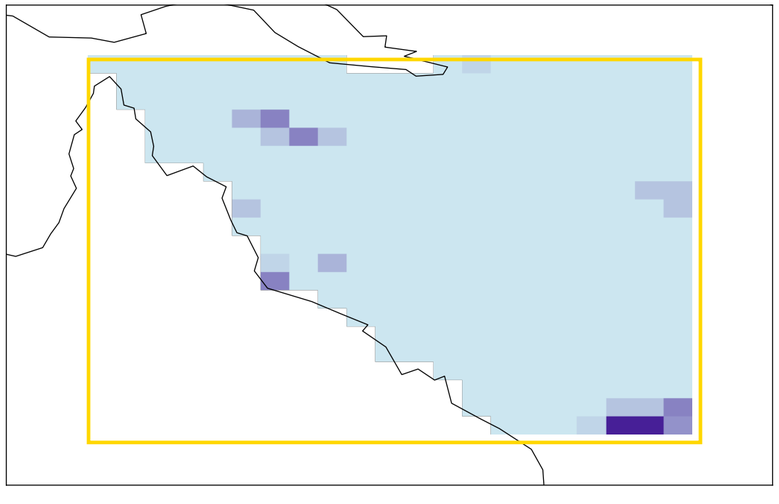
Predicted DHW - Step 46



Error (Pred - Actual) - Step 46 [scale ±0.15]



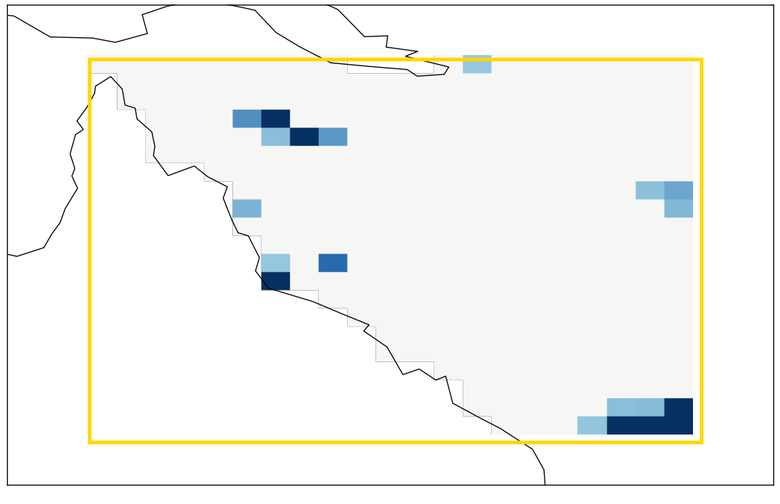
Actual DHW - Step 90

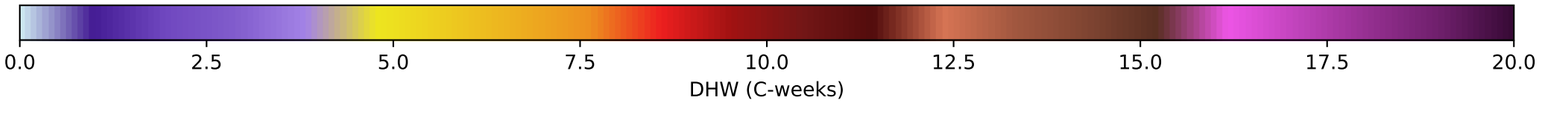


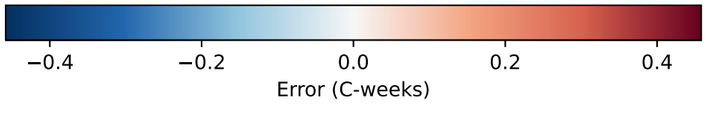
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.46]

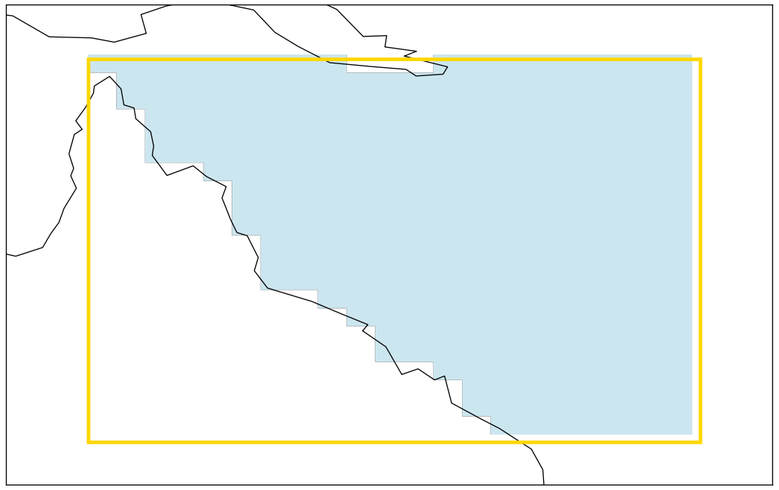




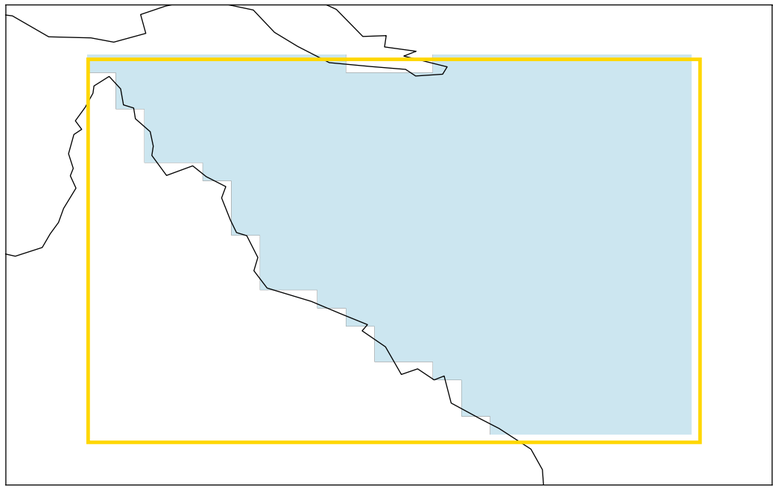


DHW Chain init: 2023-09-23 - Test - GBR (C-weeks)

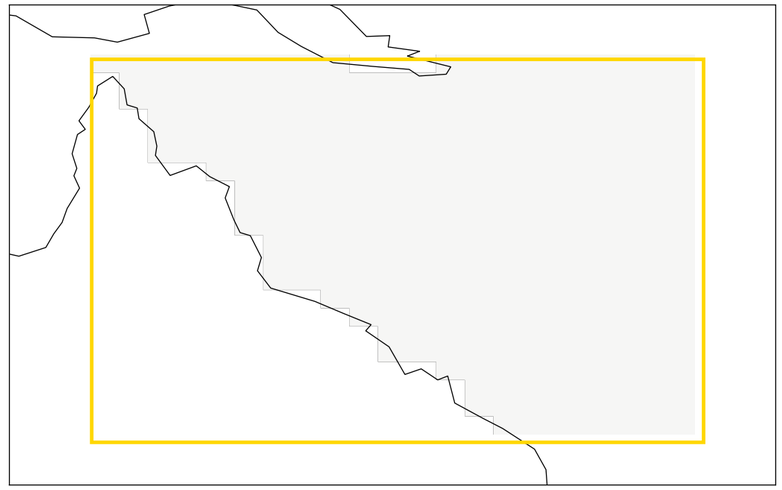
Actual DHW - Step 1



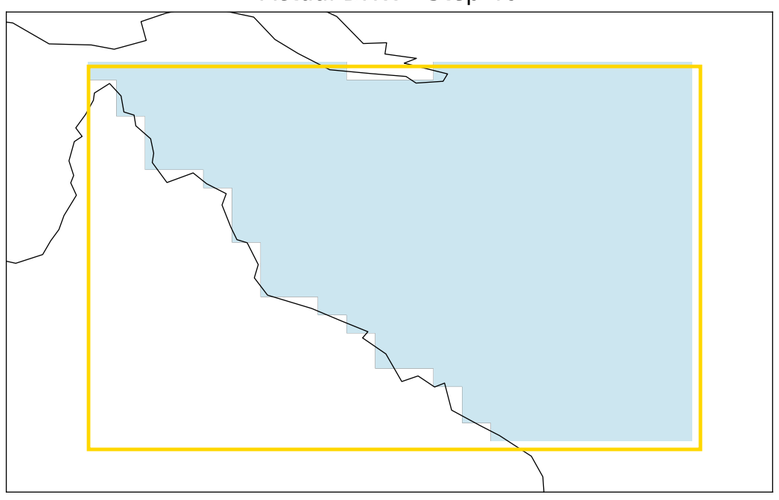
Predicted DHW - Step 1



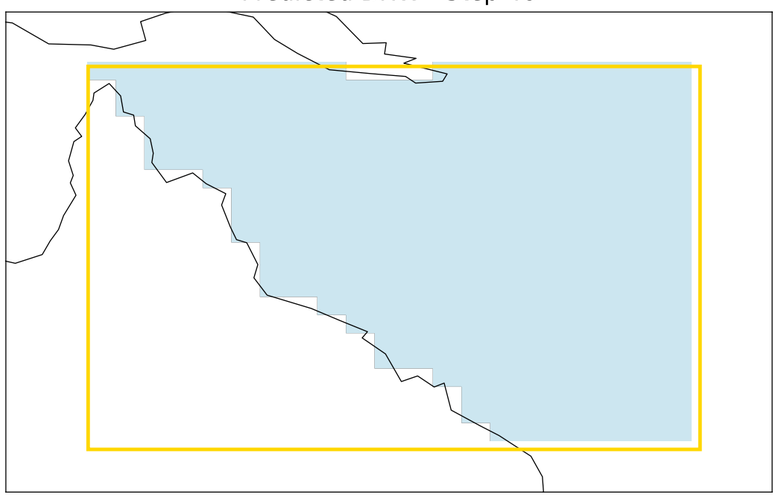
Error (Pred - Actual) - Step 1 [scale ±8]



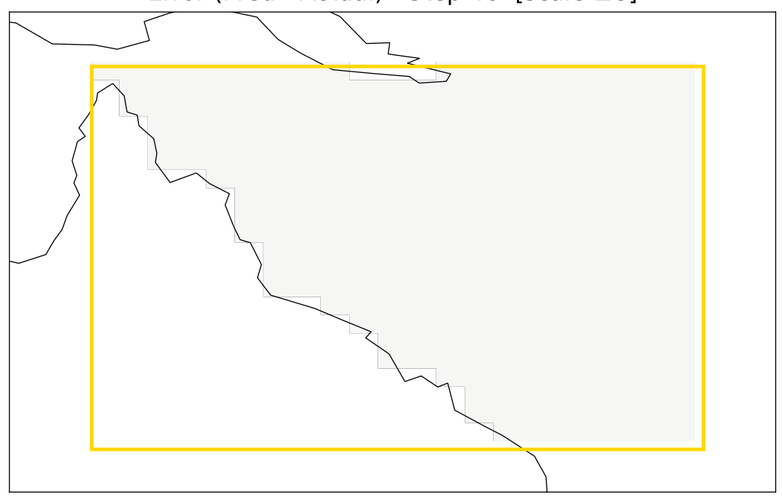
Actual DHW - Step 46



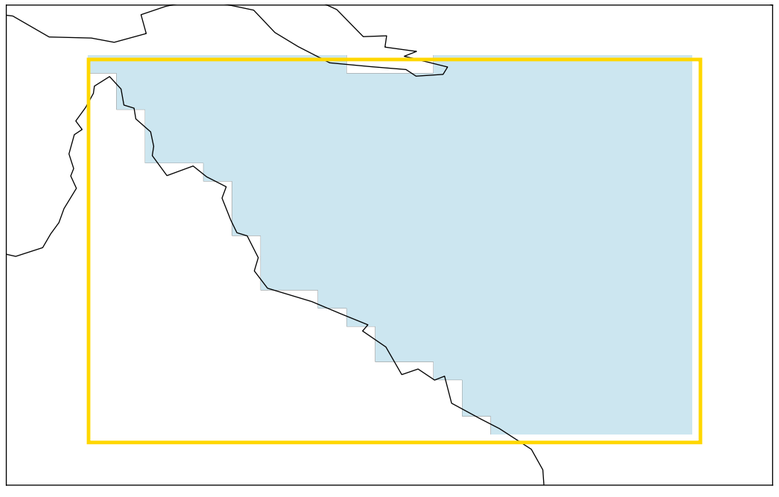
Predicted DHW - Step 46



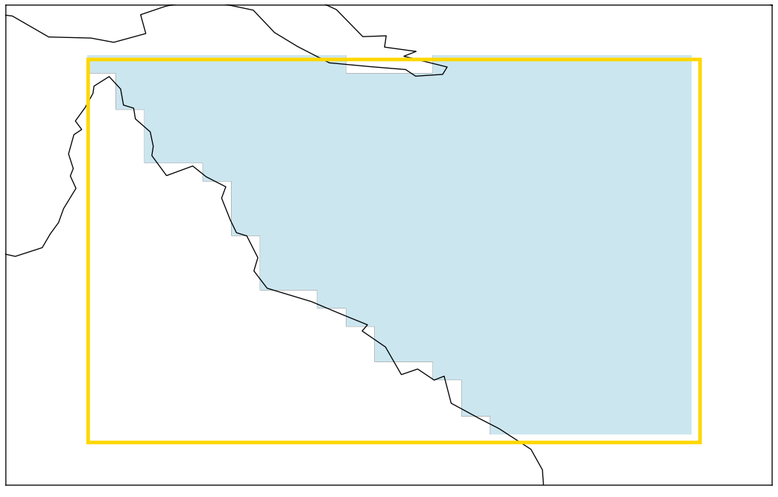
Error (Pred - Actual) - Step 46 [scale ±8]



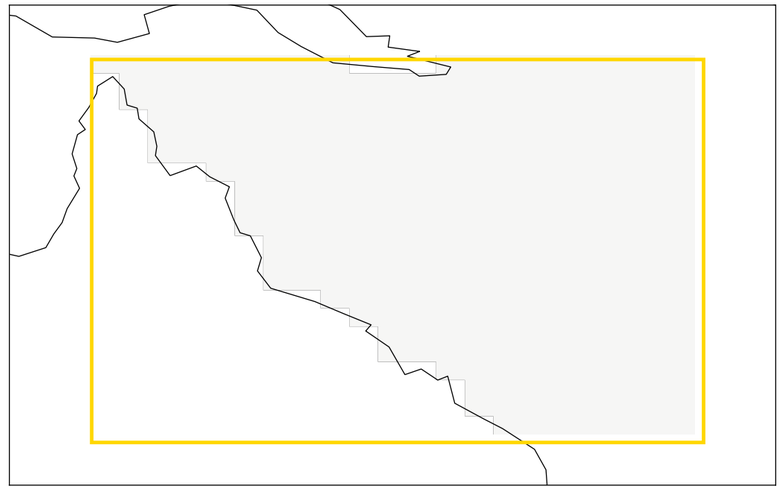
Actual DHW - Step 90

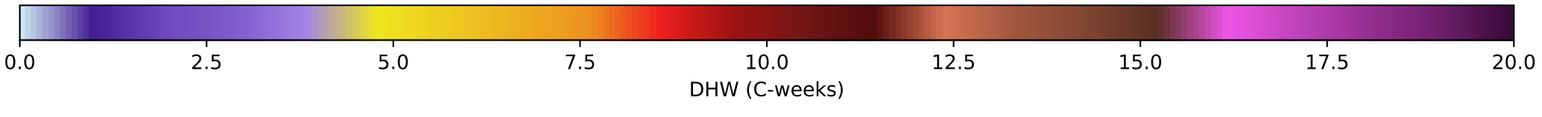


Predicted DHW - Step 90



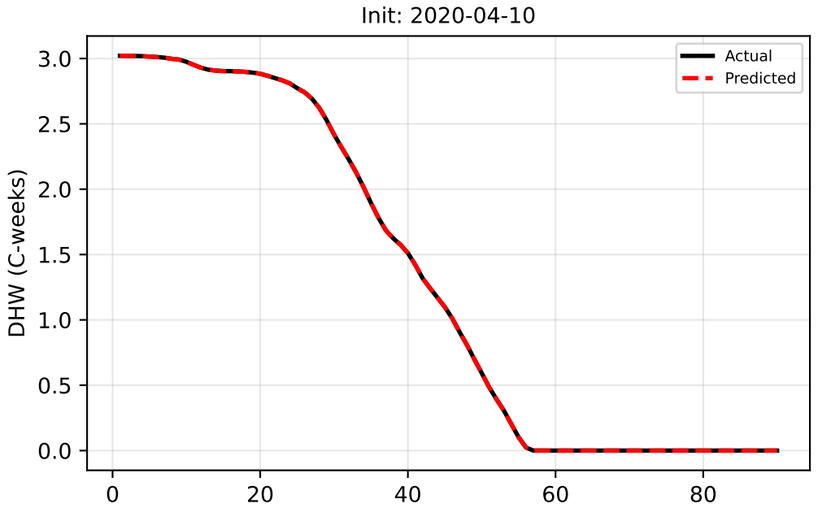
Error (Pred - Actual) - Step 90 [scale ±8]

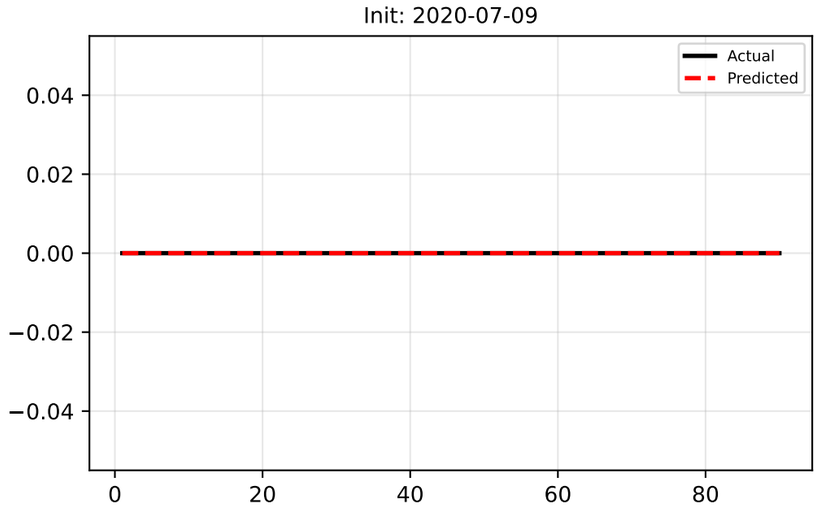


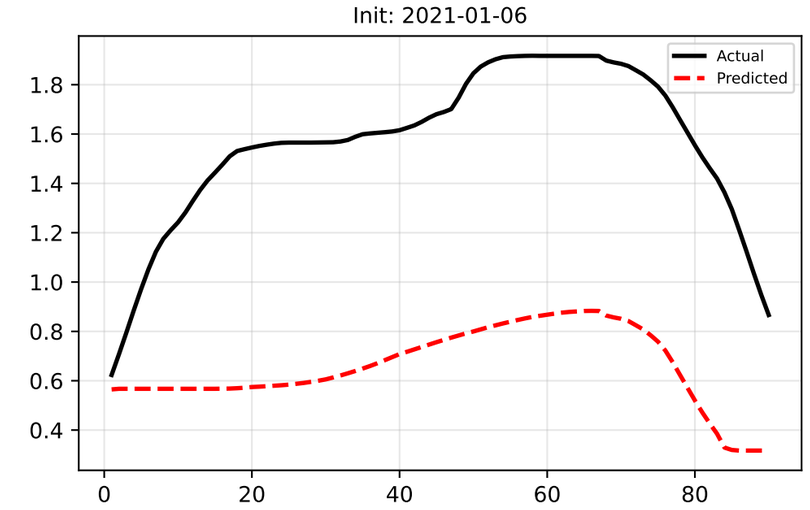


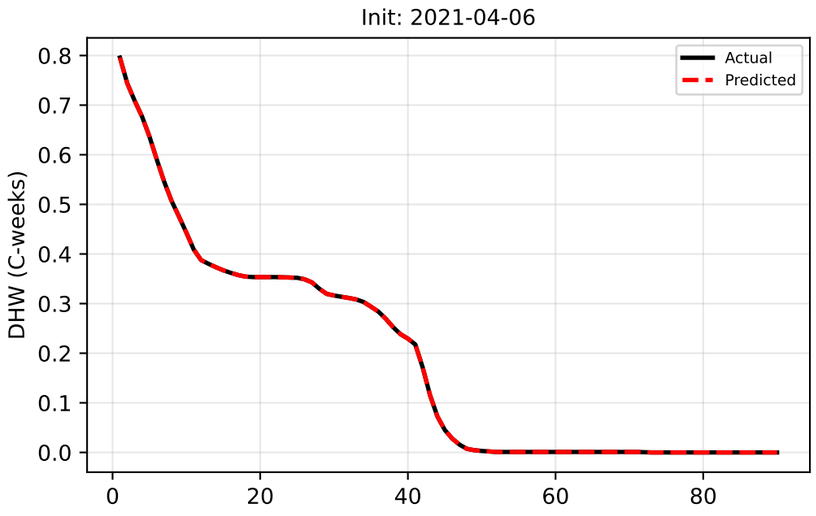


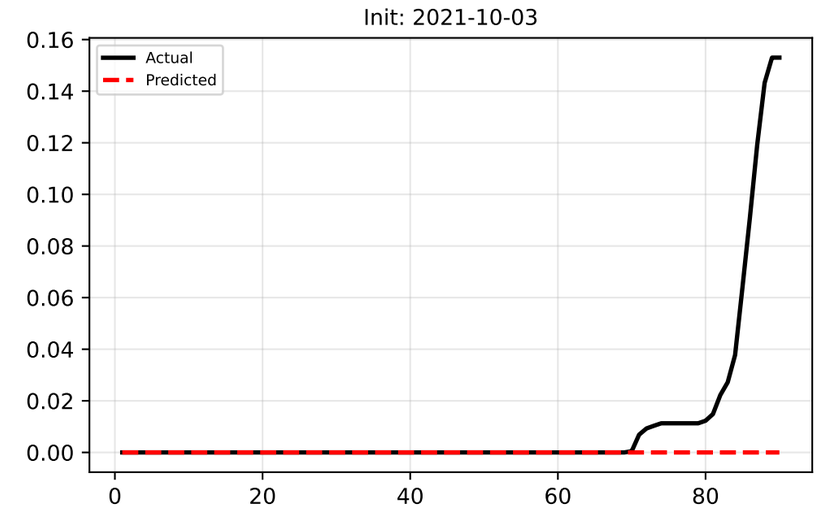
Global-Mean DHW - Test - GBR (C-weeks)

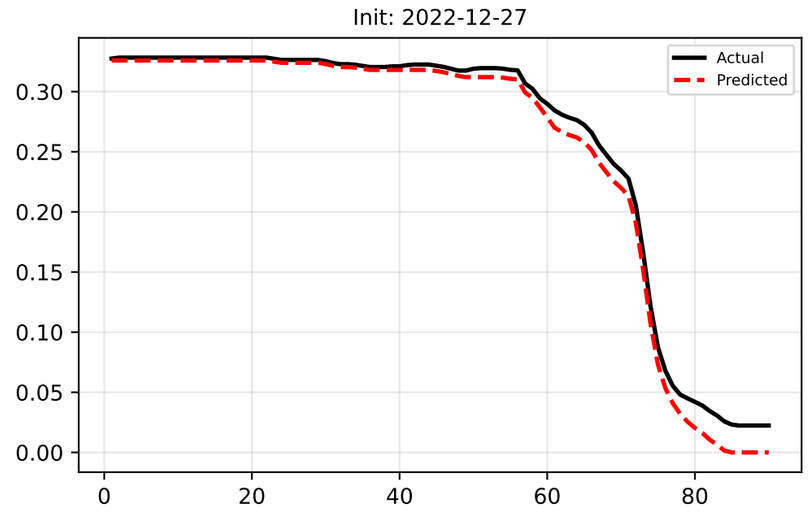


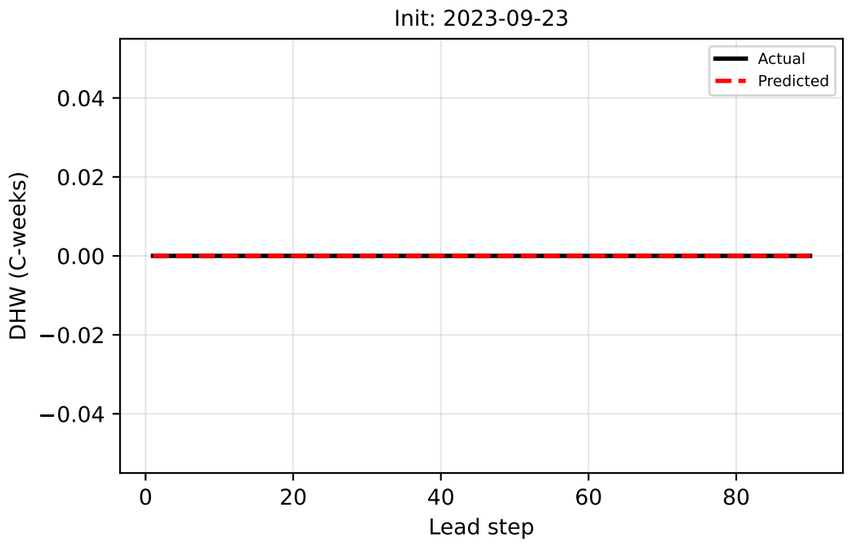


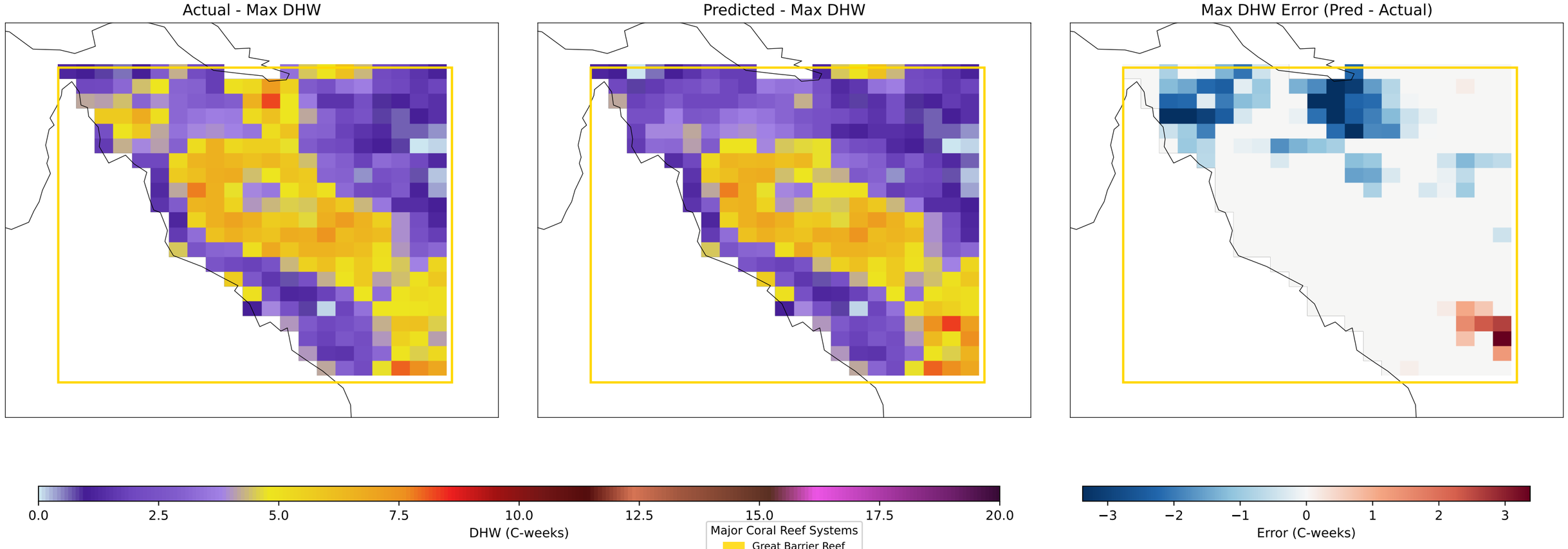


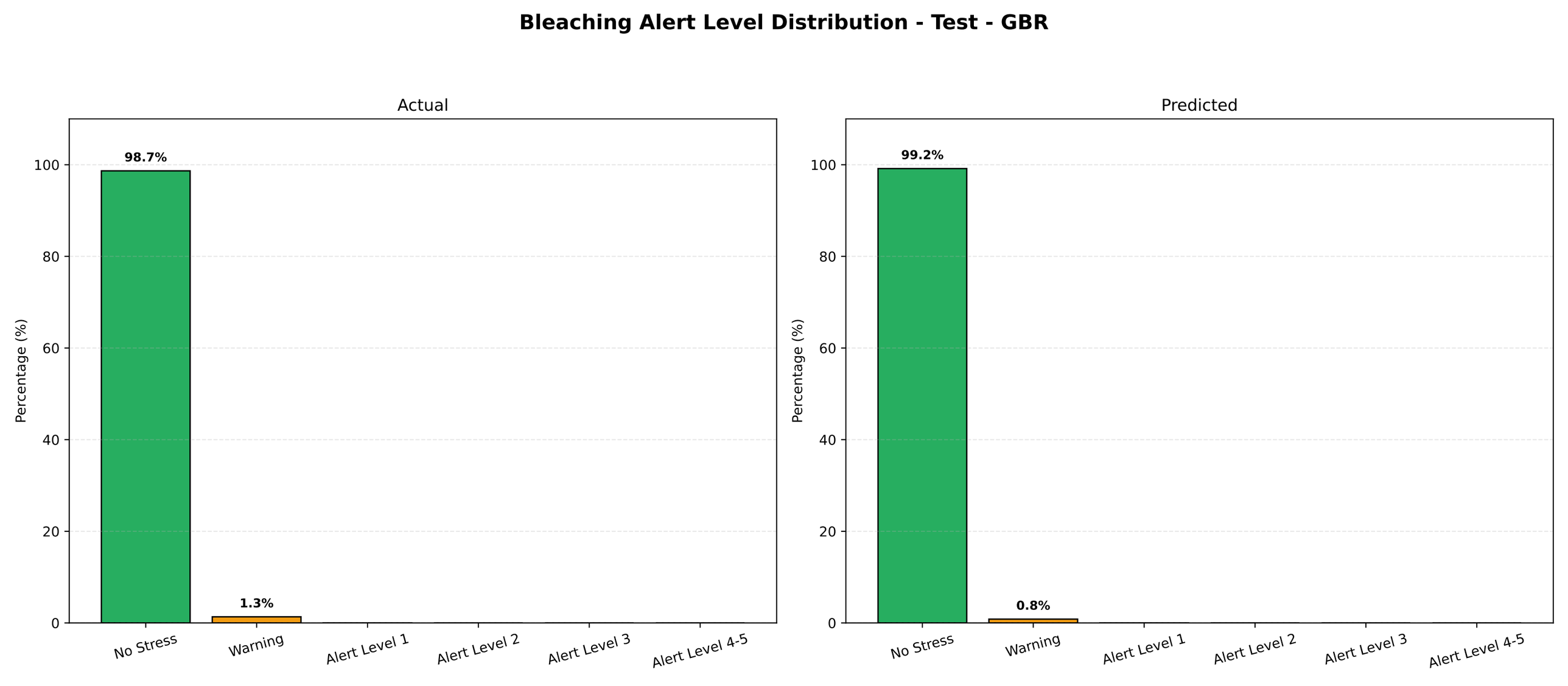






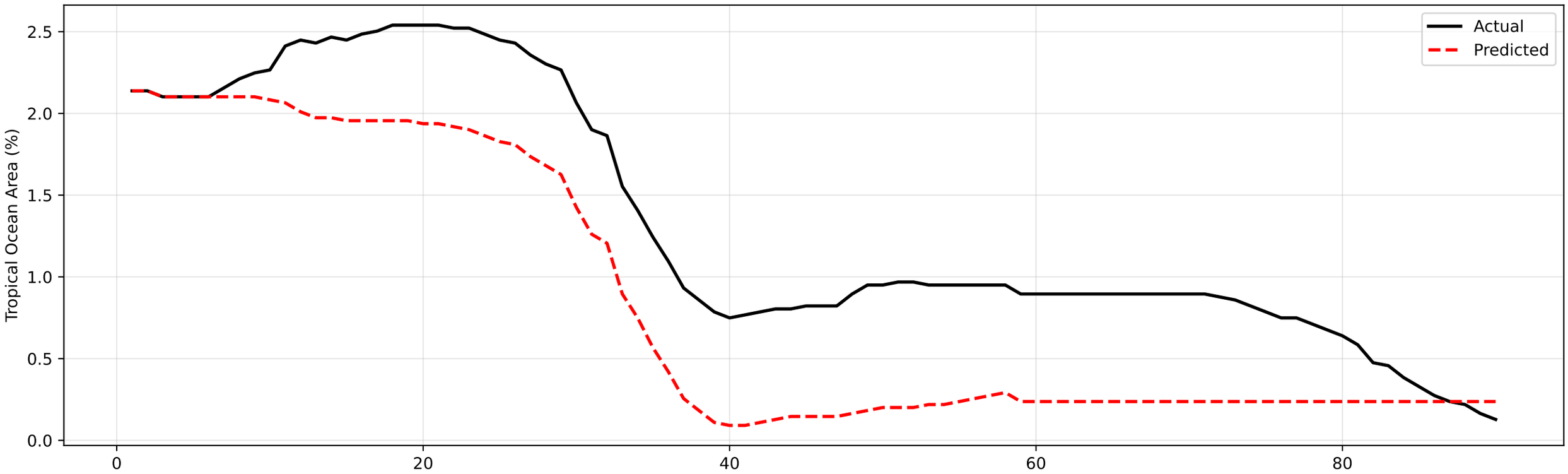



Peak DHW (max over all chains & leads) - Test - GBR (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
−3
−2
−1
0
1
2
3
Error (C-weeks)

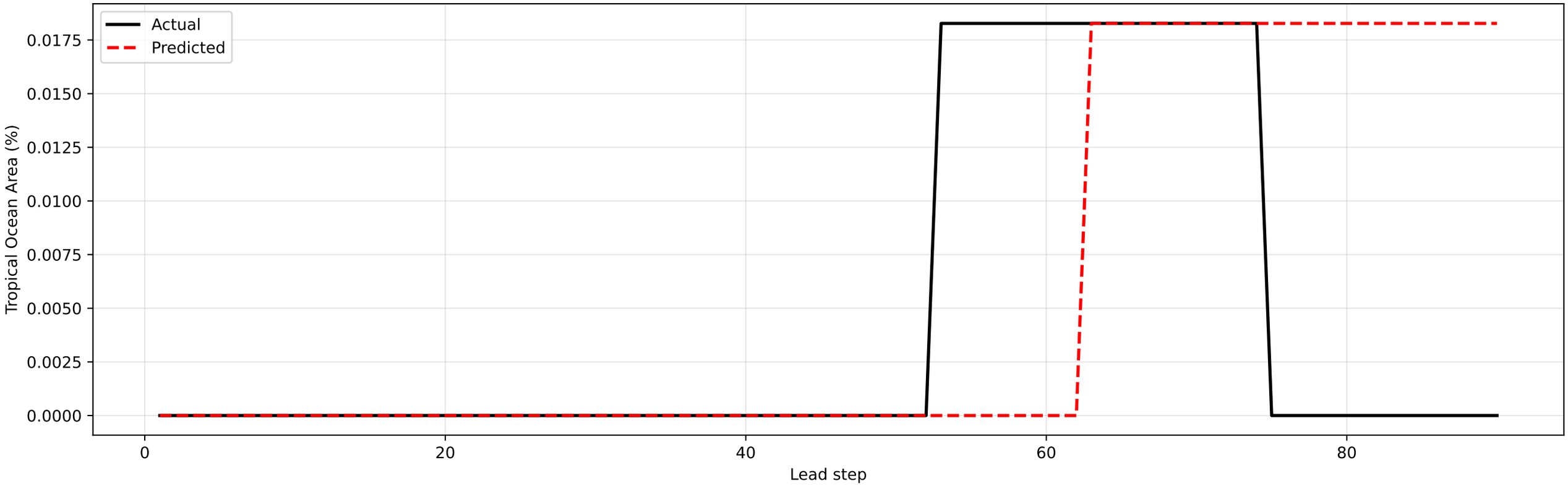

Bleaching Alert Level Distribution - Test - GBR
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
98.7%
1.3%
99.2%
0.8%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Test - GBR

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



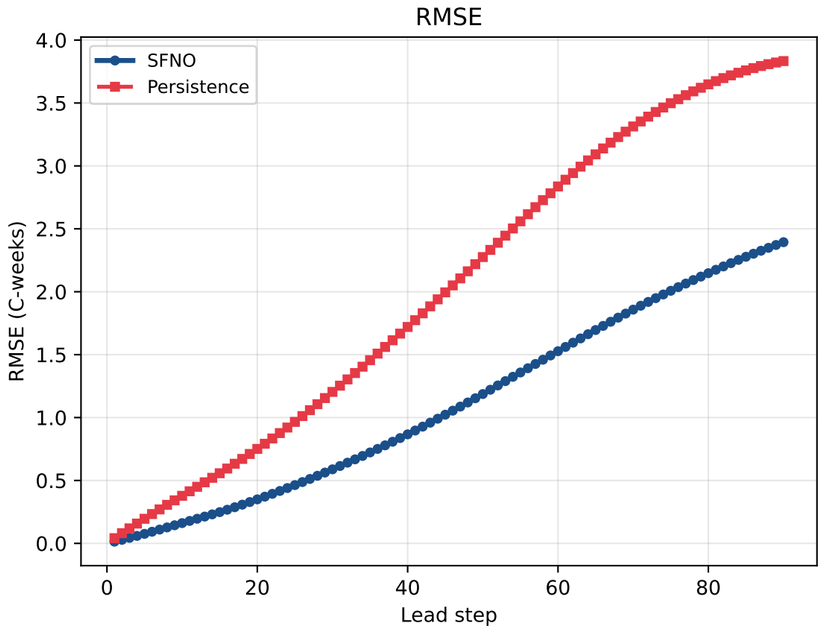
DHW >= 8 C-weeks (Alert Level 2 DHW threshold)

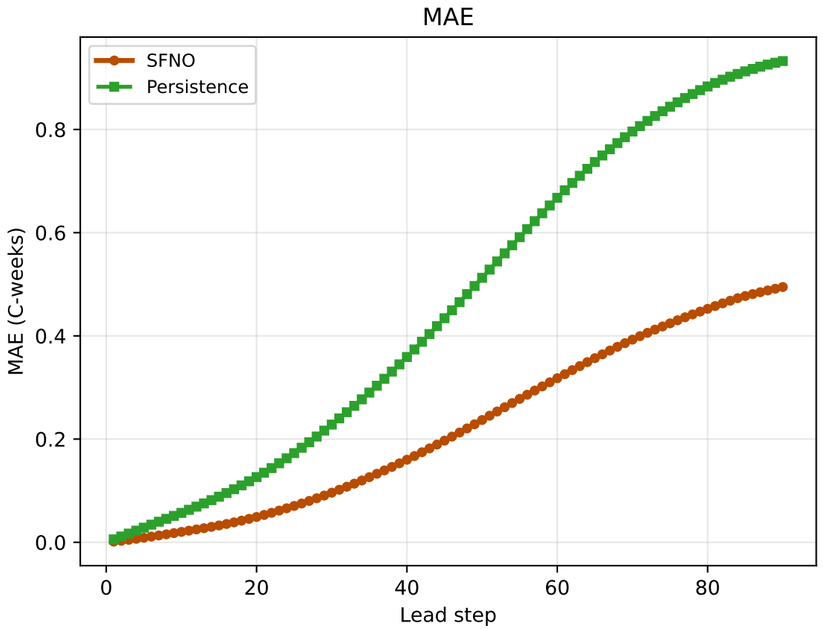


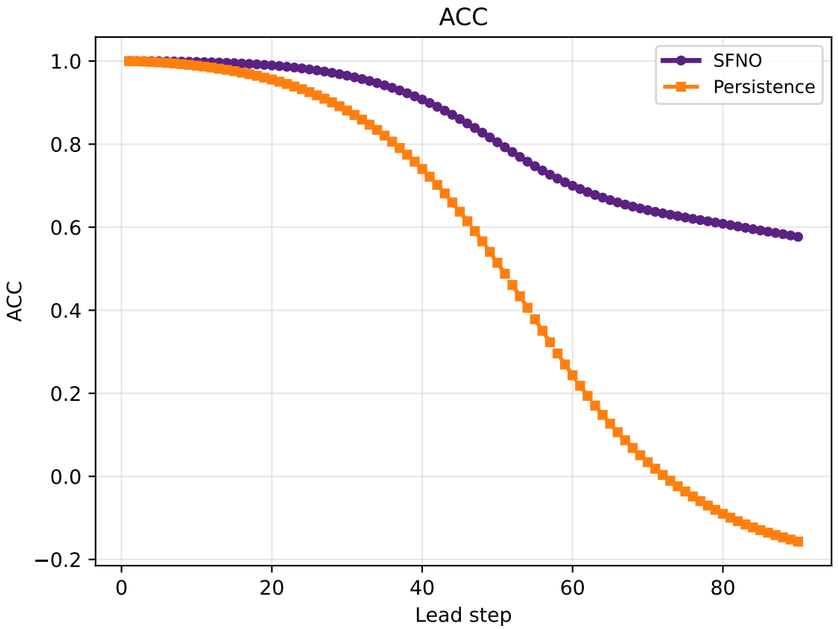
DHW Metrics by Lead Step - Val (15 chains scored at every lead)

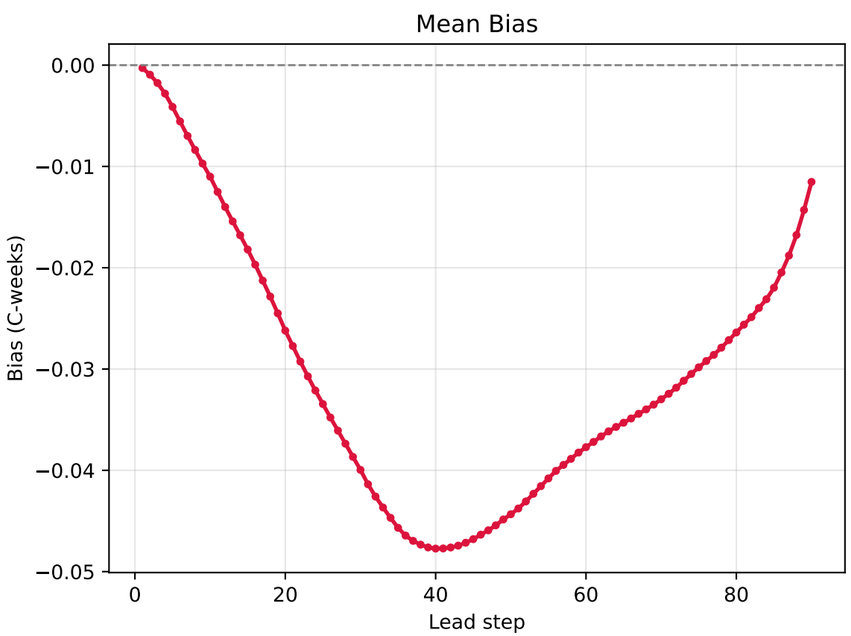
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0140	0.0415	66.3%	0.0013	0.0057	-0.0003	1.0000	0.9999	1.0000
2	0.0284	0.0810	64.9%	0.0030	0.0113	-0.0009	0.9999	0.9995	0.9999
3	0.0438	0.1194	63.3%	0.0048	0.0169	-0.0018	0.9999	0.9989	0.9998
4	0.0594	0.1574	62.3%	0.0068	0.0225	-0.0028	0.9997	0.9981	0.9995
5	0.0757	0.1956	61.3%	0.0089	0.0284	-0.0041	0.9996	0.9970	0.9993
6	0.0926	0.2331	60.3%	0.0111	0.0342	-0.0056	0.9993	0.9958	0.9989
7	0.1097	0.2700	59.4%	0.0133	0.0399	-0.0070	0.9991	0.9944	0.9985
8	0.1269	0.3062	58.5%	0.0155	0.0455	-0.0084	0.9988	0.9927	0.9979
9	0.1440	0.3421	57.9%	0.0178	0.0512	-0.0097	0.9984	0.9909	0.9973
10	0.1612	0.3780	57.3%	0.0202	0.0570	-0.0110	0.9980	0.9889	0.9967
11	0.1785	0.4140	56.9%	0.0225	0.0630	-0.0125	0.9975	0.9867	0.9959
12	0.1958	0.4497	56.5%	0.0250	0.0691	-0.0140	0.9970	0.9843	0.9950
13	0.2130	0.4855	56.1%	0.0274	0.0754	-0.0154	0.9964	0.9816	0.9941
14	0.2305	0.5213	55.8%	0.0299	0.0817	-0.0168	0.9957	0.9788	0.9930
15	0.2485	0.5575	55.4%	0.0326	0.0884	-0.0182	0.9950	0.9757	0.9918
16	0.2674	0.5948	55.0%	0.0355	0.0954	-0.0197	0.9942	0.9723	0.9905
17	0.2873	0.6328	54.6%	0.0387	0.1028	-0.0213	0.9932	0.9686	0.9889
18	0.3077	0.6714	54.2%	0.0420	0.1103	-0.0228	0.9921	0.9645	0.9872
19	0.3286	0.7107	53.8%	0.0454	0.1182	-0.0245	0.9909	0.9602	0.9852
20	0.3498	0.7507	53.4%	0.0491	0.1264	-0.0262	0.9895	0.9555	0.9831
21	0.3714	0.7917	53.1%	0.0530	0.1349	-0.0277	0.9880	0.9504	0.9807
22	0.3938	0.8337	52.8%	0.0572	0.1438	-0.0293	0.9863	0.9448	0.9780
23	0.4165	0.8766	52.5%	0.0615	0.1532	-0.0307	0.9844	0.9388	0.9751
24	0.4398	0.9206	52.2%	0.0659	0.1628	-0.0321	0.9824	0.9323	0.9719
25	0.4633	0.9658	52.0%	0.0705	0.1729	-0.0335	0.9801	0.9252	0.9684
26	0.4874	1.0118	51.8%	0.0752	0.1832	-0.0348	0.9776	0.9176	0.9645
27	0.5122	1.0587	51.6%	0.0801	0.1939	-0.0361	0.9748	0.9094	0.9603
28	0.5373	1.1062	51.4%	0.0853	0.2049	-0.0374	0.9718	0.9006	0.9557
29	0.5627	1.1549	51.3%	0.0906	0.2164	-0.0387	0.9685	0.8911	0.9507
30	0.5886	1.2039	51.1%	0.0962	0.2282	-0.0400	0.9648	0.8811	0.9453
31	0.6150	1.2534	50.9%	0.1019	0.2401	-0.0414	0.9609	0.8703	0.9394
32	0.6414	1.3033	50.8%	0.1077	0.2521	-0.0426	0.9566	0.8590	0.9331
33	0.6678	1.3536	50.7%	0.1136	0.2644	-0.0437	0.9521	0.8469	0.9264
34	0.6949	1.4043	50.5%	0.1198	0.2770	-0.0447	0.9471	0.8342	0.9192
35	0.7228	1.4559	50.4%	0.1262	0.2901	-0.0457	0.9416	0.8206	0.9113
36	0.7509	1.5083	50.2%	0.1328	0.3034	-0.0464	0.9357	0.8060	0.9028
37	0.7791	1.5611	50.1%	0.1394	0.3169	-0.0470	0.9293	0.7907	0.8939
38	0.8078	1.6141	50.0%	0.1461	0.3307	-0.0473	0.9225	0.7746	0.8843
39	0.8370	1.6671	49.8%	0.1530	0.3448	-0.0476	0.9152	0.7578	0.8741
40	0.8667	1.7202	49.6%	0.1600	0.3591	-0.0477	0.9074	0.7402	0.8632
41	0.8969	1.7738	49.4%	0.1672	0.3737	-0.0477	0.8990	0.7216	0.8517
42	0.9279	1.8284	49.2%	0.1745	0.3885	-0.0476	0.8901	0.7020	0.8394
43	0.9594	1.8837	49.1%	0.1819	0.4036	-0.0474	0.8806	0.6813	0.8264
44	0.9910	1.9395	48.9%	0.1894	0.4188	-0.0472	0.8707	0.6596	0.8129
45	1.0227	1.9951	48.7%	0.1971	0.4342	-0.0468	0.8605	0.6373	0.7989
46	1.0548	2.0504	48.6%	0.2048	0.4497	-0.0464	0.8498	0.6143	0.7844
47	1.0871	2.1061	48.4%	0.2127	0.4653	-0.0459	0.8389	0.5906	0.7694
48	1.1199	2.1623	48.2%	0.2206	0.4809	-0.0454	0.8276	0.5659	0.7539
49	1.1532	2.2188	48.0%	0.2287	0.4966	-0.0449	0.8161	0.5405	0.7380
50	1.1870	2.2756	47.8%	0.2370	0.5125	-0.0443	0.8044	0.5145	0.7217
51	1.2211	2.3325	47.6%	0.2453	0.5284	-0.0438	0.7926	0.4878	0.7049
52	1.2553	2.3893	47.5%	0.2535	0.5442	-0.0431	0.7808	0.4608	0.6880
53	1.2896	2.4458	47.3%	0.2617	0.5599	-0.0423	0.7692	0.4336	0.6711
54	1.3240	2.5028	47.1%	0.2699	0.5756	-0.0416	0.7578	0.4060	0.6542
55	1.3583	2.5595	46.9%	0.2780	0.5913	-0.0408	0.7468	0.3782	0.6377
56	1.3921	2.6160	46.8%	0.2860	0.6068	-0.0401	0.7364	0.3506	0.6216
57	1.4260	2.6721	46.6%	0.2941	0.6223	-0.0395	0.7264	0.3231	0.6061
58	1.4598	2.7277	46.5%	0.3021	0.6376	-0.0389	0.7170	0.2960	0.5909
59	1.4936	2.7824	46.3%	0.3101	0.6527	-0.0382	0.7081	0.2695	0.5762
60	1.5273	2.8367	46.2%	0.3179	0.6676	-0.0377	0.6998	0.2436	0.5621
61	1.5613	2.8903	46.0%	0.3258	0.6821	-0.0372	0.6919	0.2183	0.5483
62	1.5952	2.9429	45.8%	0.3337	0.6963	-0.0367	0.6845	0.1938	0.5350
63	1.6290	2.9940	45.6%	0.3415	0.7102	-0.0362	0.6777	0.1705	0.5222
64	1.6625	3.0438	45.4%	0.3492	0.7237	-0.0357	0.6713	0.1481	0.5099
65	1.6957	3.0919	45.2%	0.3568	0.7369	-0.0353	0.6652	0.1268	0.4978
66	1.7286	3.1389	44.9%	0.3642	0.7496	-0.0349	0.6597	0.1064	0.4862
67	1.7614	3.1847	44.7%	0.3716	0.7617	-0.0344	0.6545	0.0870	0.4750
68	1.7941	3.2293	44.4%	0.3788	0.7736	-0.0340	0.6497	0.0685	0.4640
69	1.8262	3.2721	44.2%	0.3859	0.7850	-0.0335	0.6452	0.0510	0.4533
70	1.8577	3.3135	43.9%	0.3928	0.7959	-0.0330	0.6410	0.0343	0.4430
71	1.8887	3.3532	43.7%	0.3995	0.8064	-0.0324	0.6370	0.0185	0.4328
72	1.9191	3.3916	43.4%	0.4059	0.8165	-0.0318	0.6334	0.0036	0.4232
73	1.9489	3.4286	43.2%	0.4122	0.8260	-0.0312	0.6300	-0.0105	0.4138
74	1.9783	3.4642	42.9%	0.4183	0.8352	-0.0305	0.6268	-0.0239	0.4046
75	2.0075	3.4981	42.6%	0.4243	0.8441	-0.0298	0.6235	-0.0365	0.3952
76	2.0366	3.5305	42.3%	0.4302	0.8526	-0.0292	0.6201	-0.0483	0.3858
77	2.0650	3.5621	42.0%	0.4361	0.8609	-0.0286	0.6172	-0.0595	0.3770
78	2.0928	3.5923	41.7%	0.4416	0.8687	-0.0279	0.6143	-0.0702	0.3681
79	2.1203	3.6210	41.4%	0.4471	0.8762	-0.0271	0.6113	-0.0805	0.3589
80	2.1474	3.6483	41.1%	0.4524	0.8833	-0.0264	0.6083	-0.0902	0.3496
81	2.1745	3.6740	40.8%	0.4578	0.8901	-0.0256	0.6051	-0.0994	0.3398
82	2.2011	3.6978	40.5%	0.4631	0.8964	-0.0249	0.6018	-0.1080	0.3297
83	2.2271	3.7198	40.1%	0.4681	0.9022	-0.0240	0.5986	-0.1157	0.3196
84	2.2529	3.7403	39.8%	0.4731	0.9076	-0.0231	0.5953	-0.1228	0.3095
85	2.2779	3.7593	39.4%	0.4772	0.9126	-0.0220	0.5921	-0.1294	0.2996
86	2.3021	3.7772	39.1%	0.4810	0.9172	-0.0205	0.5892	-0.1356	0.2899
87	2.3258	3.7938	38.7%	0.4846	0.9216	-0.0188	0.5862	-0.1414	0.2800
88	2.3488	3.8087	38.3%	0.4881	0.9257	-0.0168	0.5833	-0.1469	0.2695
89	2.3715	3.8217	37.9%	0.4914	0.9293	-0.0143	0.5801	-0.1521	0.2582
90	2.3938	3.8331	37.5%	0.4947	0.9325	-0.0115	0.5768	-0.1571	0.2461

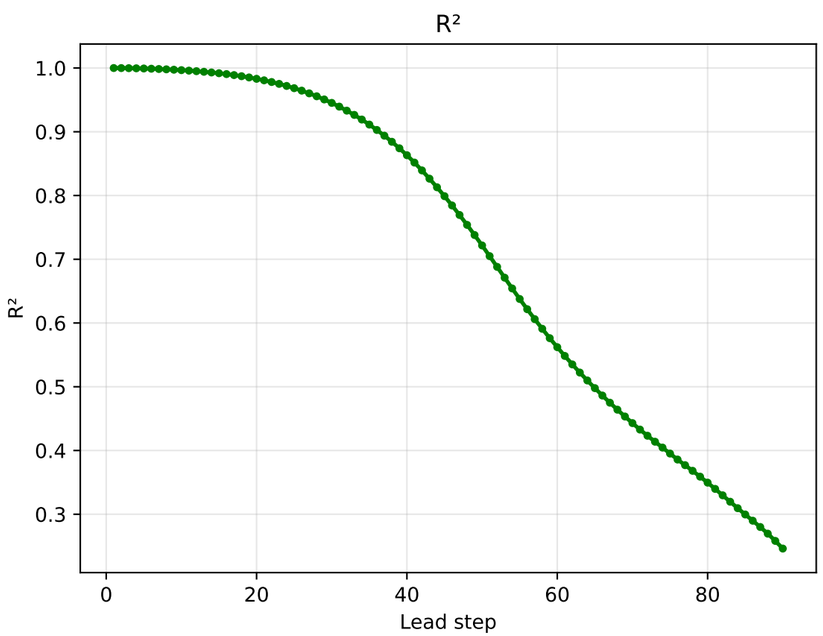
DHW Forecast Skill vs Lead Step - Val

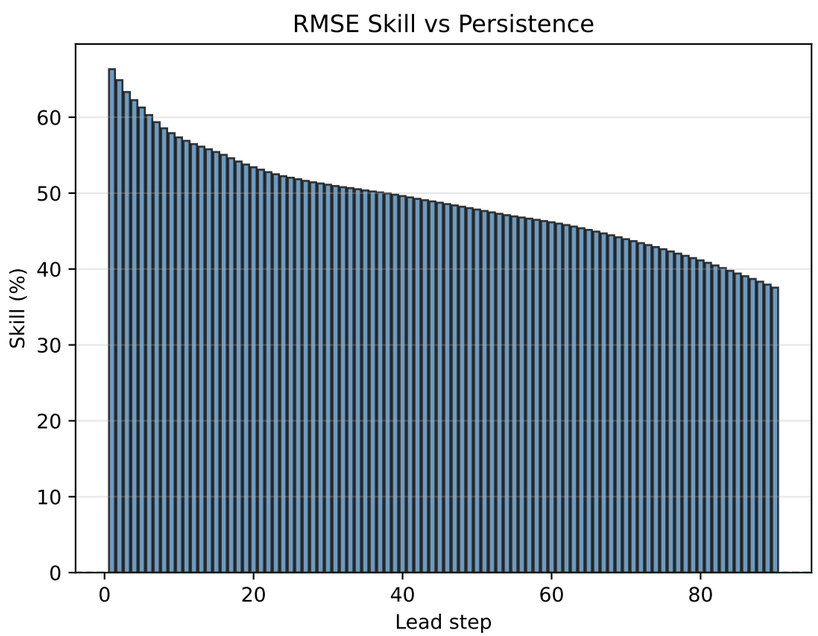


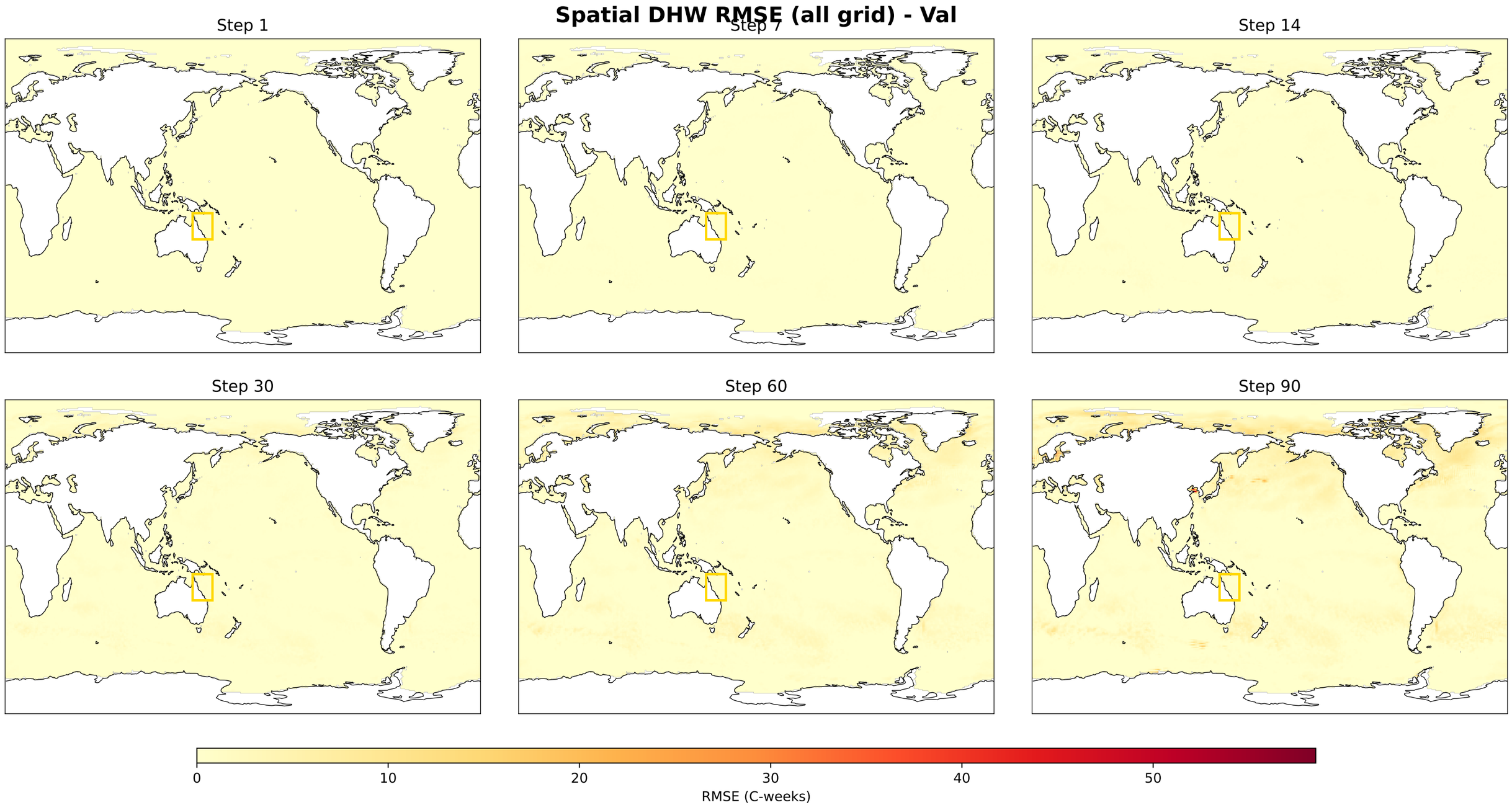


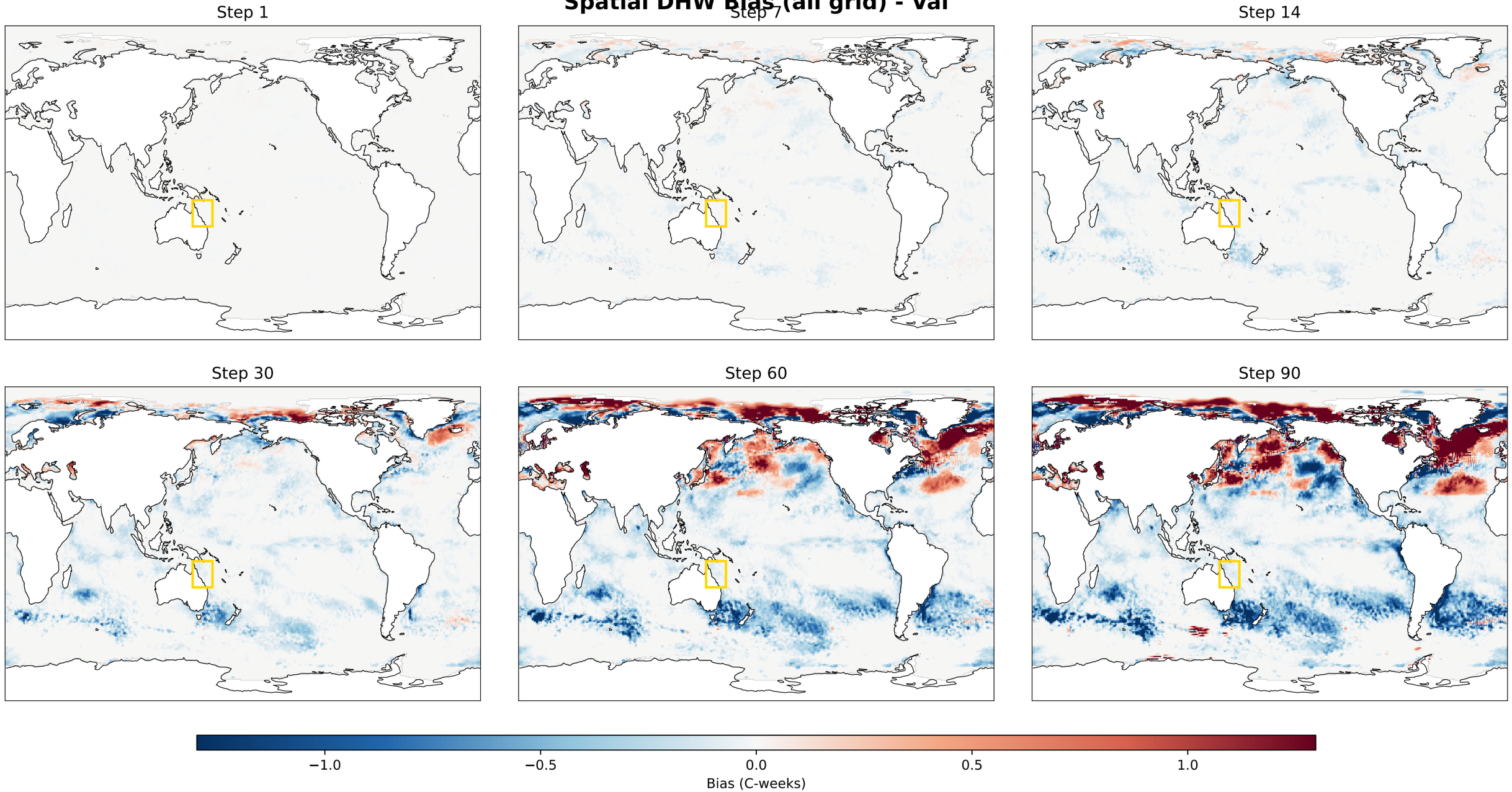


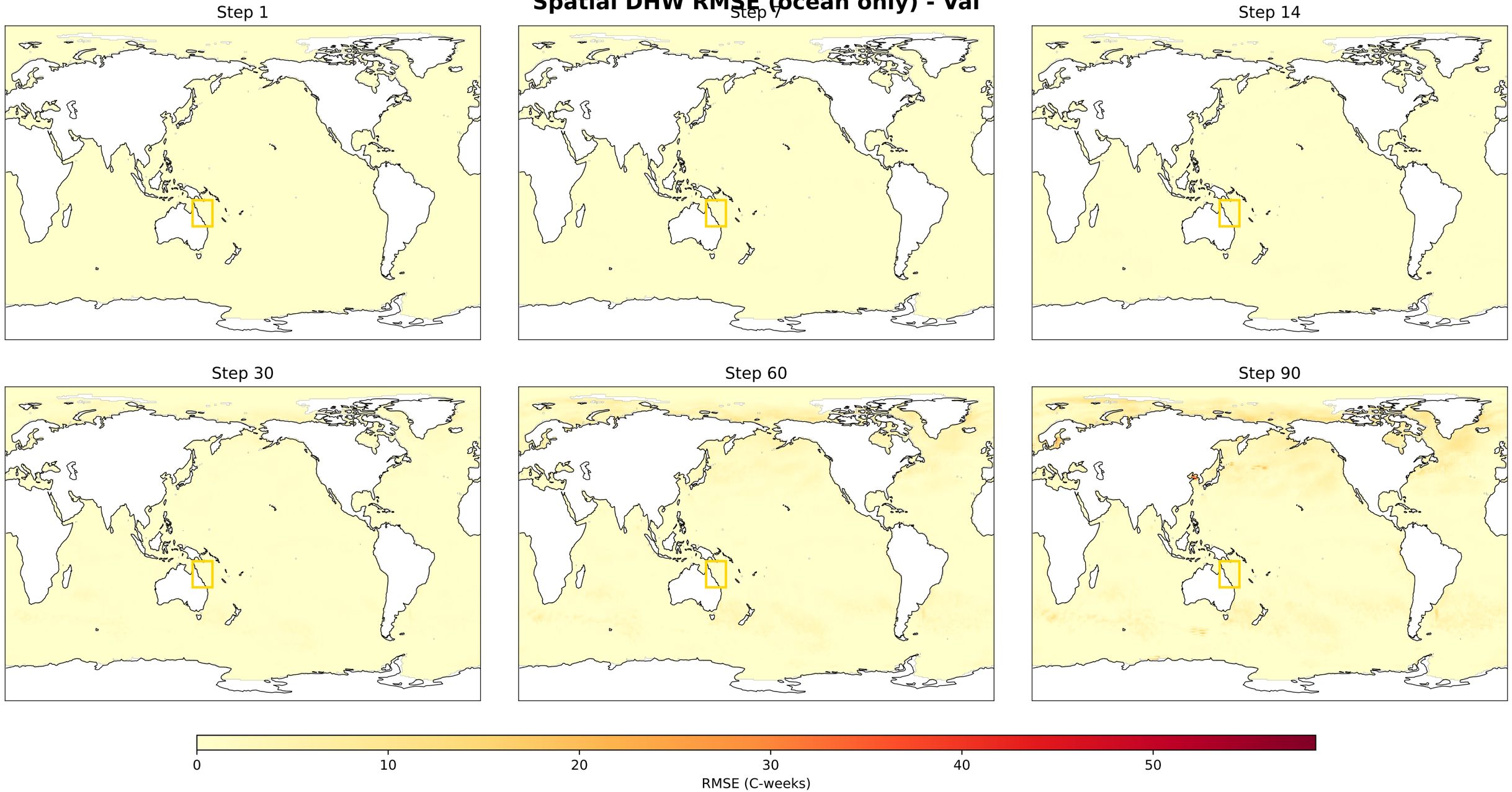


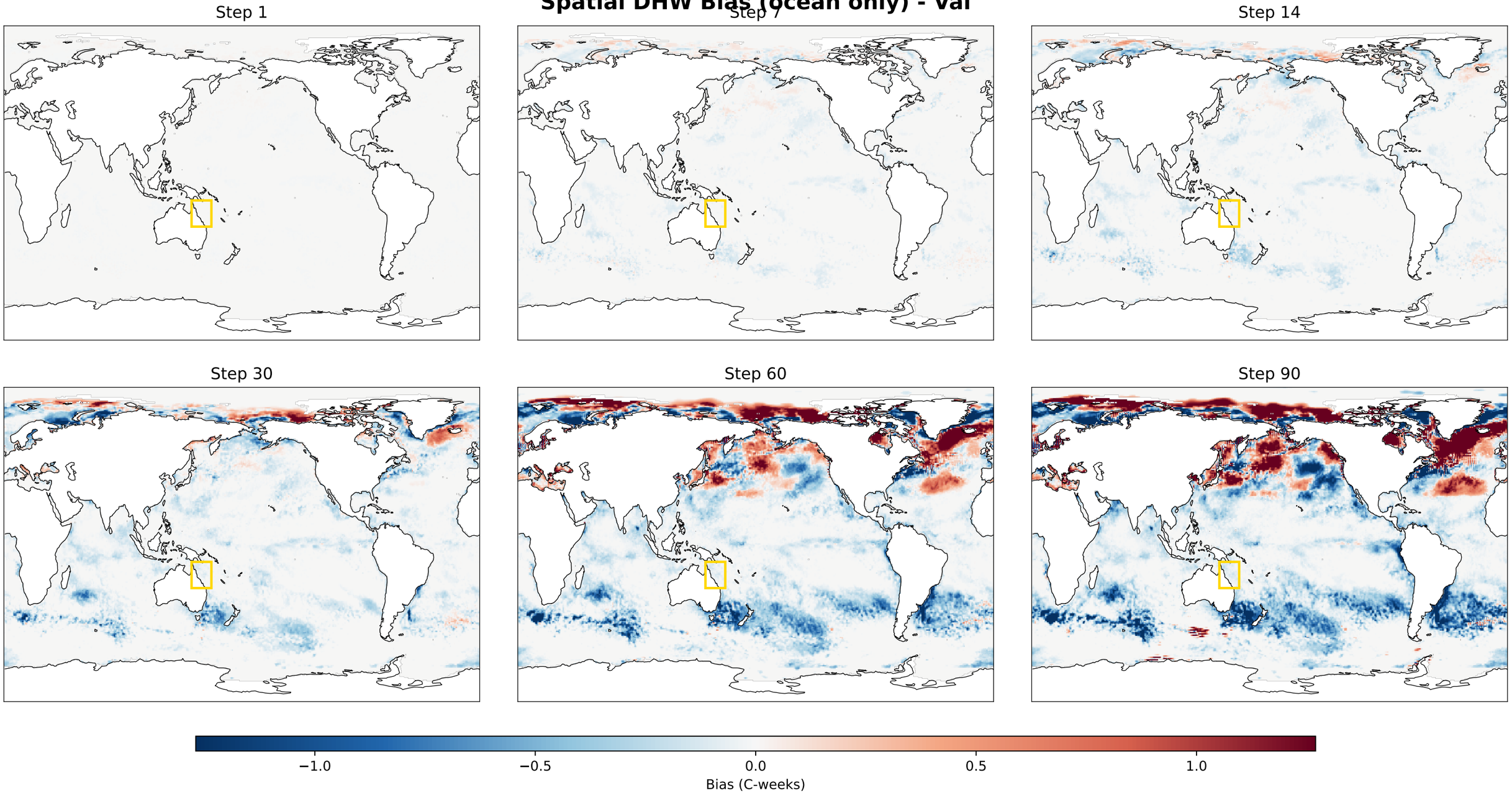





Spatial DHW RMSE (all grid) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
10
20
30
40
50
RMSE (C-weeks)

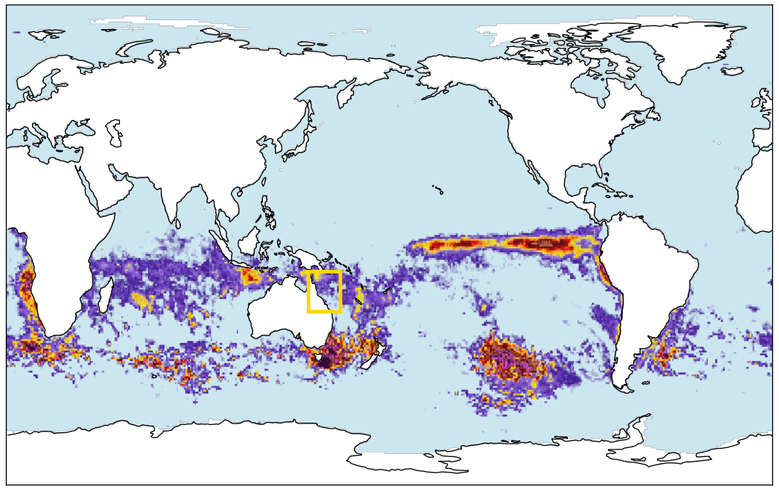

Spatial DHW Bias (all grid) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.0
−0.5
0.0
0.5
1.0
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
10
20
30
40
50
RMSE (C-weeks)

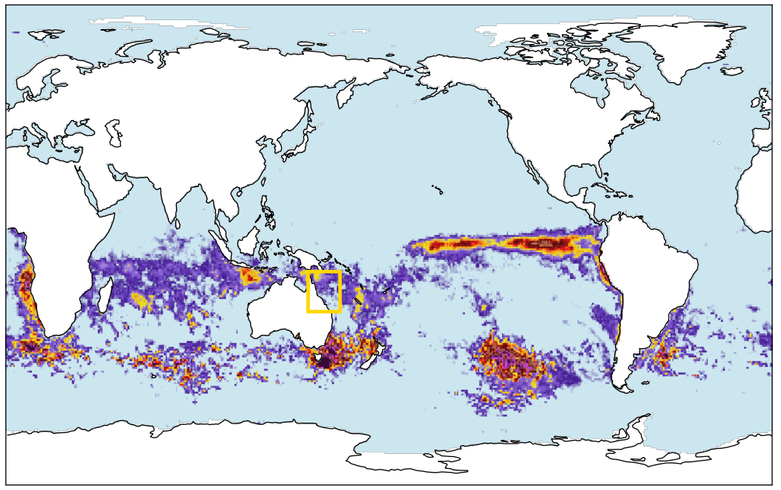

Spatial DHW Bias (ocean only) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.0
−0.5
0.0
0.5
1.0
Bias (C-weeks)

DHW Chain init: 2016-04-10 - Val (C-weeks)

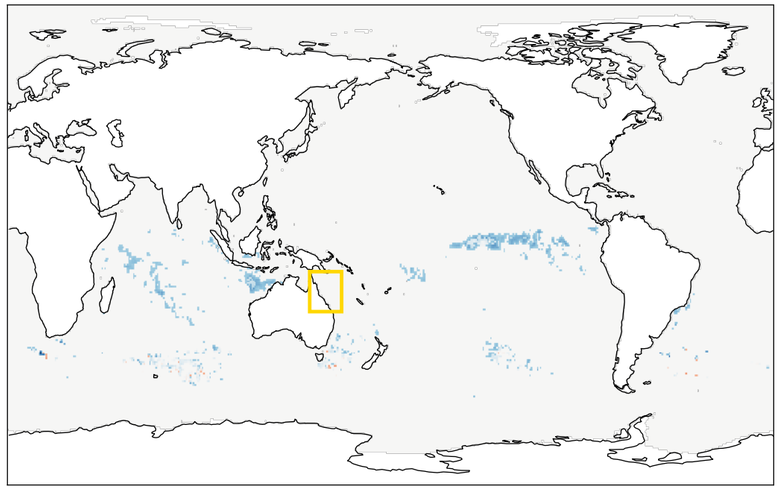
Actual DHW - Step 1



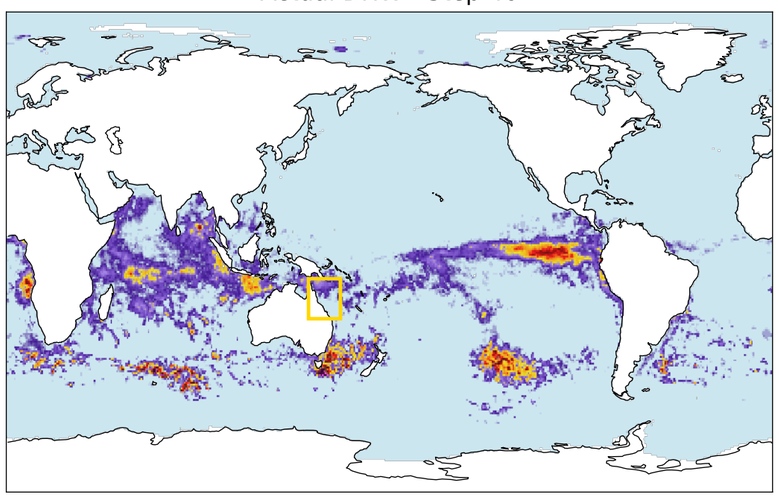
Predicted DHW - Step 1



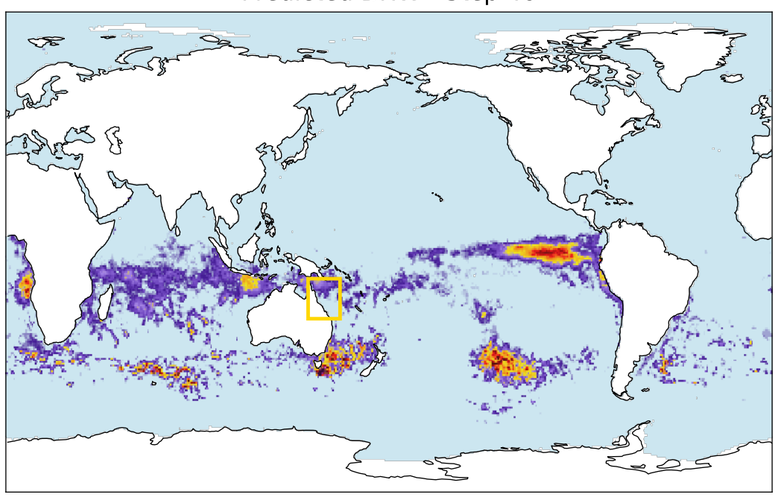
Error (Pred - Actual) - Step 1 [scale ±0.53]



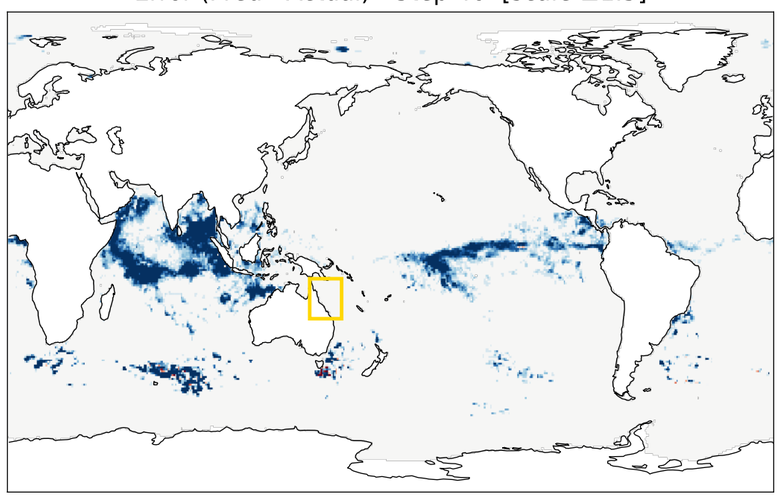
Actual DHW - Step 46



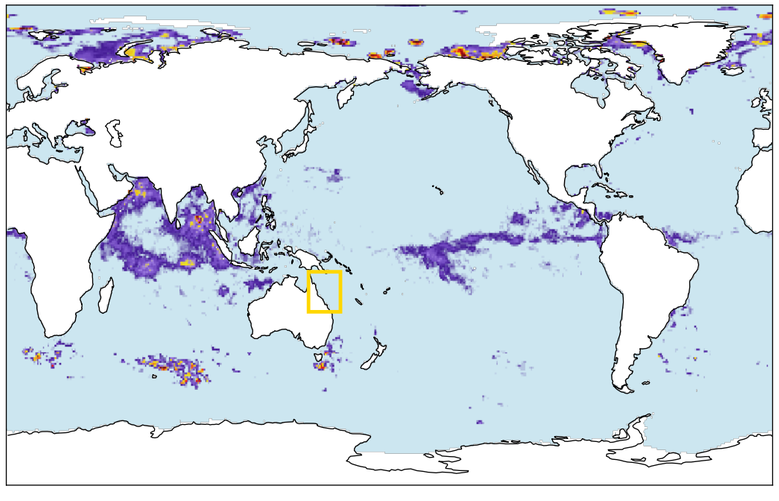
Predicted DHW - Step 46



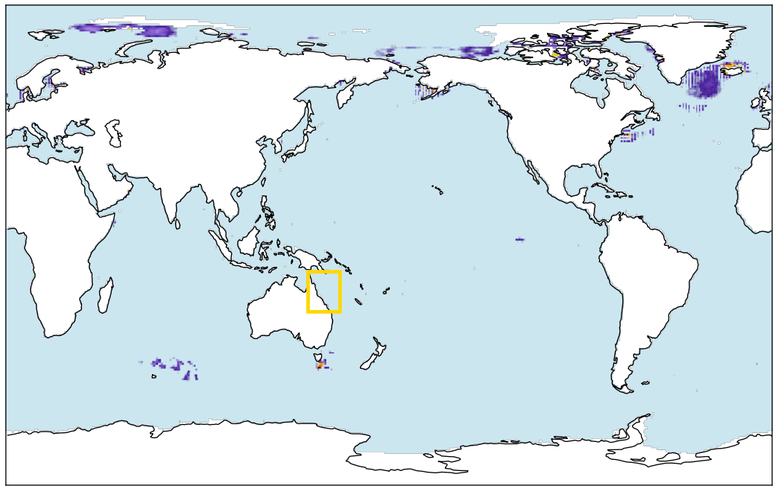
Error (Pred - Actual) - Step 46 [scale ±1.5]



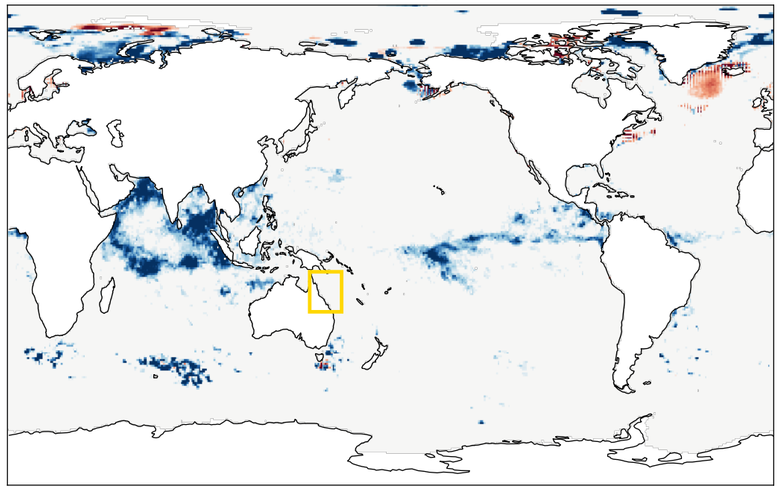
Actual DHW - Step 90

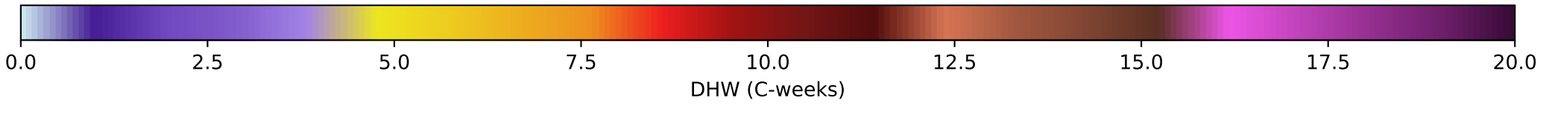


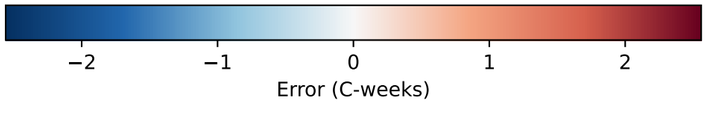
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±2.6]

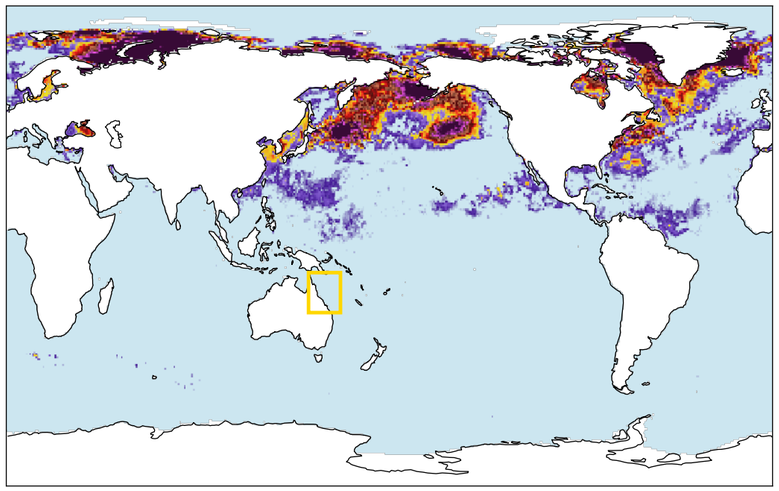




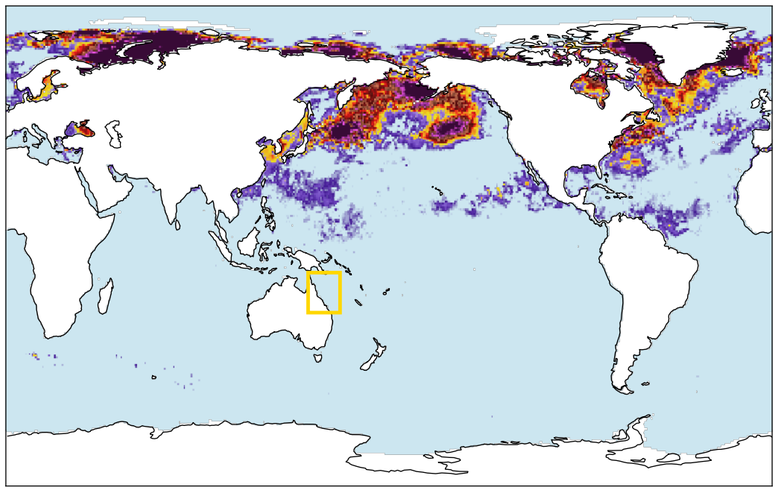


DHW Chain init: 2016-10-07 - Val (C-weeks)

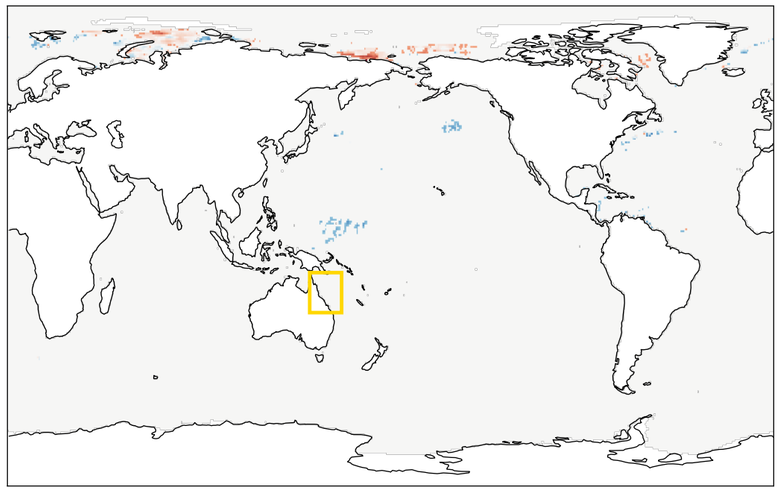
Actual DHW - Step 1



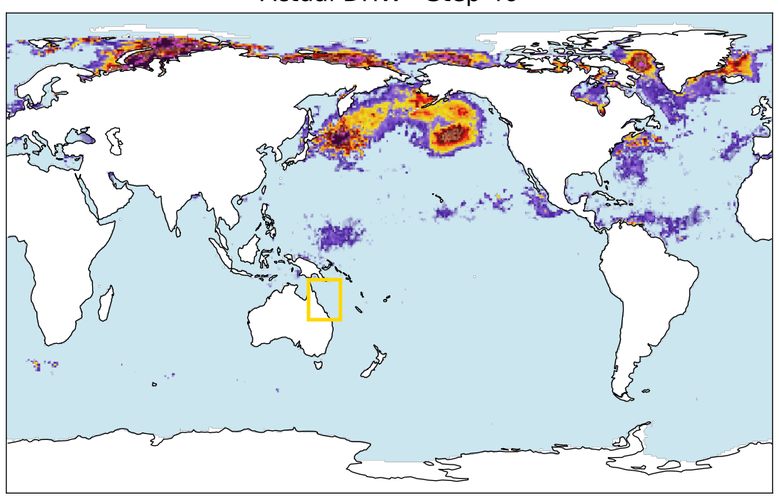
Predicted DHW - Step 1



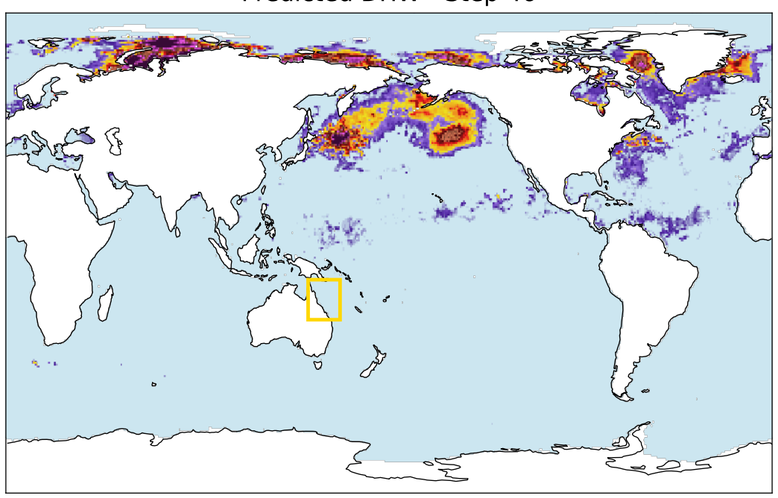
Error (Pred - Actual) - Step 1 [scale ±0.44]



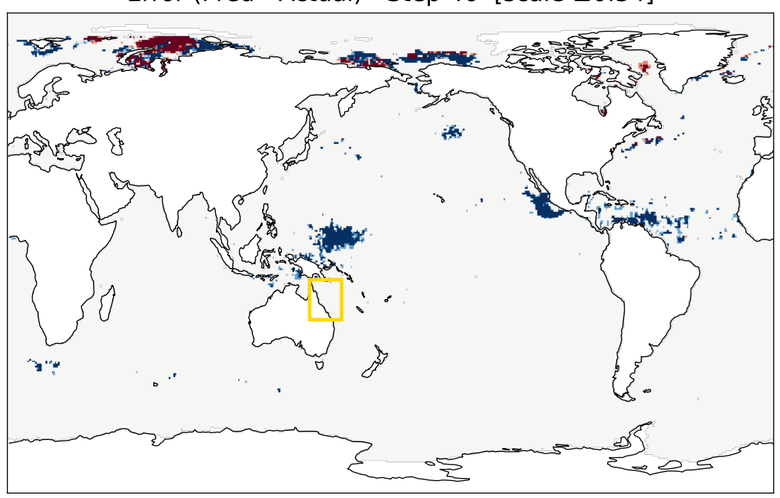
Actual DHW - Step 46



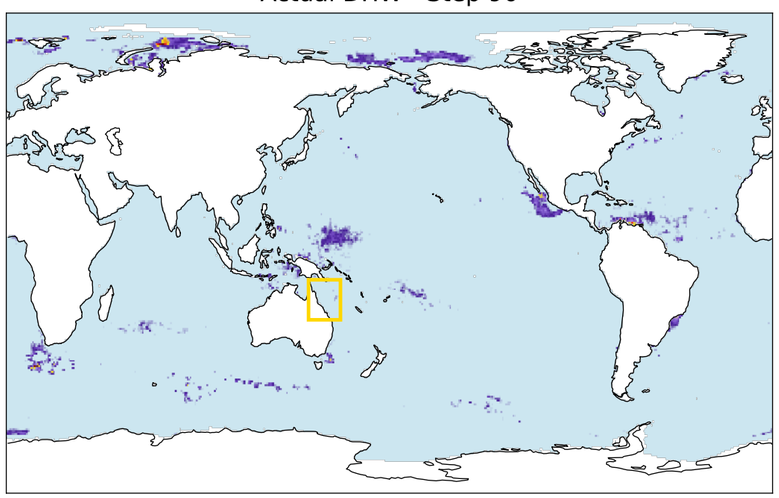
Predicted DHW - Step 46



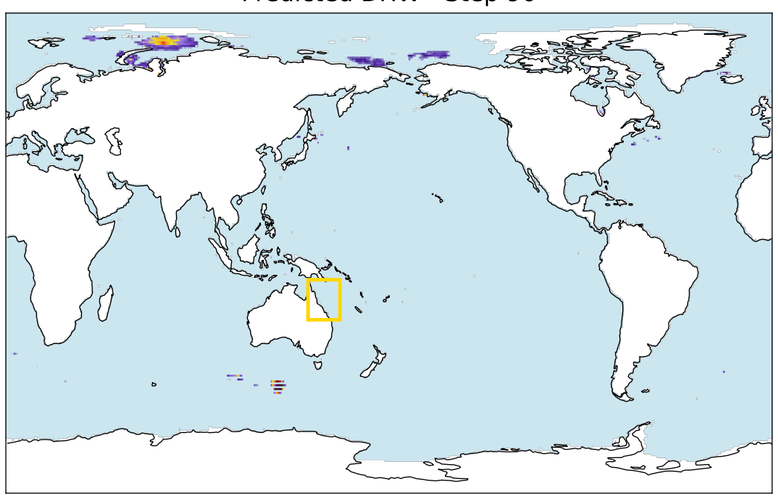
Error (Pred - Actual) - Step 46 [scale ±0.34]



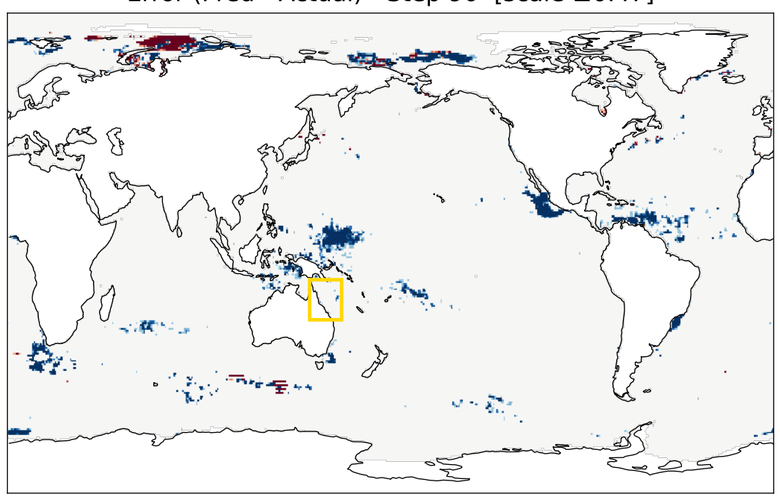
Actual DHW - Step 90

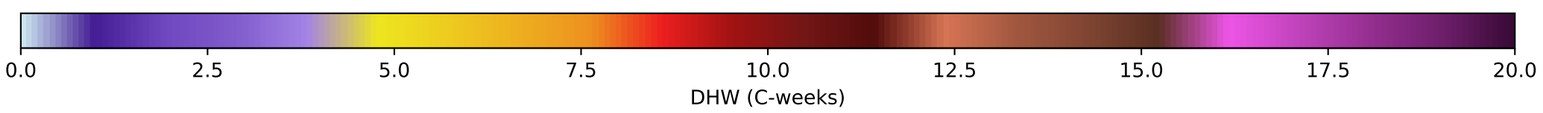


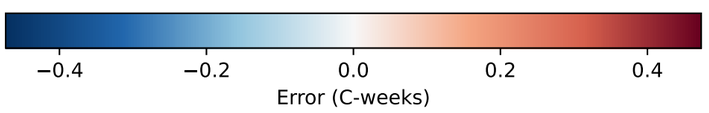
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.47]

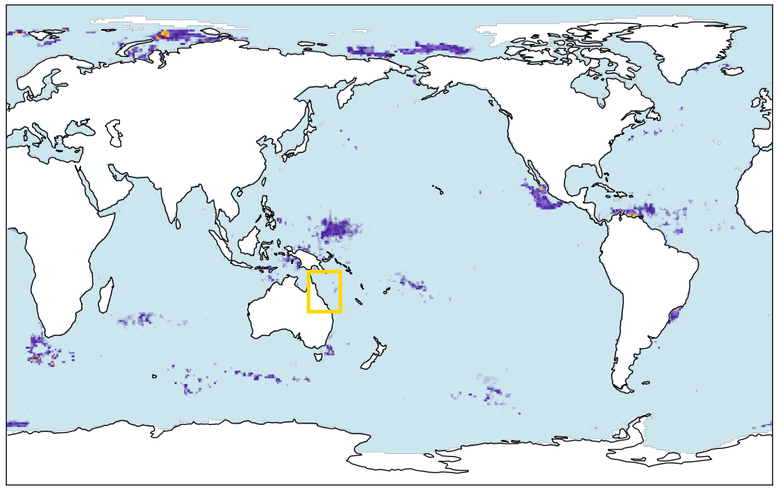




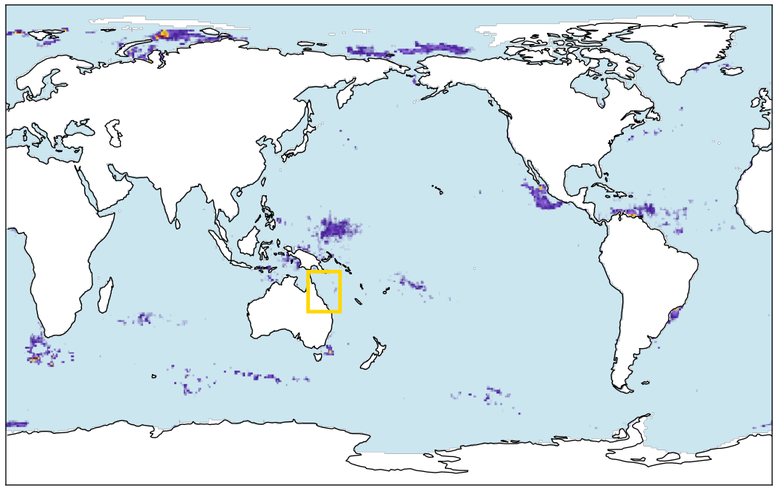


DHW Chain init: 2017-01-05 - Val (C-weeks)

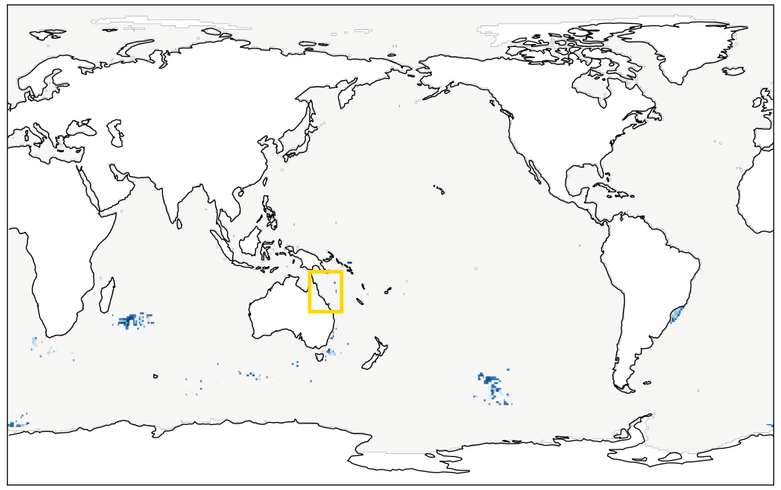
Actual DHW - Step 1



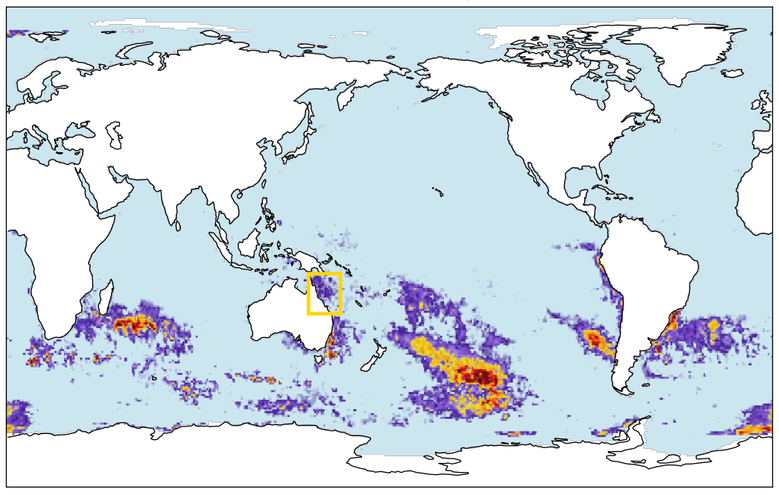
Predicted DHW - Step 1



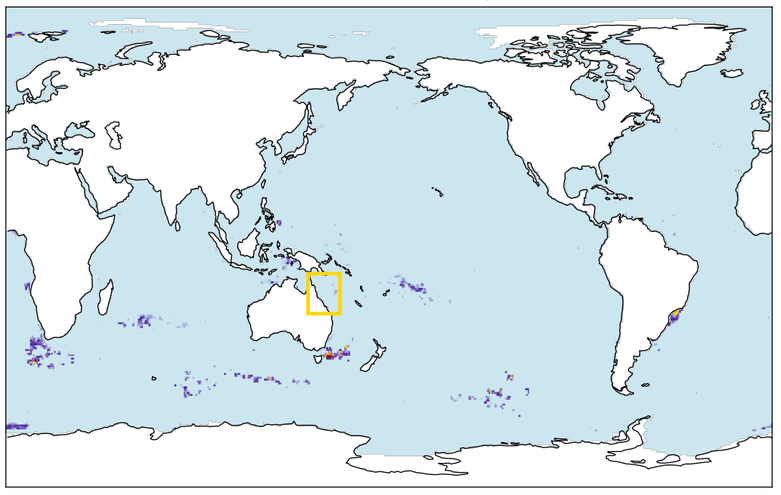
Error (Pred - Actual) - Step 1 [scale ±0.26]



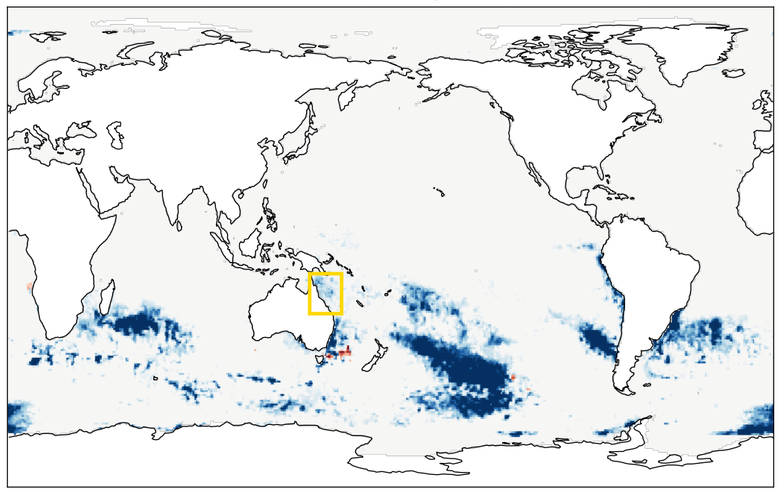
Actual DHW - Step 46



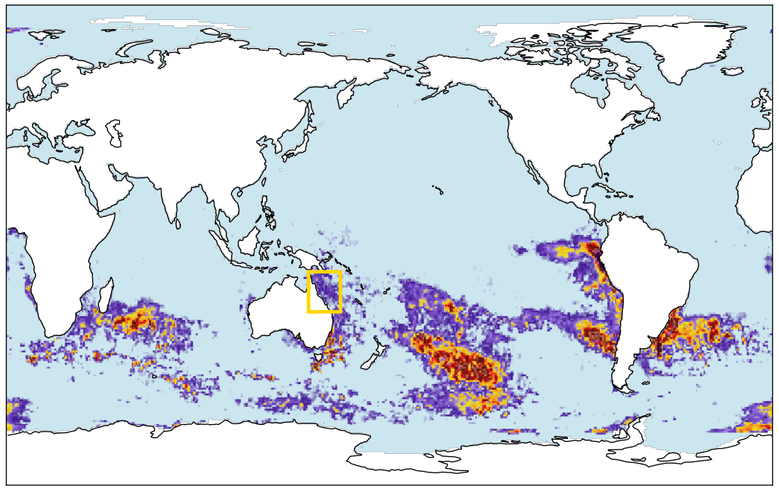
Predicted DHW - Step 46



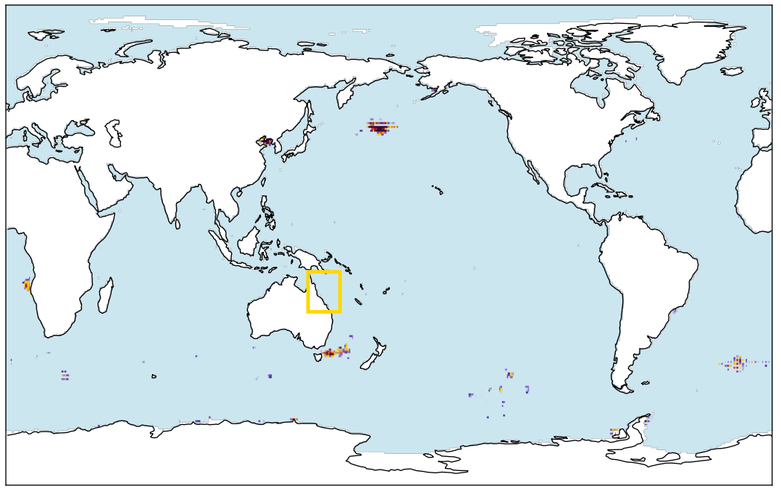
Error (Pred - Actual) - Step 46 [scale ±4.4]



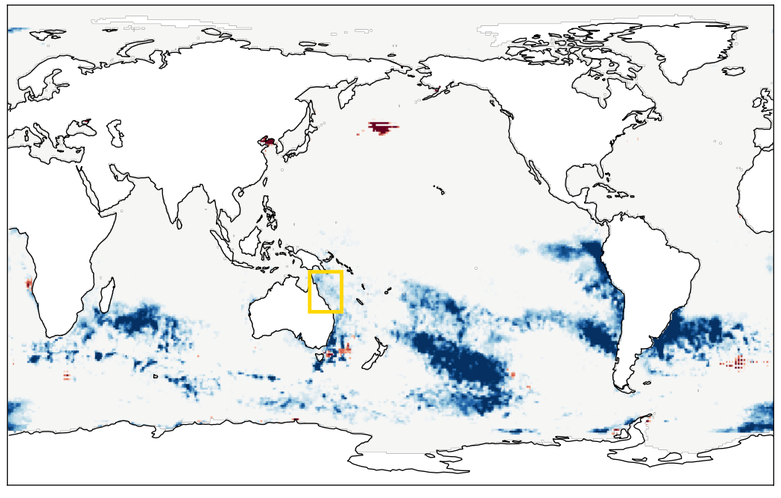
Actual DHW - Step 90

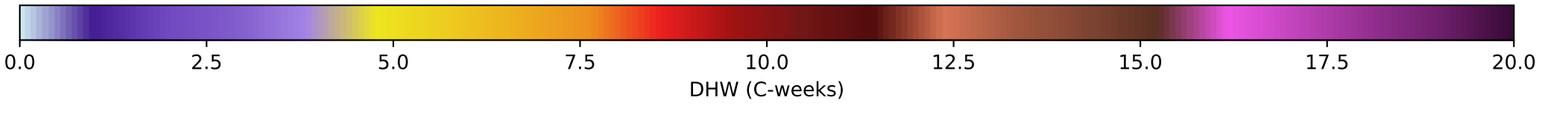


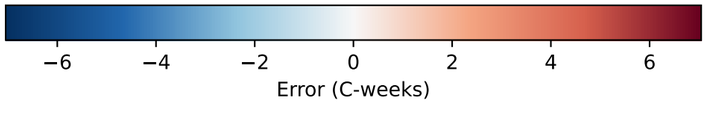
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±7]

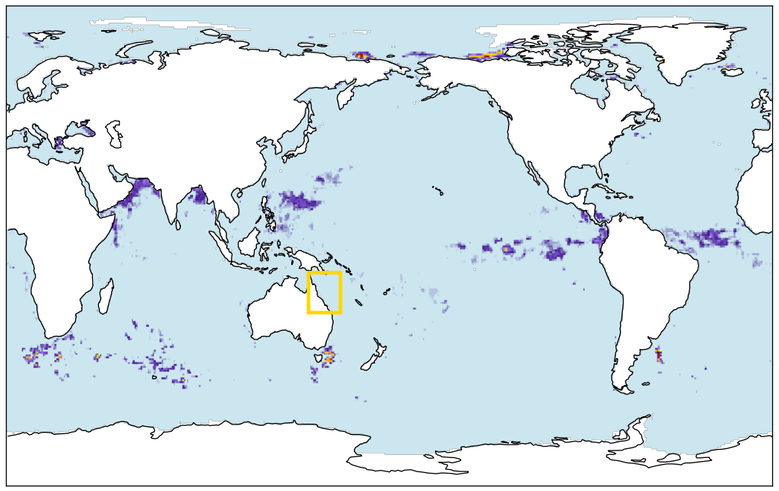




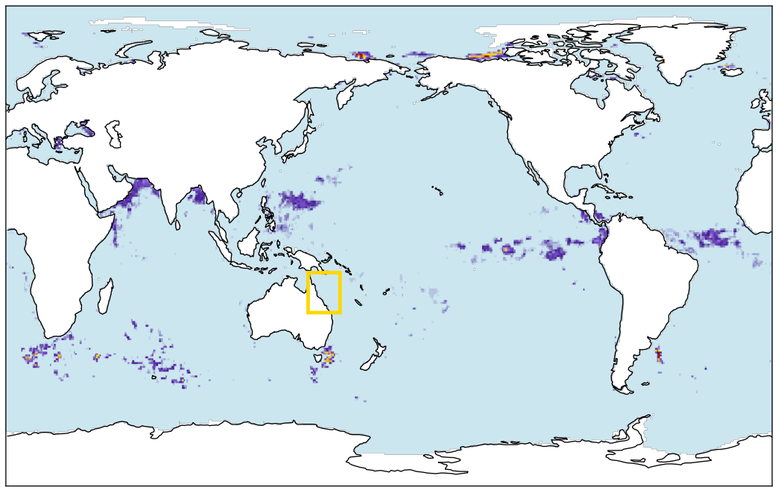


DHW Chain init: 2017-07-04 - Val (C-weeks)

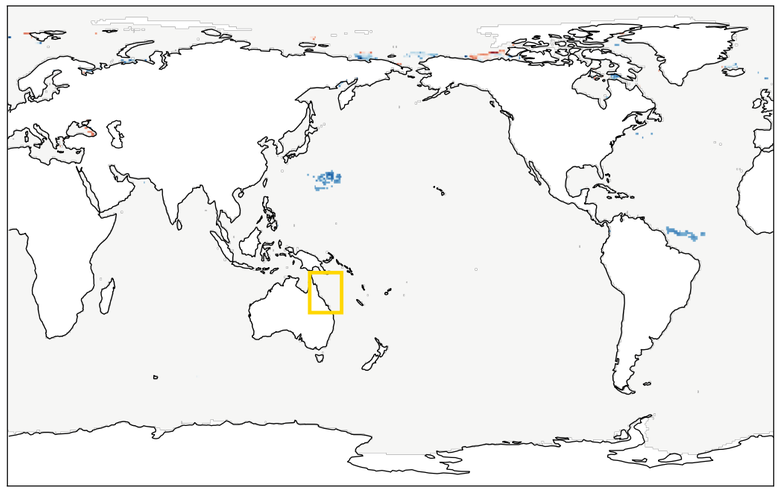
Actual DHW - Step 1



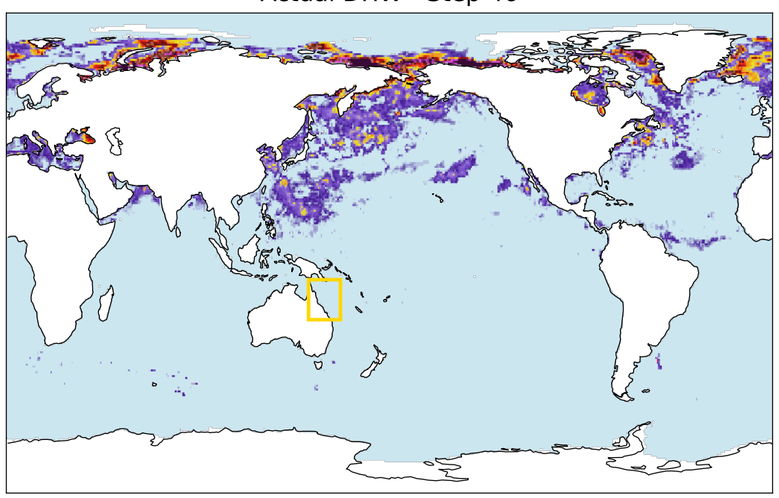
Predicted DHW - Step 1



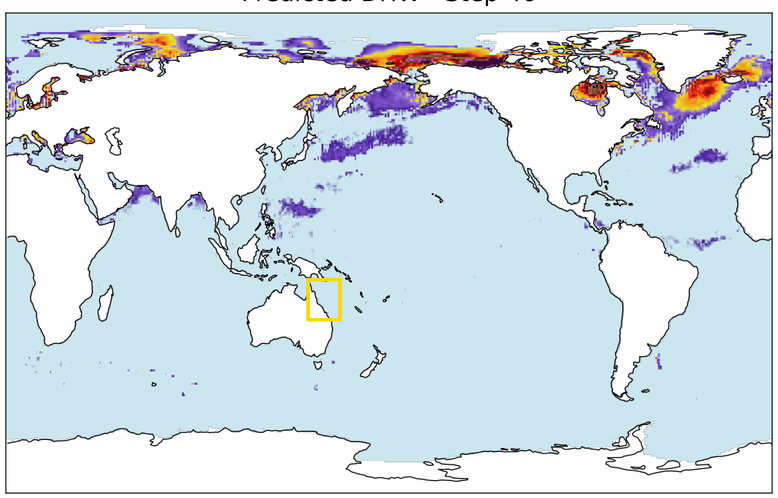
Error (Pred - Actual) - Step 1 [scale ±0.35]



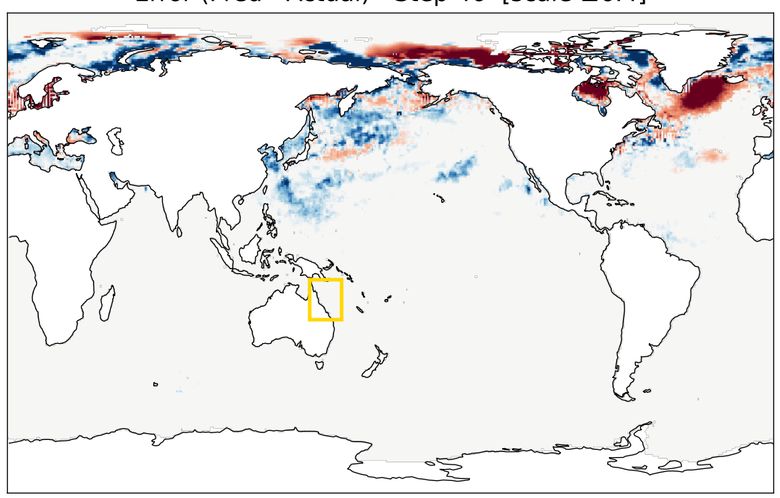
Actual DHW - Step 46



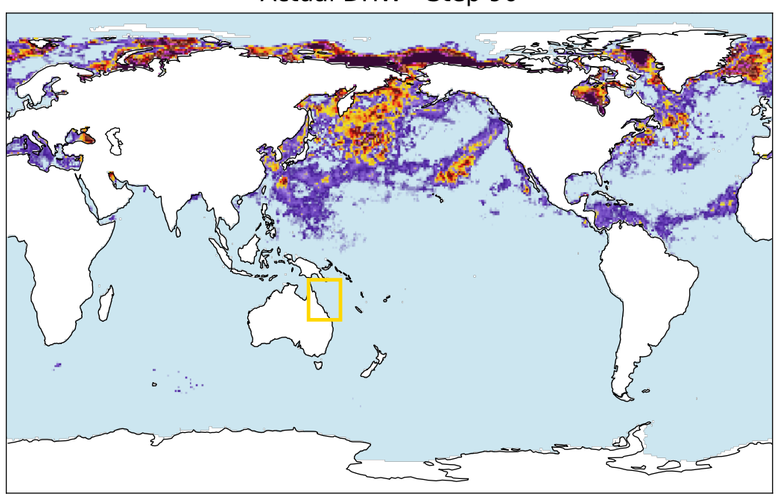
Predicted DHW - Step 46



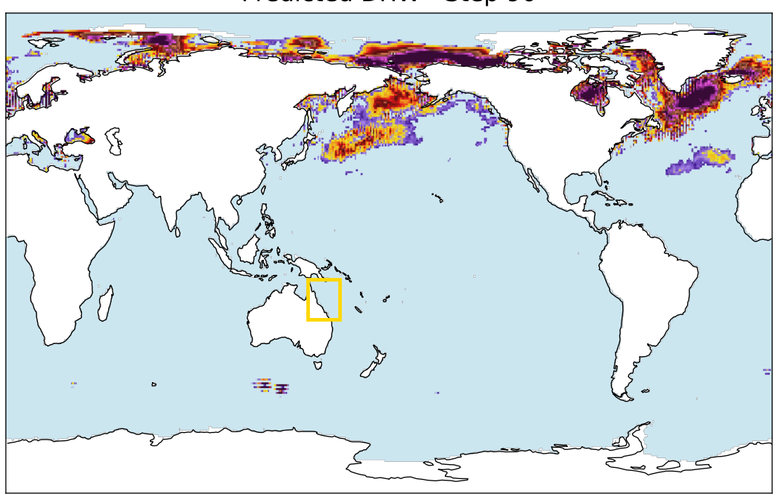
Error (Pred - Actual) - Step 46 [scale ±6.4]



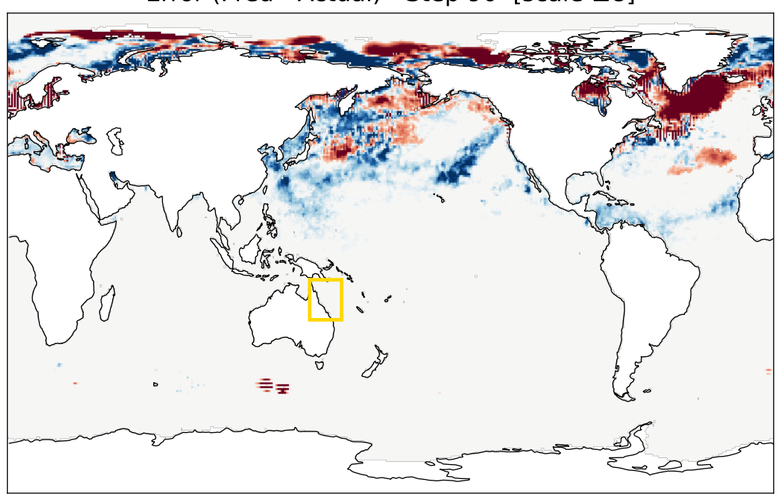
Actual DHW - Step 90

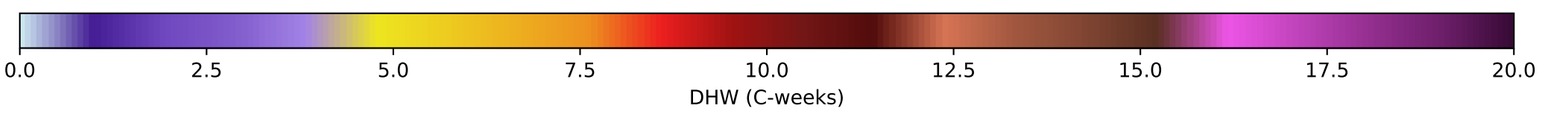


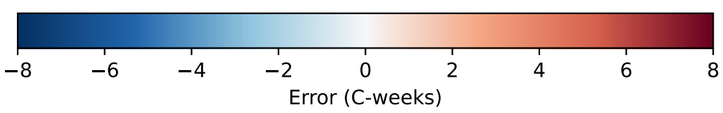
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

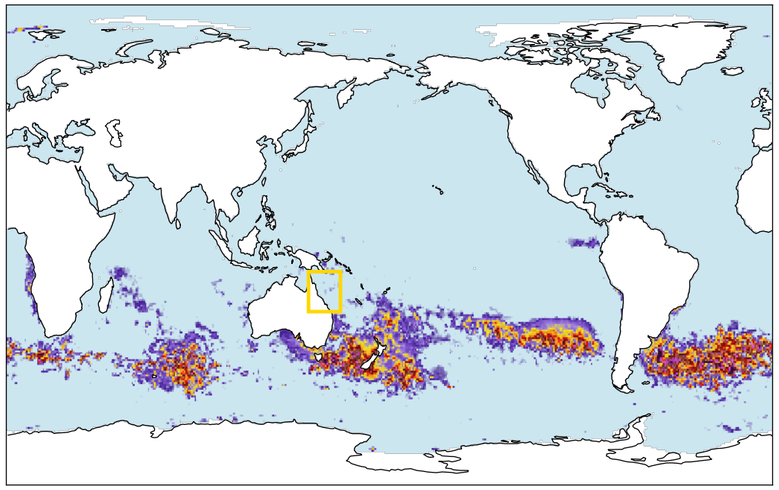




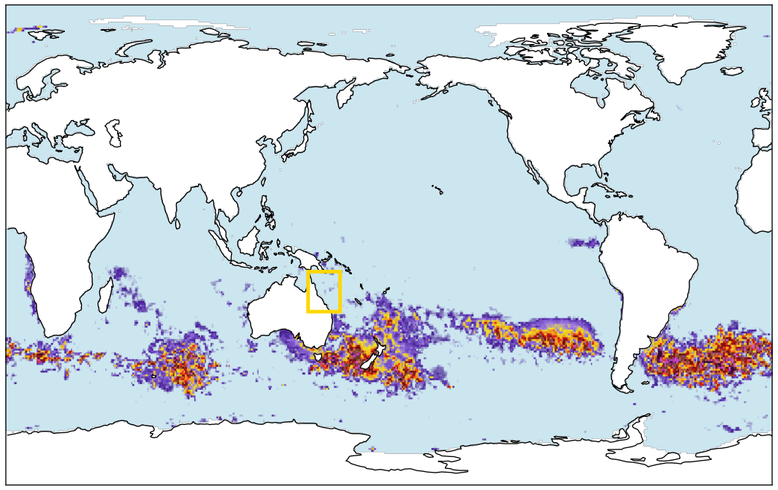


DHW Chain init: 2018-03-31 - Val (C-weeks)

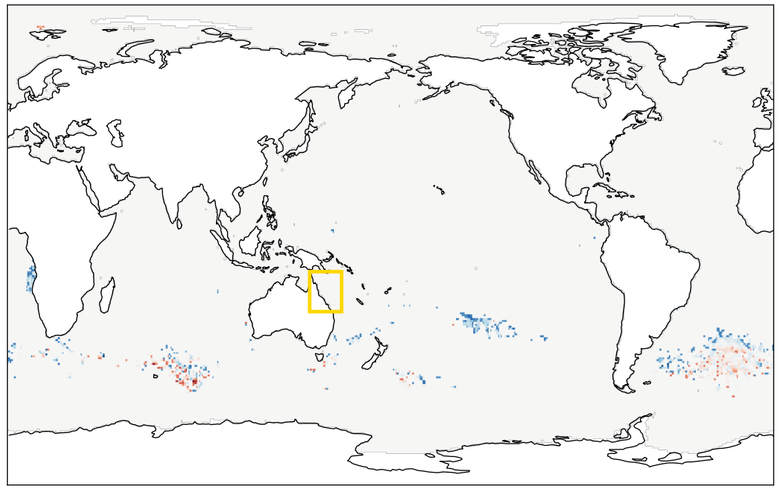
Actual DHW - Step 1



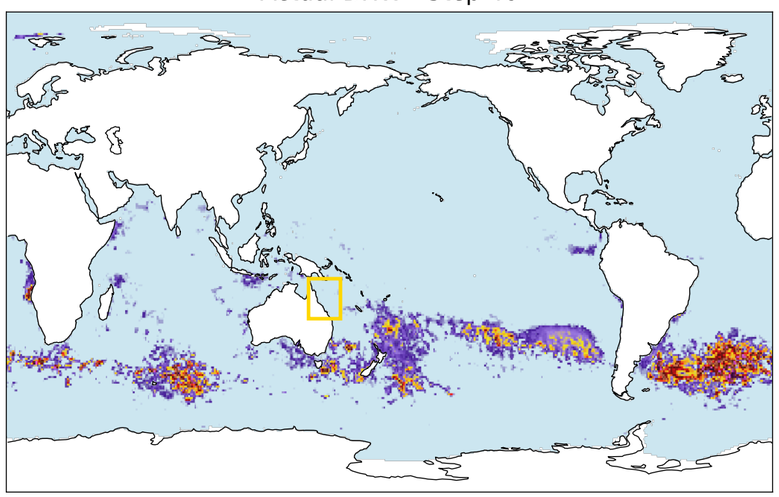
Predicted DHW - Step 1



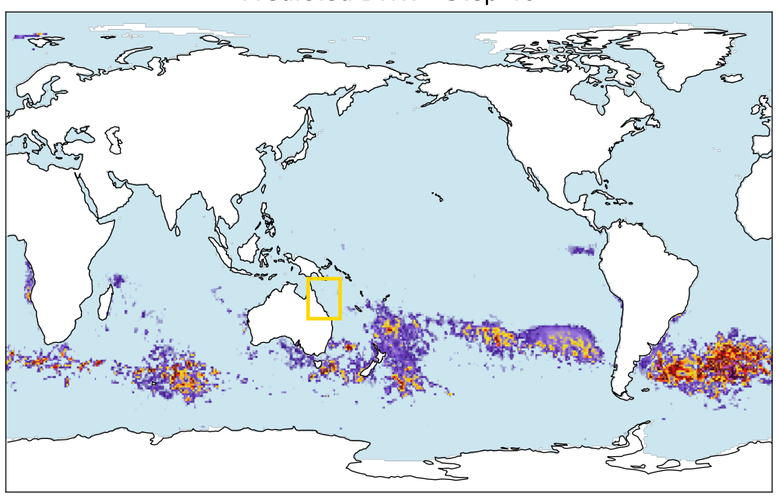
Error (Pred - Actual) - Step 1 [scale ±0.3]



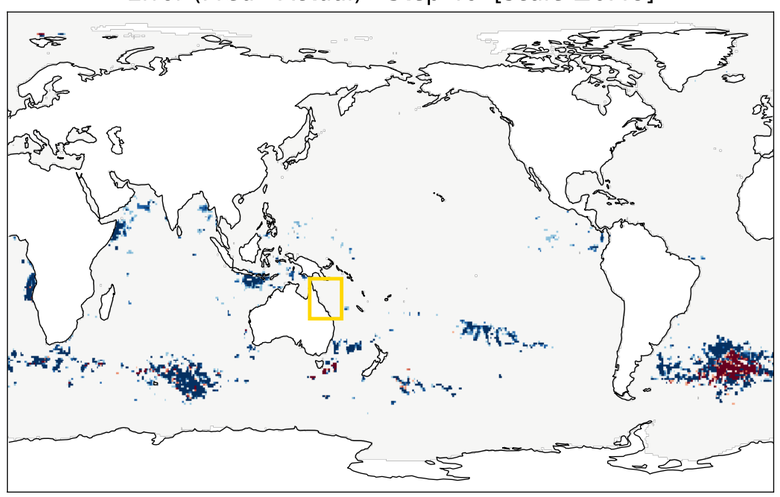
Actual DHW - Step 46



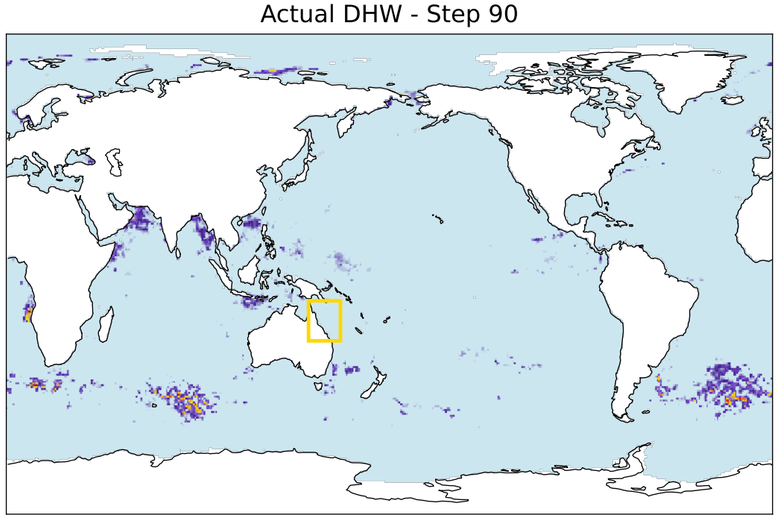
Predicted DHW - Step 46



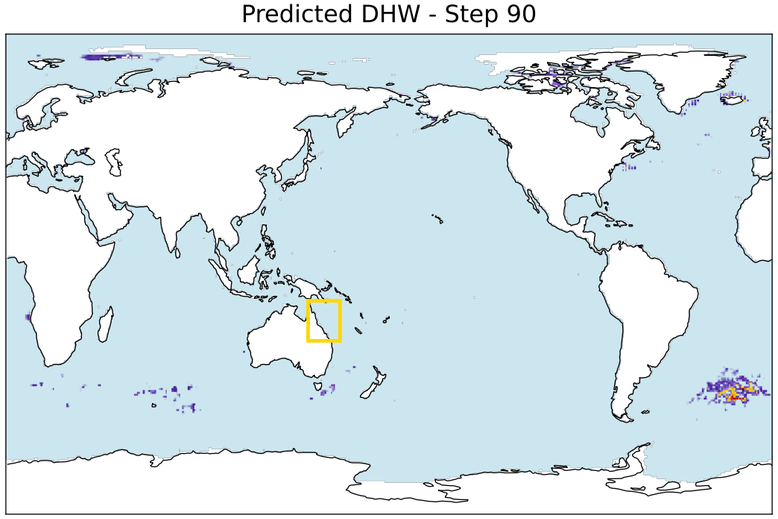
Error (Pred - Actual) - Step 46 [scale ±0.48]



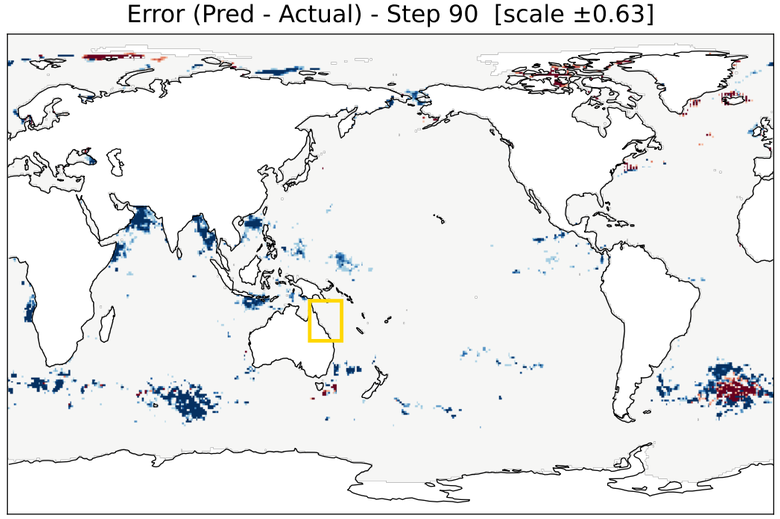
Actual DHW - Step 90

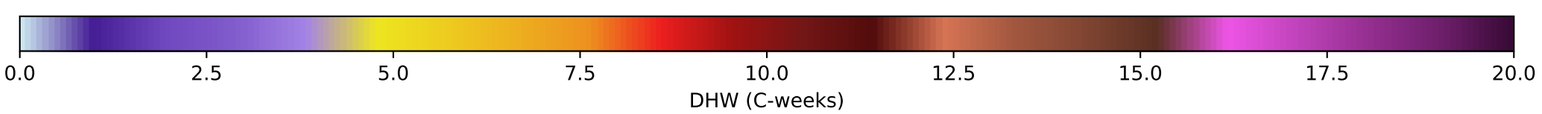


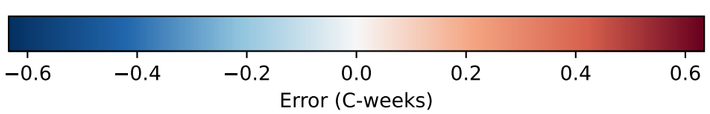
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.63]

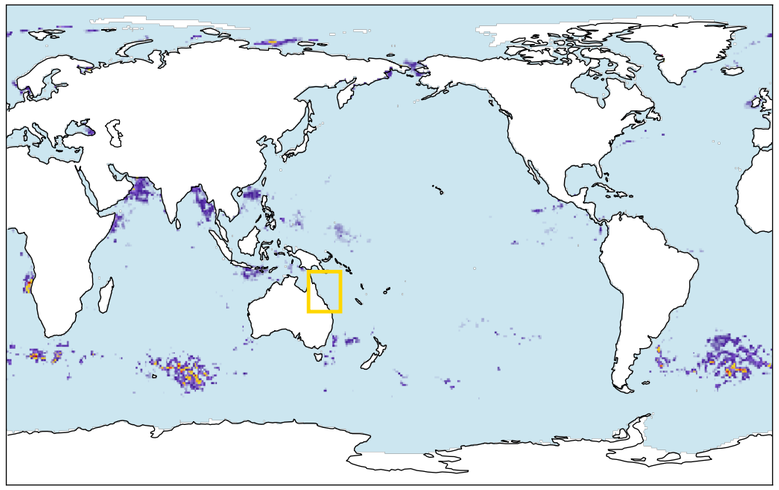




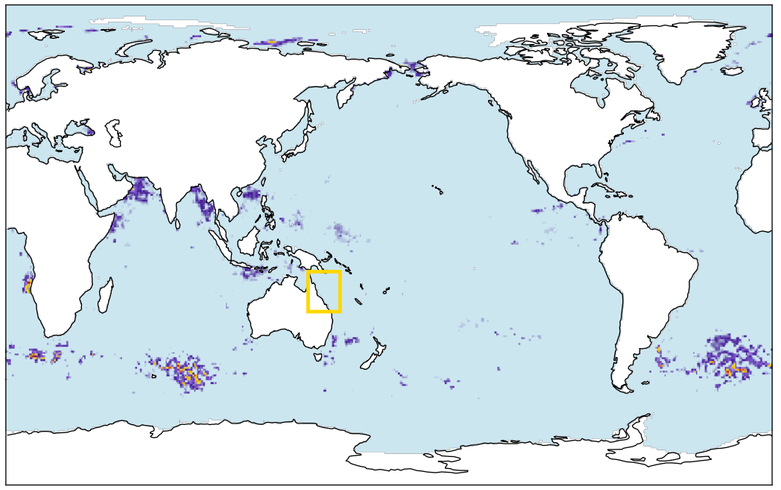


DHW Chain init: 2018-06-29 - Val (C-weeks)

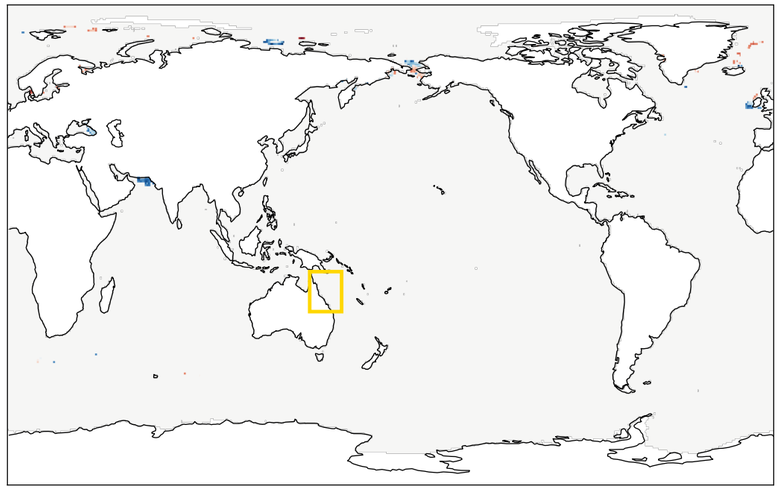
Actual DHW - Step 1



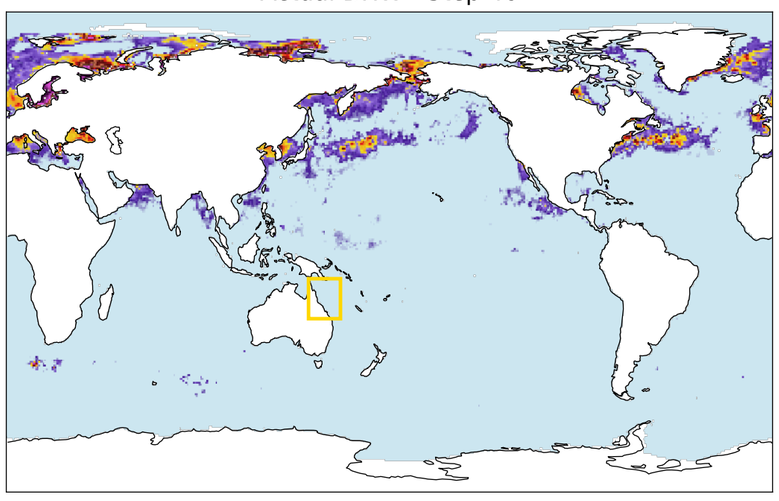
Predicted DHW - Step 1



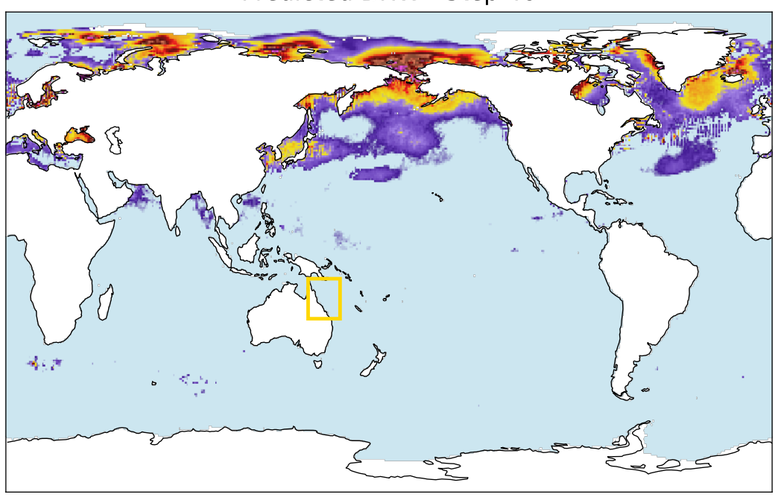
Error (Pred - Actual) - Step 1 [scale ±0.31]



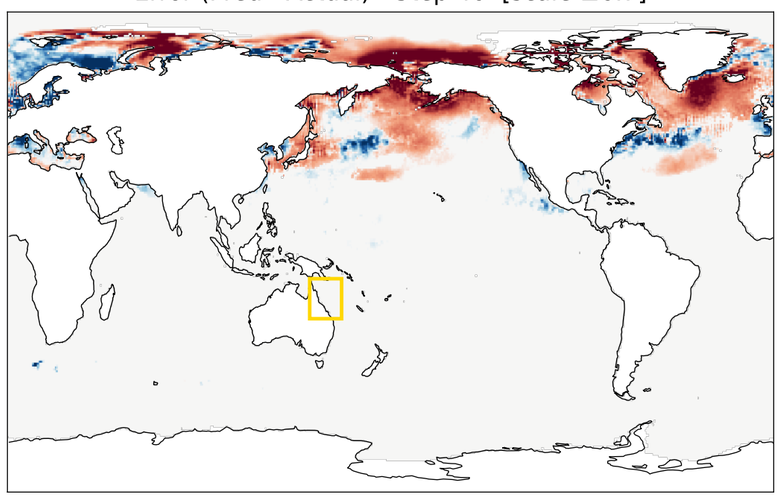
Actual DHW - Step 46



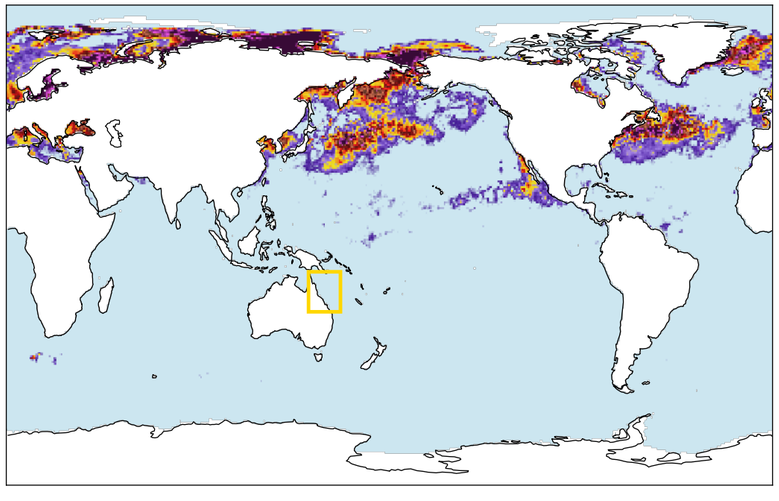
Predicted DHW - Step 46



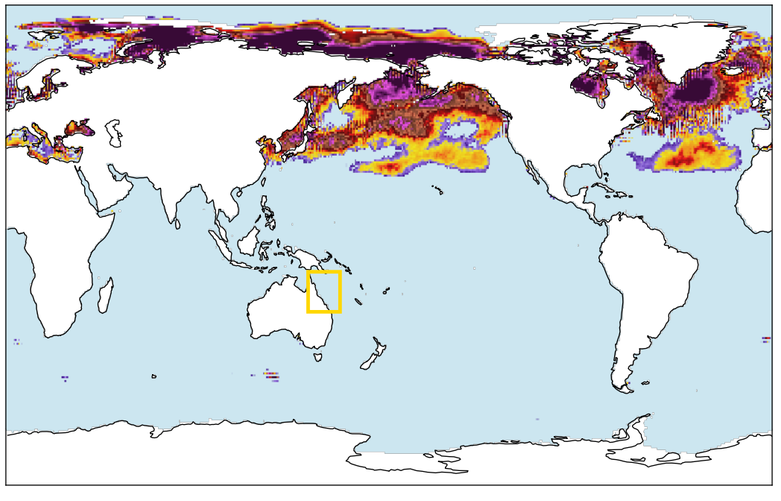
Error (Pred - Actual) - Step 46 [scale ±6.7]



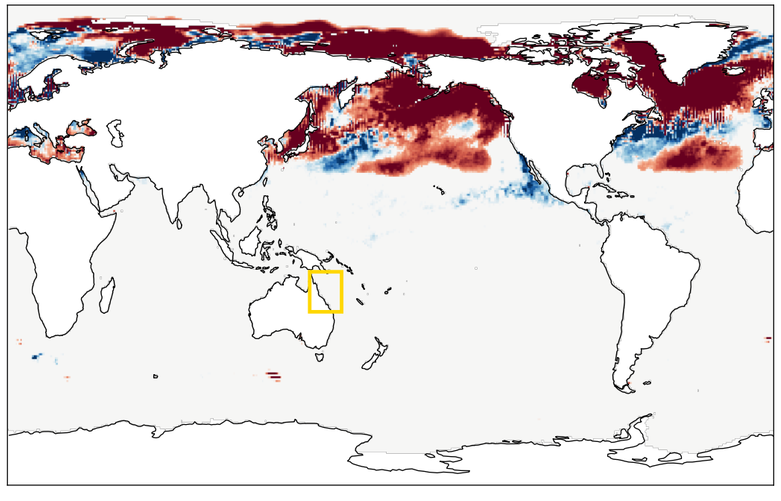
Actual DHW - Step 90

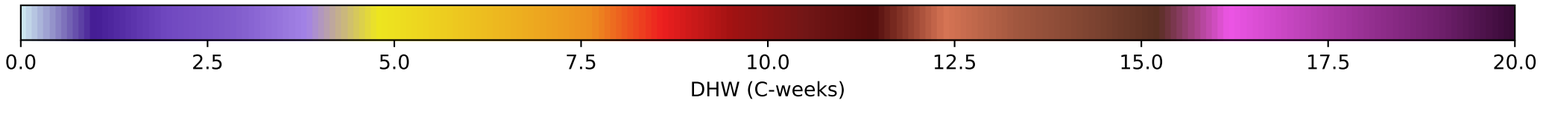


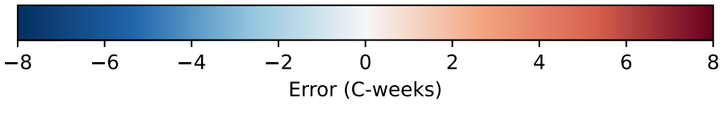
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

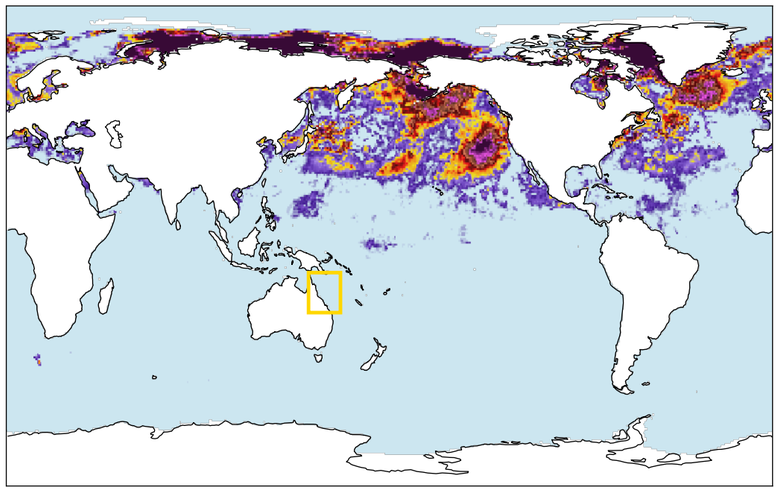




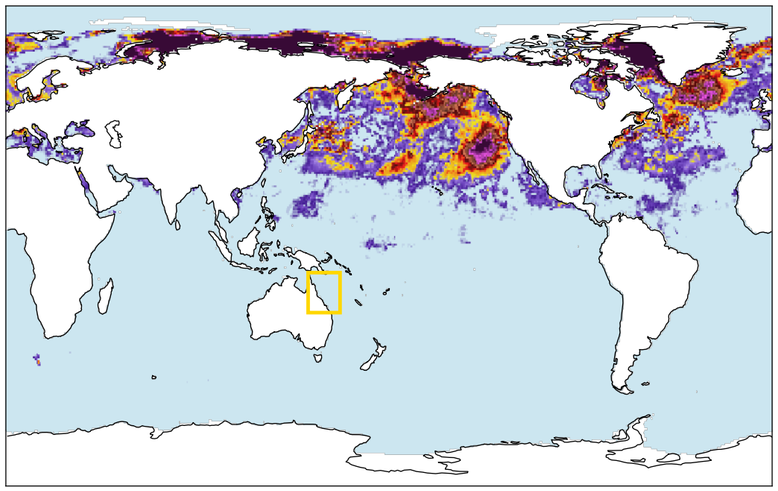


DHW Chain init: 2019-09-22 - Val (C-weeks)

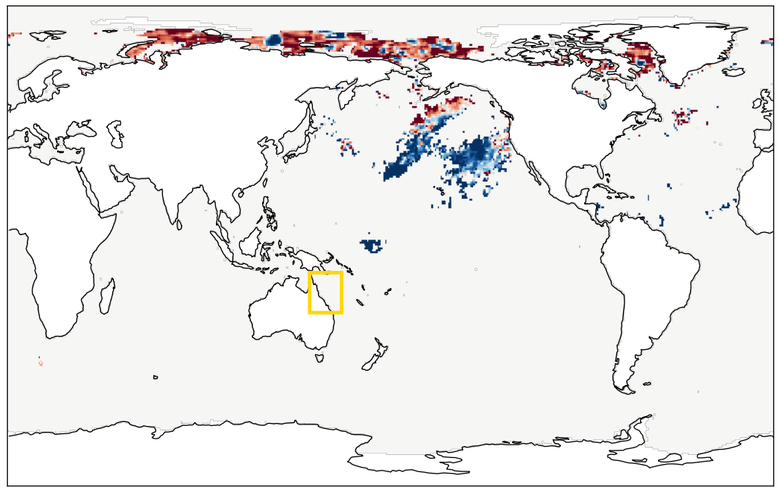
Actual DHW - Step 1



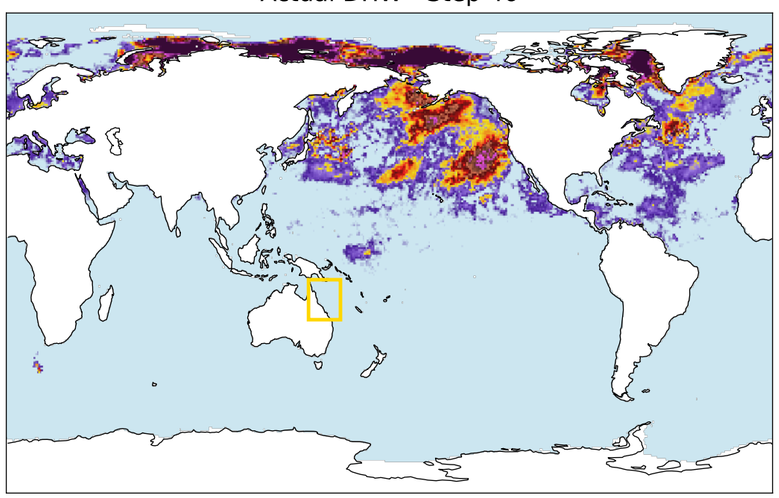
Predicted DHW - Step 1



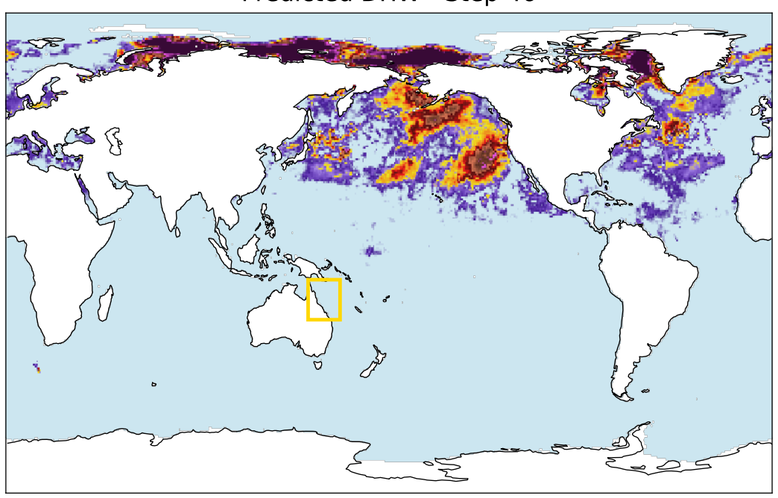
Error (Pred - Actual) - Step 1 [scale ±0.056]



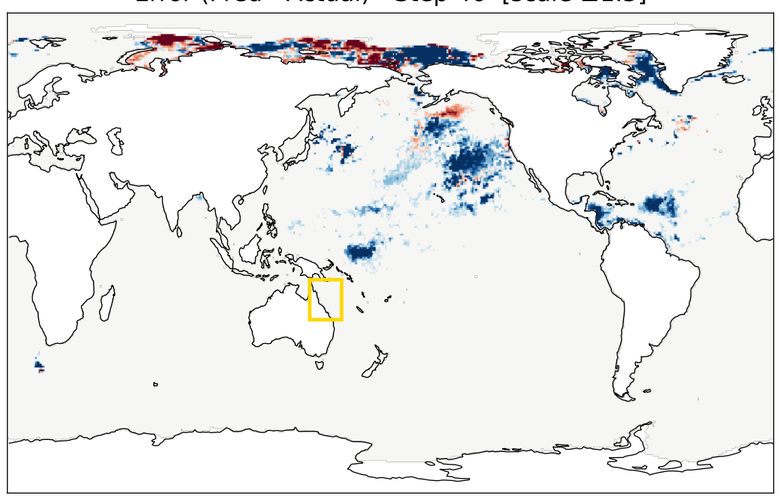
Actual DHW - Step 46



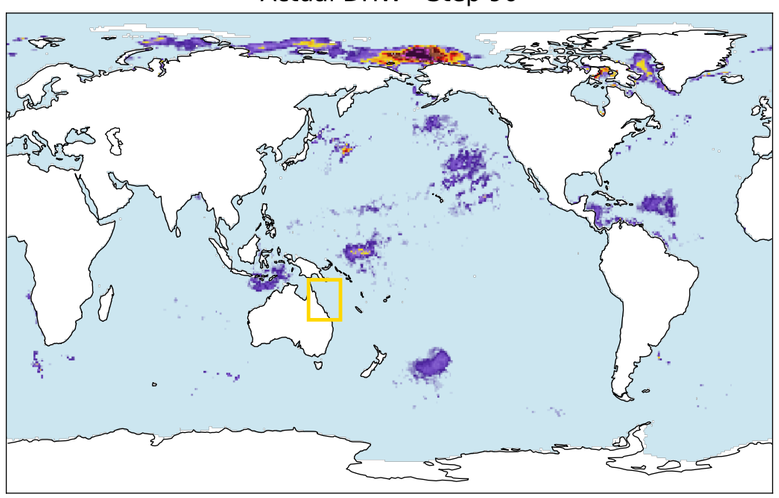
Predicted DHW - Step 46



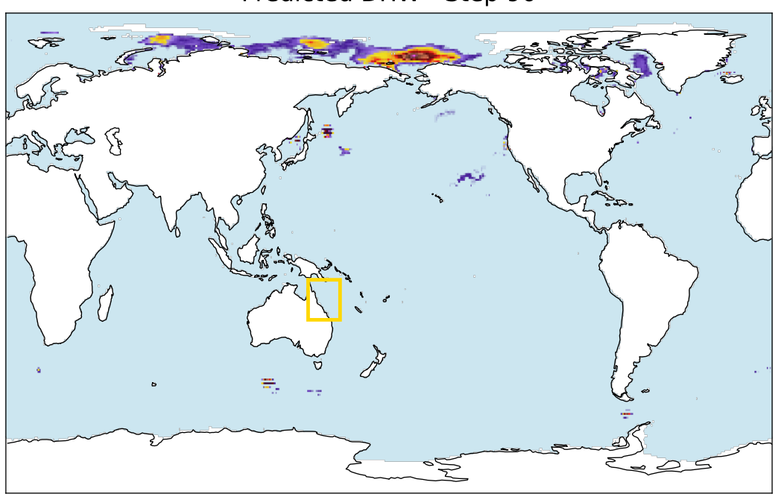
Error (Pred - Actual) - Step 46 [scale ±1.5]



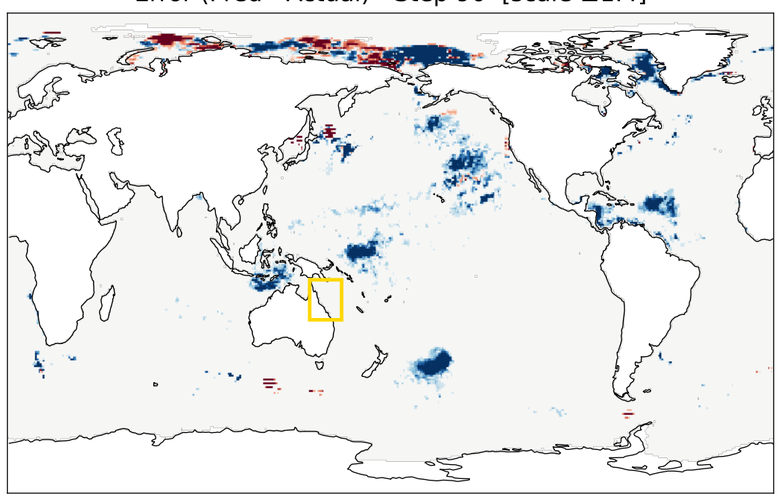
Actual DHW - Step 90

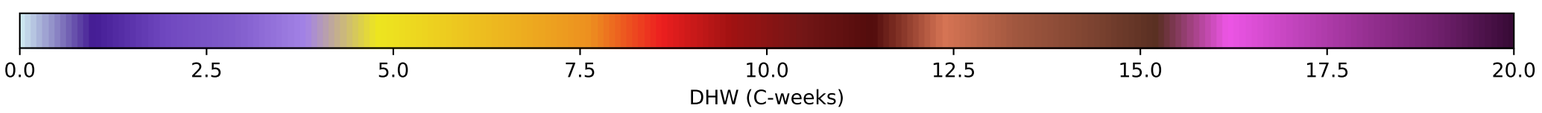


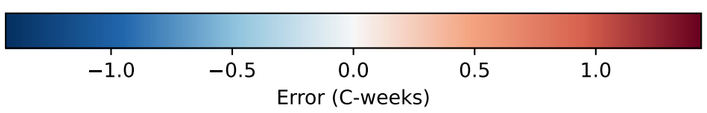
Predicted DHW - Step 90



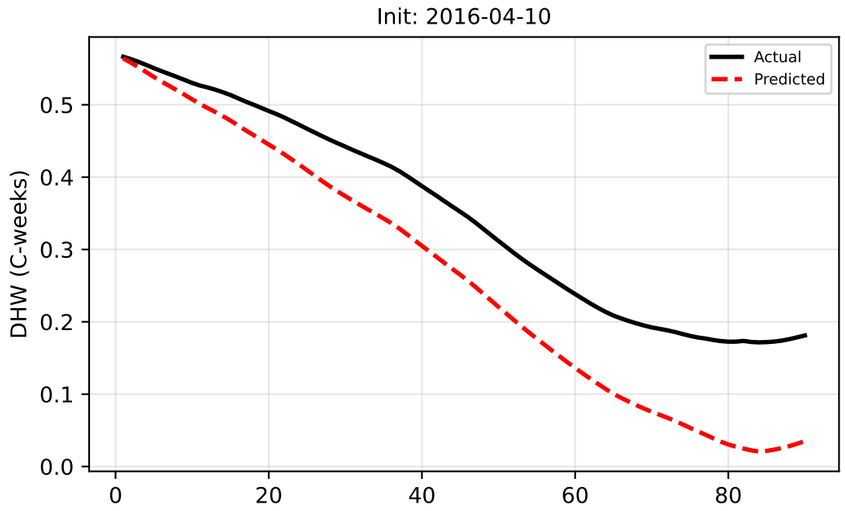
Error (Pred - Actual) - Step 90 [scale ±1.4]

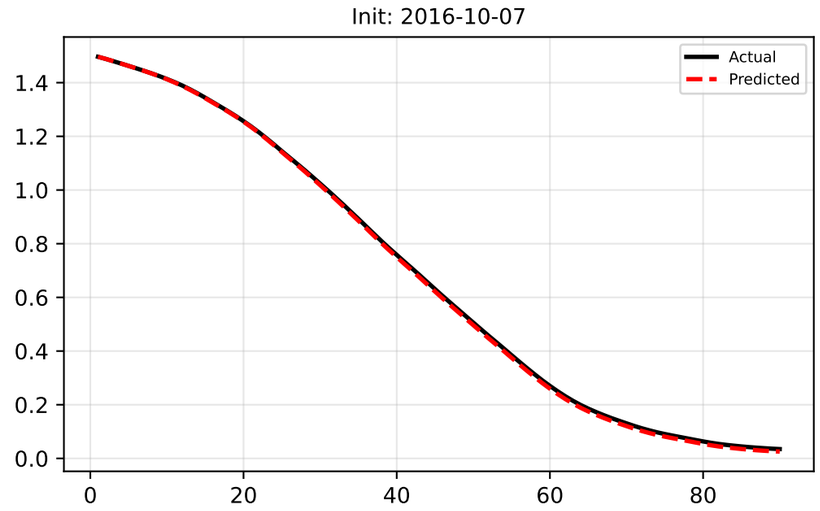


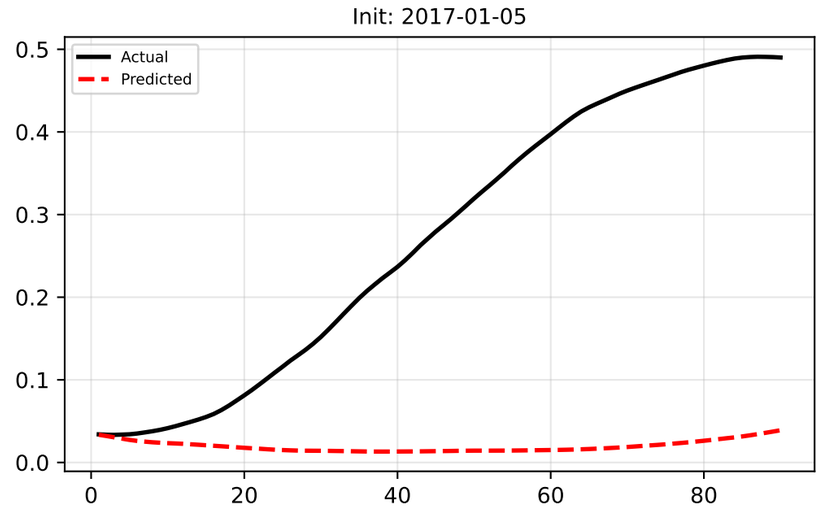


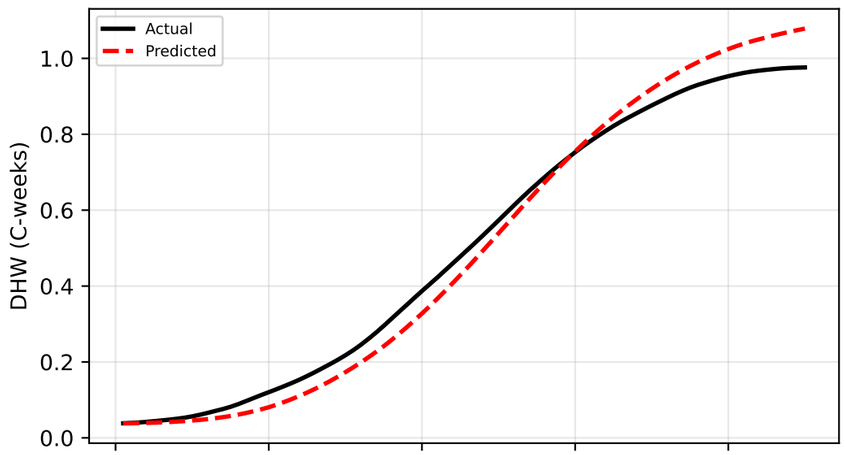


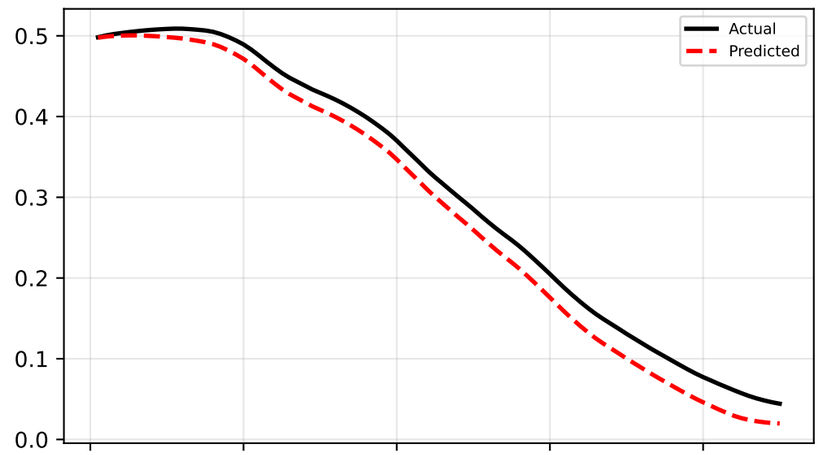
Global-Mean DHW - Val (C-weeks)

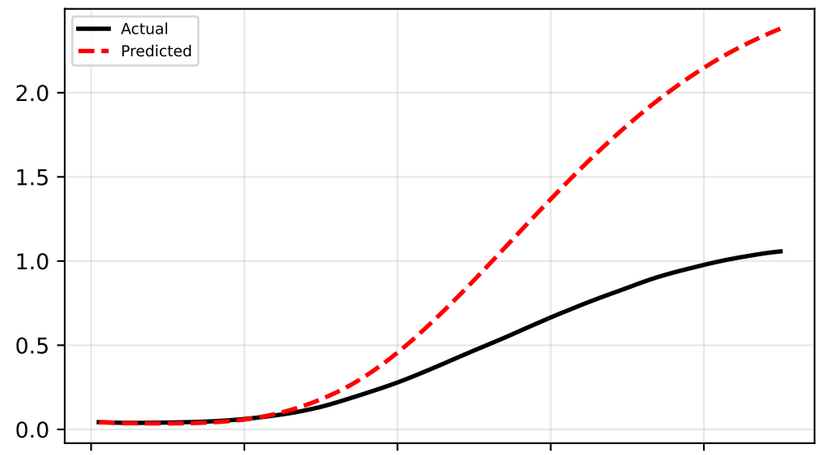


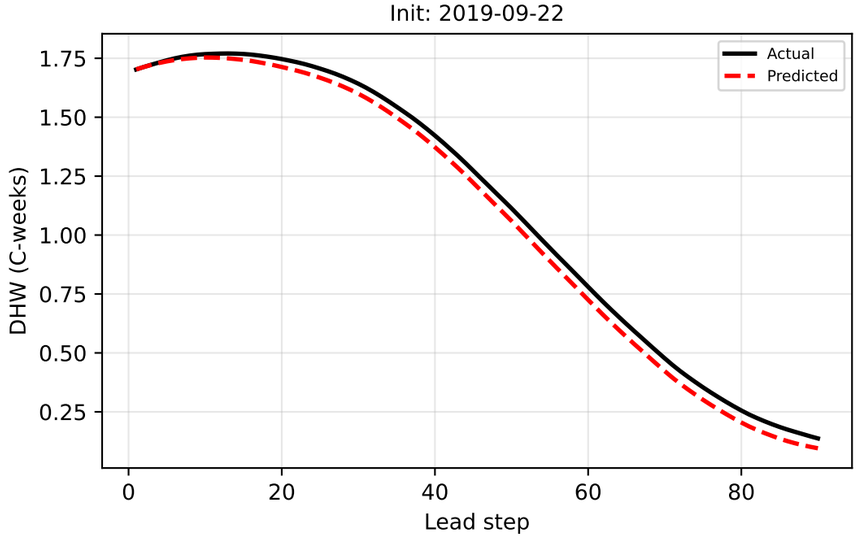




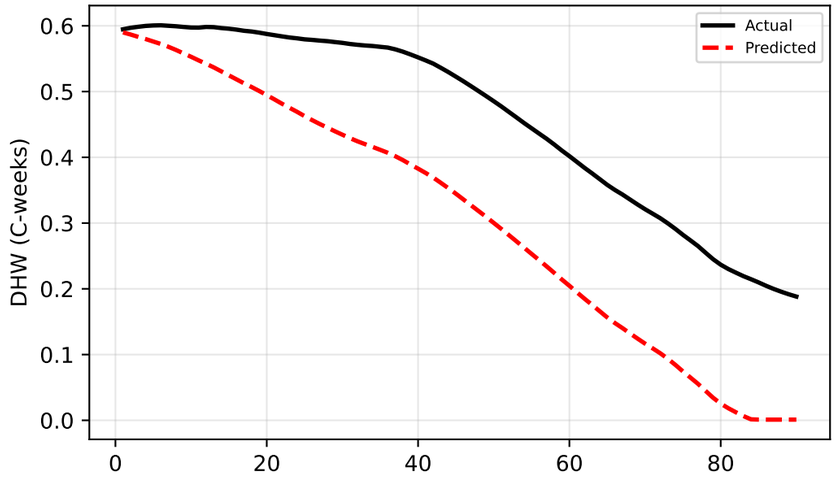


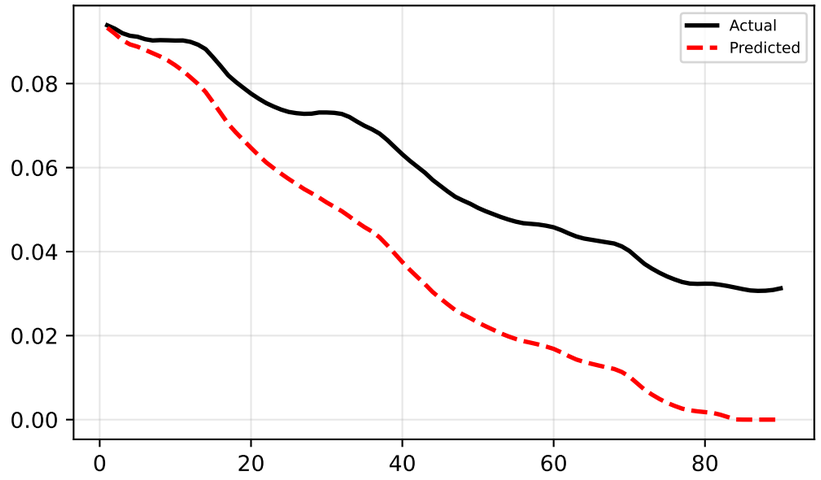


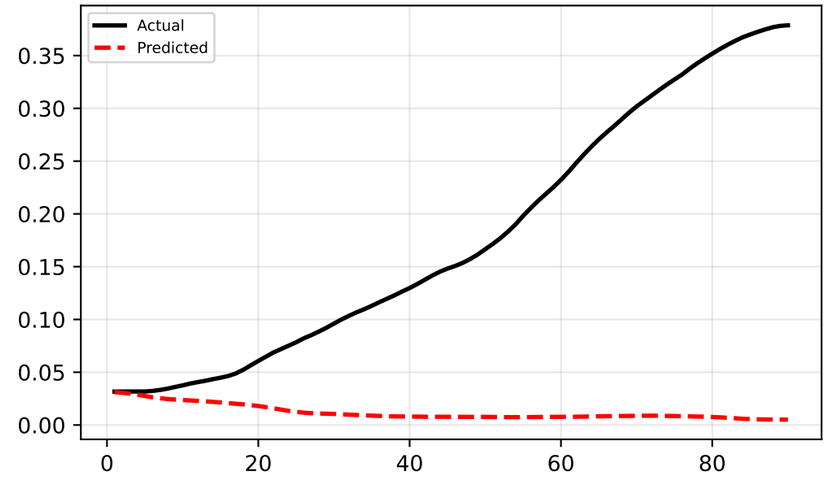


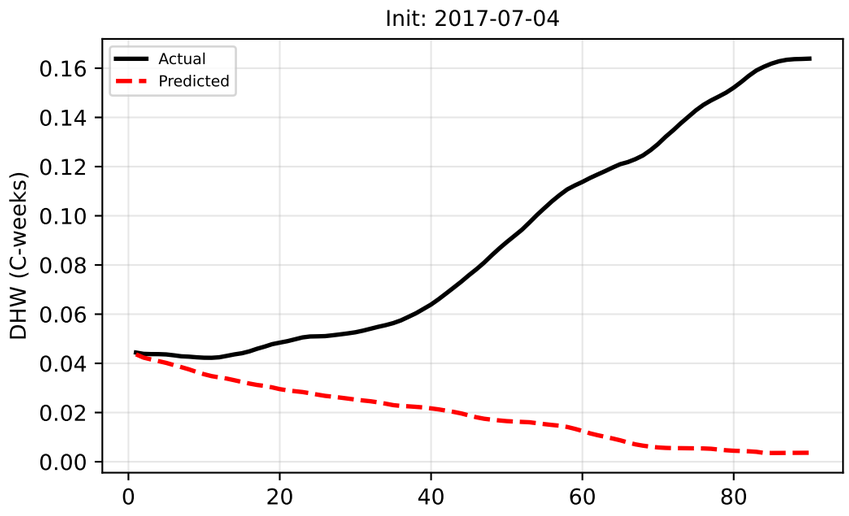


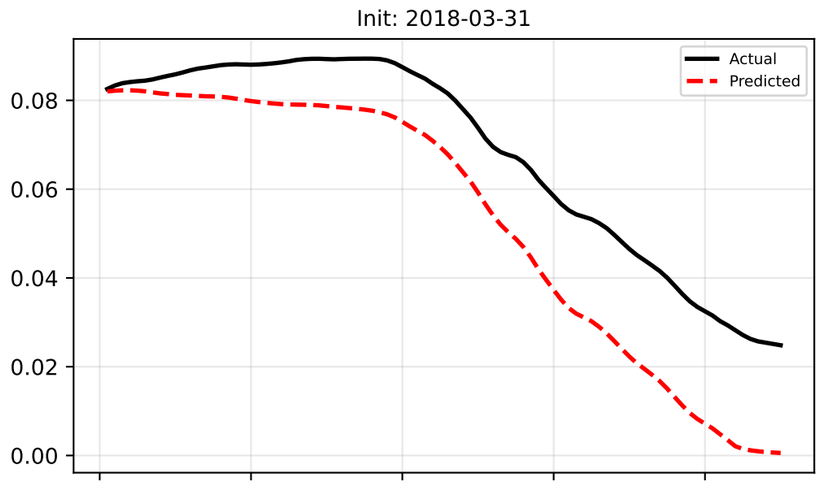
Tropical-Mean DHW (|lat|<=30) - Val (C-weeks)

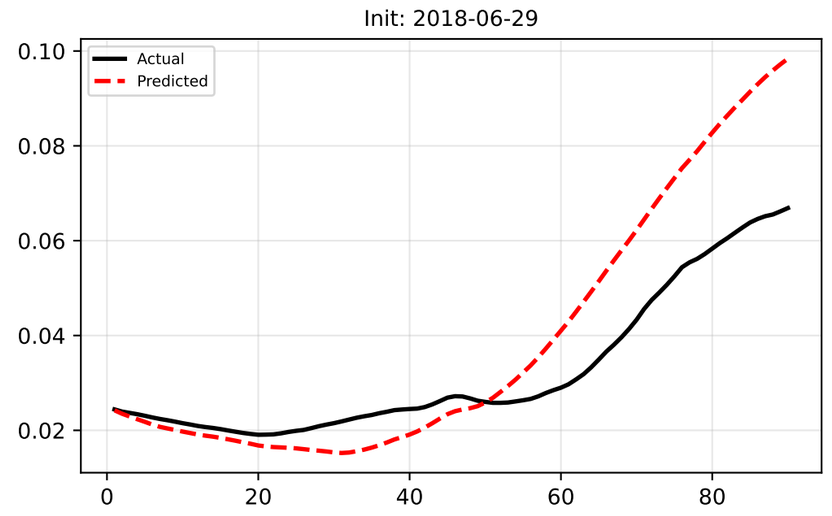


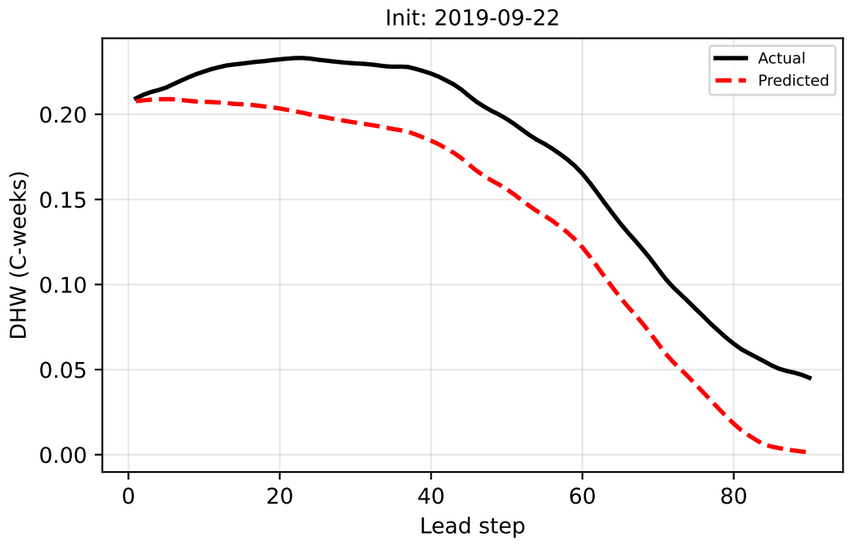




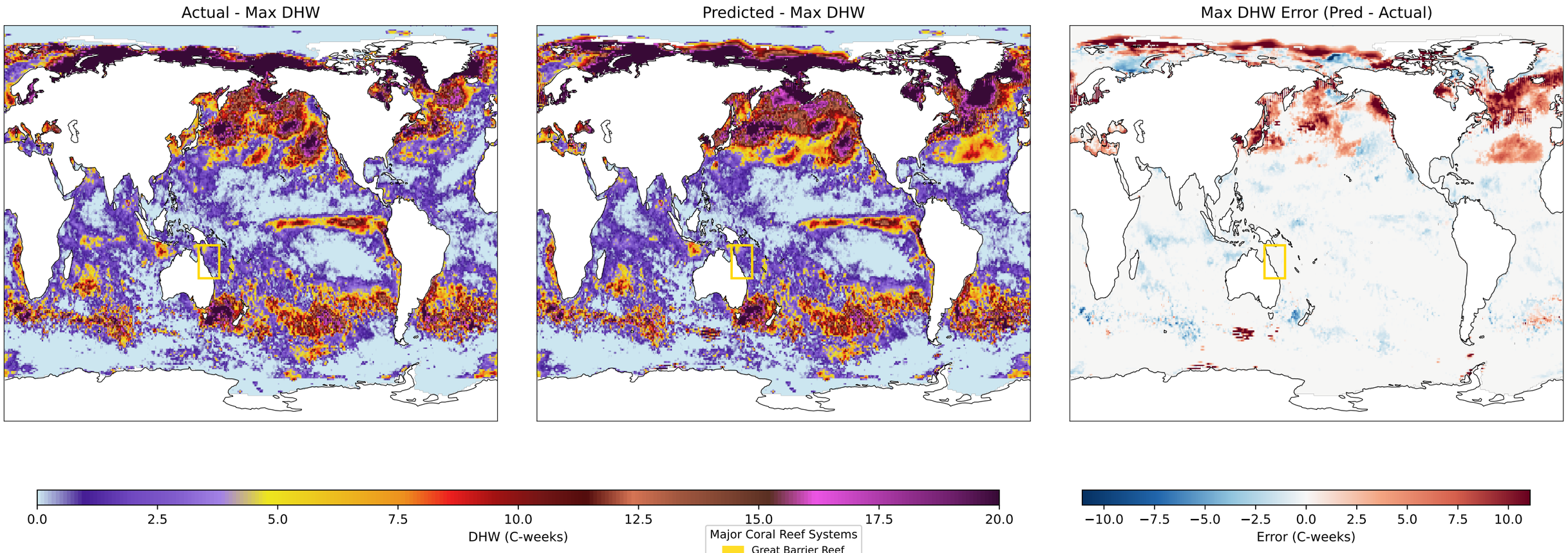


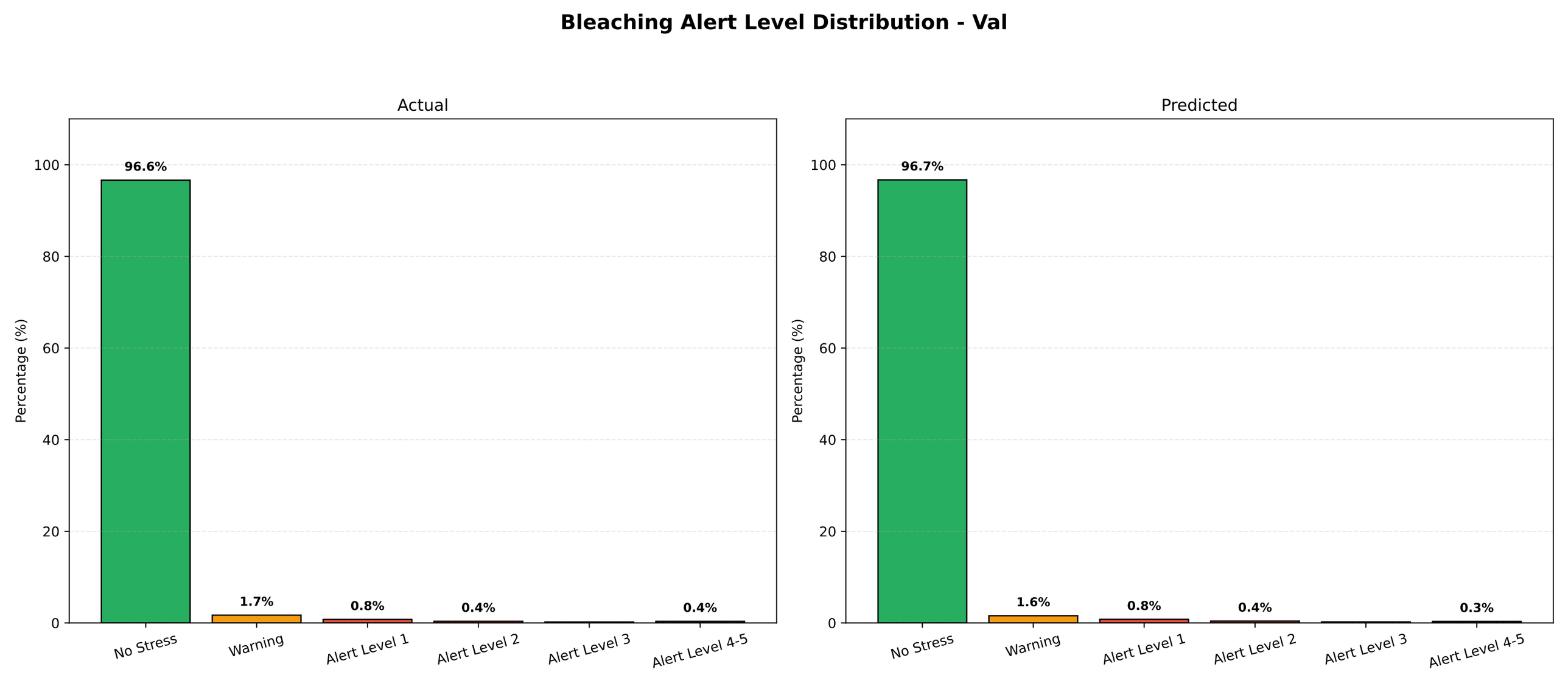






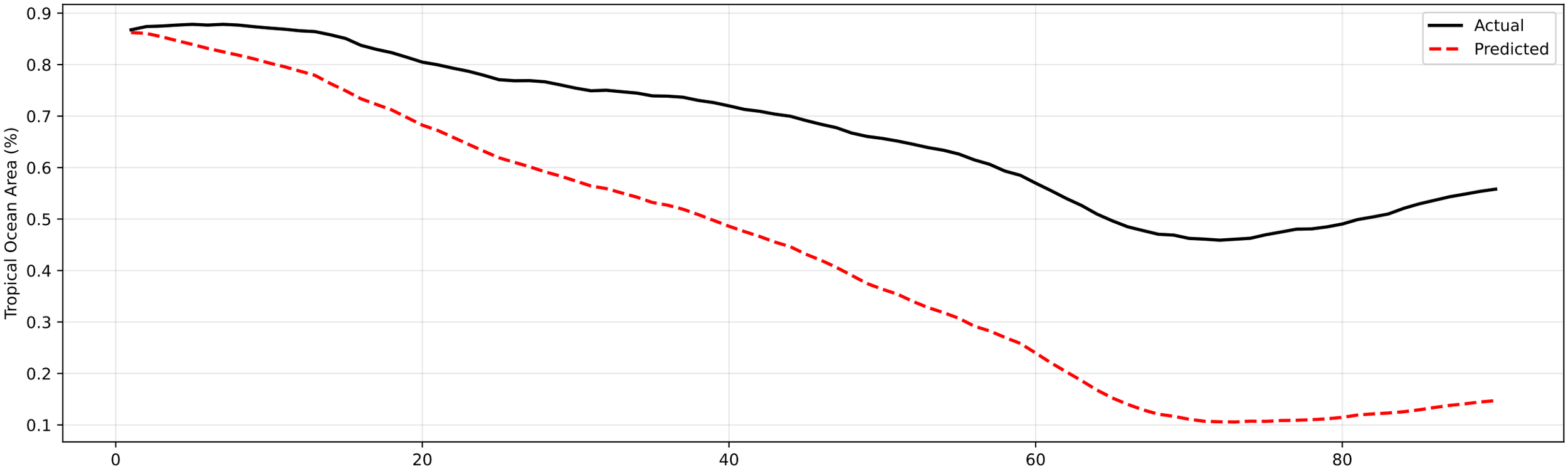
Peak DHW (max over all chains & leads) - Val (C-weeks)



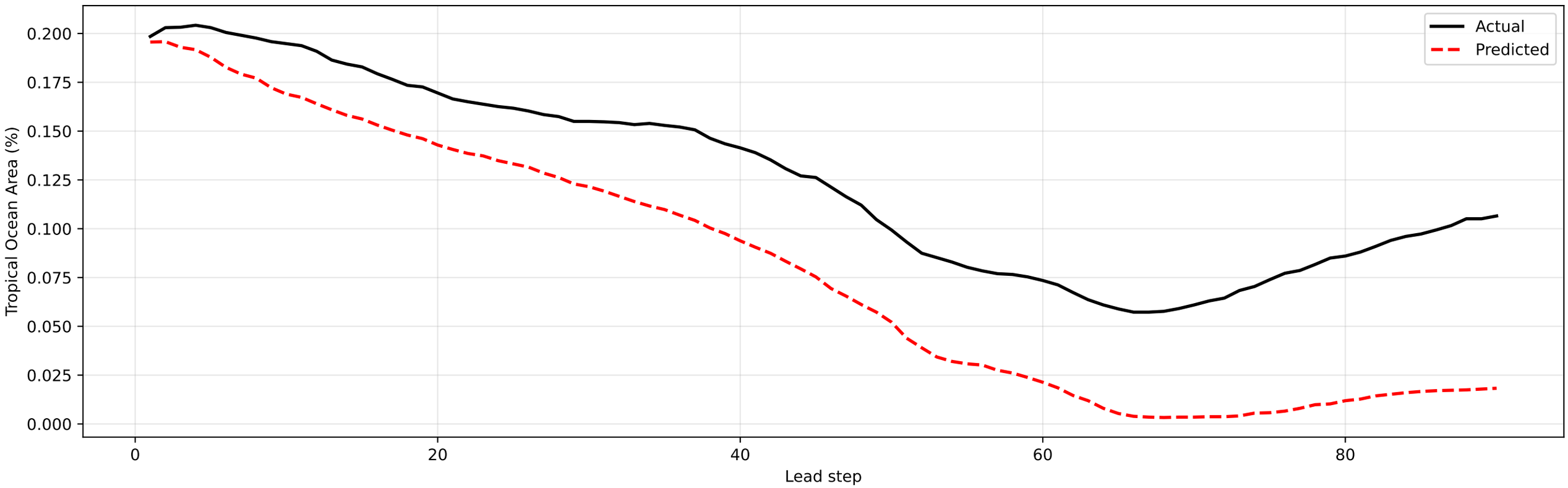

Bleaching Alert Level Distribution - Val
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
96.6%
1.7%
0.8%
0.4%
0.4%
96.7%
1.6%
0.8%
0.4%
0.3%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Val

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)



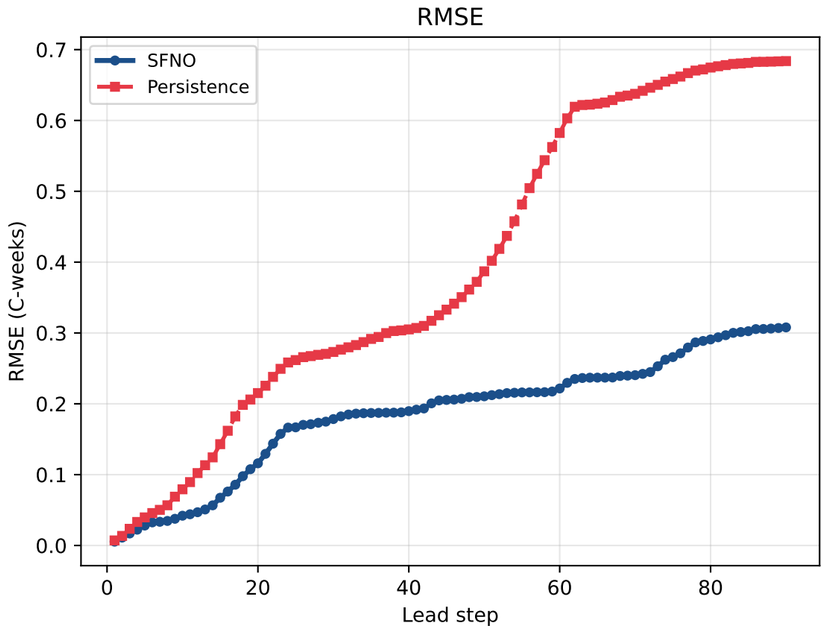
Great Barrier Reef Analysis - Val

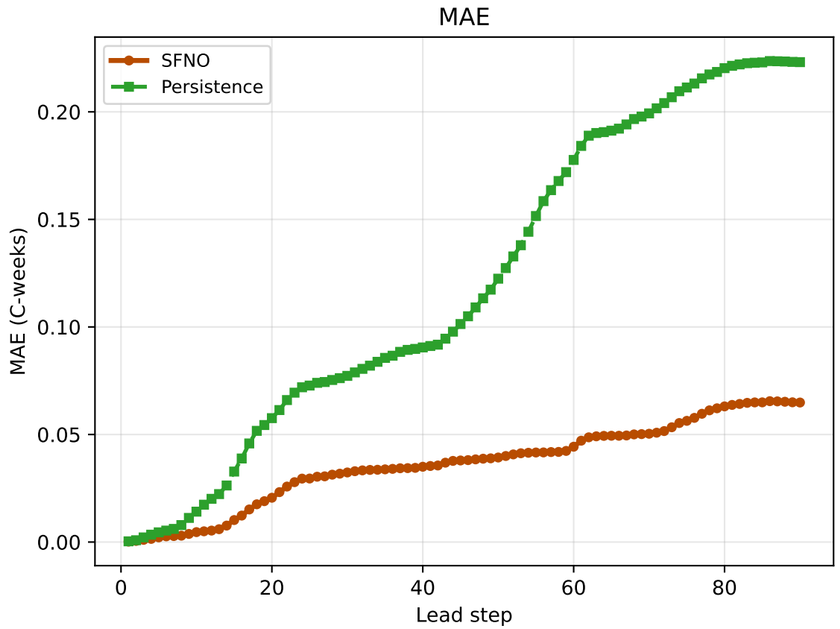
```
                    GREAT BARRIER REEF - VAL
Lat -25.0 deg to -10.0 deg  |  Lon 142.0 deg to 157.0 deg  (0-360)
                    Grid: 21 lat x 21 lon points
```

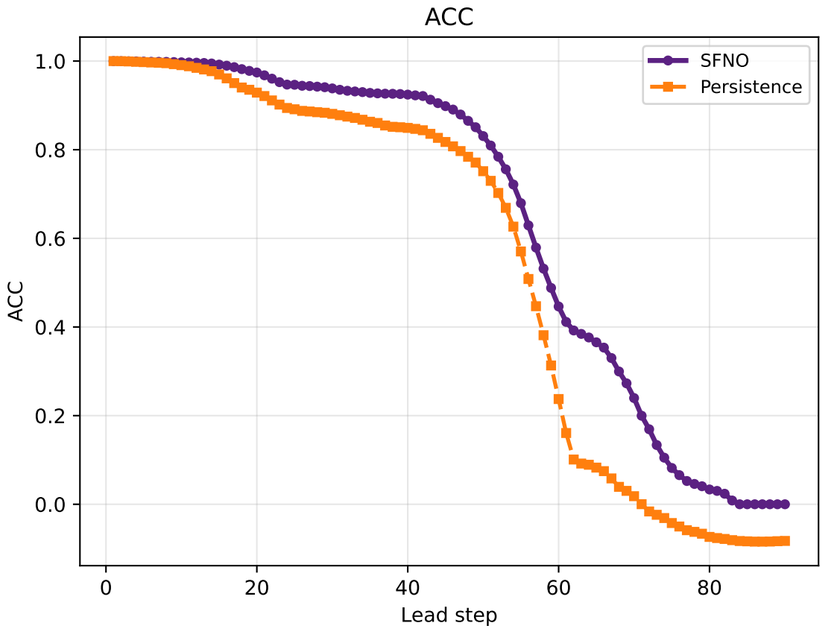
DHW Metrics by Lead Step - Val - GBR (15 chains scored at every lead)

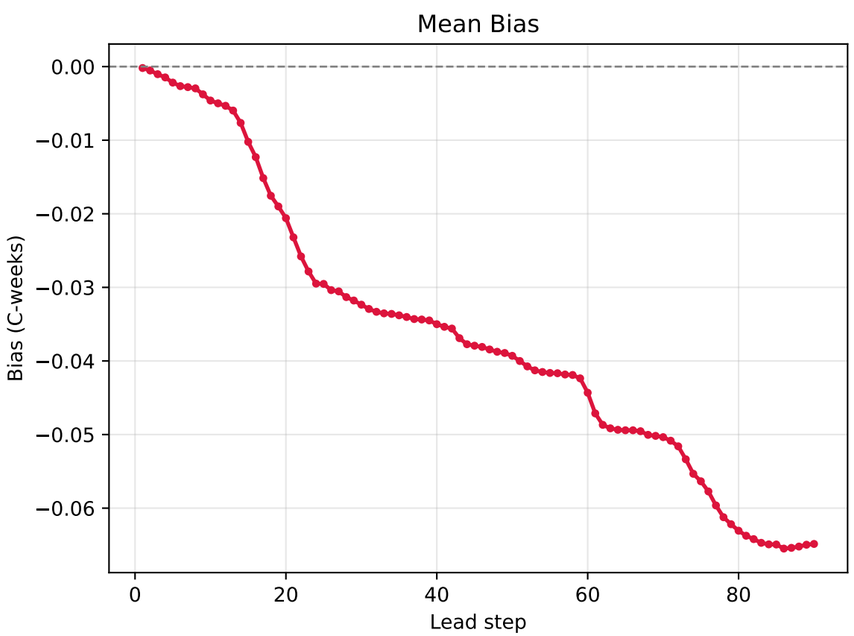
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0054	0.0071	23.9%	0.0002	0.0003	-0.0002	1.0000	0.9999	0.9999
2	0.0111	0.0135	17.5%	0.0005	0.0008	-0.0005	0.9998	0.9997	0.9996
3	0.0169	0.0235	28.1%	0.0010	0.0021	-0.0010	0.9996	0.9992	0.9992
4	0.0223	0.0332	32.8%	0.0015	0.0034	-0.0015	0.9992	0.9983	0.9986
5	0.0281	0.0396	29.0%	0.0022	0.0045	-0.0022	0.9988	0.9976	0.9977
6	0.0326	0.0458	28.8%	0.0027	0.0053	-0.0027	0.9984	0.9969	0.9969
7	0.0334	0.0503	33.5%	0.0028	0.0061	-0.0028	0.9983	0.9962	0.9967
8	0.0346	0.0567	39.0%	0.0030	0.0079	-0.0030	0.9981	0.9952	0.9964
9	0.0377	0.0689	45.3%	0.0038	0.0112	-0.0038	0.9978	0.9930	0.9957
10	0.0420	0.0793	47.0%	0.0046	0.0141	-0.0046	0.9972	0.9908	0.9946
11	0.0441	0.0895	50.8%	0.0050	0.0174	-0.0050	0.9968	0.9883	0.9940
12	0.0469	0.1022	54.1%	0.0053	0.0201	-0.0053	0.9963	0.9847	0.9931
13	0.0508	0.1133	55.2%	0.0060	0.0223	-0.0060	0.9957	0.9811	0.9918
14	0.0567	0.1244	54.4%	0.0077	0.0263	-0.0077	0.9945	0.9772	0.9896
15	0.0674	0.1430	52.9%	0.0102	0.0328	-0.0102	0.9920	0.9698	0.9849
16	0.0761	0.1619	53.0%	0.0123	0.0388	-0.0123	0.9895	0.9610	0.9802
17	0.0858	0.1822	52.9%	0.0152	0.0458	-0.0152	0.9863	0.9503	0.9740
18	0.0978	0.1985	50.7%	0.0176	0.0517	-0.0176	0.9818	0.9403	0.9656
19	0.1077	0.2060	47.7%	0.0190	0.0544	-0.0190	0.9779	0.9352	0.9582
20	0.1160	0.2151	46.1%	0.0206	0.0576	-0.0206	0.9742	0.9288	0.9514
21	0.1293	0.2255	42.6%	0.0232	0.0614	-0.0232	0.9679	0.9211	0.9397
22	0.1437	0.2380	39.6%	0.0258	0.0660	-0.0258	0.9603	0.9112	0.9259
23	0.1575	0.2494	36.8%	0.0278	0.0694	-0.0278	0.9524	0.9018	0.9116
24	0.1664	0.2583	35.6%	0.0295	0.0720	-0.0295	0.9469	0.8942	0.9017
25	0.1668	0.2617	36.3%	0.0295	0.0728	-0.0295	0.9465	0.8912	0.9009
26	0.1702	0.2658	36.0%	0.0304	0.0740	-0.0304	0.9442	0.8876	0.8969
27	0.1712	0.2673	36.0%	0.0305	0.0744	-0.0305	0.9437	0.8862	0.8958
28	0.1732	0.2691	35.6%	0.0313	0.0753	-0.0313	0.9423	0.8846	0.8933
29	0.1748	0.2705	35.4%	0.0318	0.0762	-0.0318	0.9410	0.8834	0.8910
30	0.1784	0.2732	34.7%	0.0324	0.0773	-0.0324	0.9383	0.8809	0.8863
31	0.1822	0.2765	34.1%	0.0329	0.0789	-0.0329	0.9352	0.8778	0.8808
32	0.1849	0.2797	33.9%	0.0333	0.0805	-0.0333	0.9329	0.8747	0.8769
33	0.1861	0.2829	34.2%	0.0335	0.0820	-0.0335	0.9314	0.8717	0.8743
34	0.1867	0.2871	35.0%	0.0336	0.0838	-0.0336	0.9300	0.8676	0.8718
35	0.1870	0.2916	35.9%	0.0338	0.0856	-0.0338	0.9281	0.8631	0.8686
36	0.1872	0.2943	36.4%	0.0340	0.0866	-0.0340	0.9271	0.8603	0.8668
37	0.1874	0.2998	37.5%	0.0343	0.0884	-0.0343	0.9264	0.8547	0.8655
38	0.1875	0.3027	38.1%	0.0344	0.0893	-0.0344	0.9262	0.8516	0.8652
39	0.1879	0.3036	38.1%	0.0345	0.0898	-0.0345	0.9256	0.8507	0.8641
40	0.1896	0.3050	37.8%	0.0350	0.0905	-0.0350	0.9243	0.8492	0.8619
41	0.1917	0.3071	37.6%	0.0354	0.0911	-0.0354	0.9226	0.8470	0.8589
42	0.1934	0.3100	37.6%	0.0356	0.0918	-0.0356	0.9211	0.8439	0.8563
43	0.2007	0.3172	36.7%	0.0369	0.0945	-0.0369	0.9129	0.8358	0.8418
44	0.2048	0.3250	37.0%	0.0377	0.0978	-0.0377	0.9051	0.8267	0.8283
45	0.2055	0.3329	38.3%	0.0379	0.1014	-0.0379	0.8983	0.8175	0.8167
46	0.2059	0.3414	39.7%	0.0381	0.1050	-0.0381	0.8905	0.8076	0.8035
47	0.2071	0.3506	40.9%	0.0384	0.1091	-0.0384	0.8794	0.7969	0.7848
48	0.2094	0.3613	42.0%	0.0388	0.1133	-0.0388	0.8652	0.7840	0.7613
49	0.2096	0.3722	43.7%	0.0389	0.1173	-0.0389	0.8507	0.7710	0.7377
50	0.2106	0.3870	45.6%	0.0393	0.1224	-0.0393	0.8307	0.7515	0.7057
51	0.2122	0.4019	47.2%	0.0400	0.1274	-0.0400	0.8094	0.7297	0.6722
52	0.2136	0.4187	49.0%	0.0407	0.1328	-0.0407	0.7841	0.7024	0.6338
53	0.2151	0.4370	50.8%	0.0413	0.1380	-0.0413	0.7558	0.6688	0.5921
54	0.2156	0.4577	52.9%	0.0415	0.1443	-0.0415	0.7214	0.6265	0.5440
55	0.2161	0.4814	55.1%	0.0416	0.1516	-0.0416	0.6794	0.5704	0.4887
56	0.2161	0.5045	57.2%	0.0417	0.1585	-0.0417	0.6292	0.5083	0.4259
57	0.2163	0.5246	58.8%	0.0418	0.1636	-0.0418	0.5791	0.4469	0.3663
58	0.2164	0.5439	60.2%	0.0419	0.1678	-0.0419	0.5316	0.3814	0.3131
59	0.2173	0.5625	61.4%	0.0424	0.1720	-0.0424	0.4881	0.3131	0.2669
60	0.2216	0.5823	61.9%	0.0443	0.1776	-0.0443	0.4463	0.2374	0.2239
61	0.2296	0.6030	61.9%	0.0471	0.1842	-0.0471	0.4114	0.1608	0.1899
62	0.2351	0.6193	62.0%	0.0487	0.1889	-0.0487	0.3921	0.1009	0.1726
63	0.2364	0.6218	62.0%	0.0492	0.1902	-0.0492	0.3844	0.0919	0.1664
64	0.2369	0.6223	61.9%	0.0494	0.1906	-0.0494	0.3763	0.0891	0.1598
65	0.2370	0.6236	62.0%	0.0494	0.1913	-0.0494	0.3654	0.0827	0.1506
66	0.2370	0.6253	62.1%	0.0494	0.1922	-0.0494	0.3534	0.0747	0.1410
67	0.2371	0.6288	62.3%	0.0495	0.1942	-0.0495	0.3299	0.0583	0.1229
68	0.2393	0.6334	62.2%	0.0500	0.1967	-0.0500	0.2996	0.0393	0.1008
69	0.2398	0.6351	62.2%	0.0502	0.1978	-0.0502	0.2729	0.0302	0.0831
70	0.2404	0.6376	62.3%	0.0504	0.1993	-0.0504	0.2396	0.0182	0.0625
71	0.2423	0.6418	62.3%	0.0508	0.2016	-0.0508	0.1998	0.0002	0.0398
72	0.2448	0.6462	62.1%	0.0516	0.2041	-0.0516	0.1692	-0.0162	0.0239
73	0.2530	0.6503	61.1%	0.0534	0.2068	-0.0534	0.1336	-0.0238	0.0076
74	0.2623	0.6549	60.0%	0.0553	0.2096	-0.0553	0.1051	-0.0311	-0.0043
75	0.2659	0.6585	59.6%	0.0564	0.2113	-0.0564	0.0818	-0.0424	-0.0138
76	0.2713	0.6620	59.0%	0.0577	0.2132	-0.0577	0.0657	-0.0502	-0.0204
77	0.2794	0.6668	58.1%	0.0596	0.2156	-0.0596	0.0525	-0.0585	-0.0259
78	0.2867	0.6704	57.2%	0.0612	0.2174	-0.0612	0.0459	-0.0622	-0.0286
79	0.2887	0.6721	57.0%	0.0622	0.2185	-0.0622	0.0407	-0.0664	-0.0313
80	0.2909	0.6747	56.9%	0.0631	0.2203	-0.0631	0.0334	-0.0737	-0.0341
81	0.2939	0.6765	56.6%	0.0637	0.2214	-0.0637	0.0302	-0.0762	-0.0350
82	0.2968	0.6781	56.2%	0.0642	0.2221	-0.0642	0.0238	-0.0782	-0.0374
83	0.3002	0.6798	55.8%	0.0647	0.2226	-0.0647	0.0085	-0.0809	-0.0443
84	0.3013	0.6805	55.7%	0.0649	0.2228	-0.0649	-0.0000	-0.0830	-0.0487
85	0.3025	0.6812	55.6%	0.0649	0.2230	-0.0649	0.0000	-0.0836	-0.0483
86	0.3055	0.6827	55.3%	0.0655	0.2236	-0.0655	0.0000	-0.0843	-0.0482
87	0.3056	0.6828	55.2%	0.0654	0.2235	-0.0654	0.0000	-0.0843	-0.0480
88	0.3062	0.6830	55.2%	0.0652	0.2234	-0.0652	-0.0000	-0.0841	-0.0475
89	0.3070	0.6834	55.1%	0.0650	0.2232	-0.0650	0.0000	-0.0833	-0.0469
90	0.3079	0.6838	55.0%	0.0649	0.2231	-0.0649	0.0000	-0.0827	-0.0465

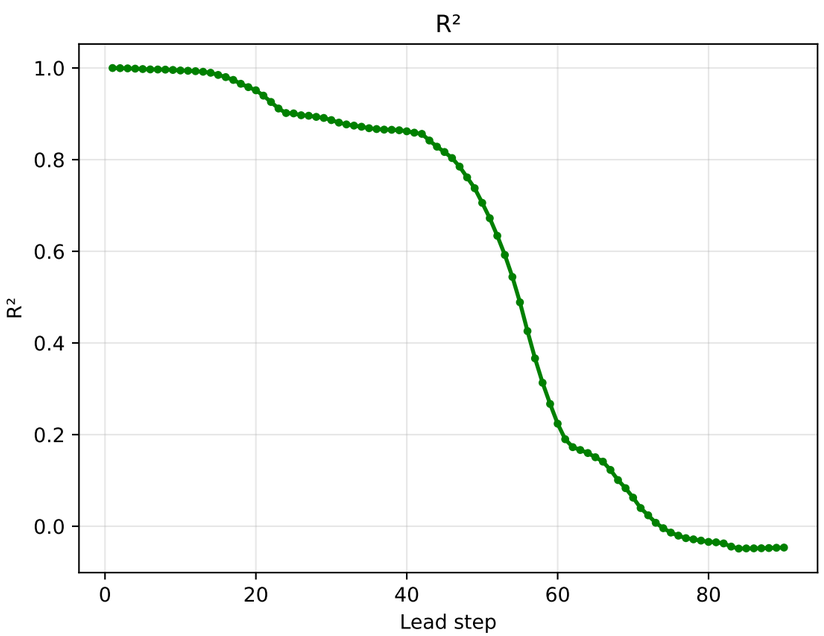
DHW Forecast Skill vs Lead Step - Val - GBR

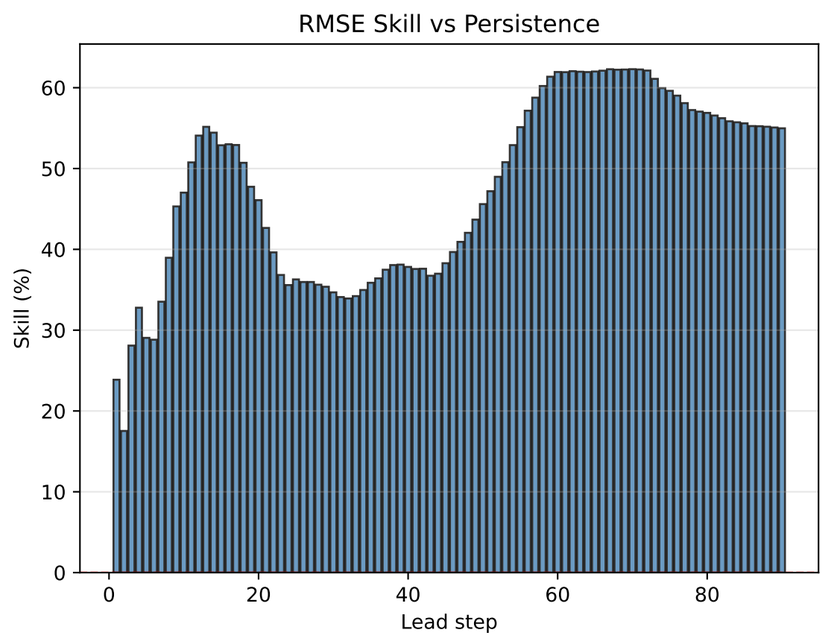


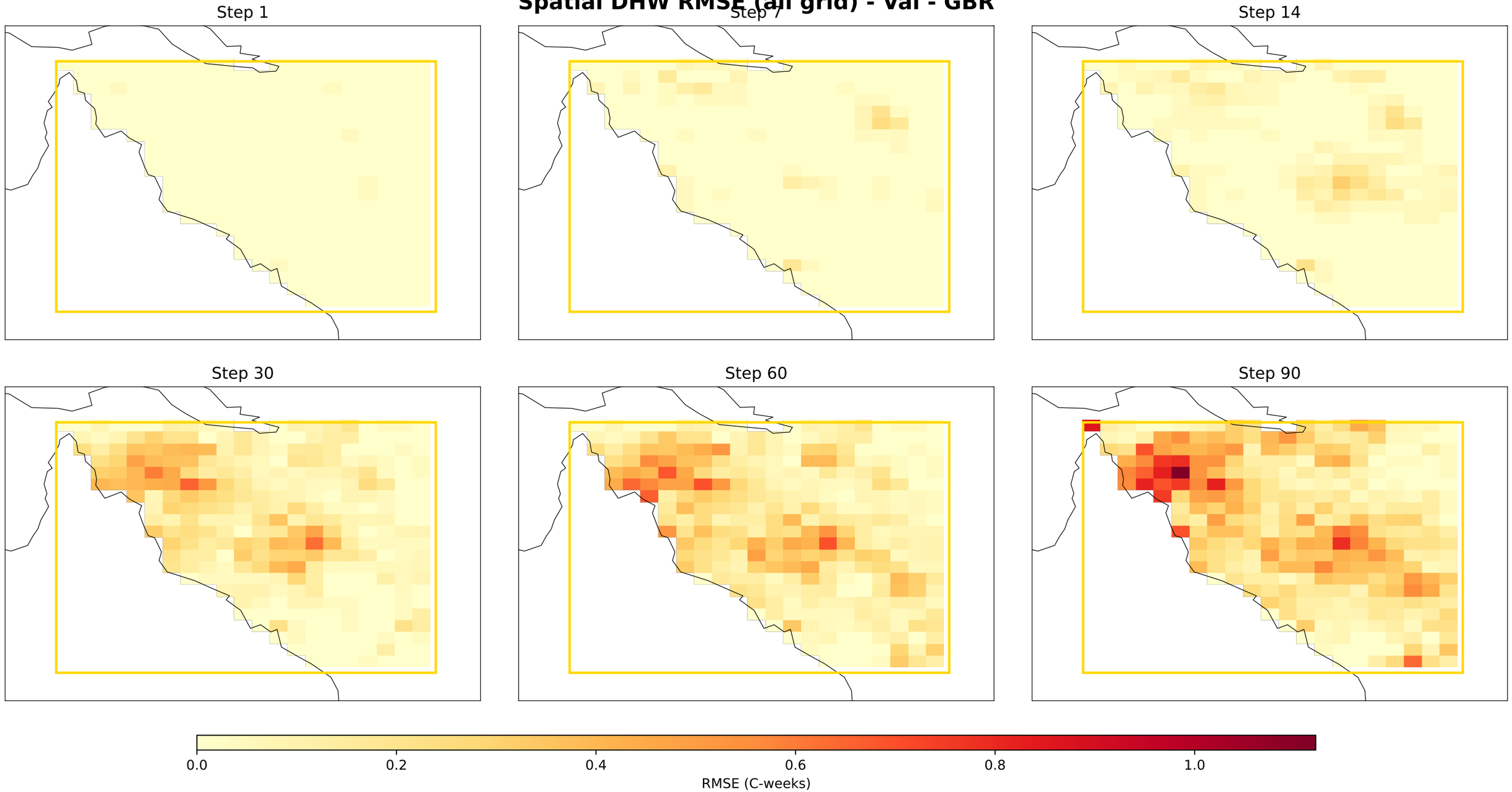


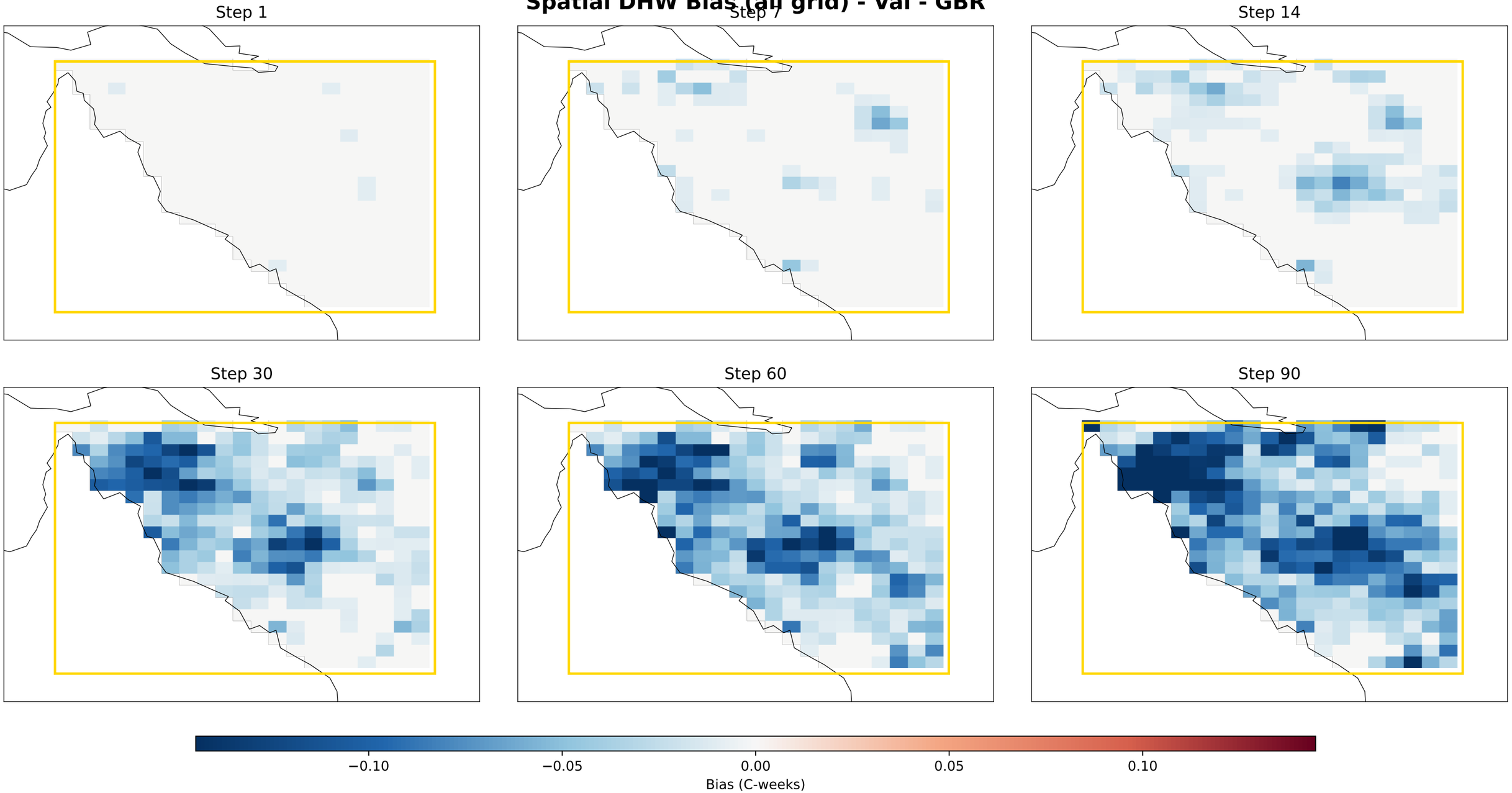


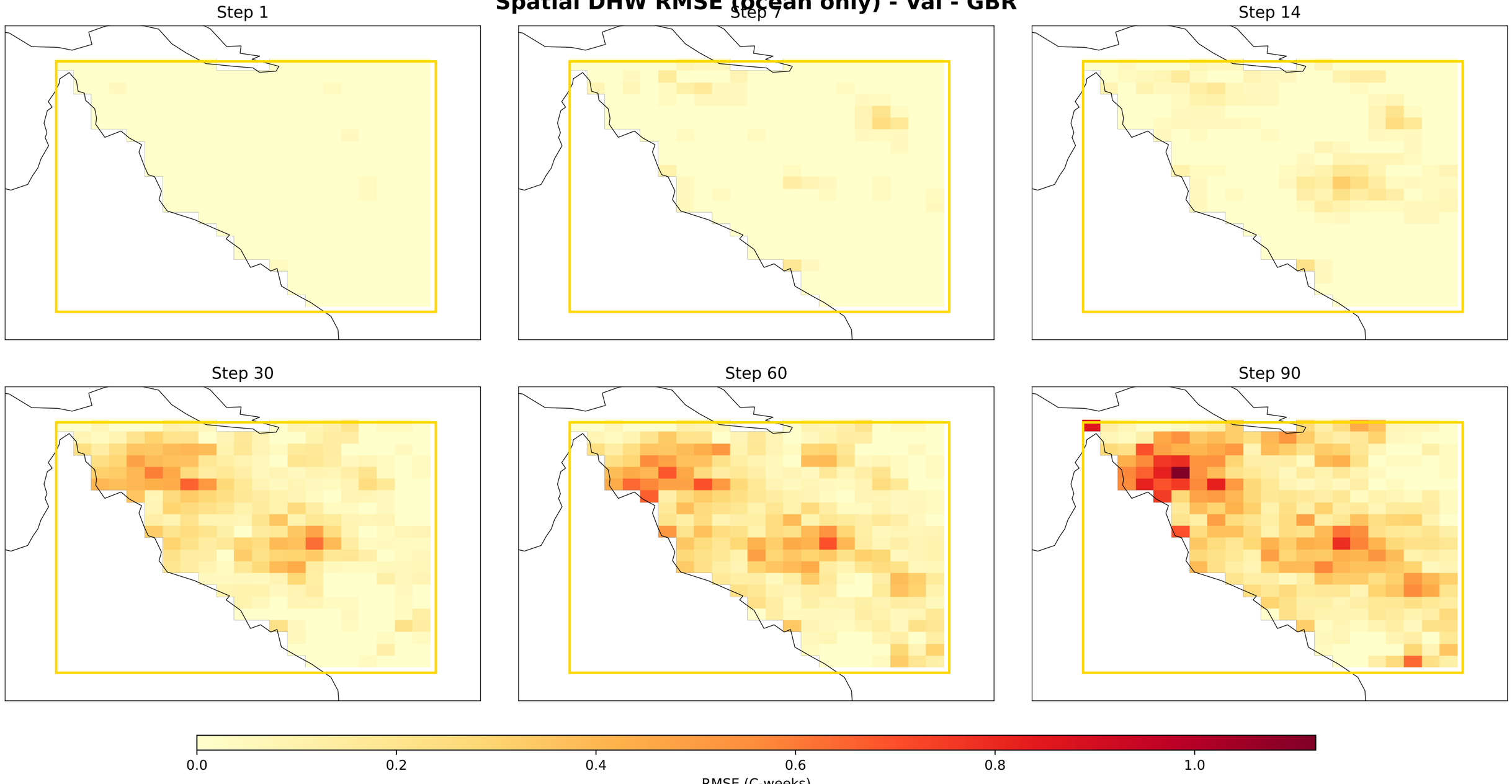


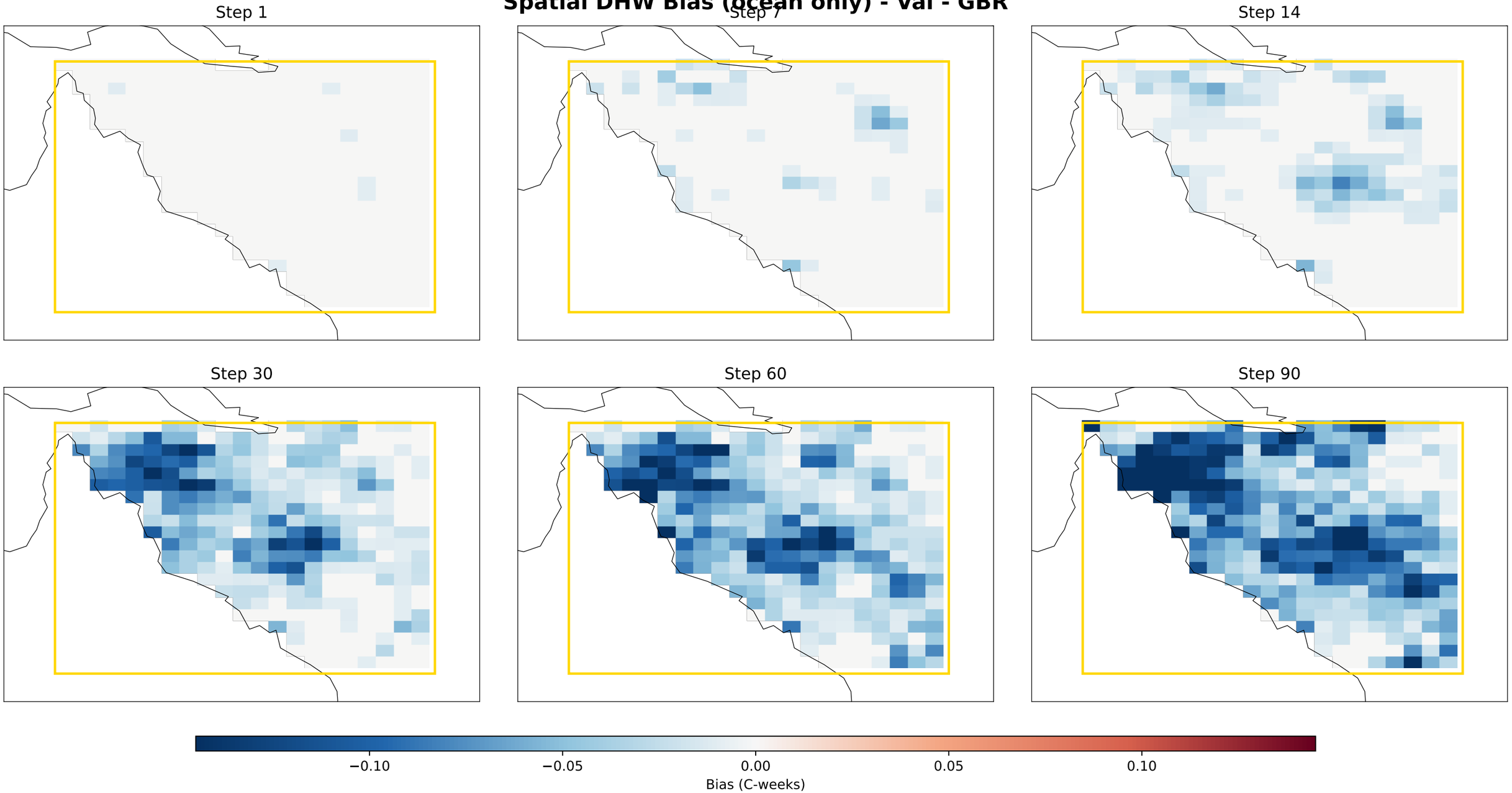





Spatial DHW RMSE (all grid) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.2
0.4
0.6
0.8
1.0
RMSE (C-weeks)

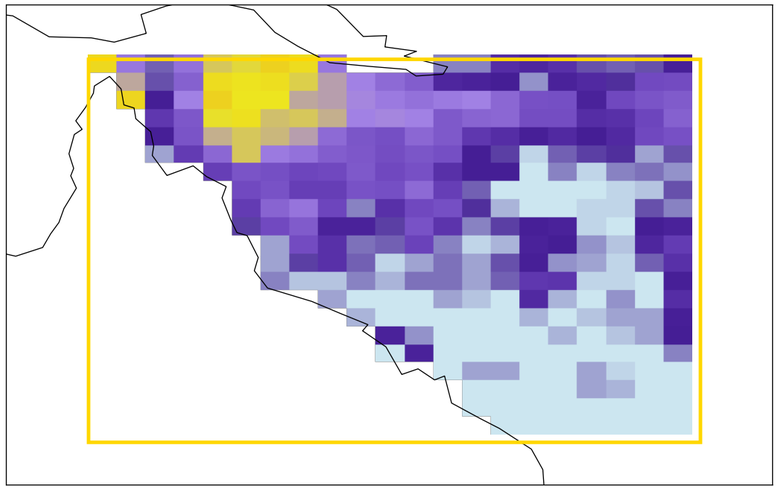

Spatial DHW Bias (all grid) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.10
−0.05
0.00
0.05
0.10
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.2
0.4
0.6
0.8
1.0
RMSE (C-weeks)

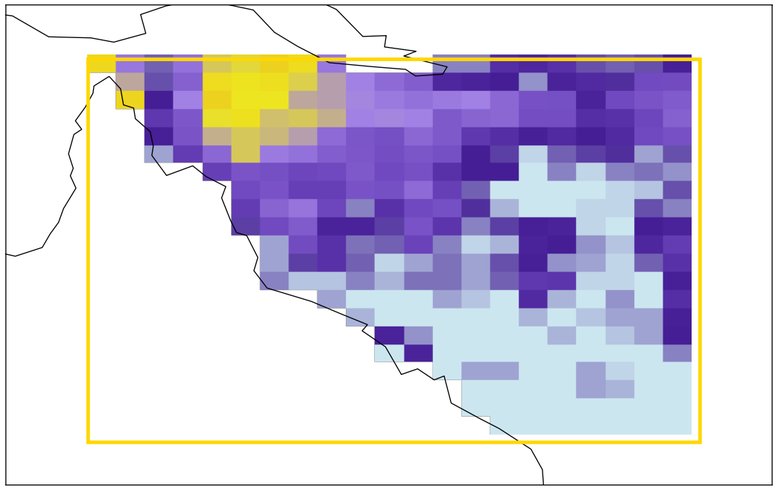

Spatial DHW Bias (ocean only) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.10
−0.05
0.00
0.05
0.10
Bias (C-weeks)

DHW Chain init: 2016-04-10 - Val - GBR (C-weeks)

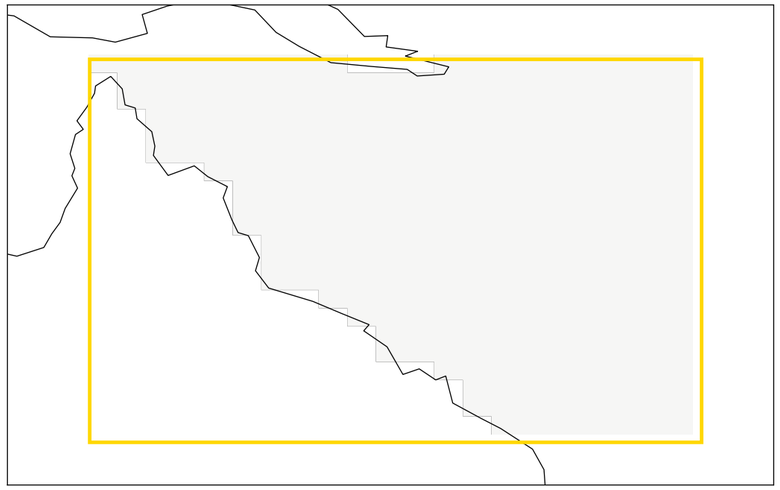
Actual DHW - Step 1



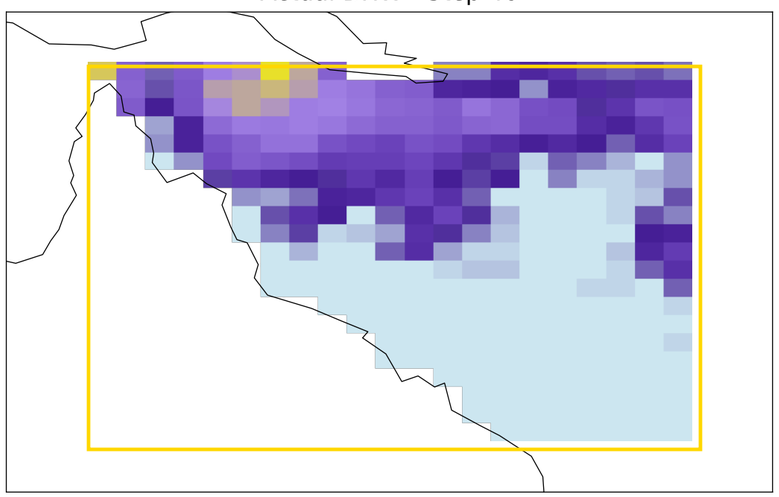
Predicted DHW - Step 1



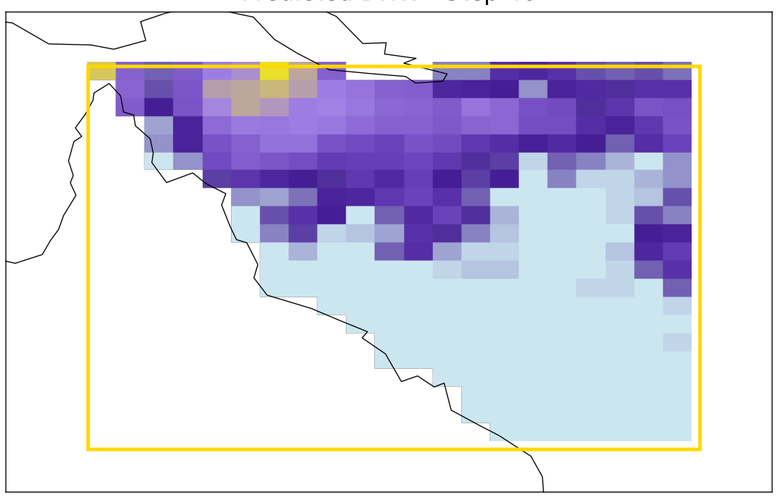
Error (Pred - Actual) - Step 1 [scale ±8]



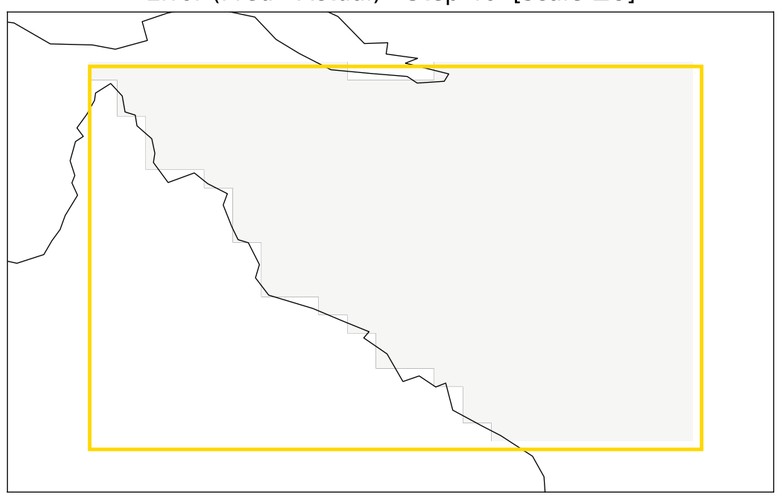
Actual DHW - Step 46



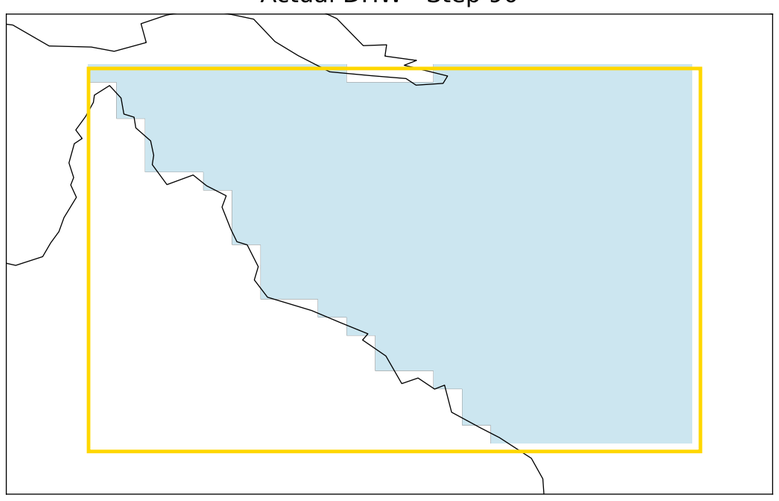
Predicted DHW - Step 46



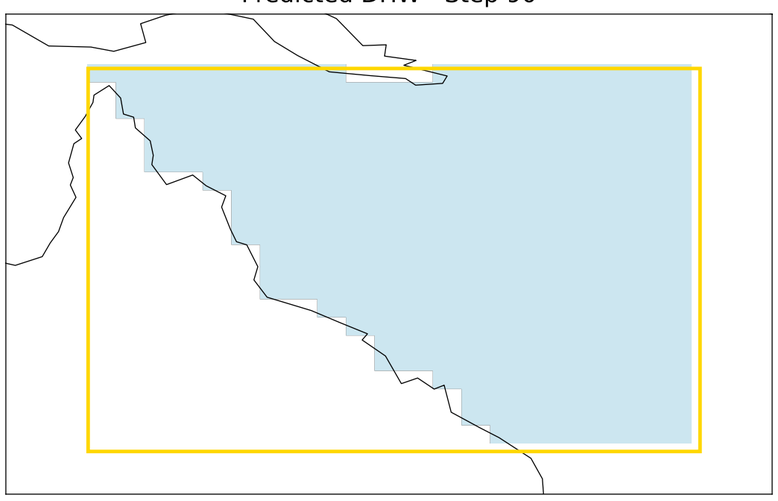
Error (Pred - Actual) - Step 46 [scale ±8]



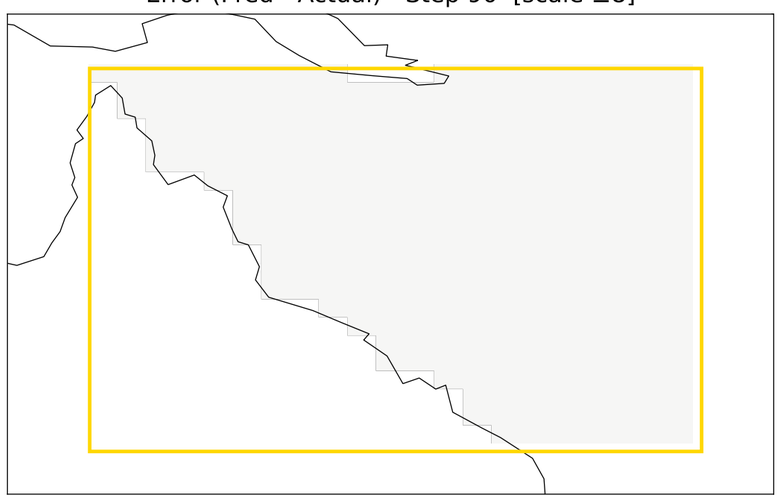
Actual DHW - Step 90

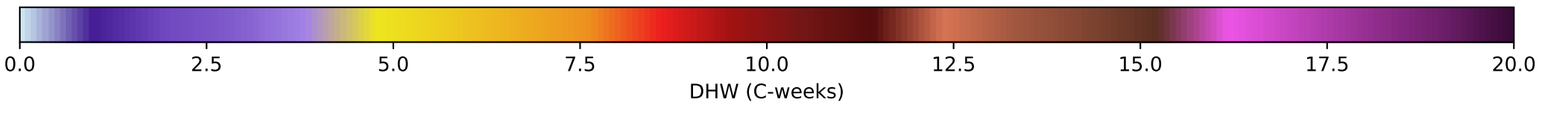


Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

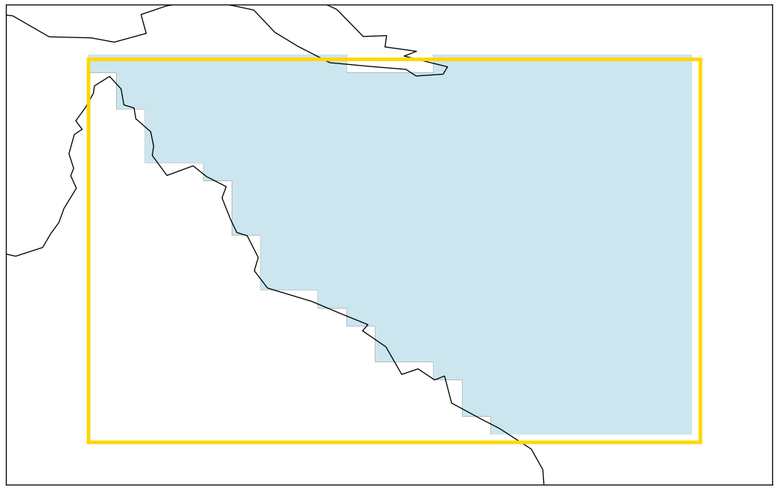




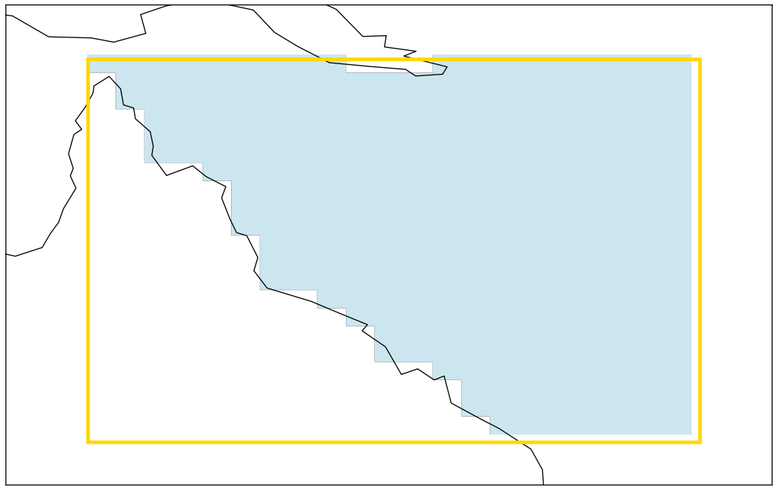


DHW Chain init: 2016-10-07 - Val - GBR (C-weeks)

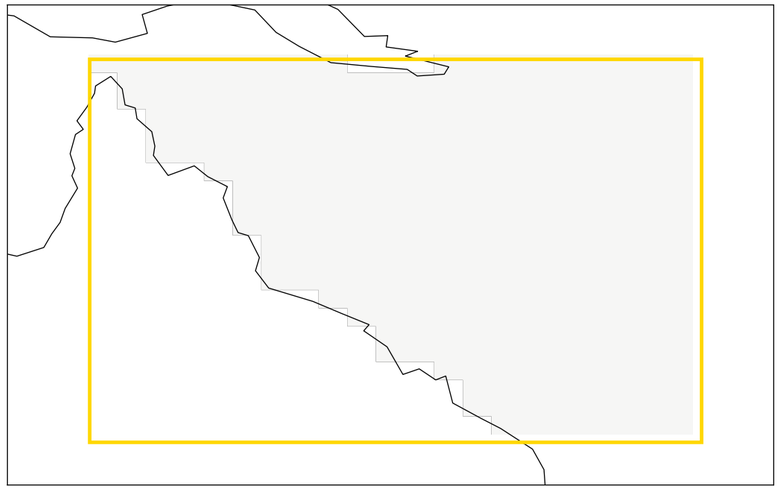
Actual DHW - Step 1



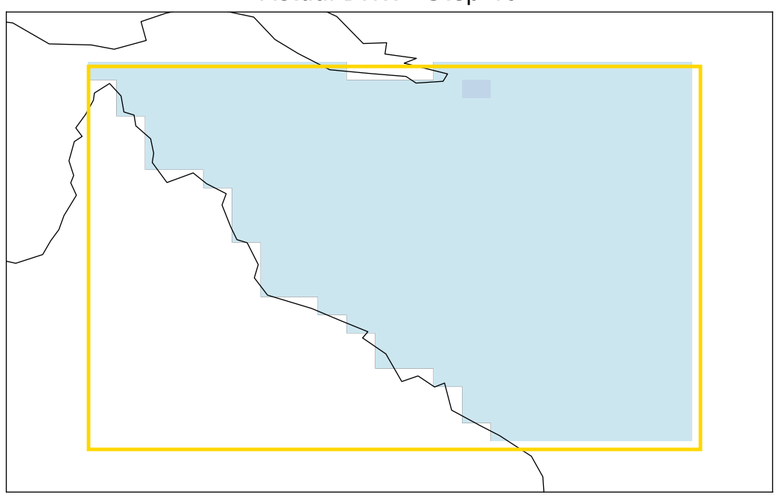
Predicted DHW - Step 1



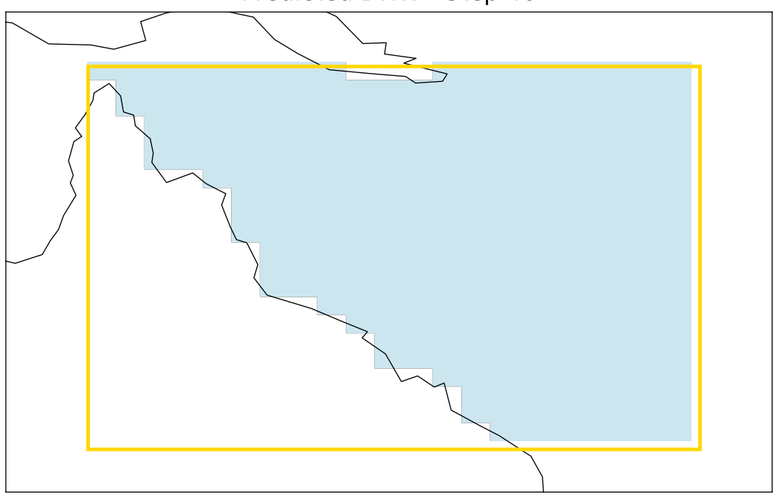
Error (Pred - Actual) - Step 1 [scale ±8]



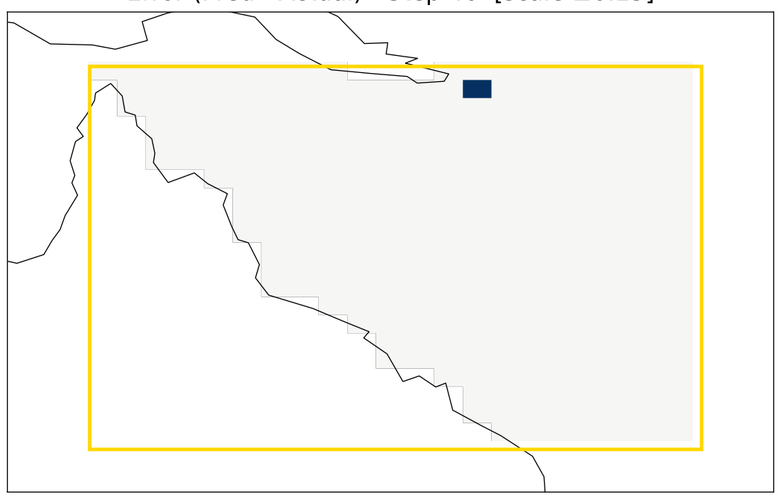
Actual DHW - Step 46



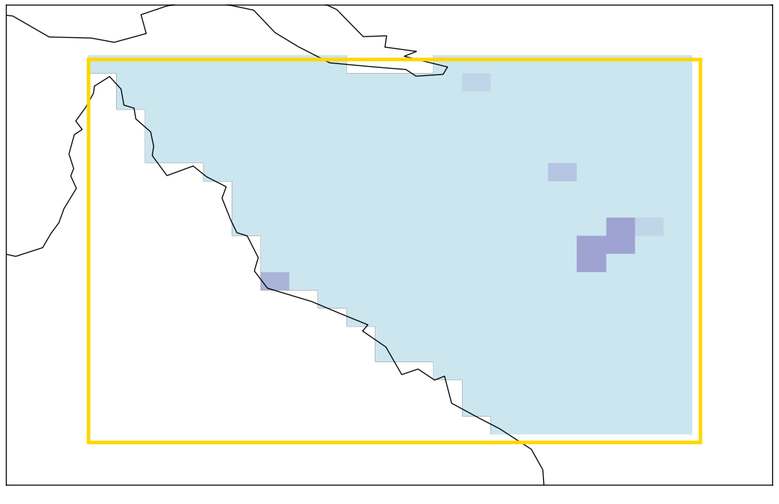
Predicted DHW - Step 46



Error (Pred - Actual) - Step 46 [scale ±0.15]



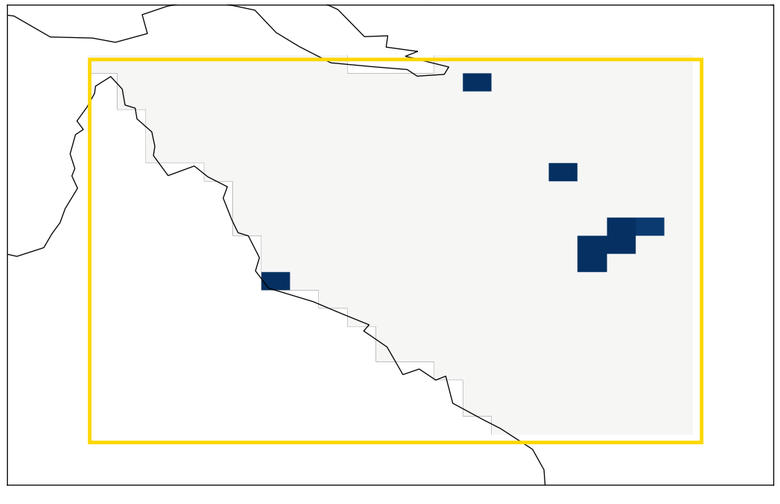
Actual DHW - Step 90

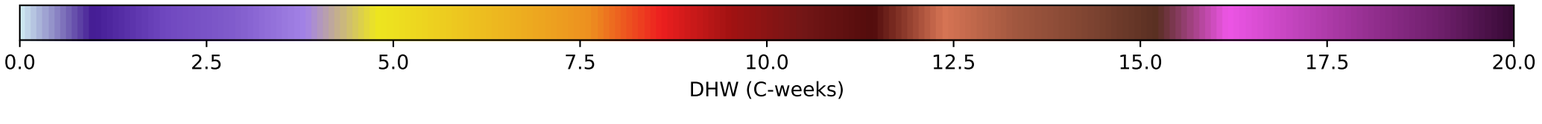


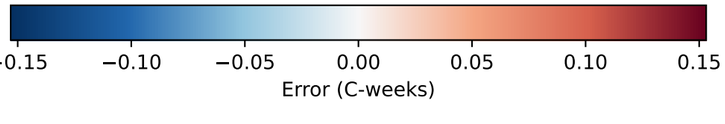
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.15]

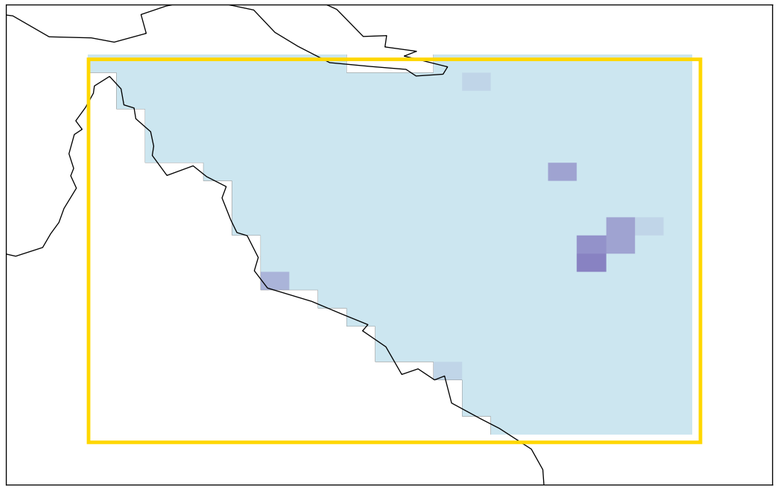




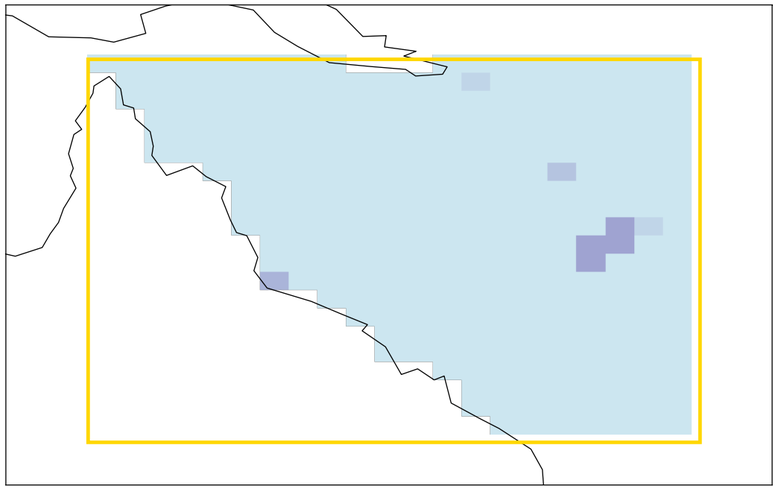


DHW Chain init: 2017-01-05 - Val - GBR (C-weeks)

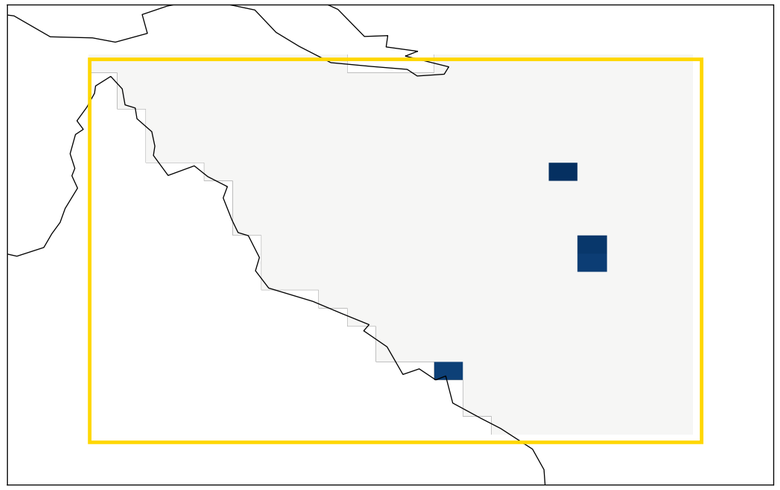
Actual DHW - Step 1



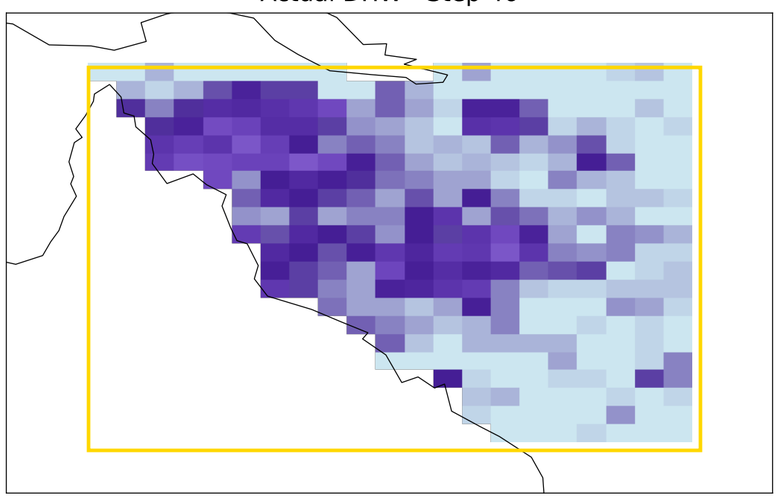
Predicted DHW - Step 1



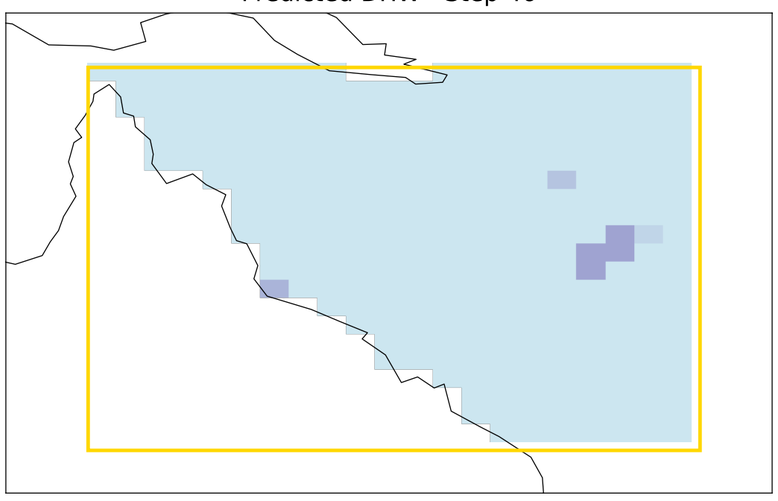
Error (Pred - Actual) - Step 1 [scale ±0.16]



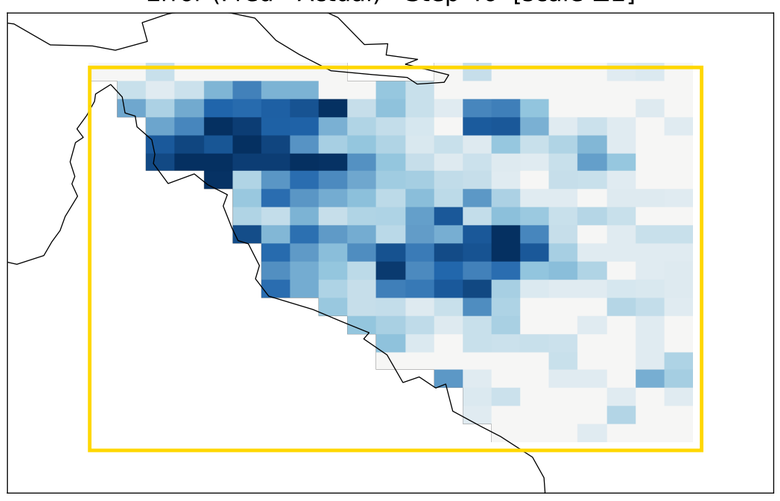
Actual DHW - Step 46



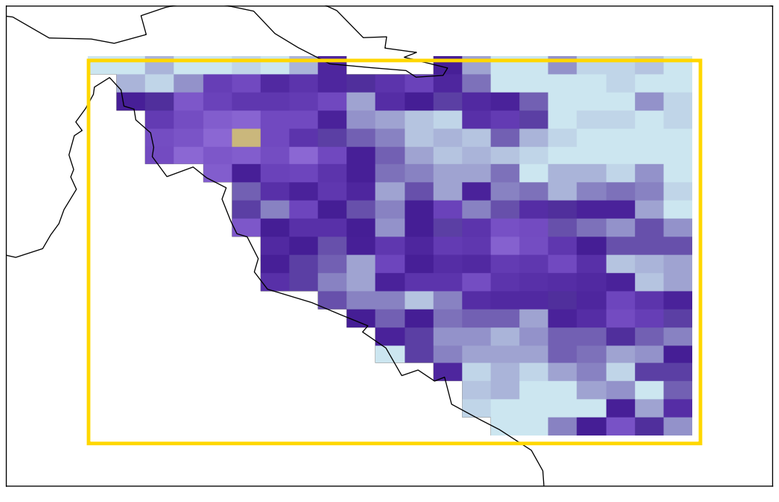
Predicted DHW - Step 46



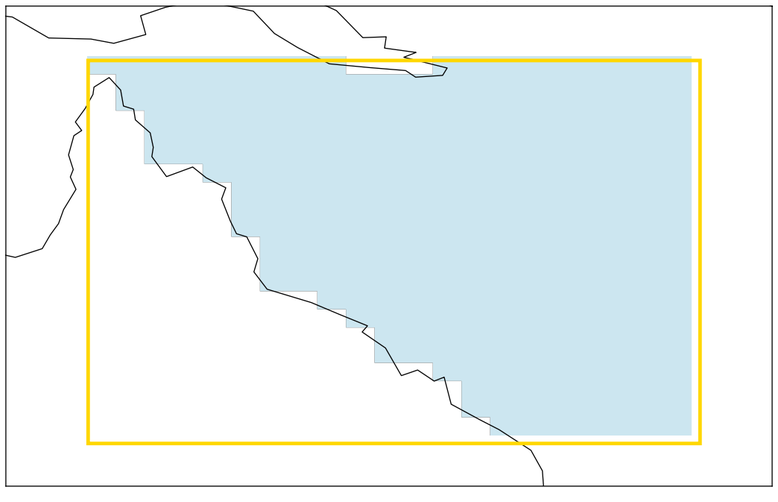
Error (Pred - Actual) - Step 46 [scale ±2]



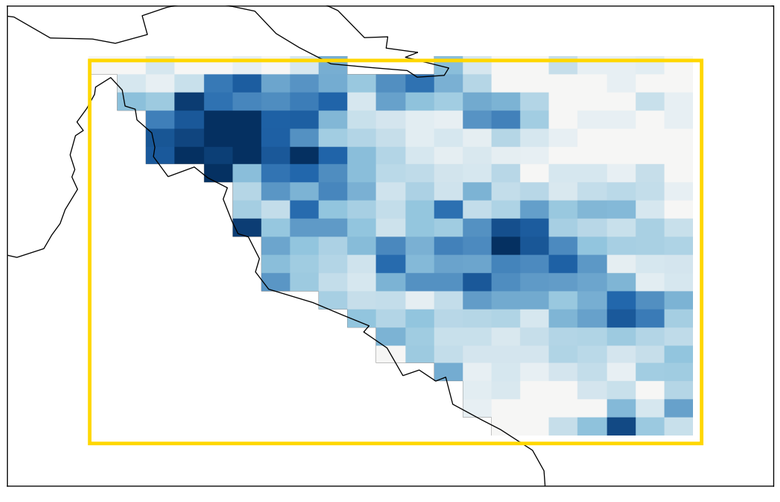
Actual DHW - Step 90

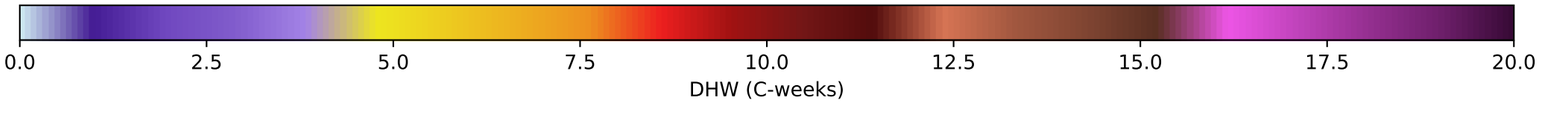


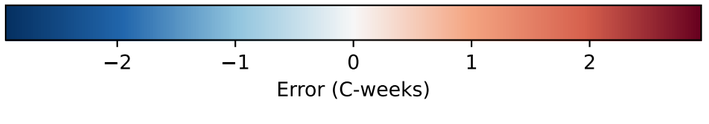
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±2.9]

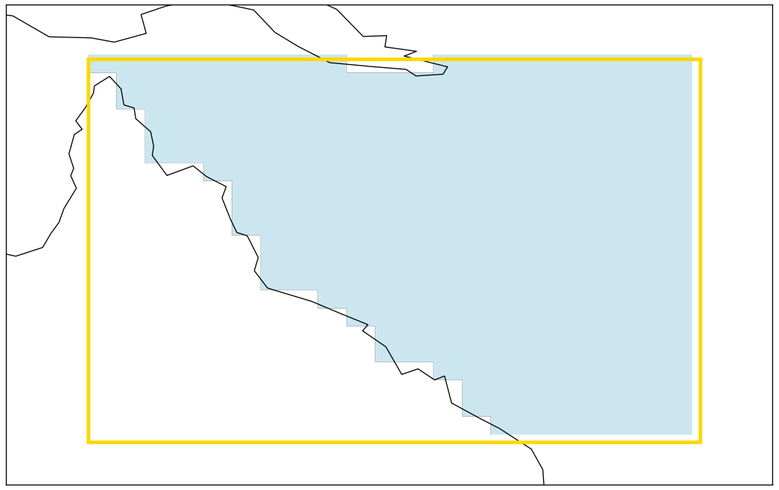




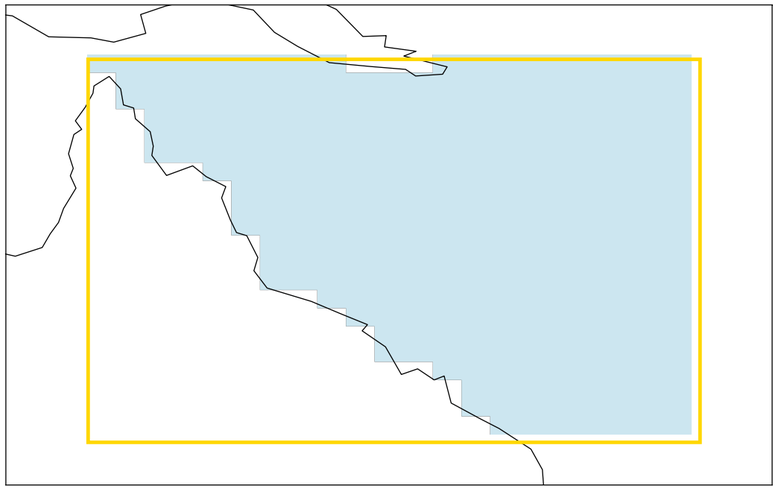


DHW Chain init: 2017-07-04 - Val - GBR (C-weeks)

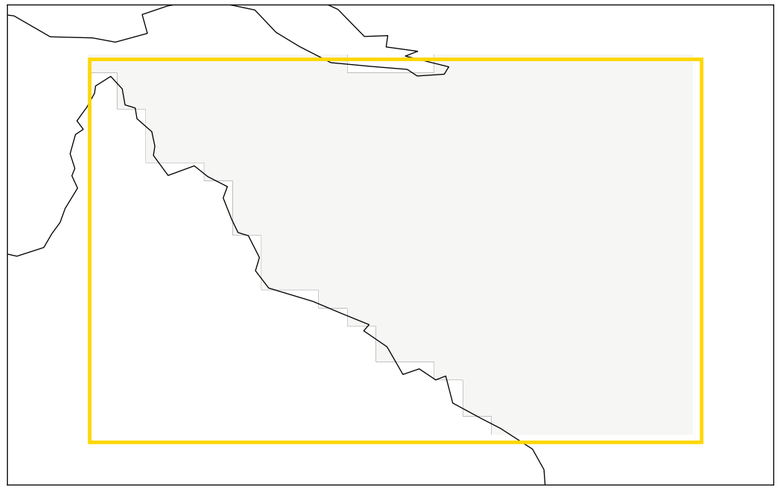
Actual DHW - Step 1



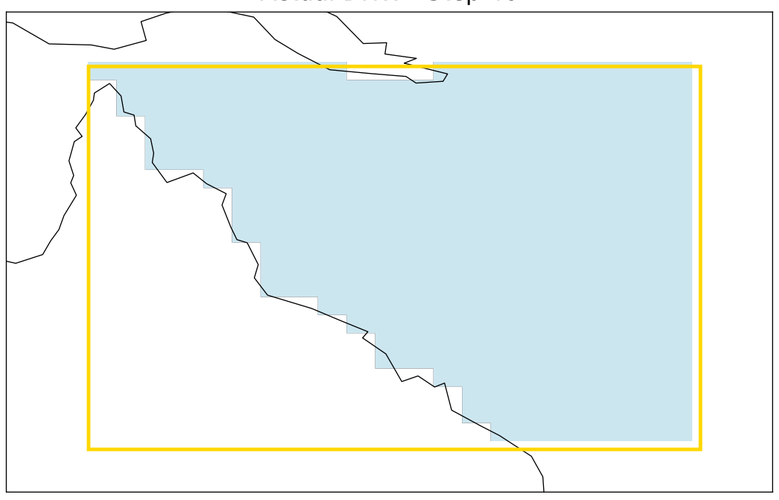
Predicted DHW - Step 1



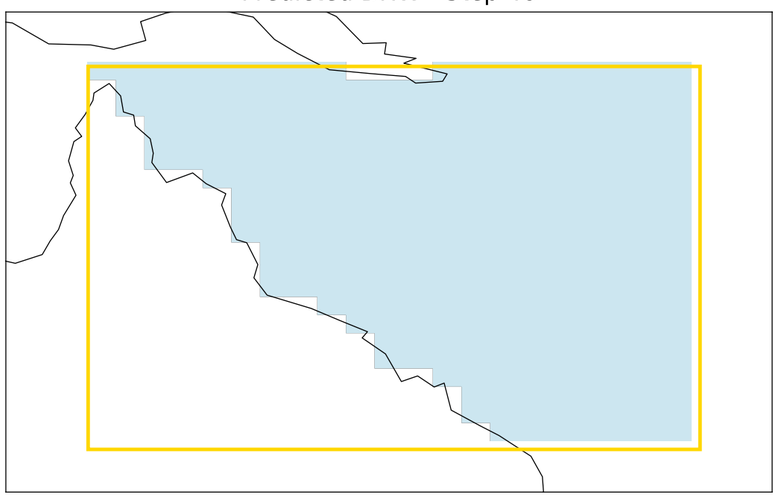
Error (Pred - Actual) - Step 1 [scale ±8]



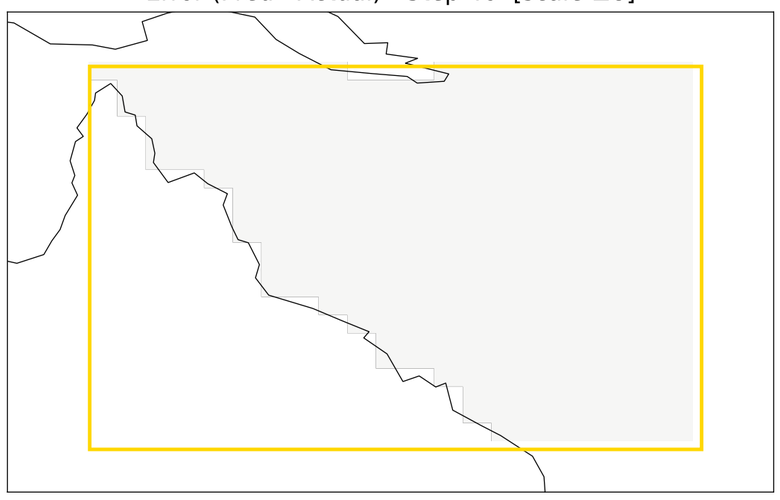
Actual DHW - Step 46



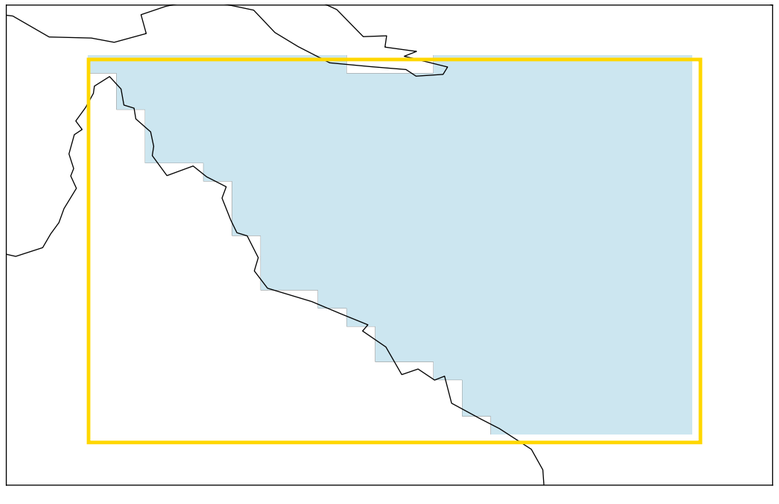
Predicted DHW - Step 46



Error (Pred - Actual) - Step 46 [scale ±8]



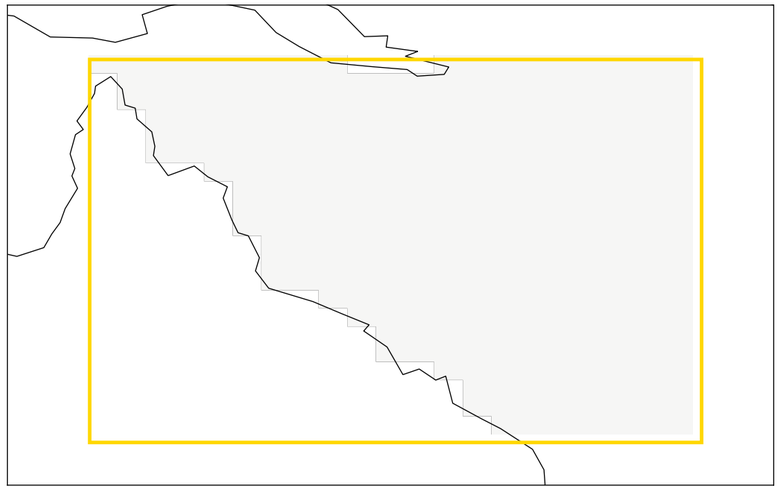
Actual DHW - Step 90

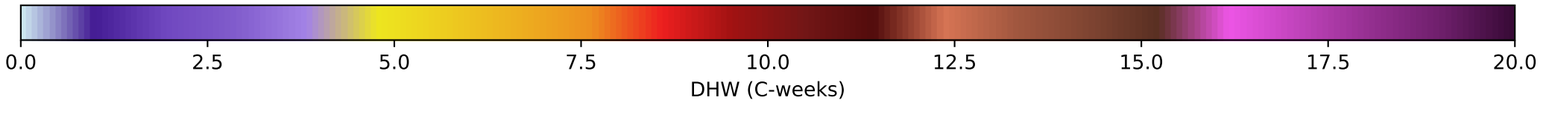


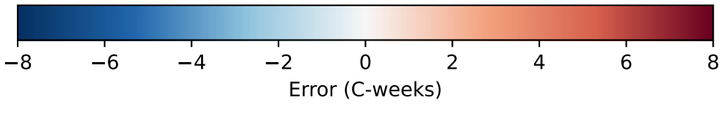
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

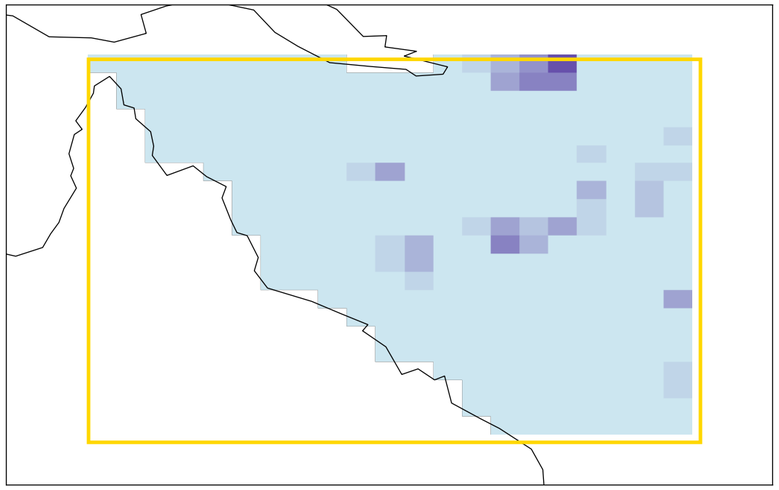




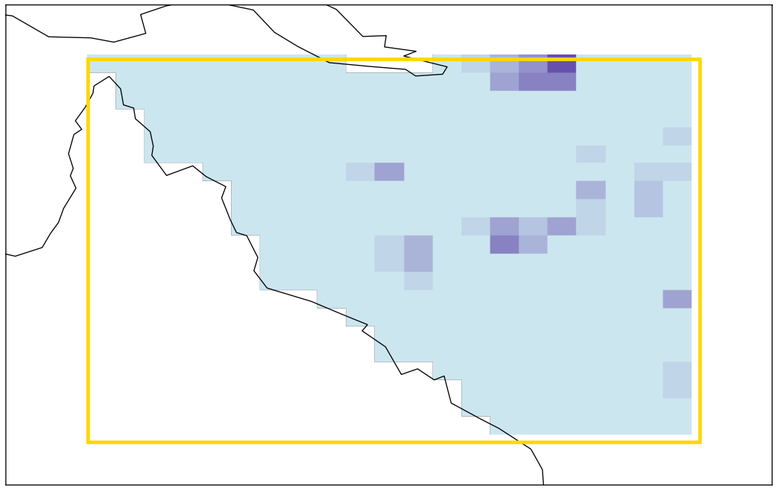


DHW Chain init: 2018-03-31 - Val - GBR (C-weeks)

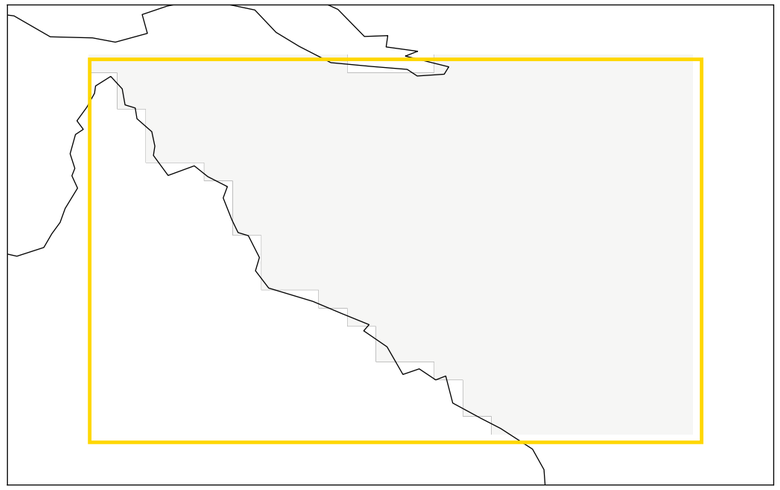
Actual DHW - Step 1



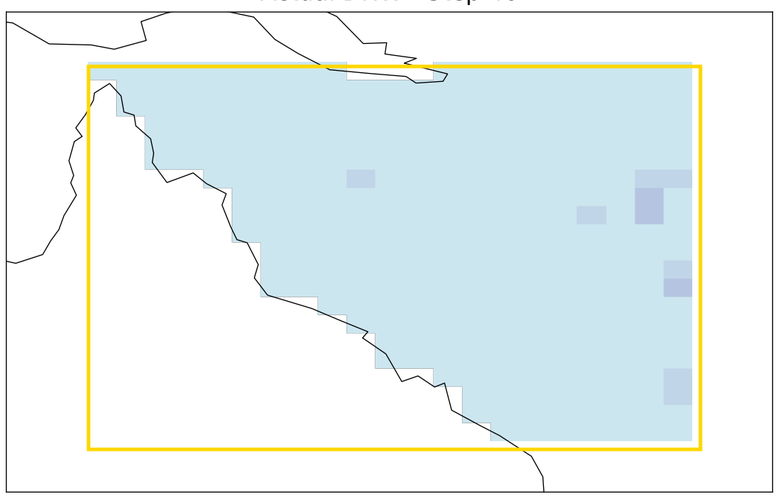
Predicted DHW - Step 1



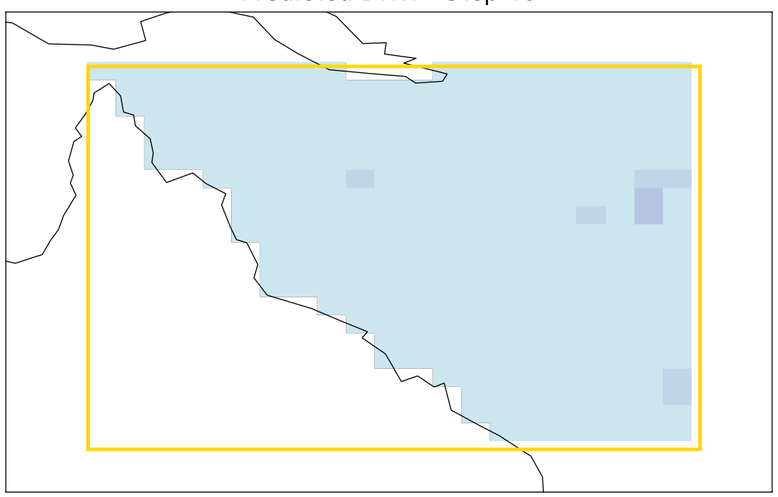
Error (Pred - Actual) - Step 1 [scale ±8]



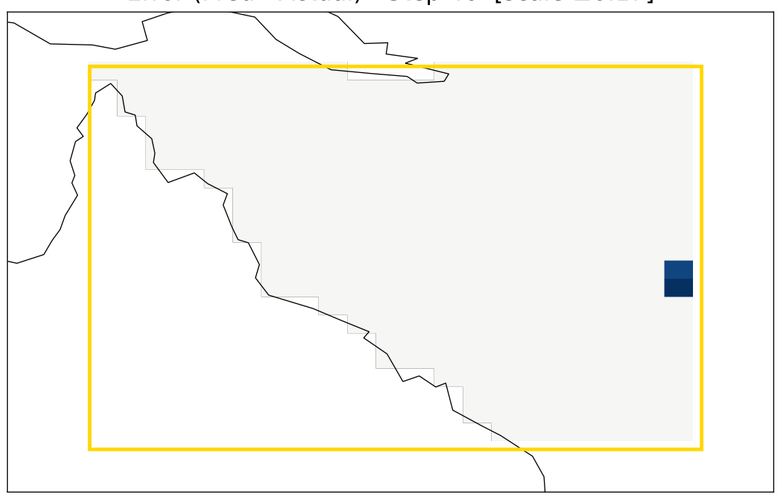
Actual DHW - Step 46



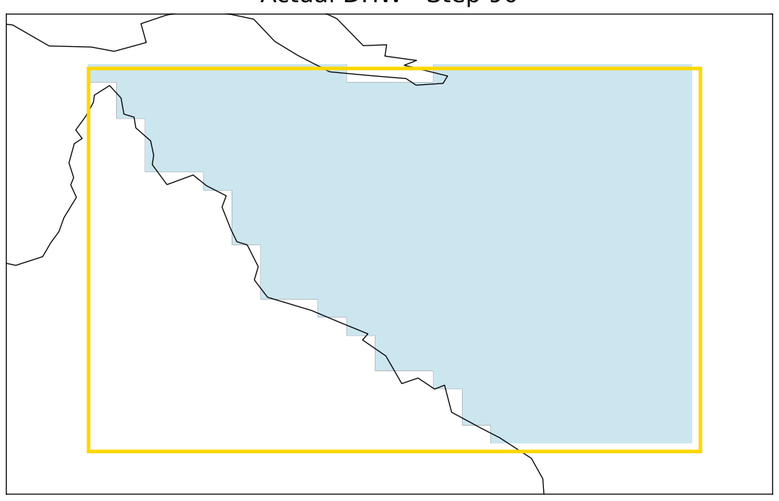
Predicted DHW - Step 46



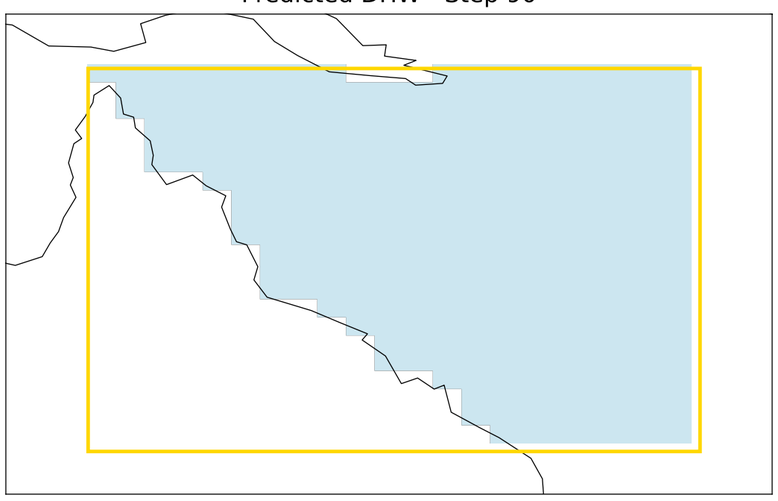
Error (Pred - Actual) - Step 46 [scale ±0.17]



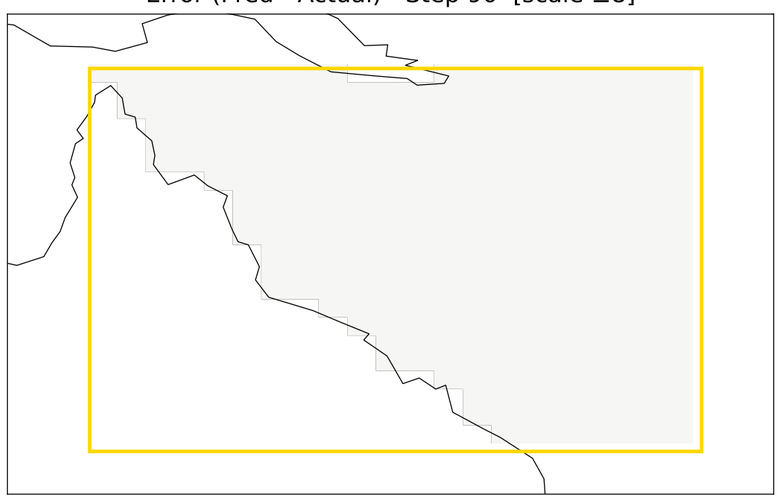
Actual DHW - Step 90

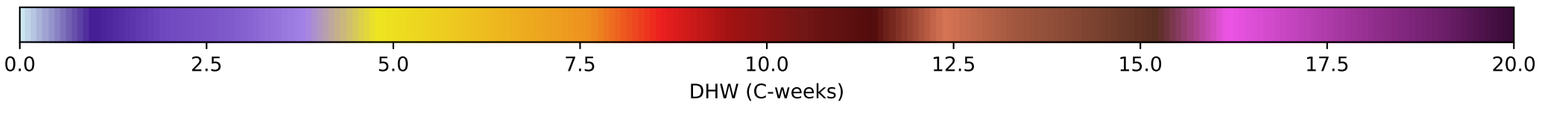


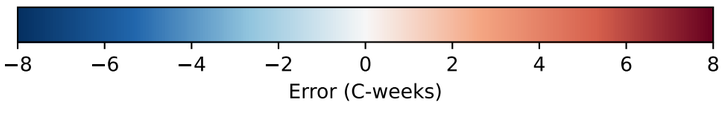
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

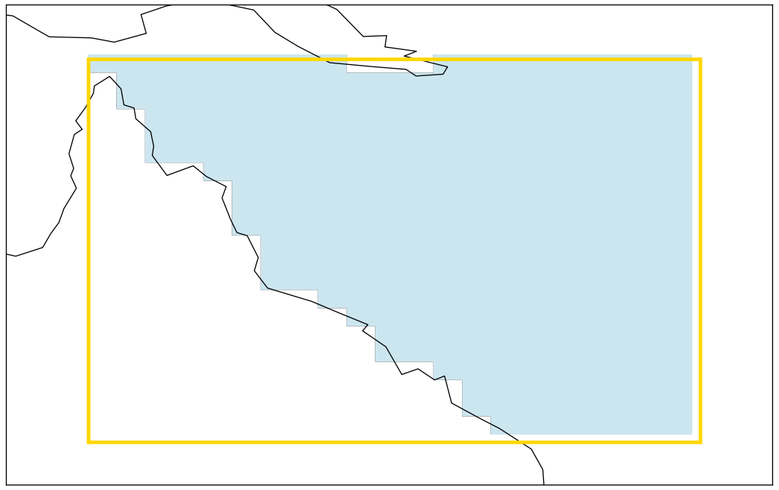




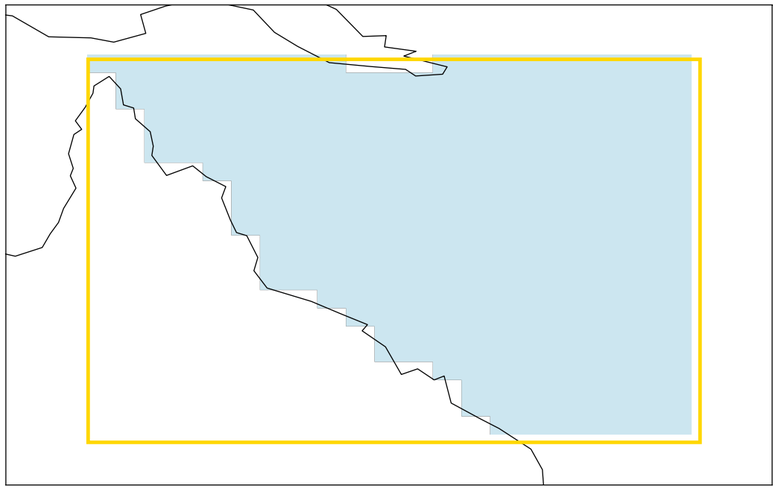


DHW Chain init: 2018-06-29 - Val - GBR (C-weeks)

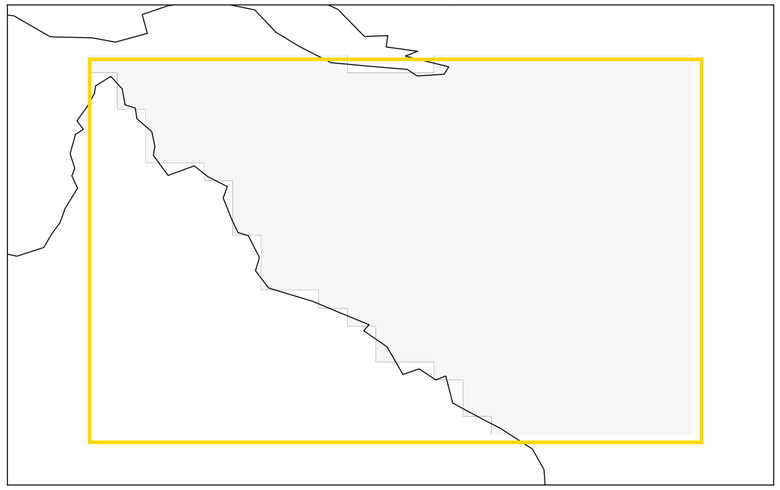
Actual DHW - Step 1



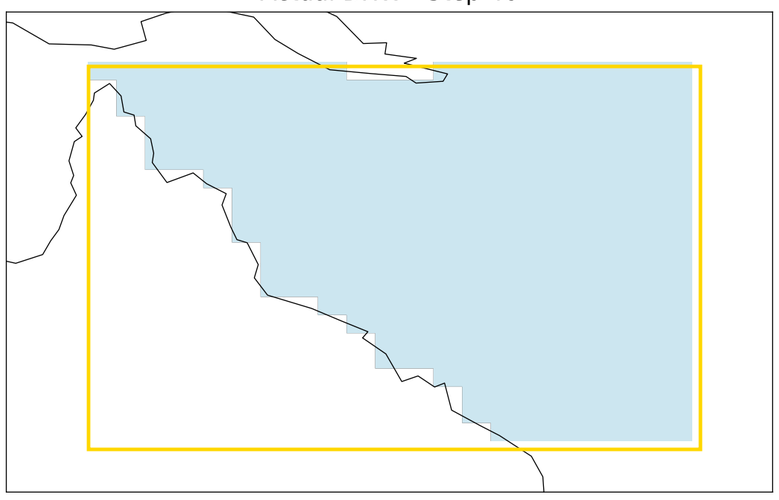
Predicted DHW - Step 1



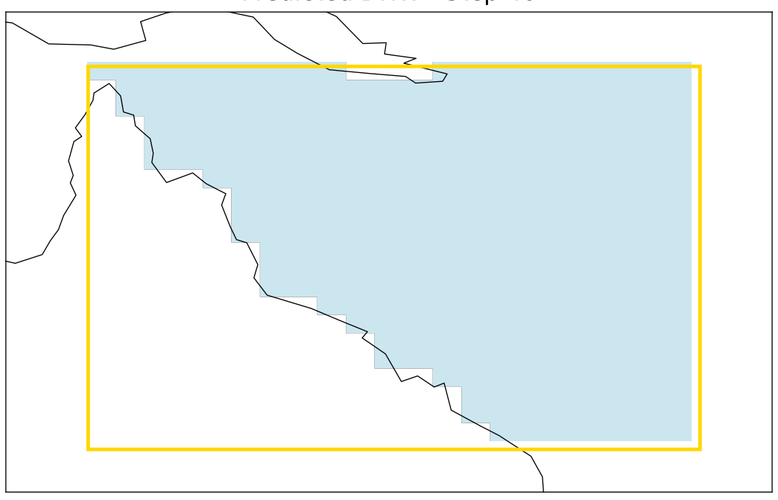
Error (Pred - Actual) - Step 1 [scale ±8]



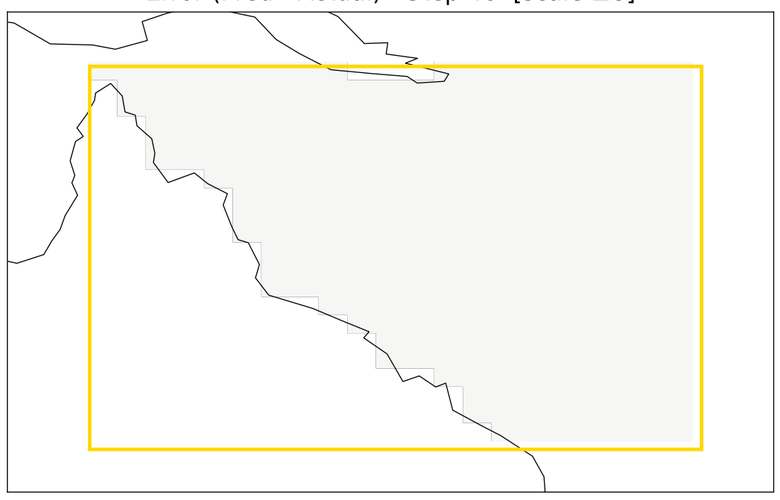
Actual DHW - Step 46



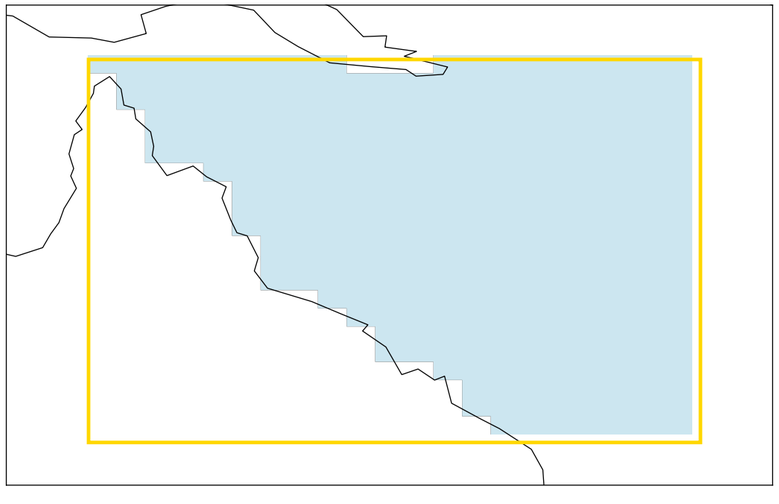
Predicted DHW - Step 46



Error (Pred - Actual) - Step 46 [scale ±8]



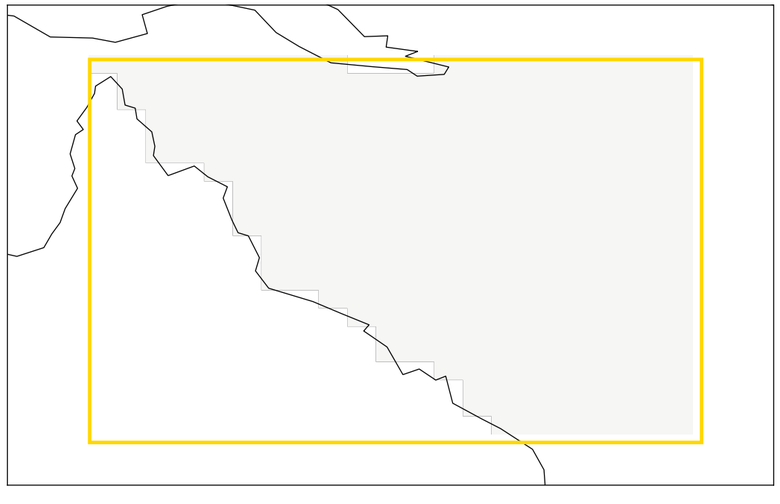
Actual DHW - Step 90

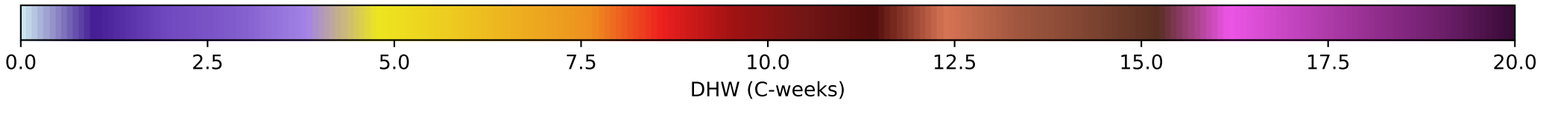


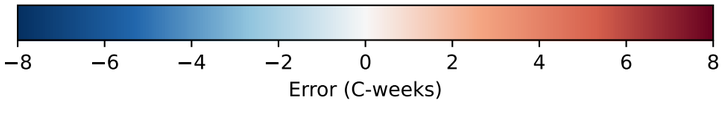
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

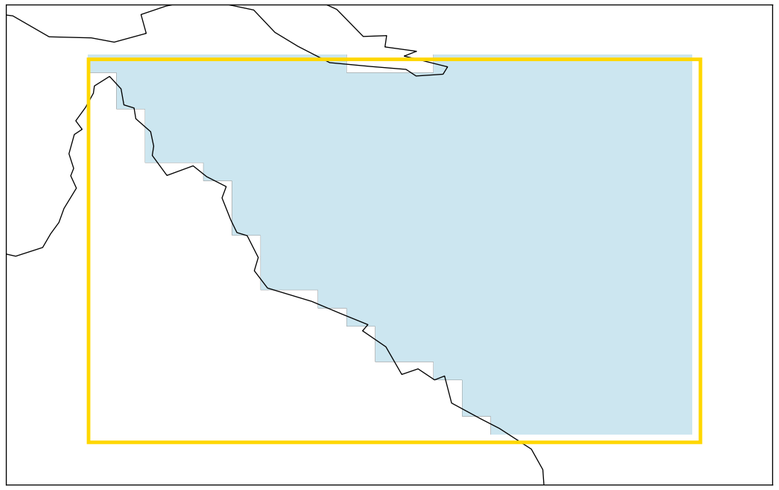




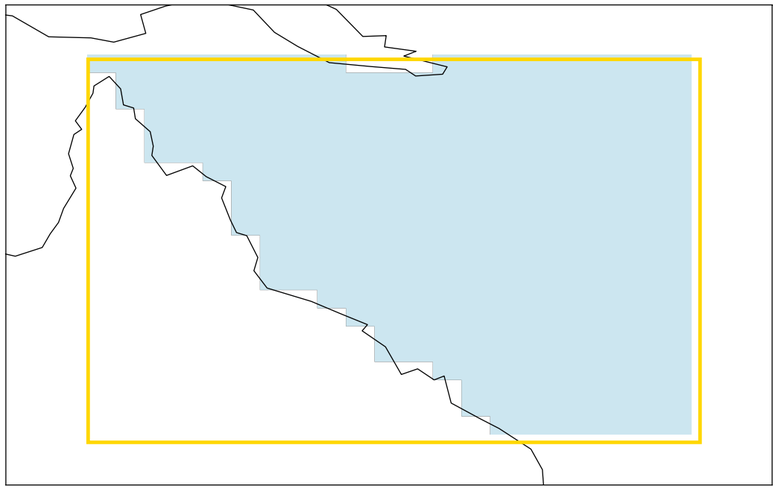


DHW Chain init: 2019-09-22 - Val - GBR (C-weeks)

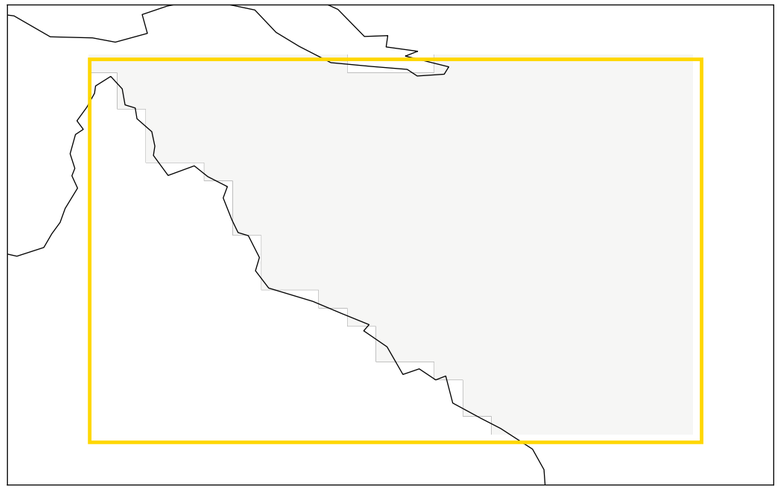
Actual DHW - Step 1



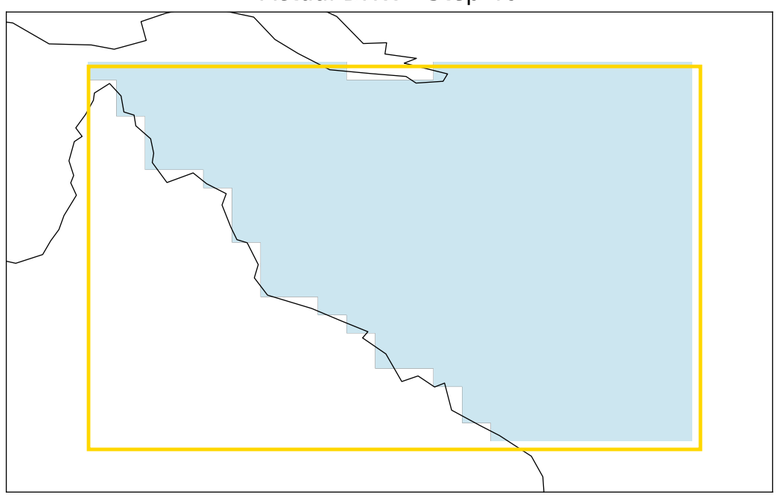
Predicted DHW - Step 1



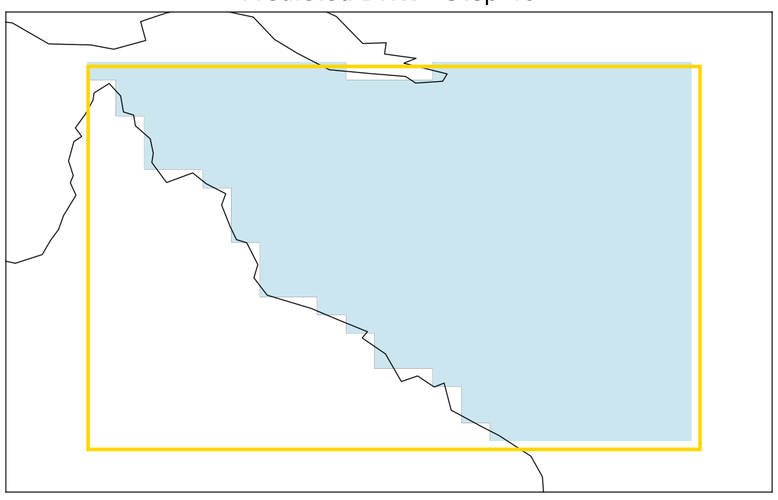
Error (Pred - Actual) - Step 1 [scale ±8]



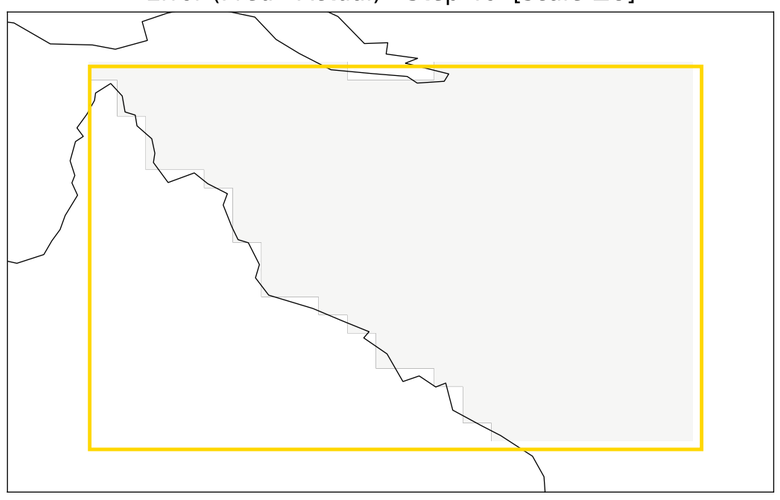
Actual DHW - Step 46



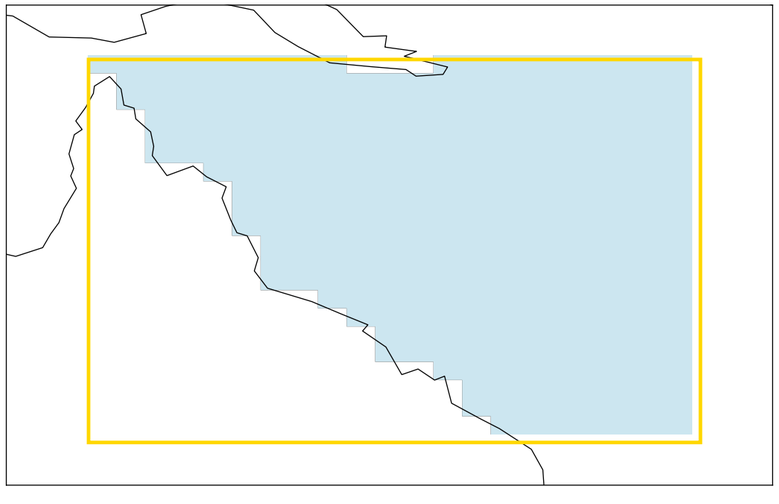
Predicted DHW - Step 46



Error (Pred - Actual) - Step 46 [scale ±8]



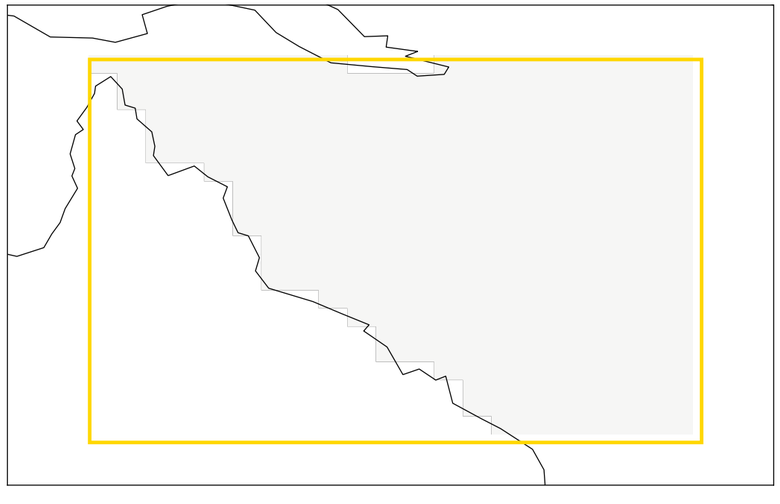
Actual DHW - Step 90

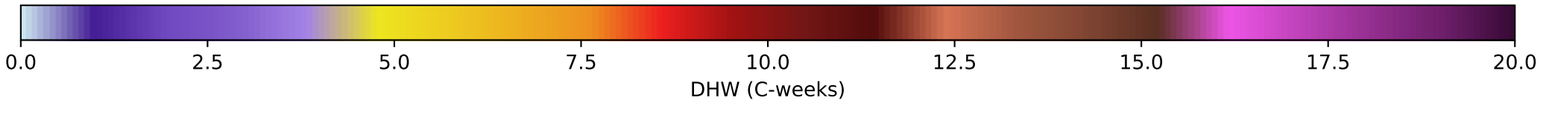


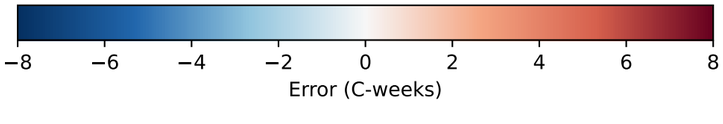
Predicted DHW - Step 90



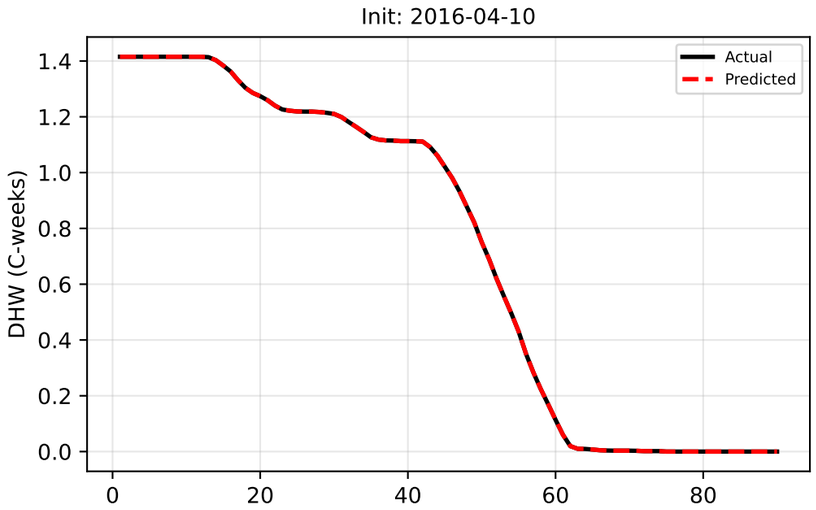
Error (Pred - Actual) - Step 90 [scale ±8]

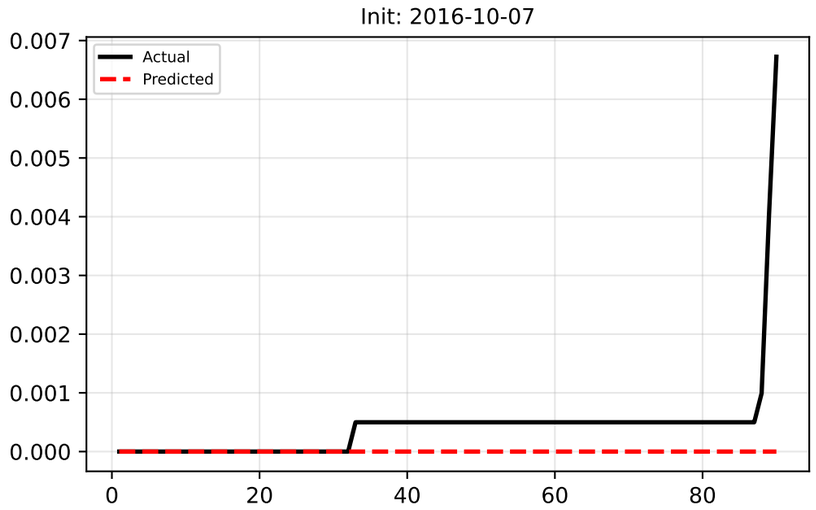


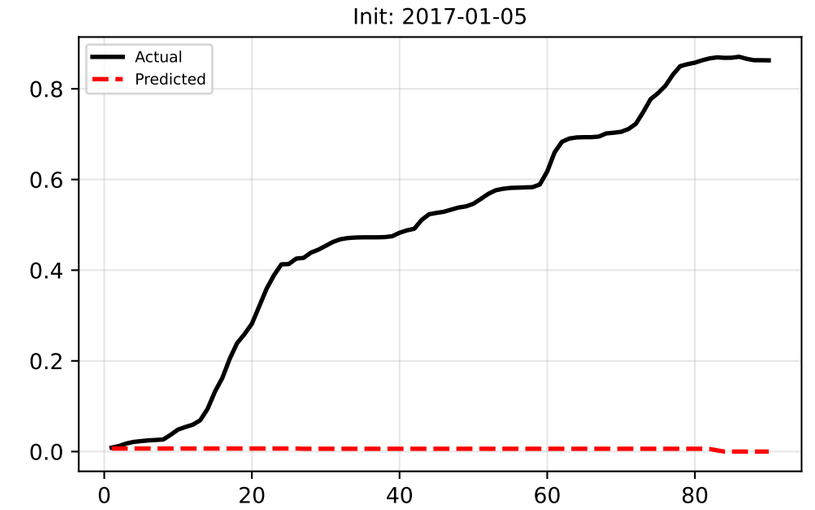


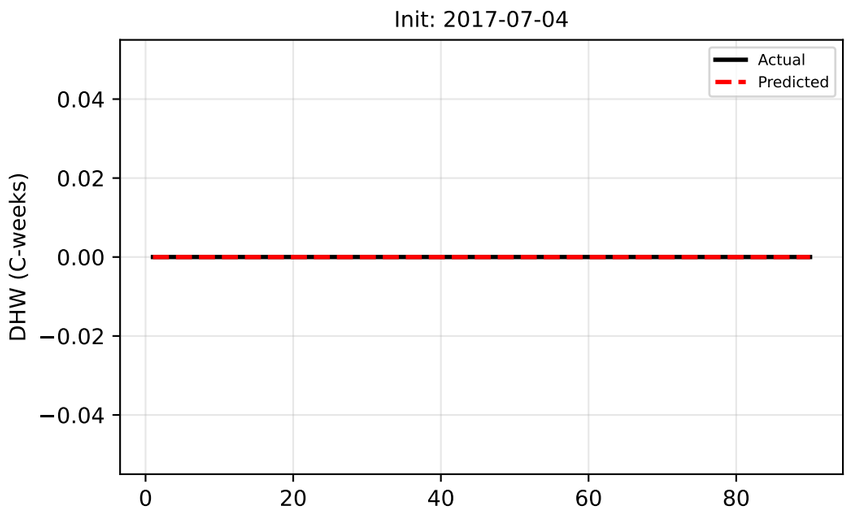


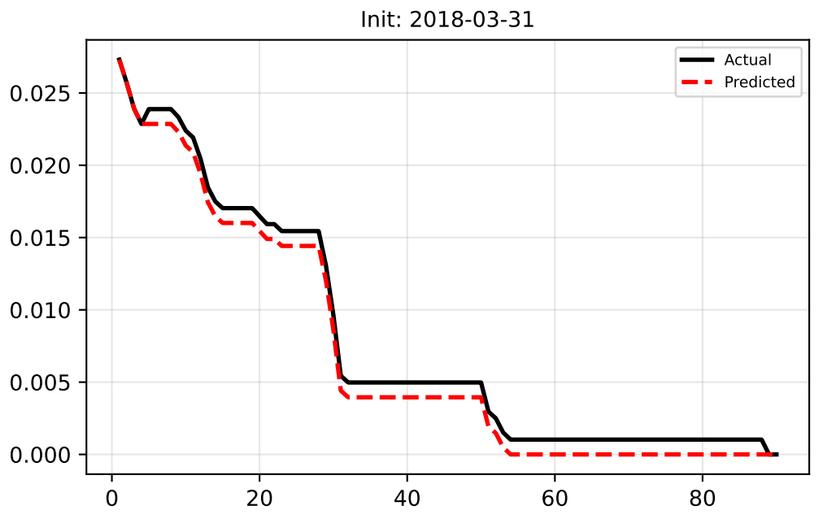
Global-Mean DHW - Val - GBR (C-weeks)

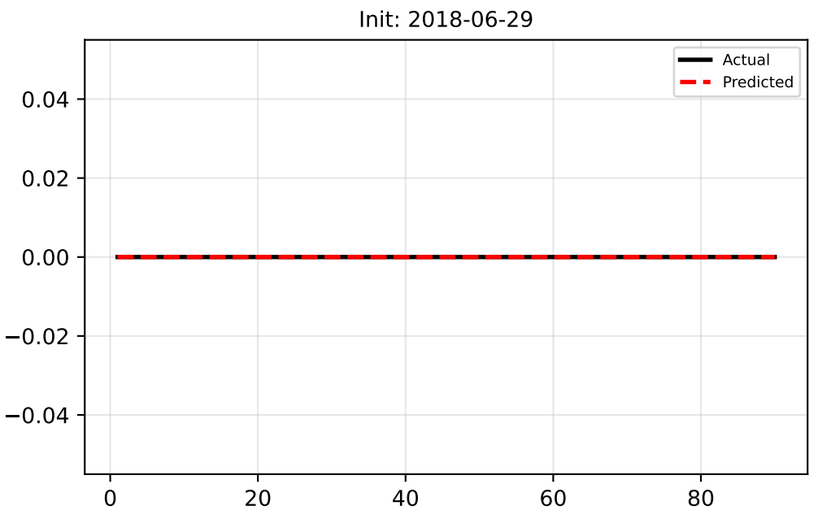


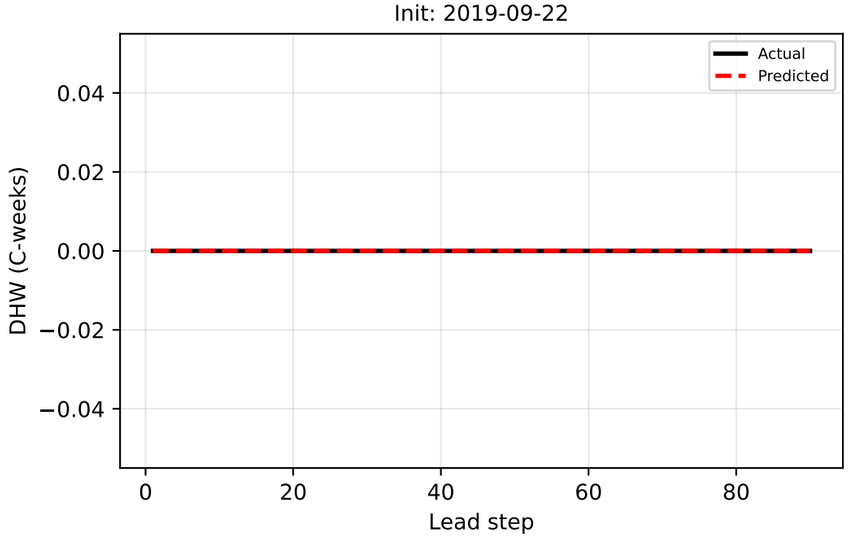


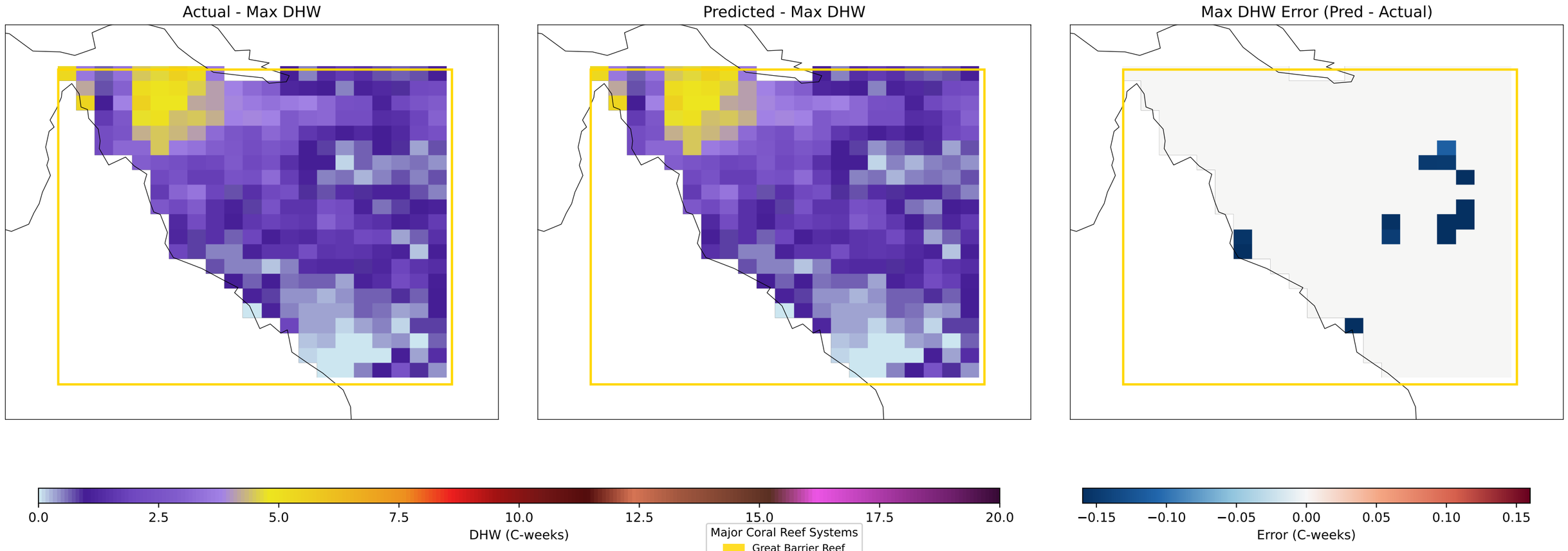


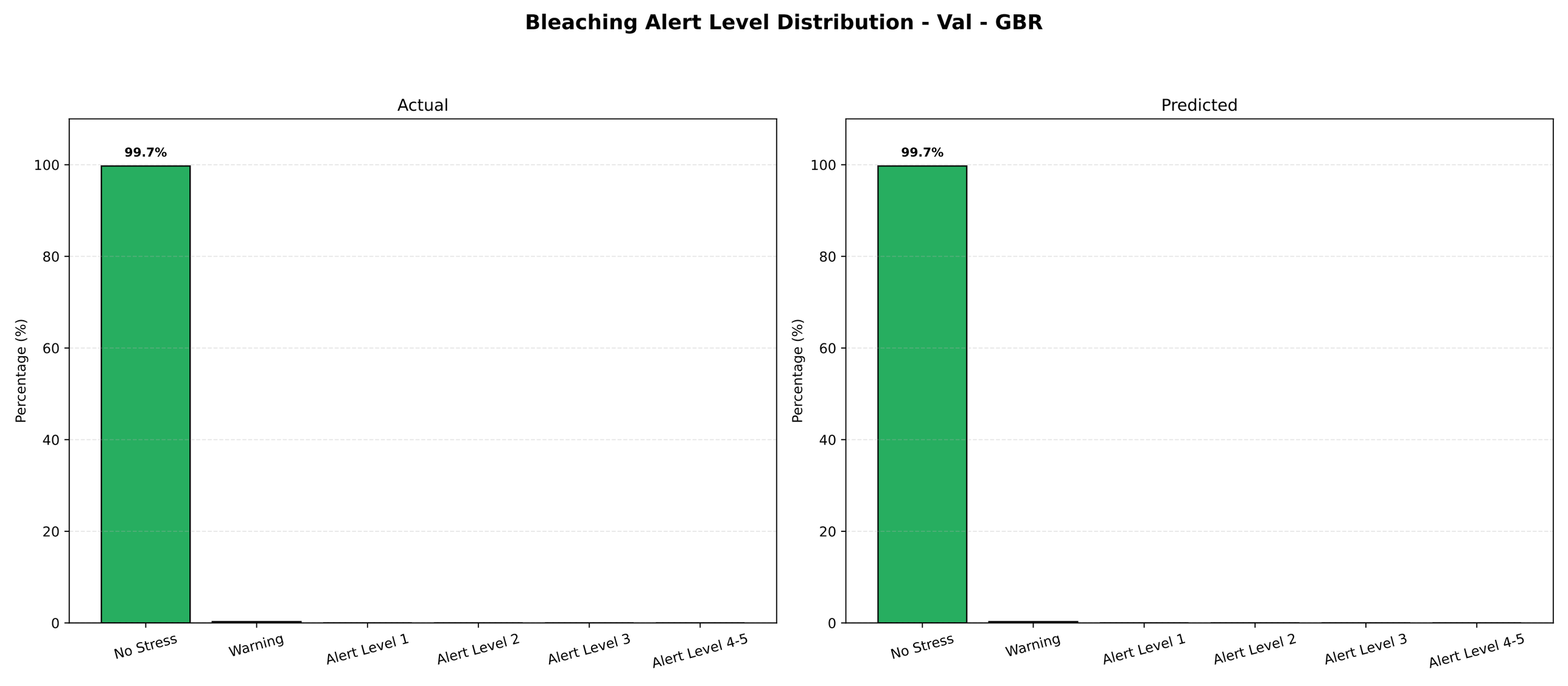






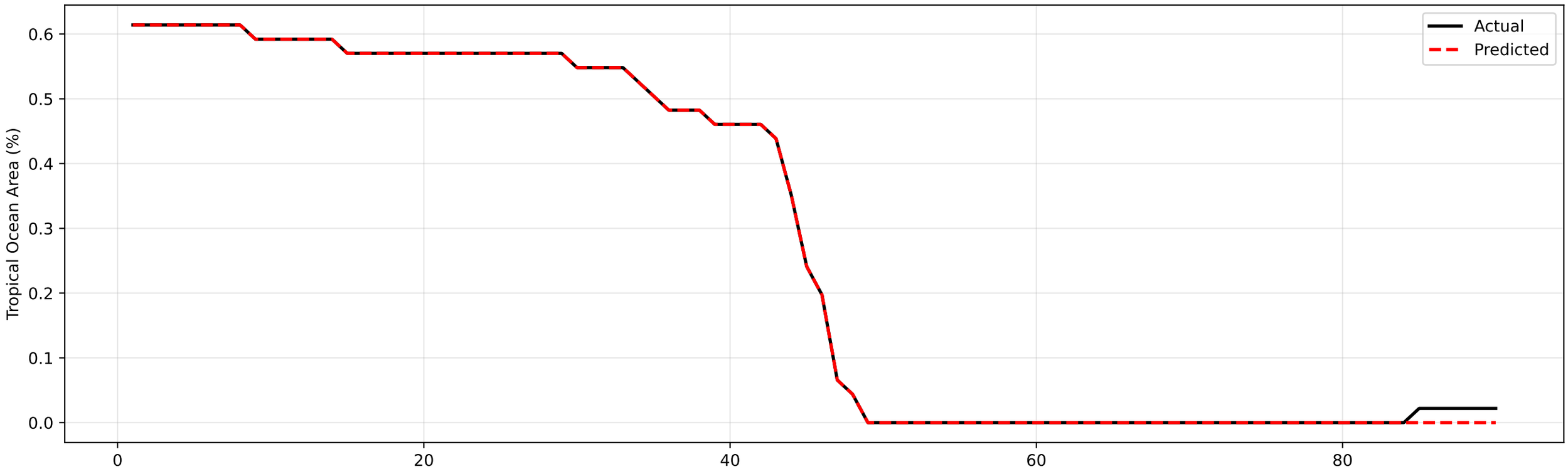



Peak DHW (max over all chains & leads) - Val - GBR (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
Error (C-weeks)


Bleaching Alert Level Distribution - Val - GBR
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
99.7%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Val - GBR

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)

